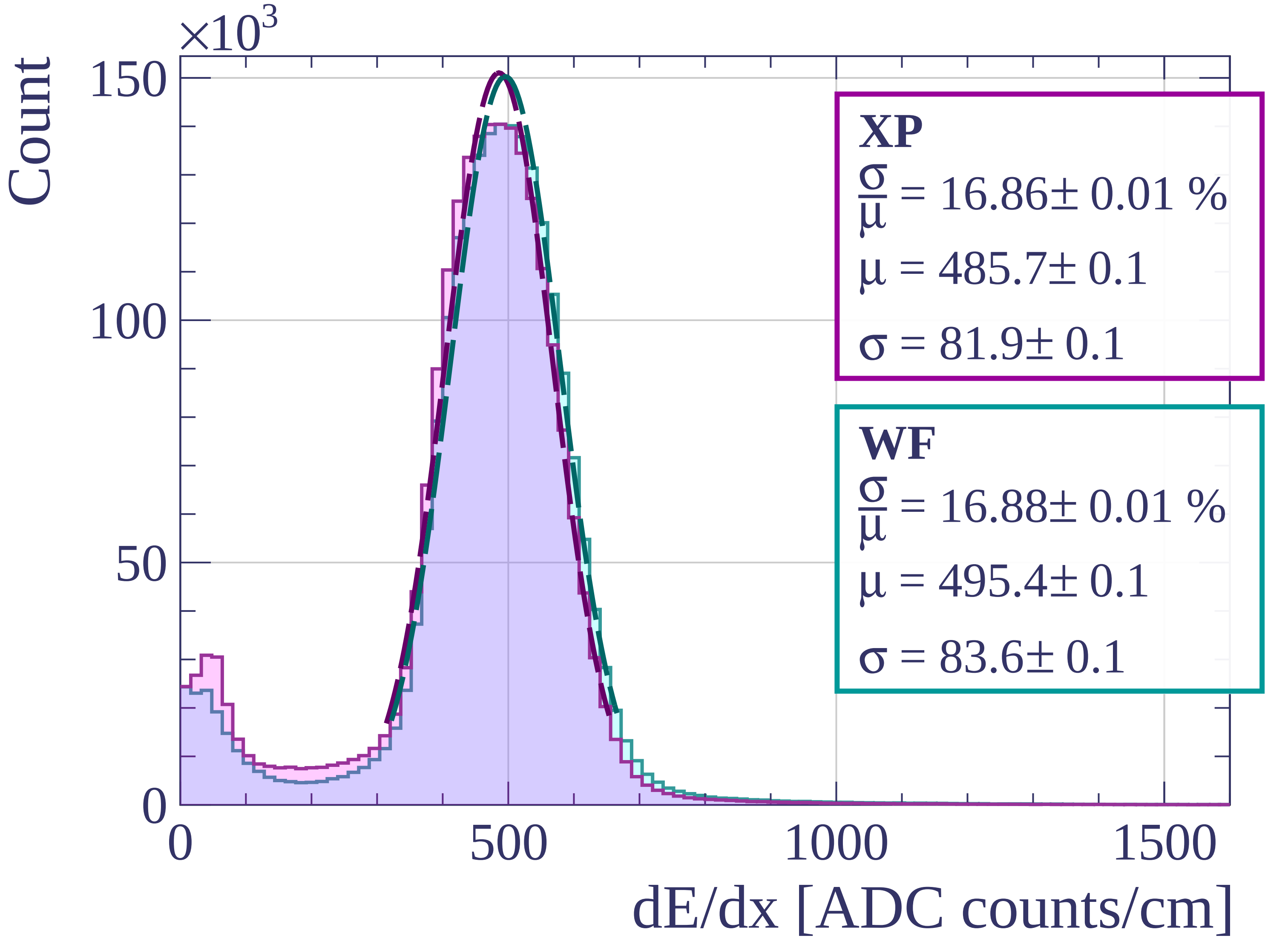
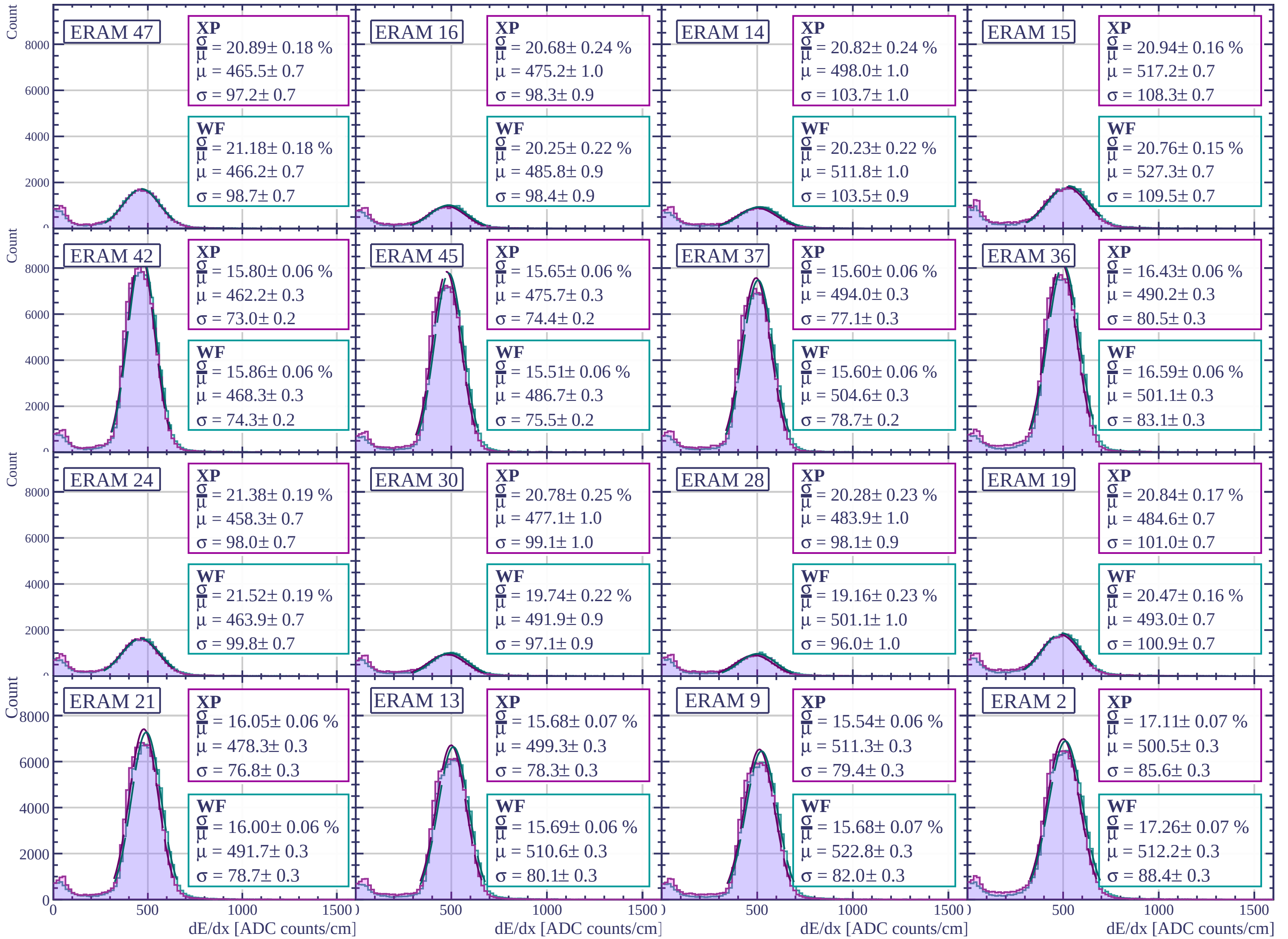
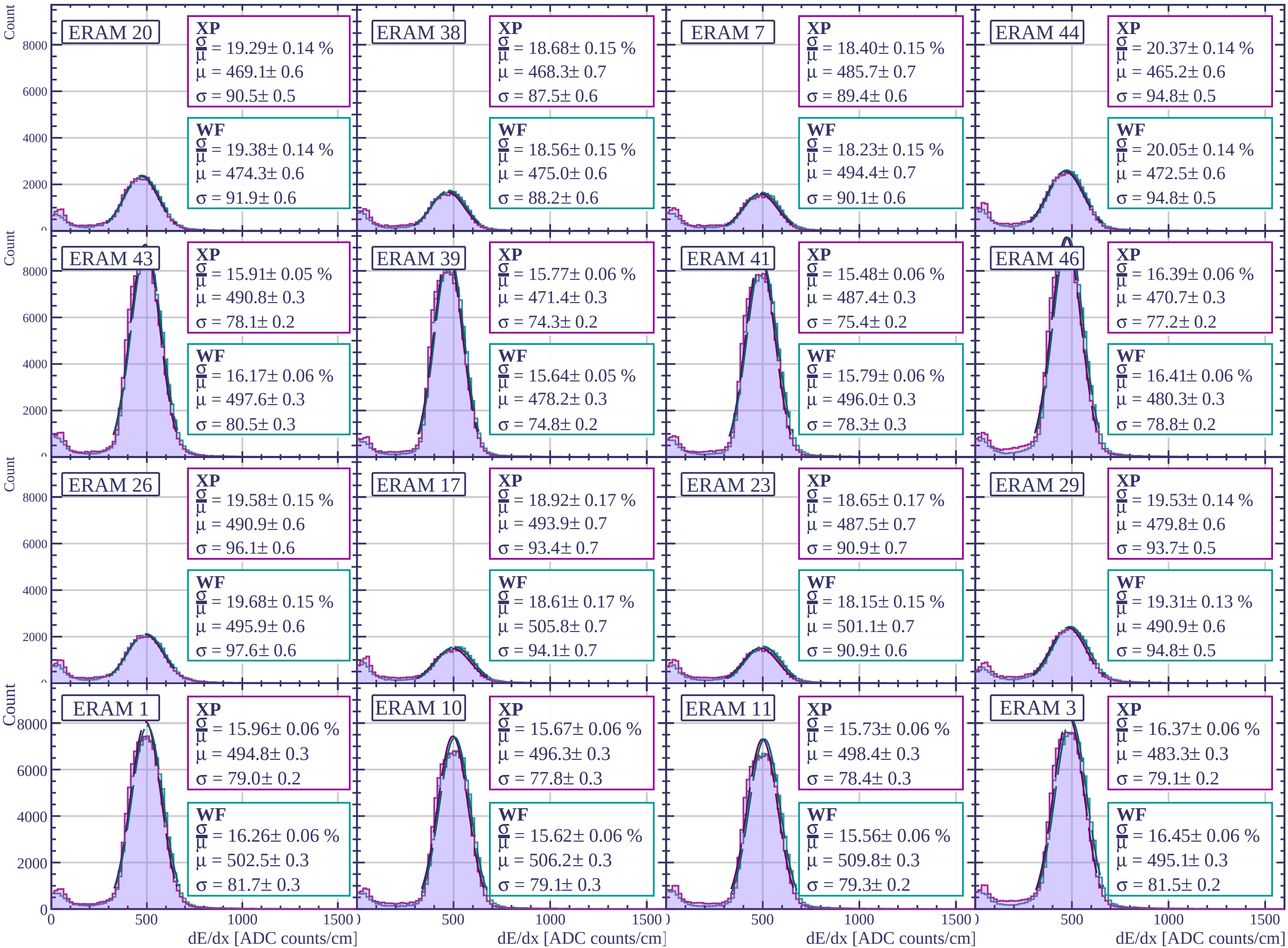


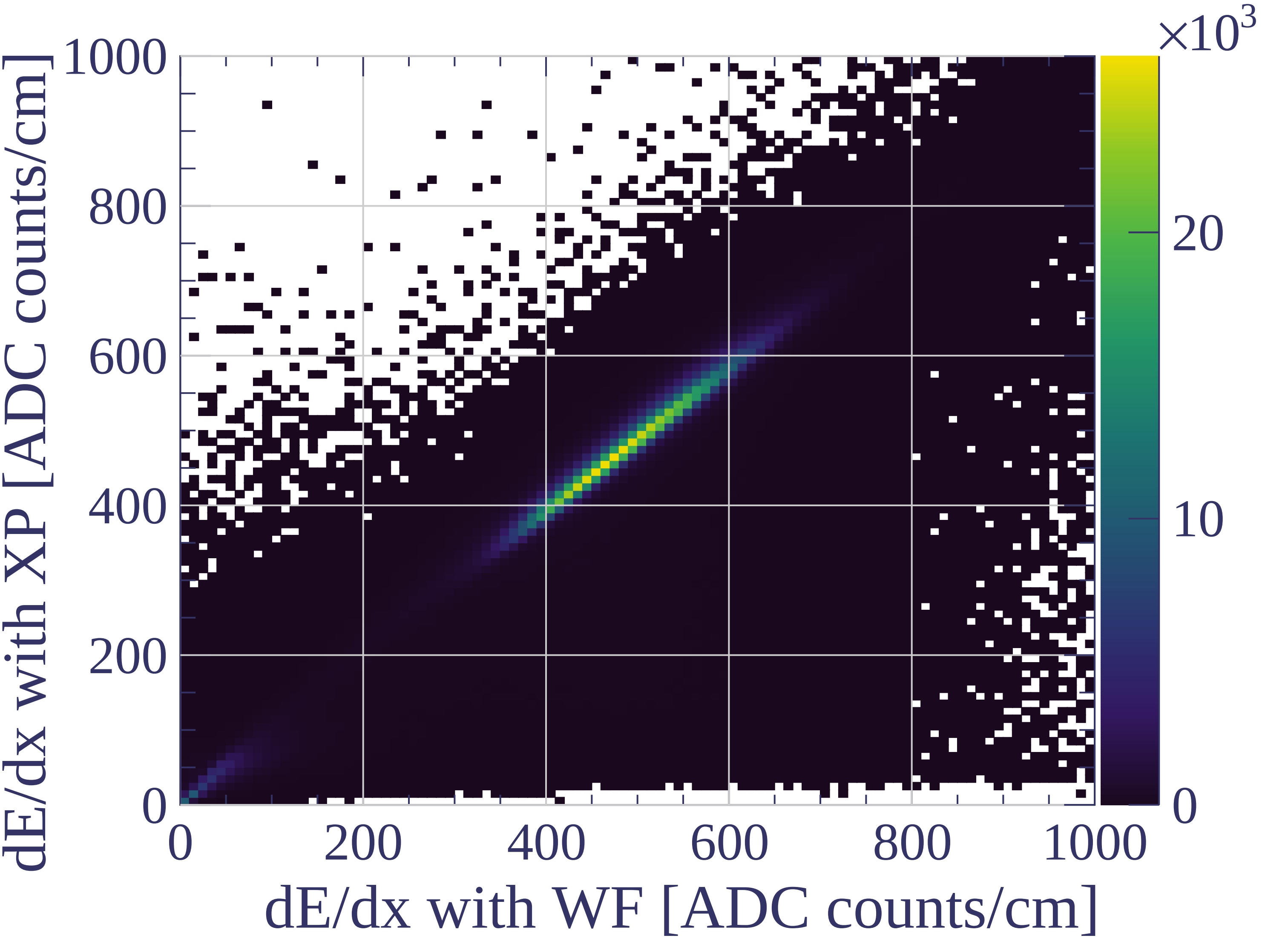
×10³
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 16.86± 0.01 %
μ = 485.7± 0.1
σ = 81.9± 0.1
WF
σ/μ = 16.88± 0.01 %
μ = 495.4± 0.1
σ = 83.6± 0.1



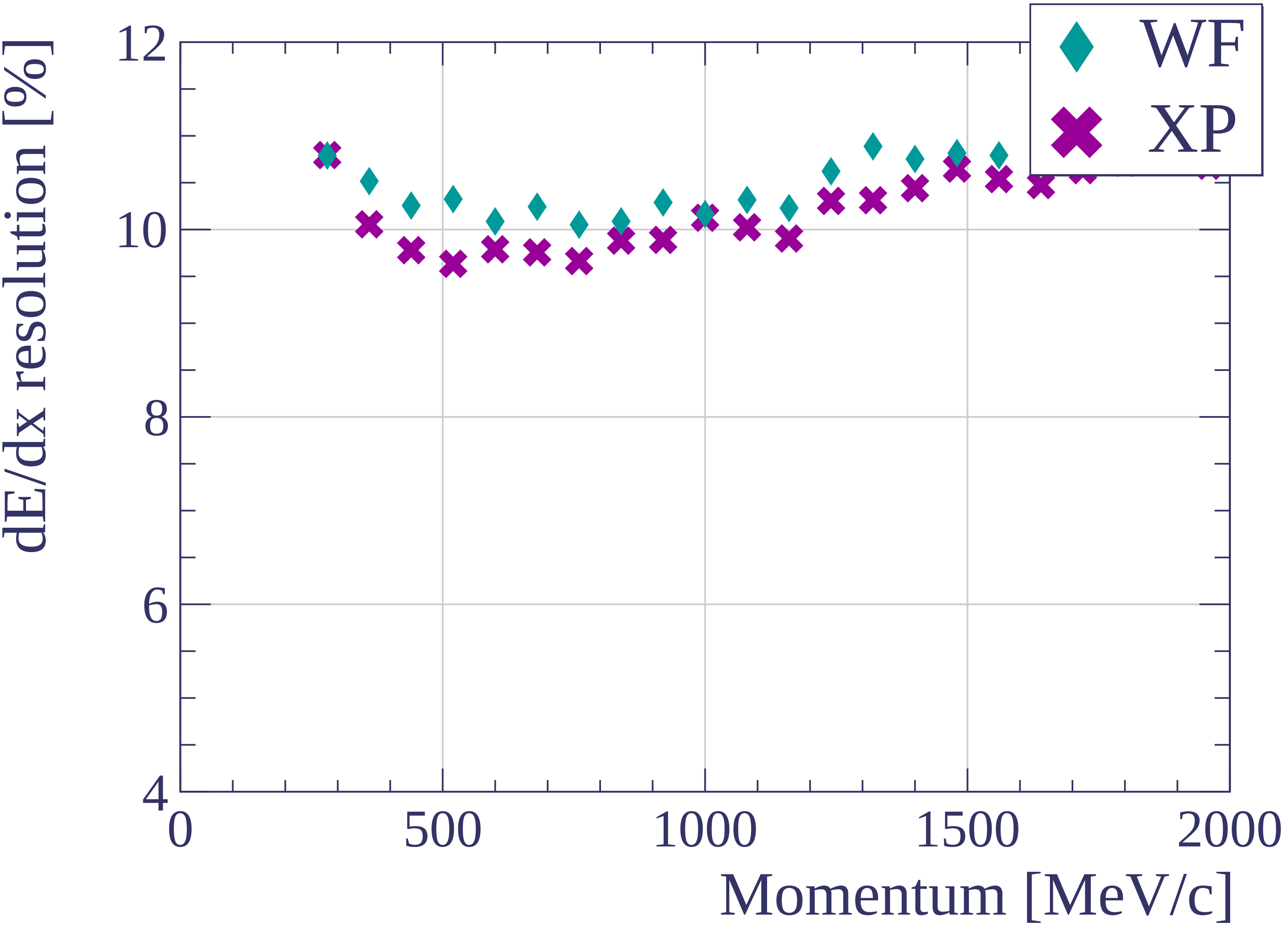
ERAM 47
XP
σ/μ = 20.89± 0.18 %
μ = 465.5± 0.7
σ = 97.2± 0.7
WF
σ/μ = 21.18± 0.18 %
μ = 466.2± 0.7
σ = 98.7± 0.7
ERAM 16
XP
σ/μ = 20.68± 0.24 %
μ = 475.2± 1.0
σ = 98.3± 0.9
WF
σ/μ = 20.25± 0.22 %
μ = 485.8± 0.9
σ = 98.4± 0.9
ERAM 14
XP
σ/μ = 20.82± 0.24 %
μ = 498.0± 1.0
σ = 103.7± 1.0
WF
σ/μ = 20.23± 0.22 %
μ = 511.8± 1.0
σ = 103.5± 0.9
ERAM 15
XP
σ/μ = 20.94± 0.16 %
μ = 517.2± 0.7
σ = 108.3± 0.7
WF
σ/μ = 20.76± 0.15 %
μ = 527.3± 0.7
σ = 109.5± 0.7
ERAM 42
XP
σ/μ = 15.80± 0.06 %
μ = 462.2± 0.3
σ = 73.0± 0.2
WF
σ/μ = 15.86± 0.06 %
μ = 468.3± 0.3
σ = 74.3± 0.2
ERAM 45
XP
σ/μ = 15.65± 0.06 %
μ = 475.7± 0.3
σ = 74.4± 0.2
WF
σ/μ = 15.51± 0.06 %
μ = 486.7± 0.3
σ = 75.5± 0.2
ERAM 37
XP
σ/μ = 15.60± 0.06 %
μ = 494.0± 0.3
σ = 77.1± 0.3
WF
σ/μ = 15.60± 0.06 %
μ = 504.6± 0.3
σ = 78.7± 0.2
ERAM 36
XP
σ/μ = 16.43± 0.06 %
μ = 490.2± 0.3
σ = 80.5± 0.3
WF
σ/μ = 16.59± 0.06 %
μ = 501.1± 0.3
σ = 83.1± 0.3
ERAM 24
XP
σ/μ = 21.38± 0.19 %
μ = 458.3± 0.7
σ = 98.0± 0.7
WF
σ/μ = 21.52± 0.19 %
μ = 463.9± 0.7
σ = 99.8± 0.7
ERAM 30
XP
σ/μ = 20.78± 0.25 %
μ = 477.1± 1.0
σ = 99.1± 1.0
WF
σ/μ = 19.74± 0.22 %
μ = 491.9± 0.9
σ = 97.1± 0.9
ERAM 28
XP
σ/μ = 20.28± 0.23 %
μ = 483.9± 1.0
σ = 98.1± 0.9
WF
σ/μ = 19.16± 0.23 %
μ = 501.1± 1.0
σ = 96.0± 1.0
ERAM 19
XP
σ/μ = 20.84± 0.17 %
μ = 484.6± 0.7
σ = 101.0± 0.7
WF
σ/μ = 20.47± 0.16 %
μ = 493.0± 0.7
σ = 100.9± 0.7
ERAM 21
XP
σ/μ = 16.05± 0.06 %
μ = 478.3± 0.3
σ = 76.8± 0.3
WF
σ/μ = 16.00± 0.06 %
μ = 491.7± 0.3
σ = 78.7± 0.3
ERAM 13
XP
σ/μ = 15.68± 0.07 %
μ = 499.3± 0.3
σ = 78.3± 0.3
WF
σ/μ = 15.69± 0.06 %
μ = 510.6± 0.3
σ = 80.1± 0.3
ERAM 9
XP
σ/μ = 15.54± 0.06 %
μ = 511.3± 0.3
σ = 79.4± 0.3
WF
σ/μ = 15.68± 0.07 %
μ = 522.8± 0.3
σ = 82.0± 0.3
ERAM 2
XP
σ/μ = 17.11± 0.07 %
μ = 500.5± 0.3
σ = 85.6± 0.3
WF
σ/μ = 17.26± 0.07 %
μ = 512.2± 0.3
σ = 88.4± 0.3
Count
dE/dx [ADC counts/cm]

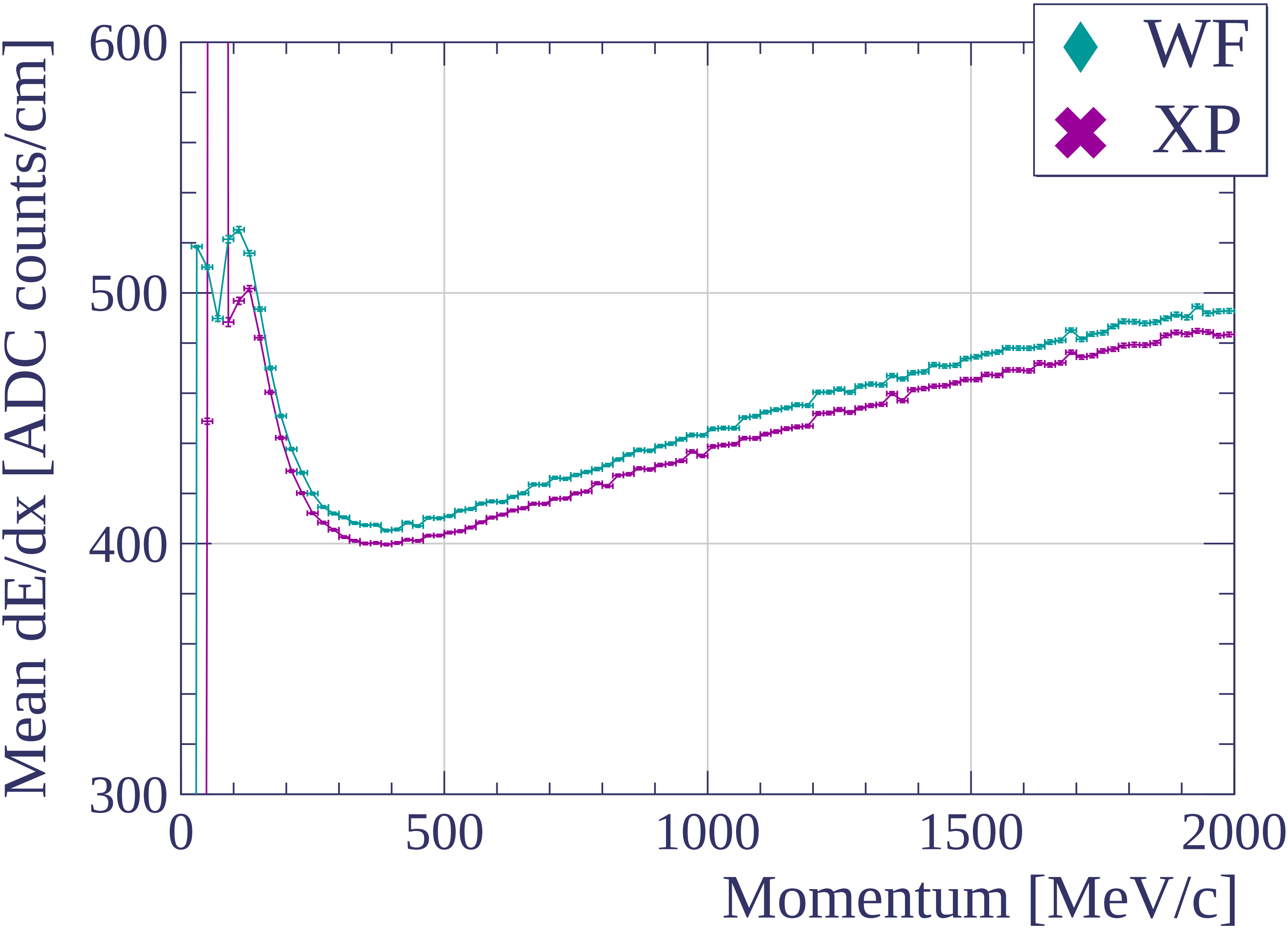


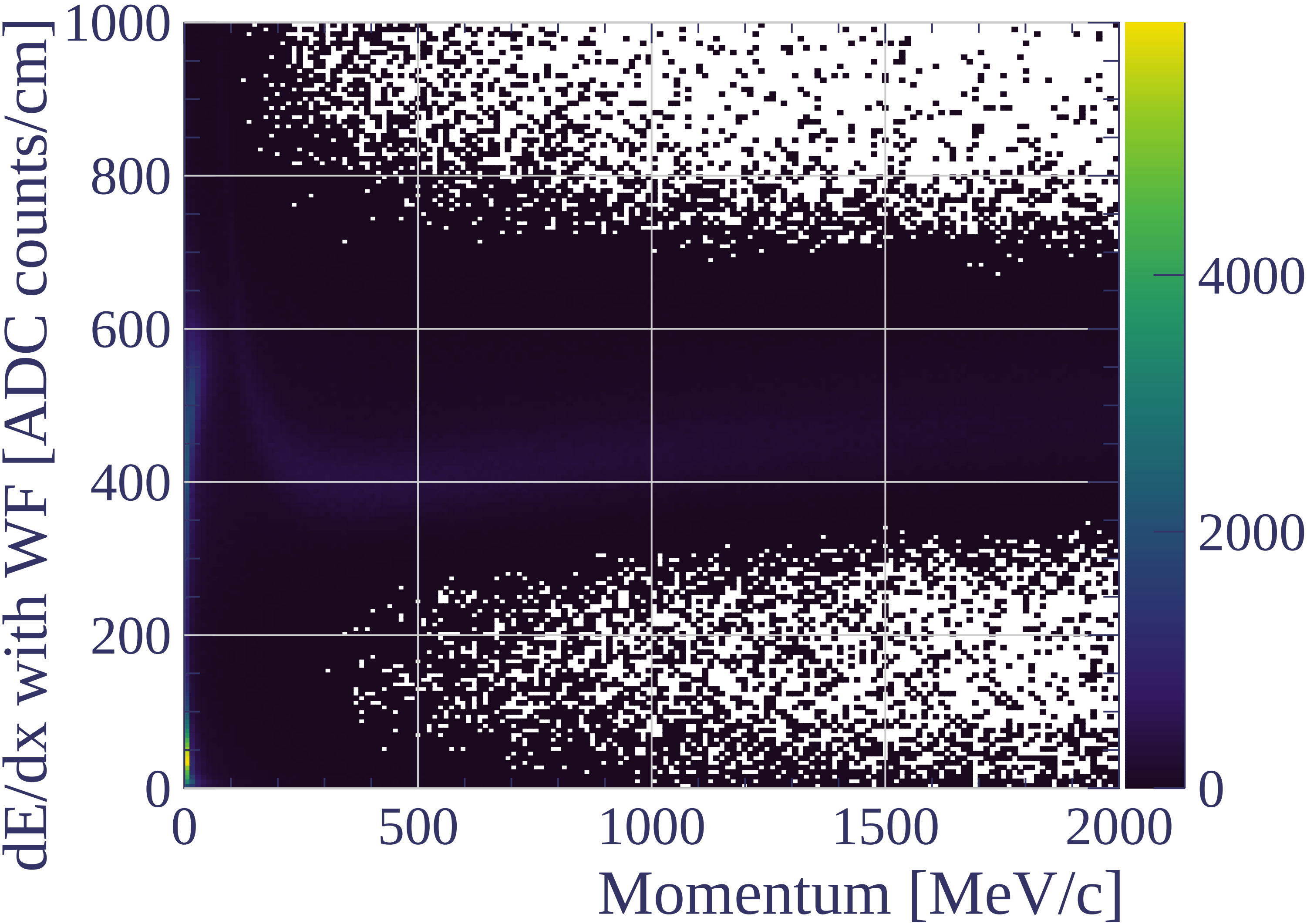
Count
ERAM 20
XP
σ/μ = 19.29± 0.14 %
μ = 469.1± 0.6
σ = 90.5± 0.5
WF
σ/μ = 19.38± 0.14 %
μ = 474.3± 0.6
σ = 91.9± 0.6
ERAM 38
XP
σ/μ = 18.68± 0.15 %
μ = 468.3± 0.7
σ = 87.5± 0.6
WF
σ/μ = 18.56± 0.15 %
μ = 475.0± 0.6
σ = 88.2± 0.6
ERAM 7
XP
σ/μ = 18.40± 0.15 %
μ = 485.7± 0.7
σ = 89.4± 0.6
WF
σ/μ = 18.23± 0.15 %
μ = 494.4± 0.7
σ = 90.1± 0.6
ERAM 44
XP
σ/μ = 20.37± 0.14 %
μ = 465.2± 0.6
σ = 94.8± 0.5
WF
σ/μ = 20.05± 0.14 %
μ = 472.5± 0.6
σ = 94.8± 0.5
ERAM 43
XP
σ/μ = 15.91± 0.05 %
μ = 490.8± 0.3
σ = 78.1± 0.2
WF
σ/μ = 16.17± 0.06 %
μ = 497.6± 0.3
σ = 80.5± 0.3
ERAM 39
XP
σ/μ = 15.77± 0.06 %
μ = 471.4± 0.3
σ = 74.3± 0.2
WF
σ/μ = 15.64± 0.05 %
μ = 478.2± 0.3
σ = 74.8± 0.2
ERAM 41
XP
σ/μ = 15.48± 0.06 %
μ = 487.4± 0.3
σ = 75.4± 0.2
WF
σ/μ = 15.79± 0.06 %
μ = 496.0± 0.3
σ = 78.3± 0.3
ERAM 46
XP
σ/μ = 16.39± 0.06 %
μ = 470.7± 0.3
σ = 77.2± 0.2
WF
σ/μ = 16.41± 0.06 %
μ = 480.3± 0.3
σ = 78.8± 0.2
ERAM 26
XP
σ/μ = 19.58± 0.15 %
μ = 490.9± 0.6
σ = 96.1± 0.6
WF
σ/μ = 19.68± 0.15 %
μ = 495.9± 0.6
σ = 97.6± 0.6
ERAM 17
XP
σ/μ = 18.92± 0.17 %
μ = 493.9± 0.7
σ = 93.4± 0.7
WF
σ/μ = 18.61± 0.17 %
μ = 505.8± 0.7
σ = 94.1± 0.7
ERAM 23
XP
σ/μ = 18.65± 0.17 %
μ = 487.5± 0.7
σ = 90.9± 0.7
WF
σ/μ = 18.15± 0.15 %
μ = 501.1± 0.7
σ = 90.9± 0.6
ERAM 29
XP
σ/μ = 19.53± 0.14 %
μ = 479.8± 0.6
σ = 93.7± 0.5
WF
σ/μ = 19.31± 0.13 %
μ = 490.9± 0.6
σ = 94.8± 0.5
ERAM 1
XP
σ/μ = 15.96± 0.06 %
μ = 494.8± 0.3
σ = 79.0± 0.2
WF
σ/μ = 16.26± 0.06 %
μ = 502.5± 0.3
σ = 81.7± 0.3
ERAM 10
XP
σ/μ = 15.67± 0.06 %
μ = 496.3± 0.3
σ = 77.8± 0.3
WF
σ/μ = 15.62± 0.06 %
μ = 506.2± 0.3
σ = 79.1± 0.3
ERAM 11
XP
σ/μ = 15.73± 0.06 %
μ = 498.4± 0.3
σ = 78.4± 0.3
WF
σ/μ = 15.56± 0.06 %
μ = 509.8± 0.3
σ = 79.3± 0.2
ERAM 3
XP
σ/μ = 16.37± 0.06 %
μ = 483.3± 0.3
σ = 79.1± 0.2
WF
σ/μ = 16.45± 0.06 %
μ = 495.1± 0.3
σ = 81.5± 0.2
dE/dx [ADC counts/cm]



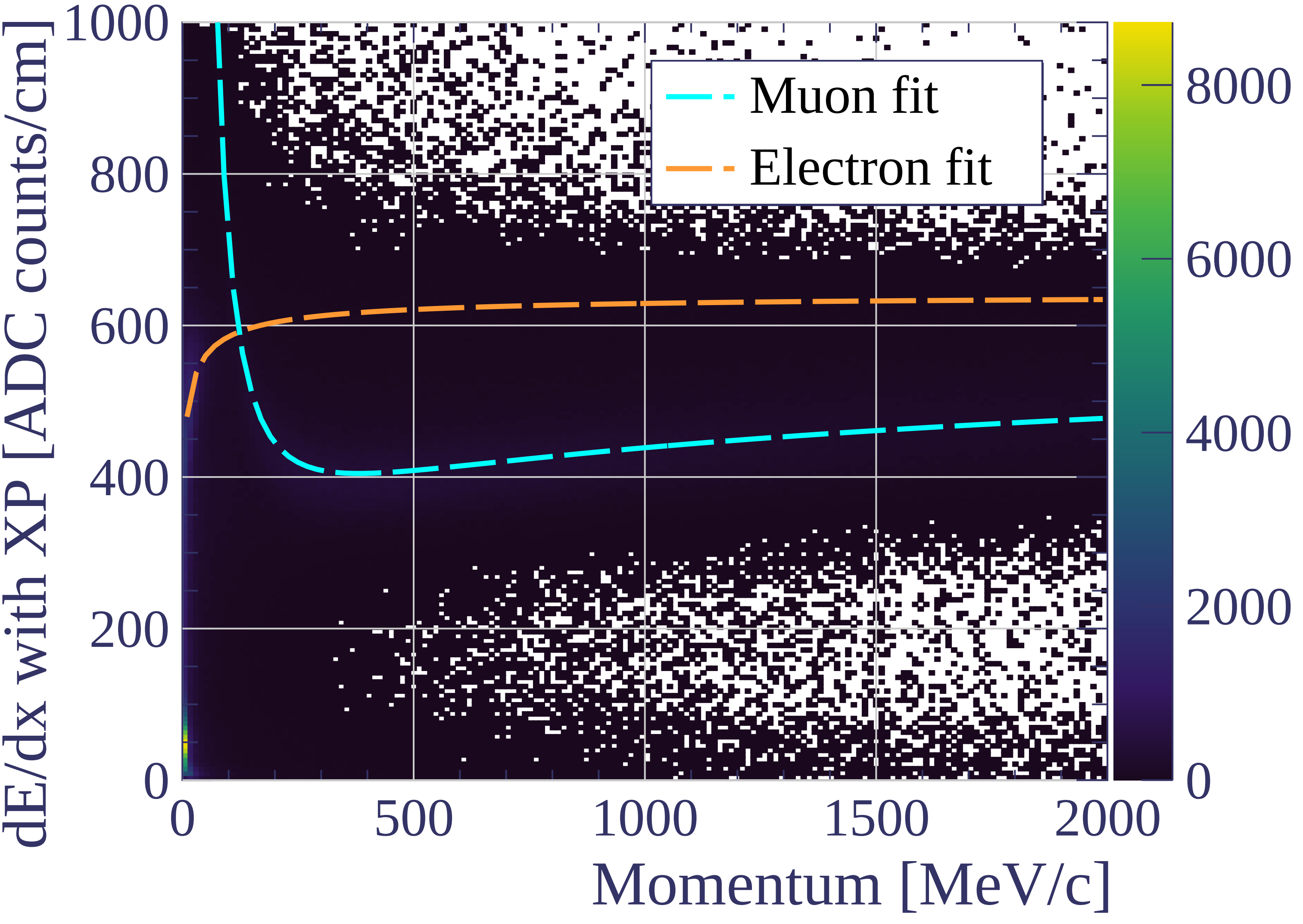
dE/dx with XP [ADC counts/cm]
dE/dx with WF [ADC counts/cm]
0
200
400
600
800
1000
×10^3
0
10
20


WF
XP
dE/dx resolution [%]
Momentum [MeV/c]
12
10
8
6
4
0
500
1000
1500
2000

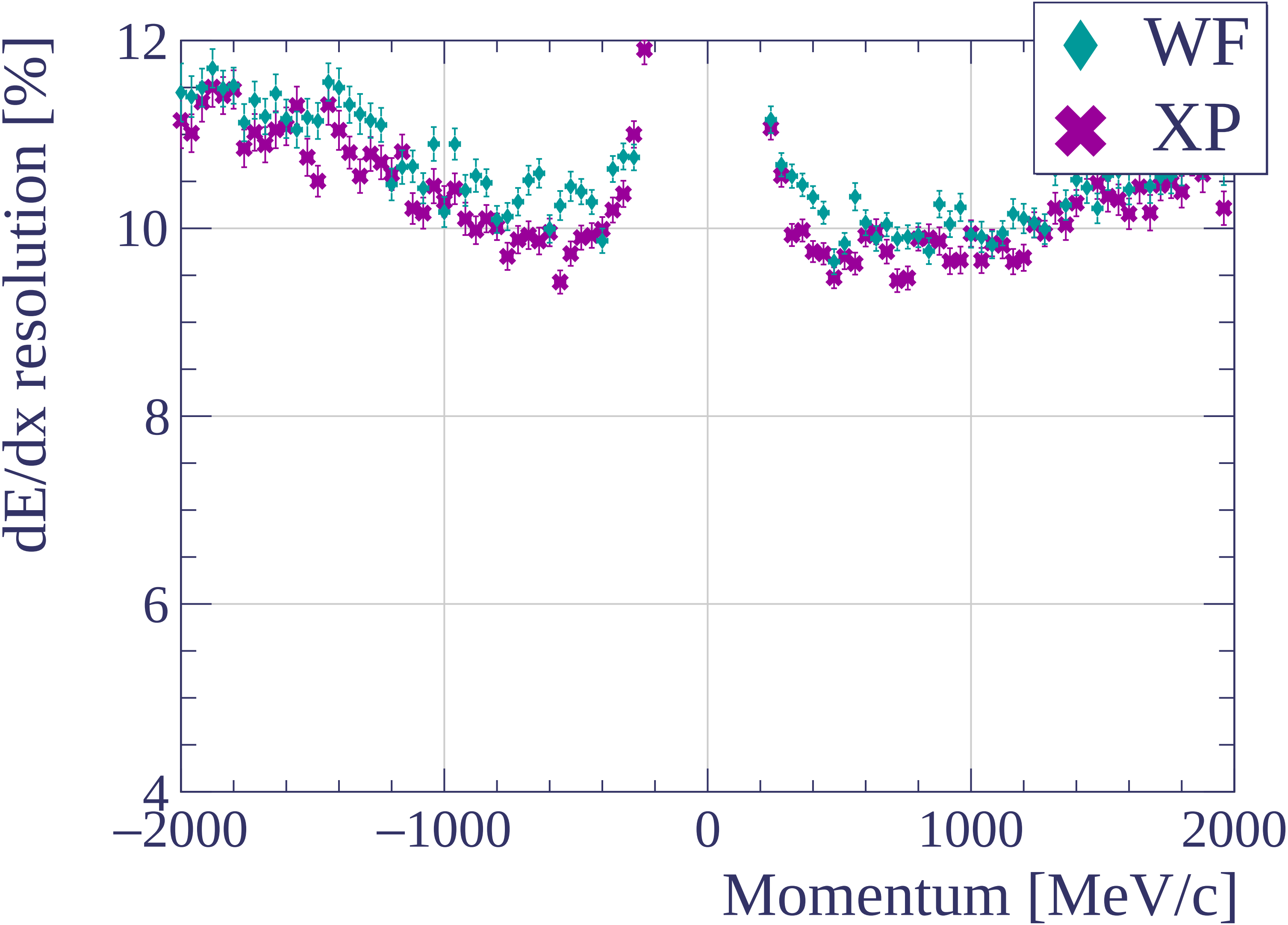

Mean dE/dx [ADC counts/cm]
600
500
400
300
0
500
1000
1500
2000
Momentum [MeV/c]
WF
XP



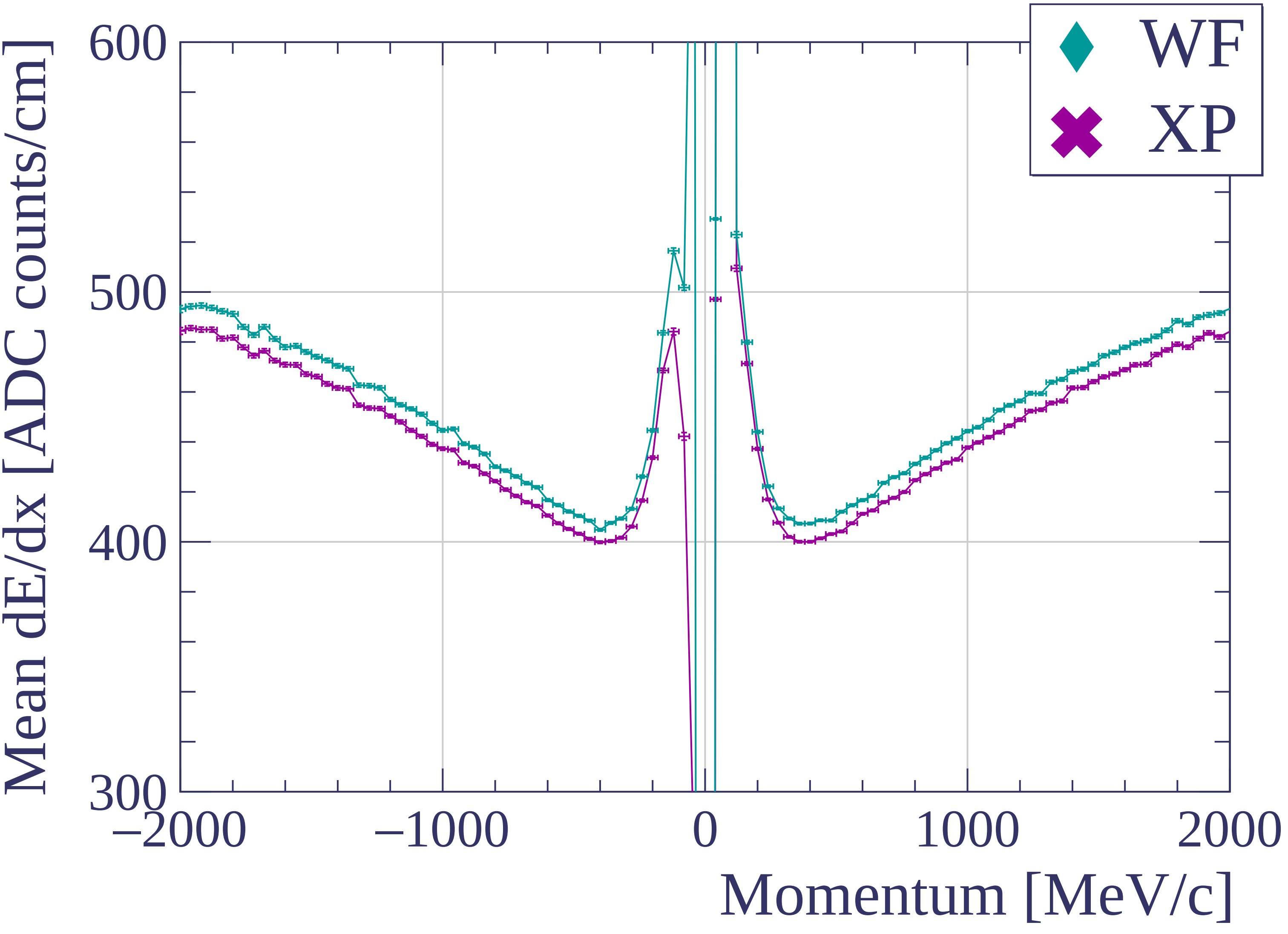
dE/dx with WF [ADC counts/cm]
Momentum [MeV/c]
0
200
400
600
800
1000
500
1000
1500
2000
4000



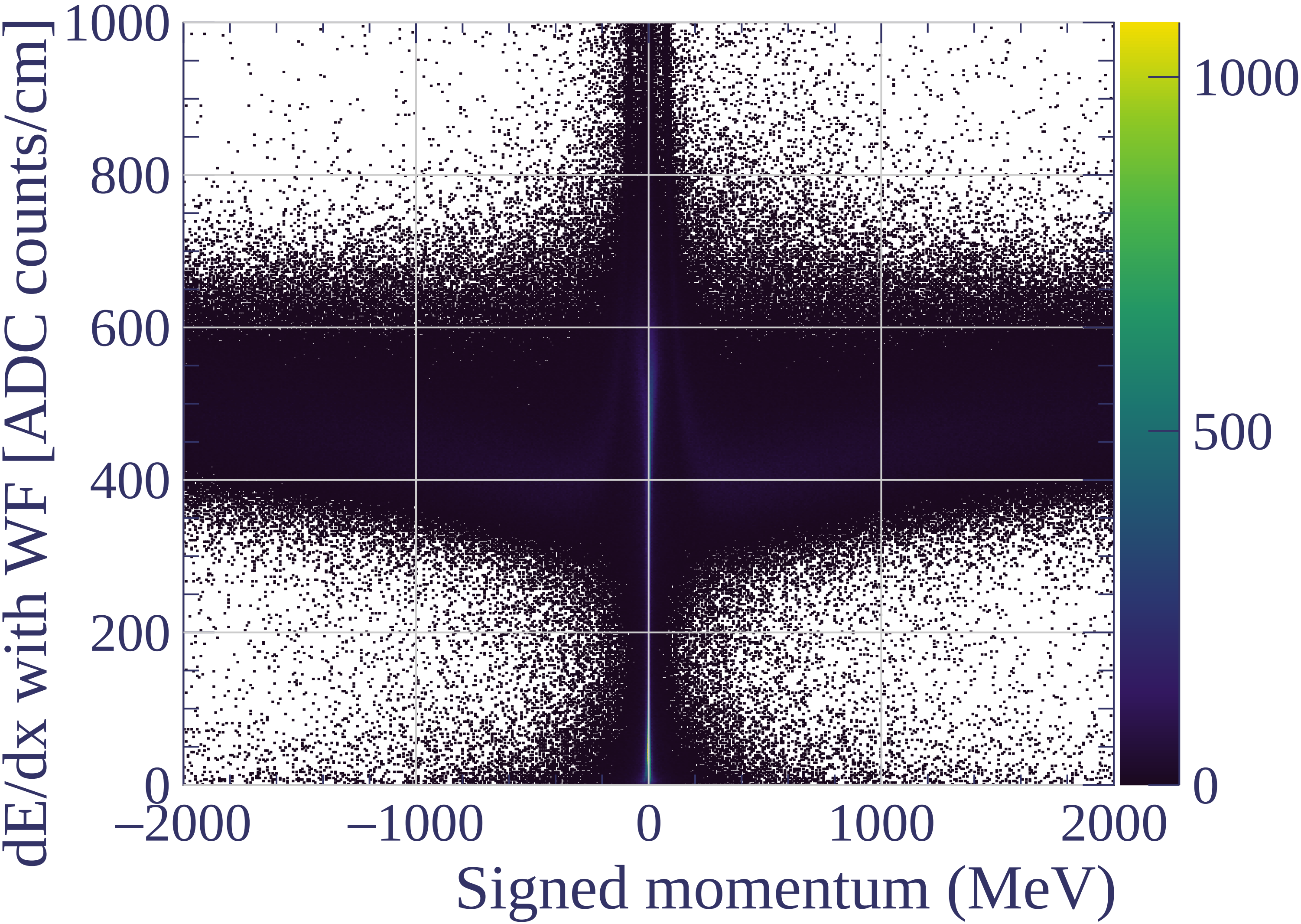
Muon fit
Electron fit
dE/dx with XP [ADC counts/cm]
Momentum [MeV/c]
0
200
400
600
800
1000
500
1000
1500
2000
2000
4000
6000
8000



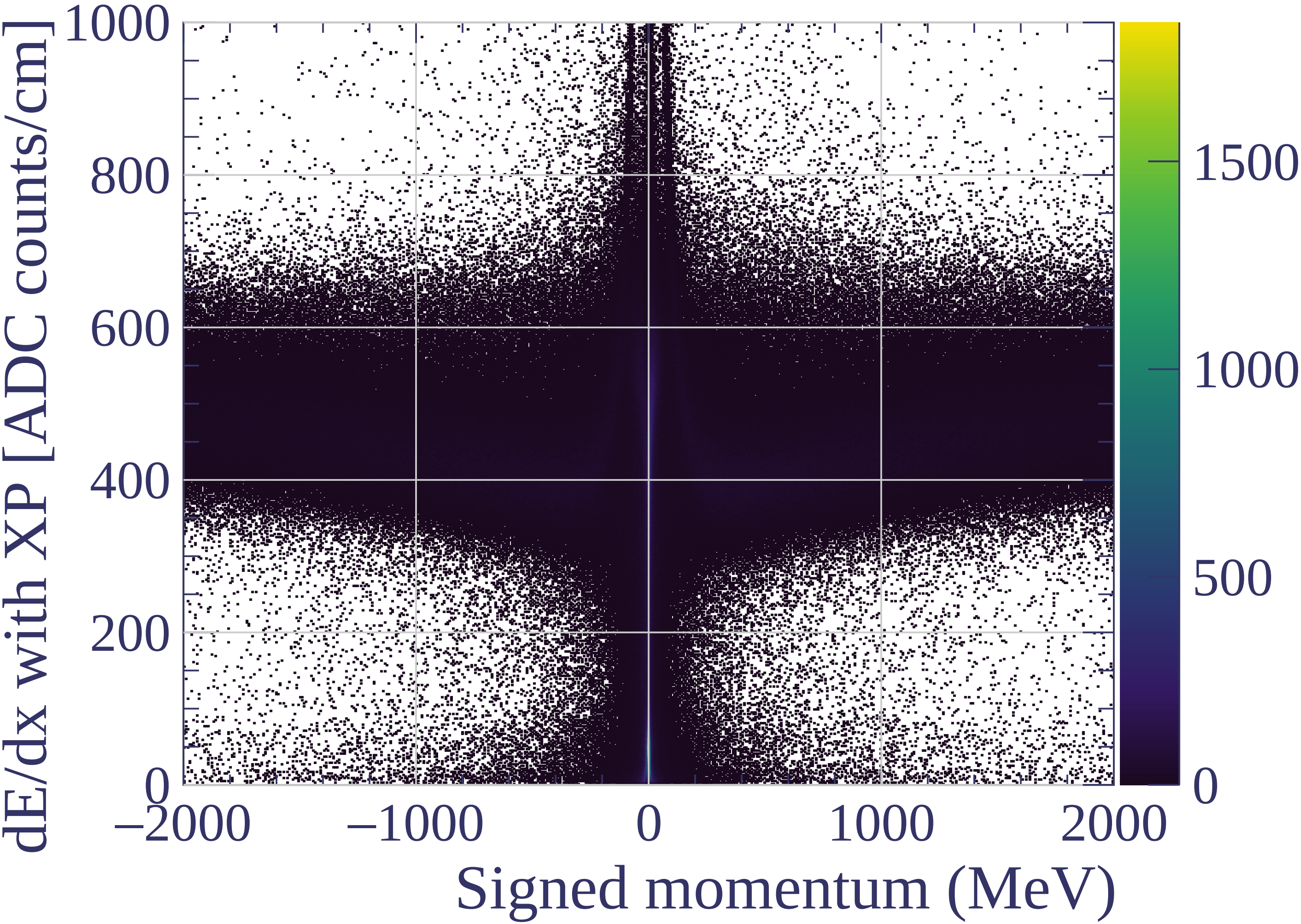
WF
XP
dE/dx resolution [%]
12
10
8
6
4
−2000
−1000
0
1000
2000
Momentum [MeV/c]



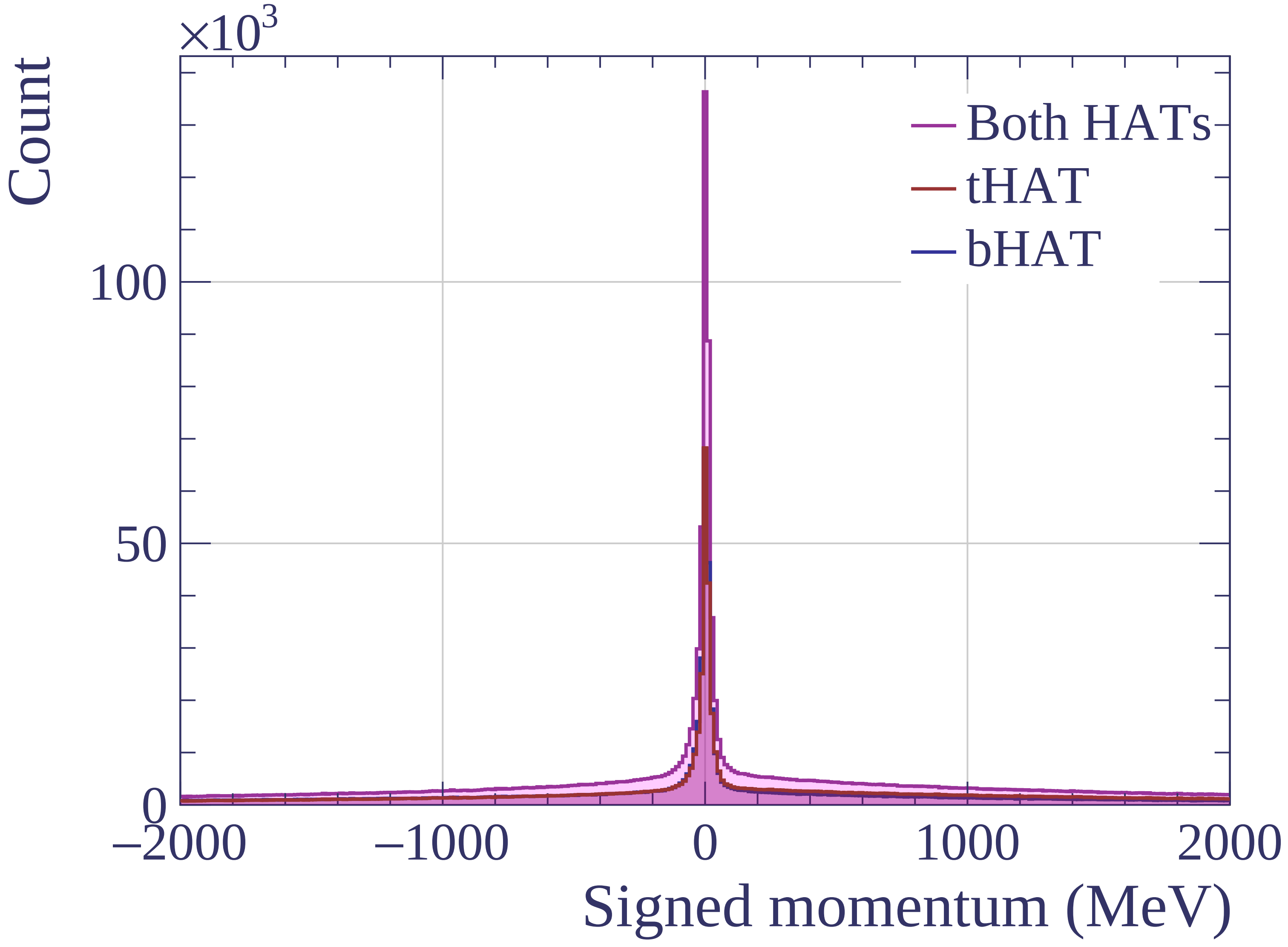
WF
XP
600
500
400
300
−2000
−1000
0
1000
2000
Mean dE/dx [ADC counts/cm]
Momentum [MeV/c]



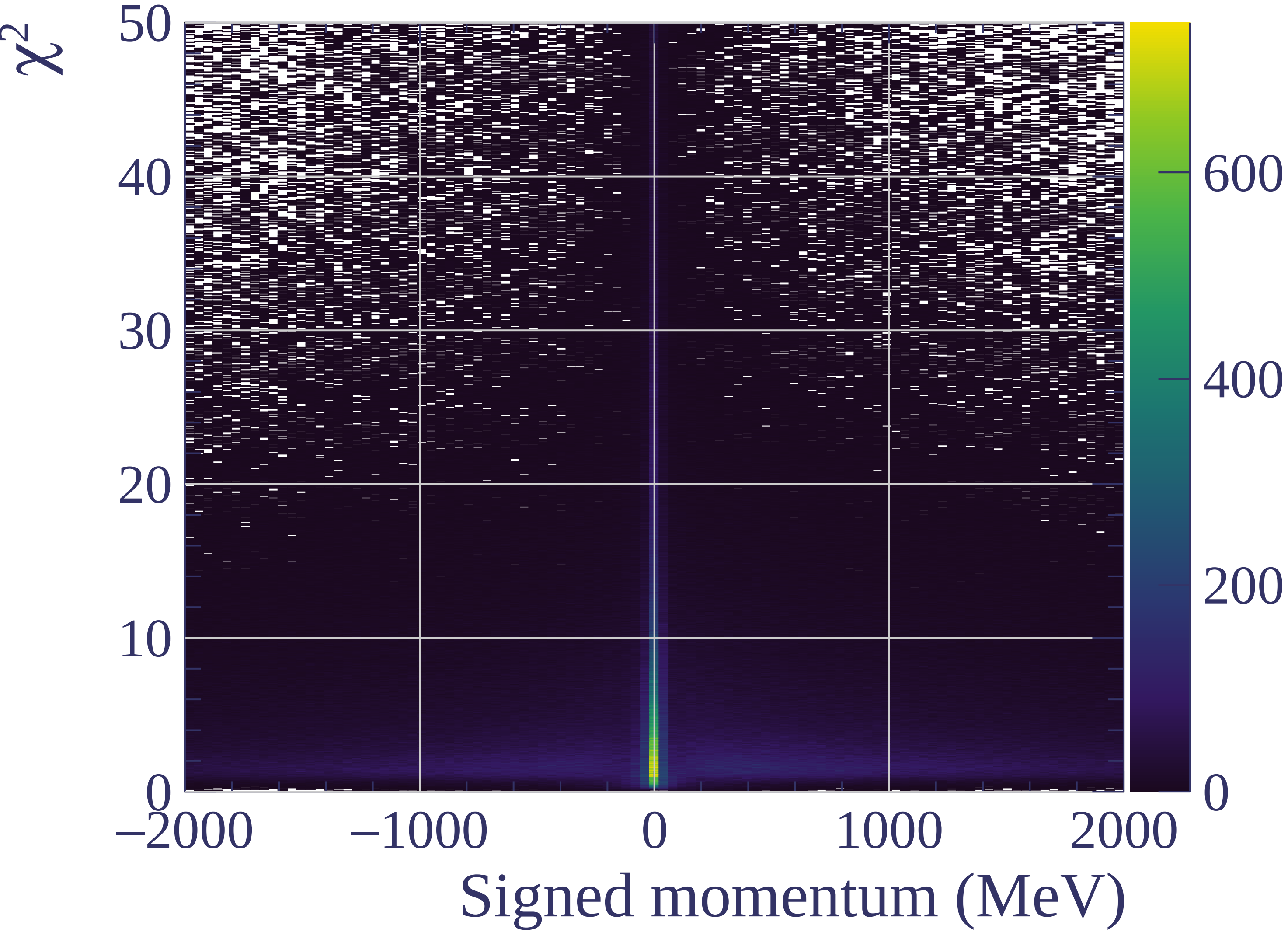
dE/dx with WF [ADC counts/cm]
1000
800
600
400
200
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
1000
500
0



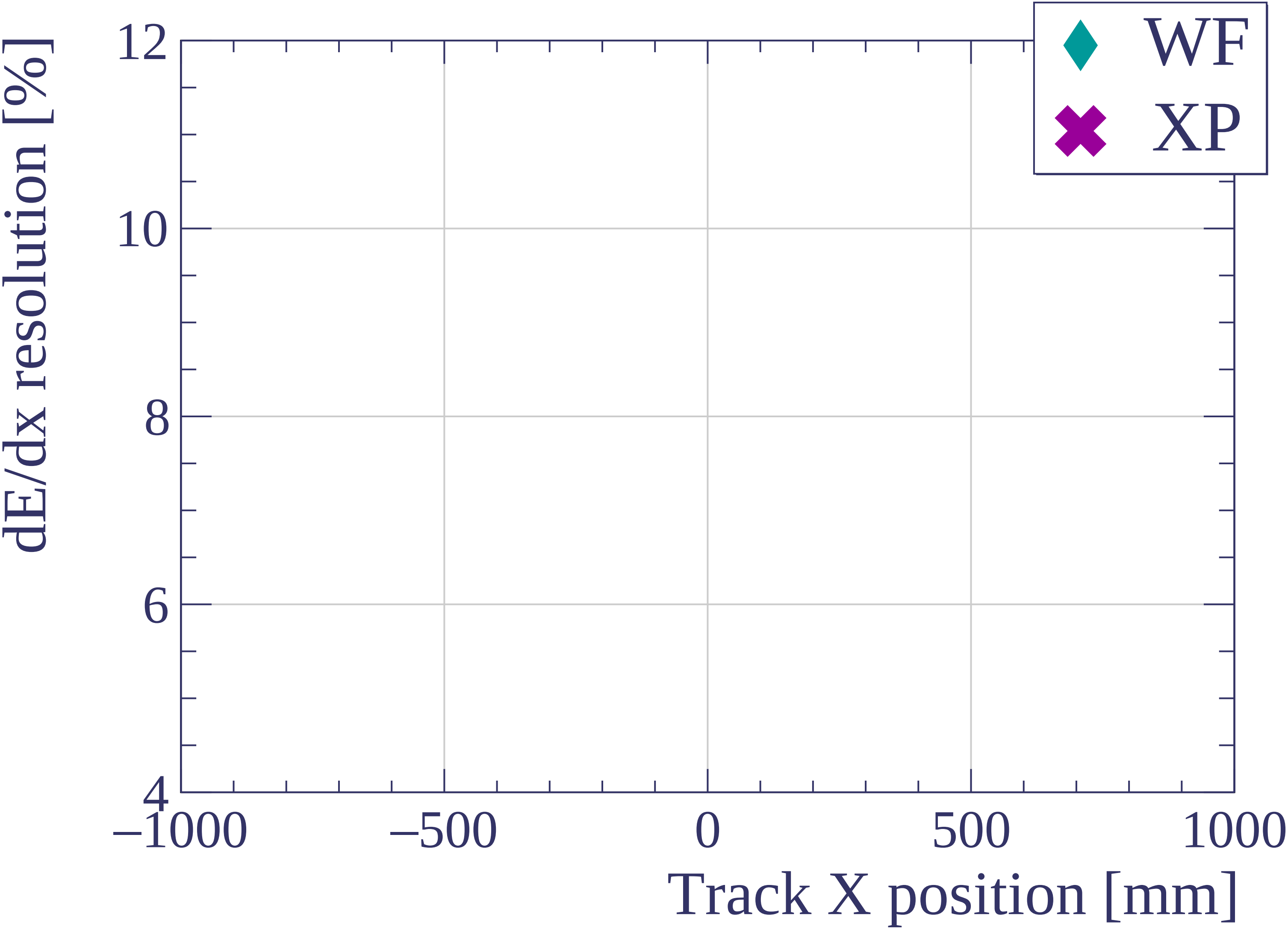
1000
800
600
400
200
0
dE/dx with XP [ADC counts/cm]
−2000
−1000
0
1000
2000
Signed momentum (MeV)
1500
1000
500
0

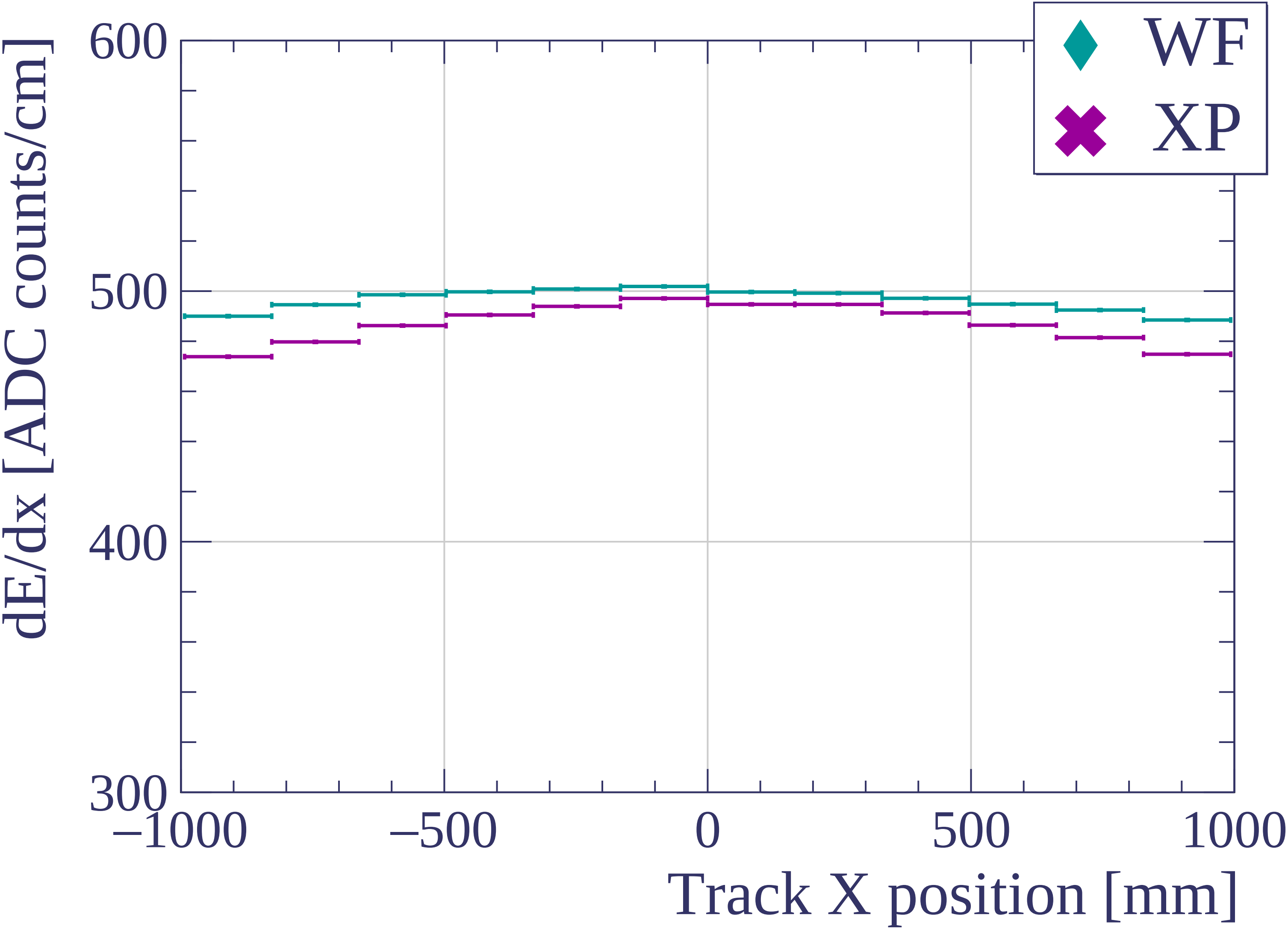


×10^3
Count
Both HATs
tHAT
bHAT
100
50
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)

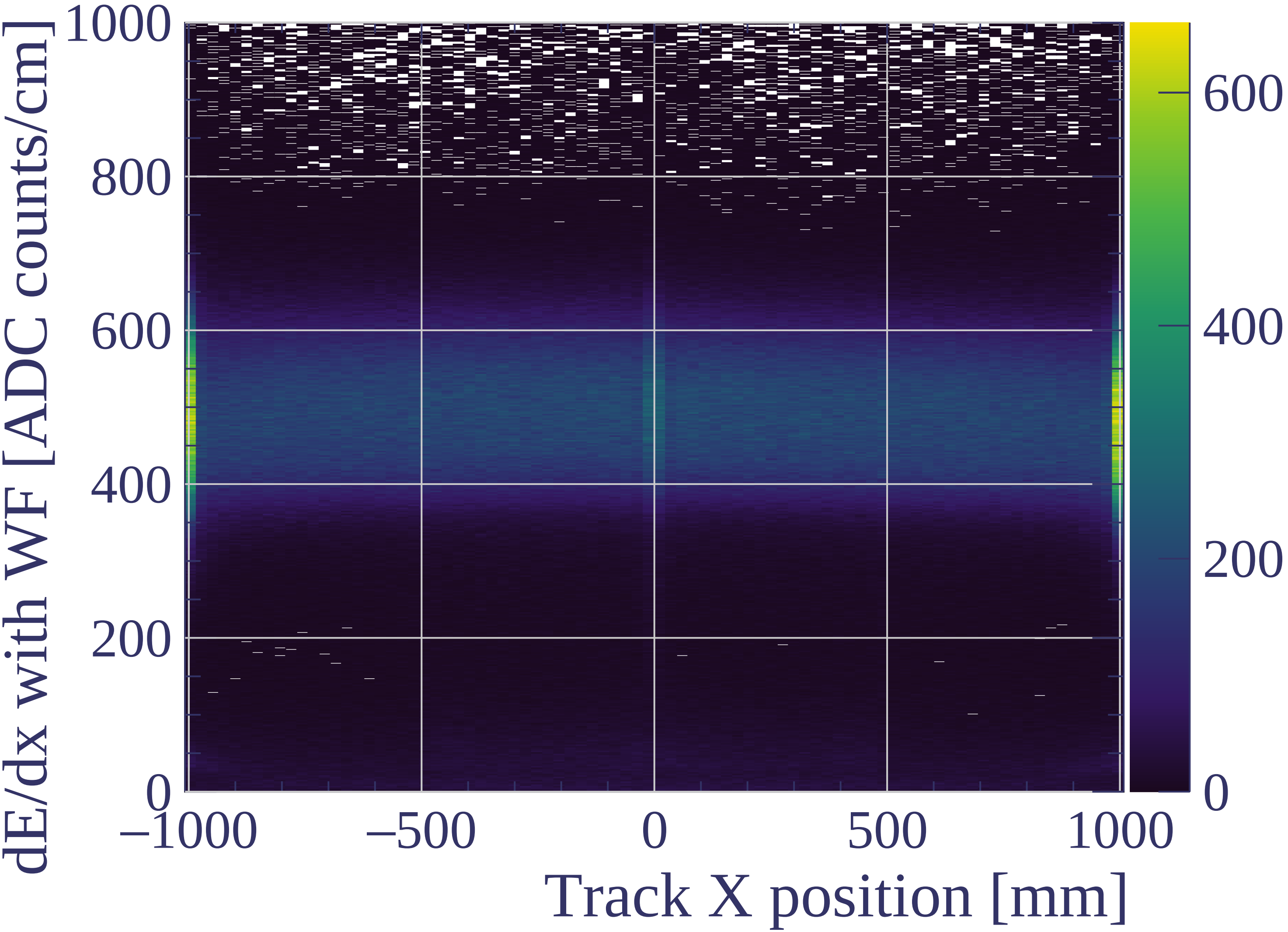


χ^2
50
40
30
20
10
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
600
400
200
0

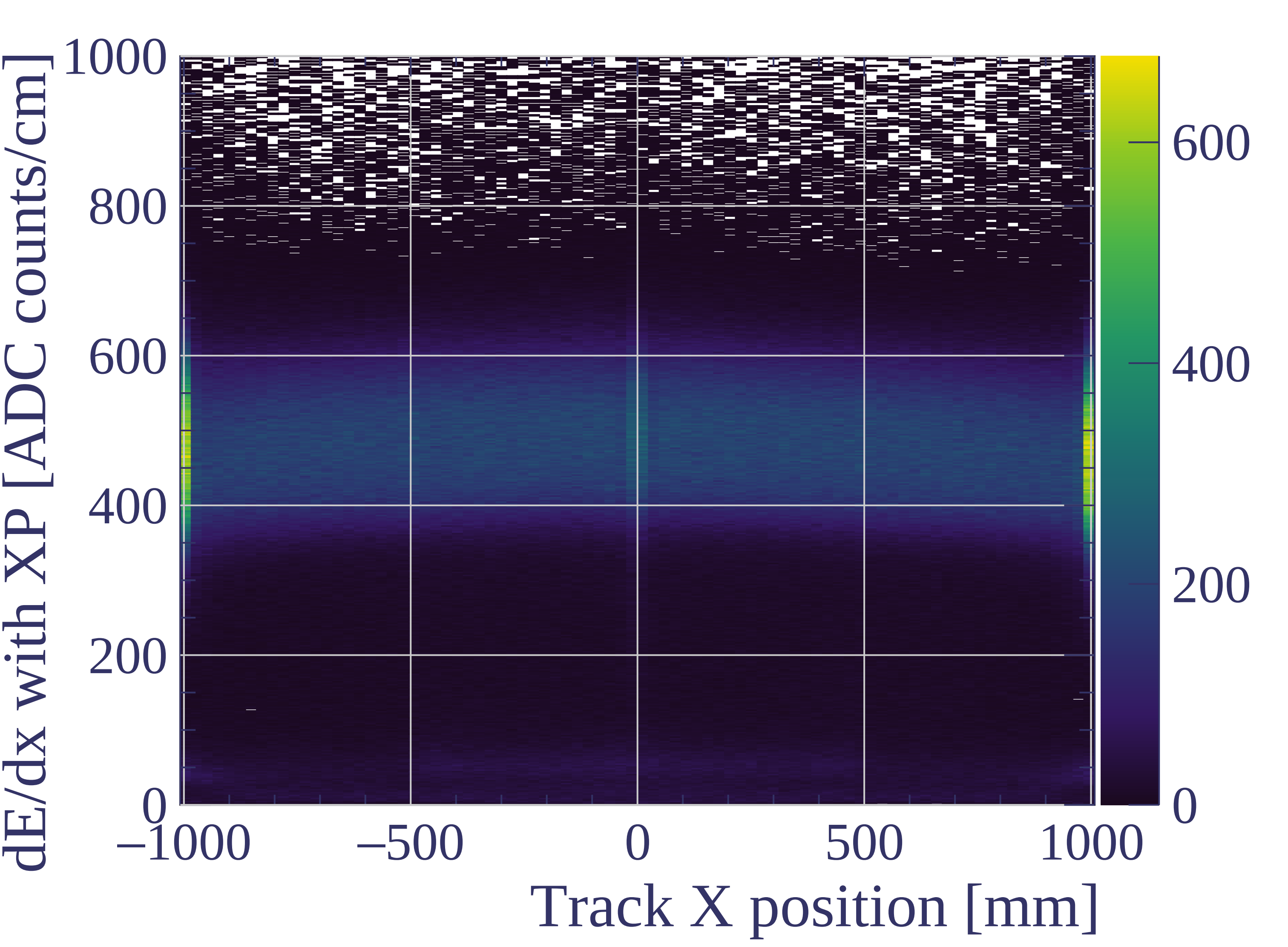

dE/dx resolution [%]
12
10
8
6
4
−1000
−500
0
500
1000
Track X position [mm]
WF
XP



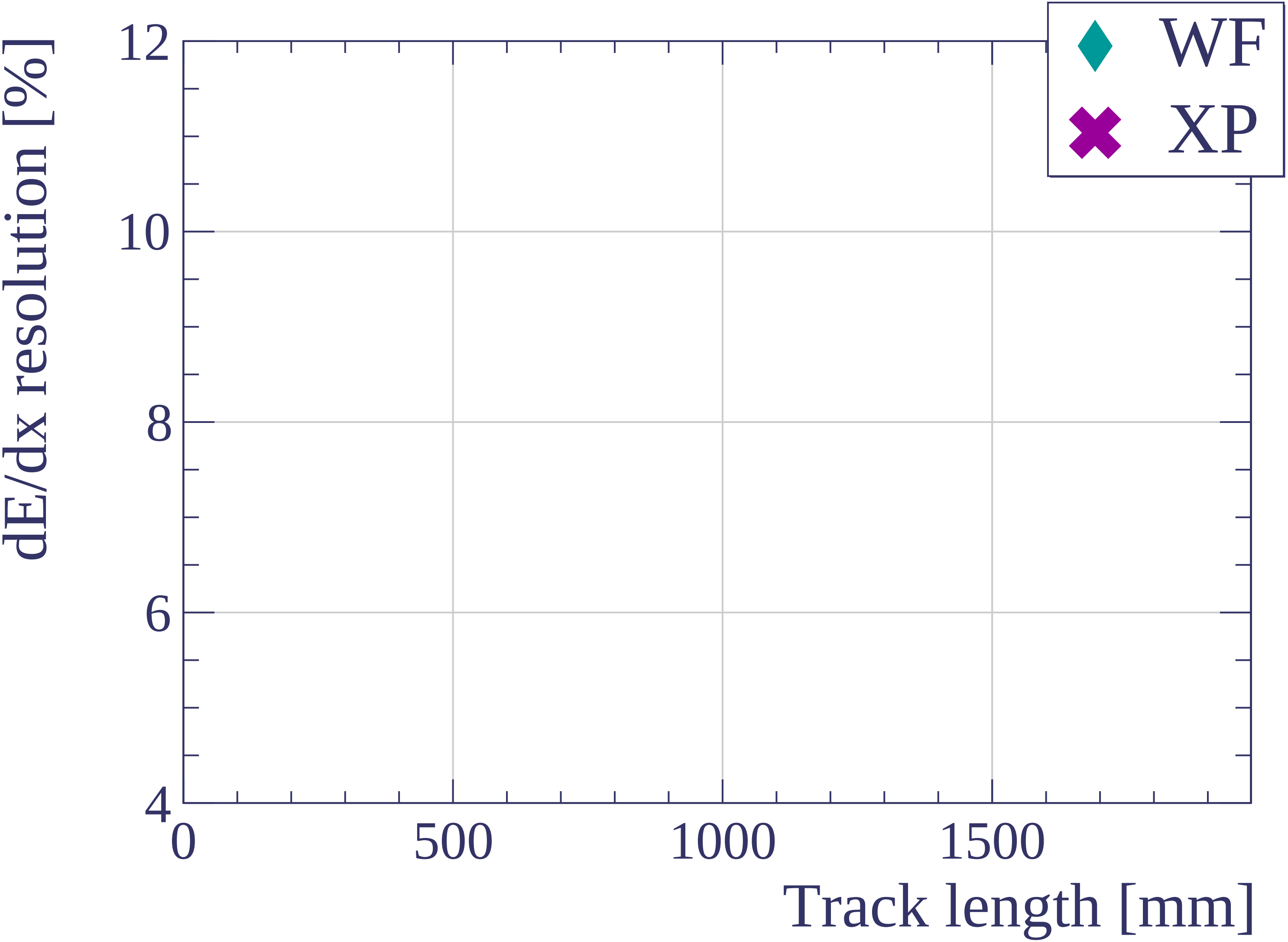
WF
XP
600
500
400
300
−1000
−500
0
500
1000
dE/dx [ADC counts/cm]
Track X position [mm]



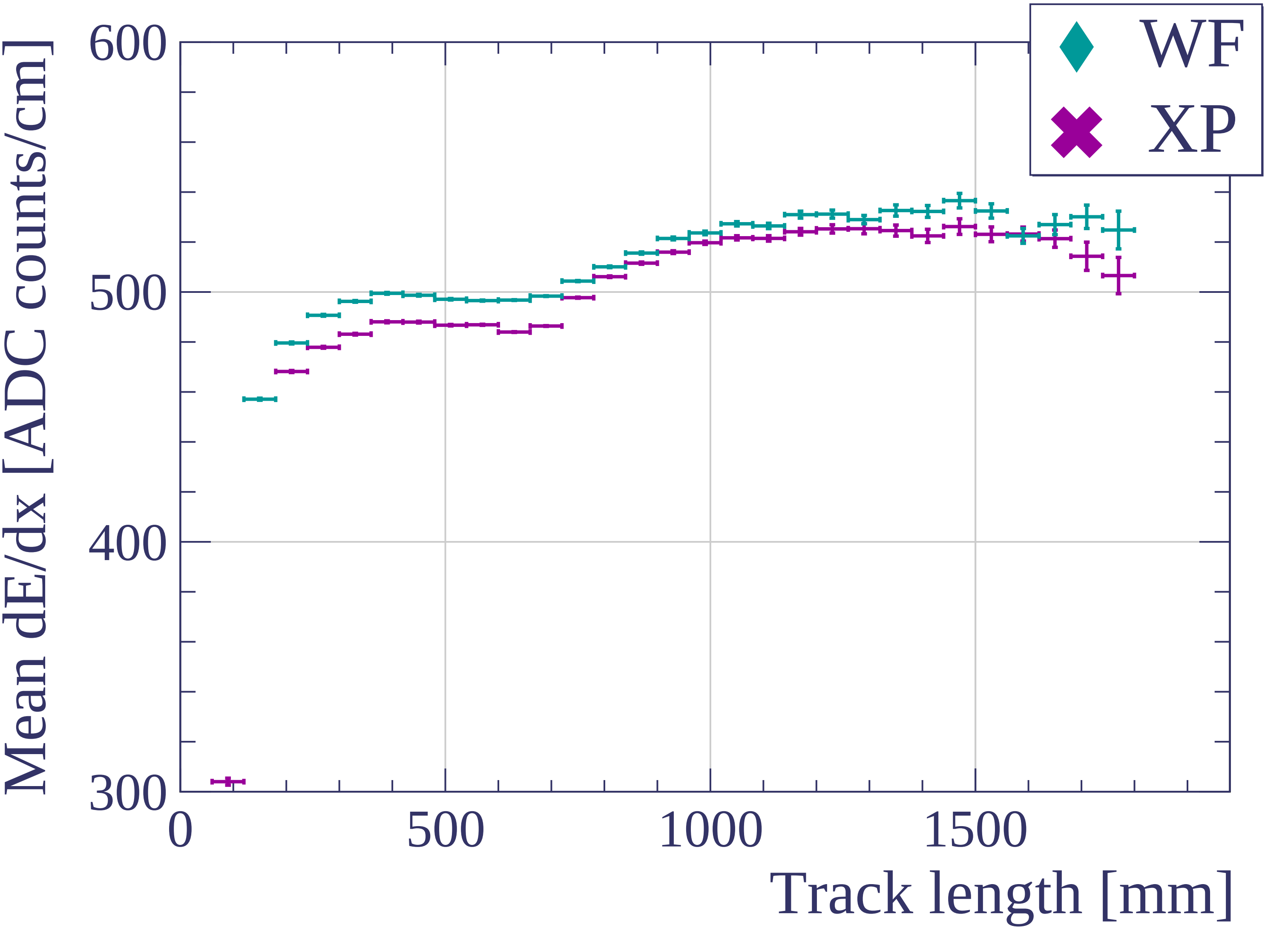
dE/dx with WF [ADC counts/cm]
1000
800
600
400
200
0
−1000
−500
0
500
1000
Track X position [mm]
600
400
200
0



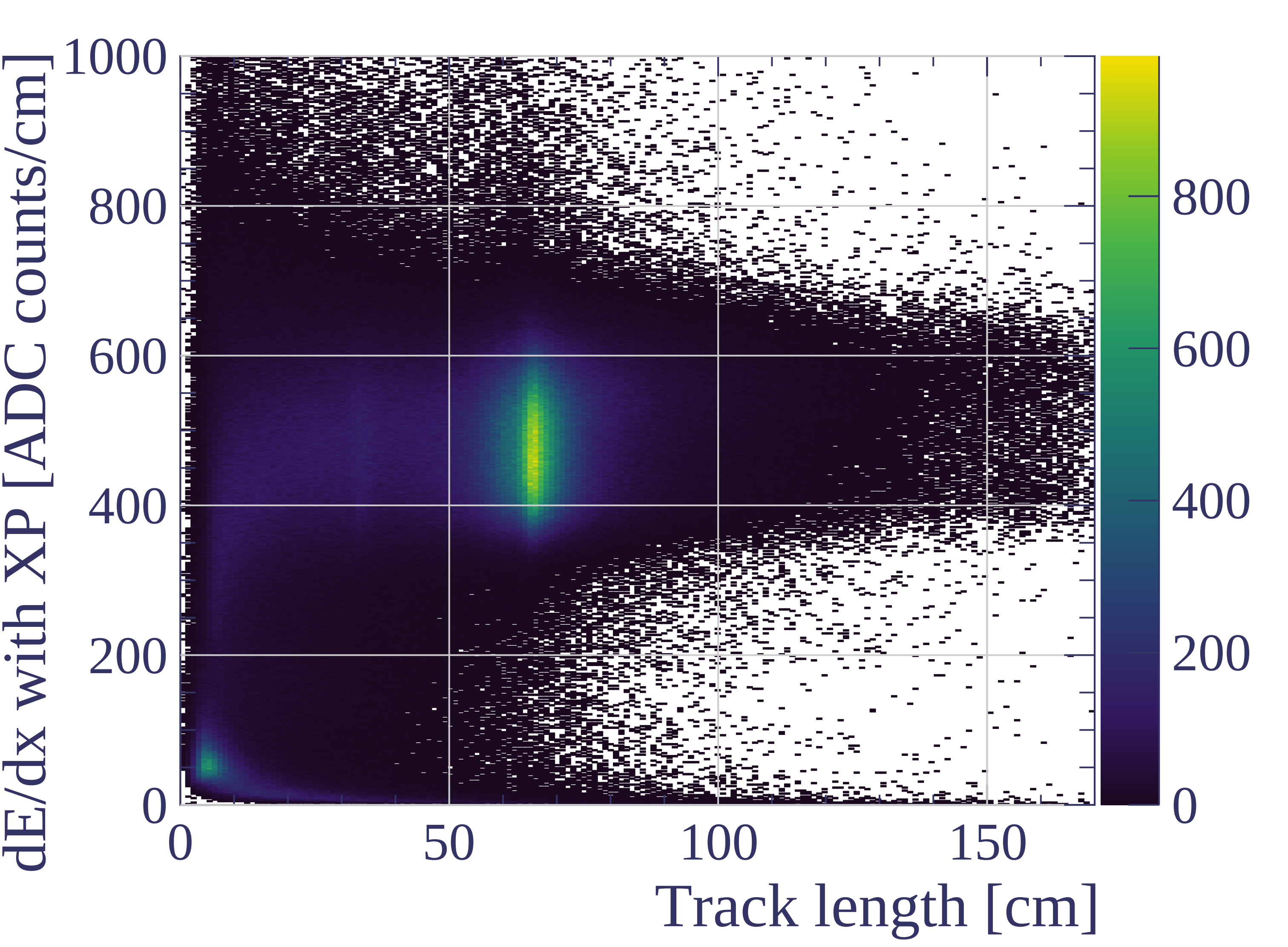
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−1000
−500
0
500
1000
Track X position [mm]
600
400
200
0



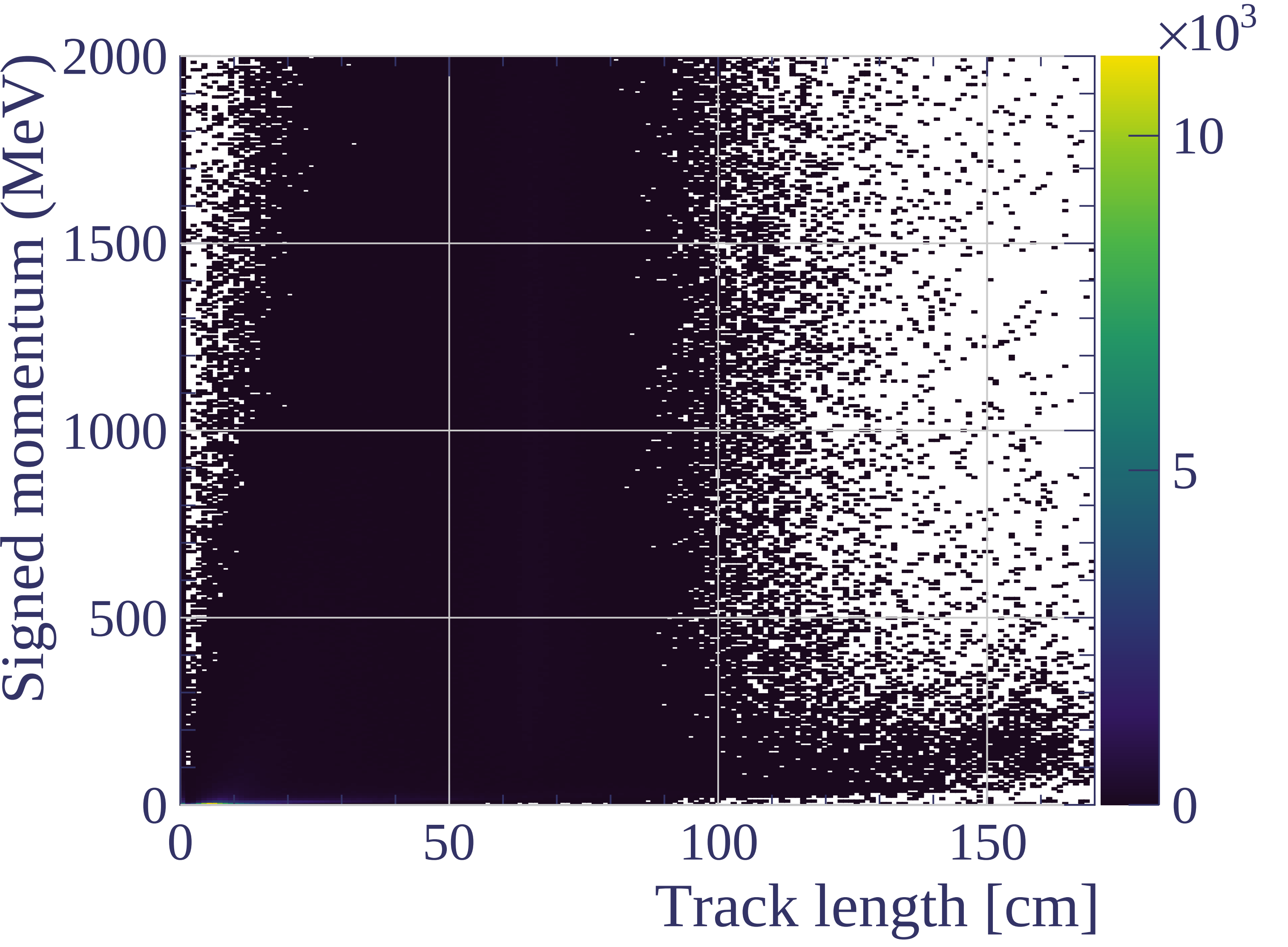
WF
XP
12
10
8
6
4
0
500
1000
1500
dE/dx resolution [%]
Track length [mm]

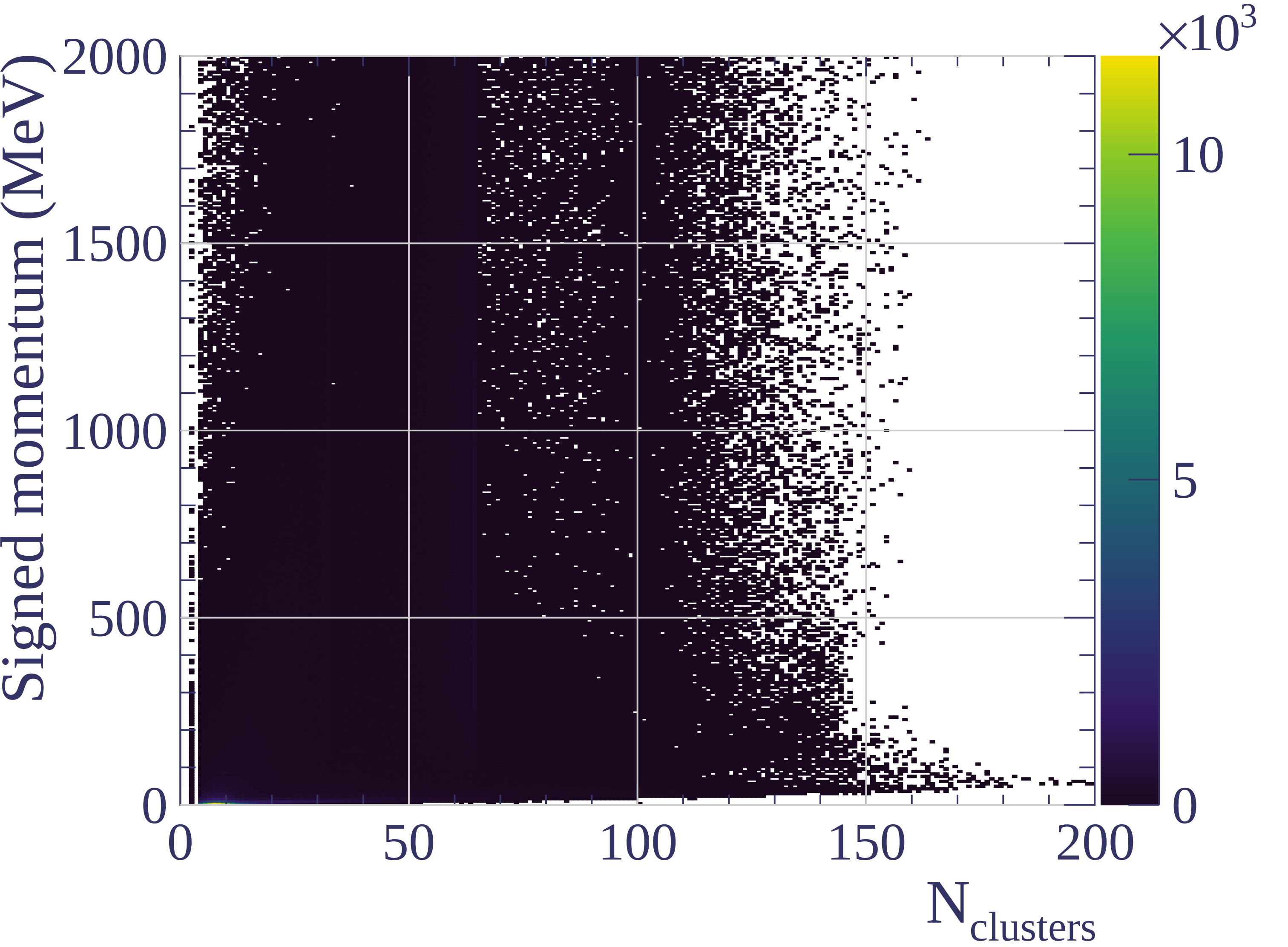


Mean dE/dx [ADC counts/cm]
Track length [mm]
WF
XP
600
500
400
300
0
500
1000
1500

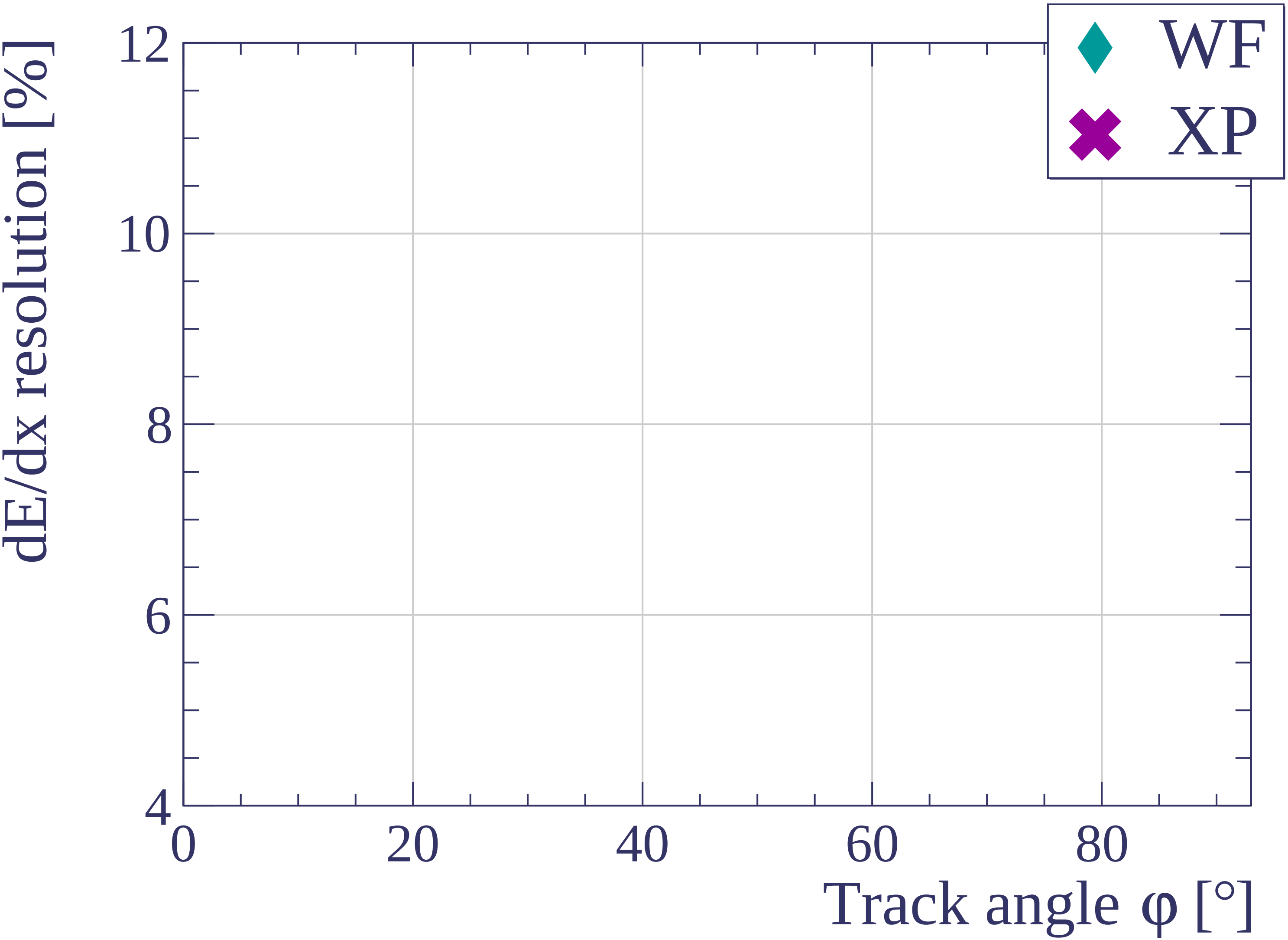


dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
0
50
100
150
Track length [cm]
800
600
400
200
0

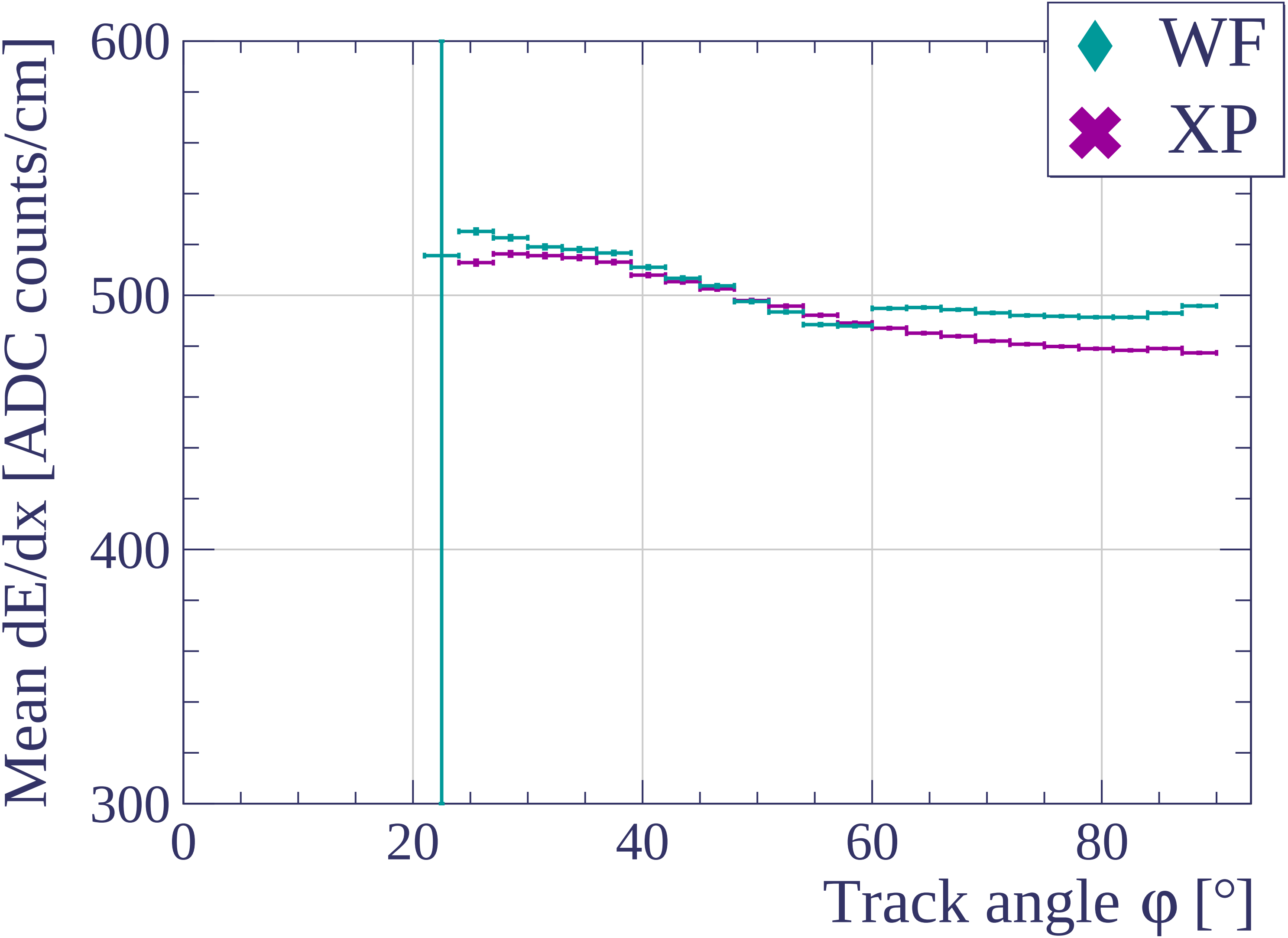

Signed momentum (MeV)
Track length [cm]
×10^3
2000
1500
1000
500
0
0
50
100
150
10
5
0



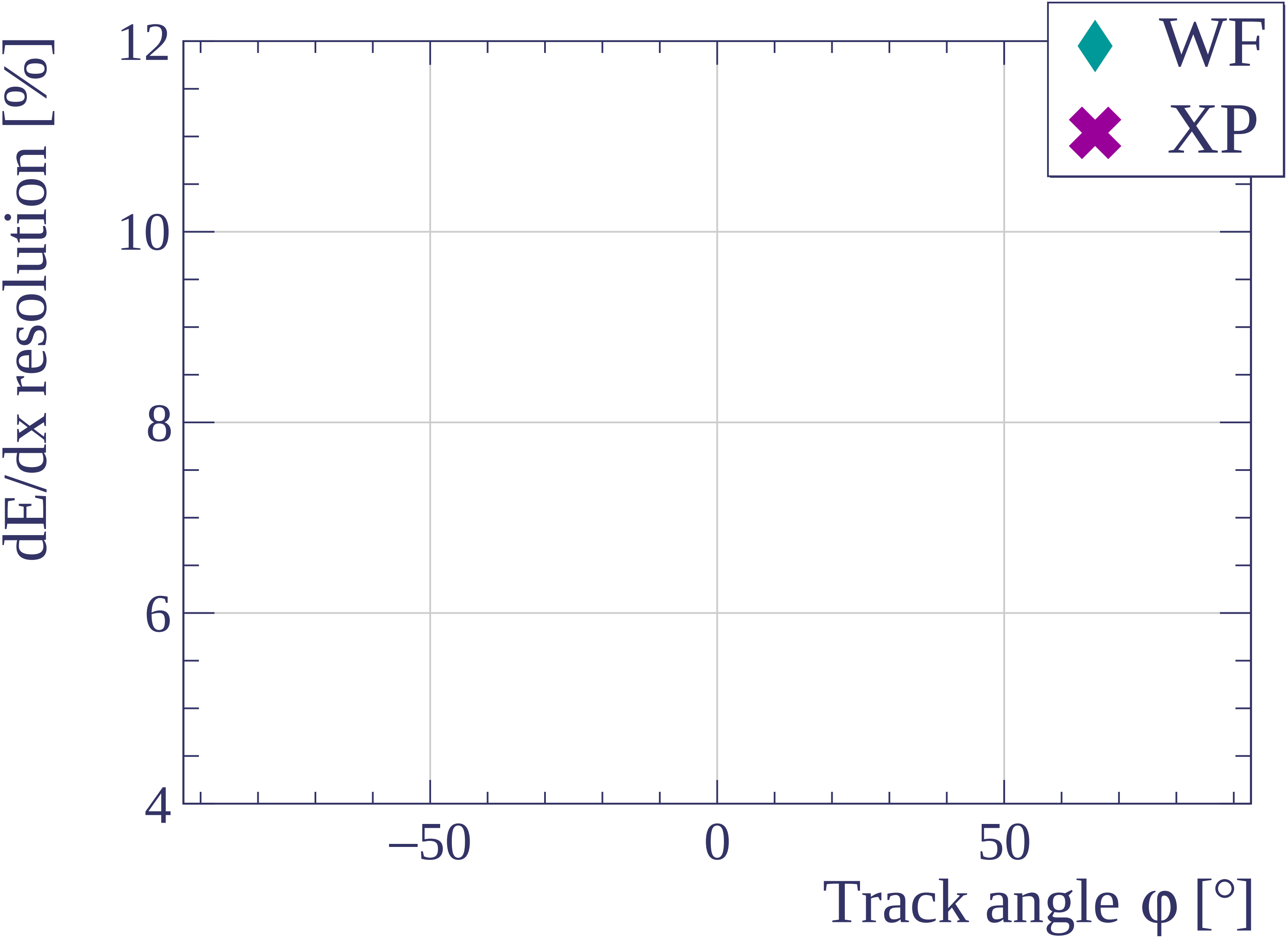
×10^3
Signed momentum (MeV)
2000
1500
1000
500
0
0
50
100
150
200
N_clusters
10
5
0



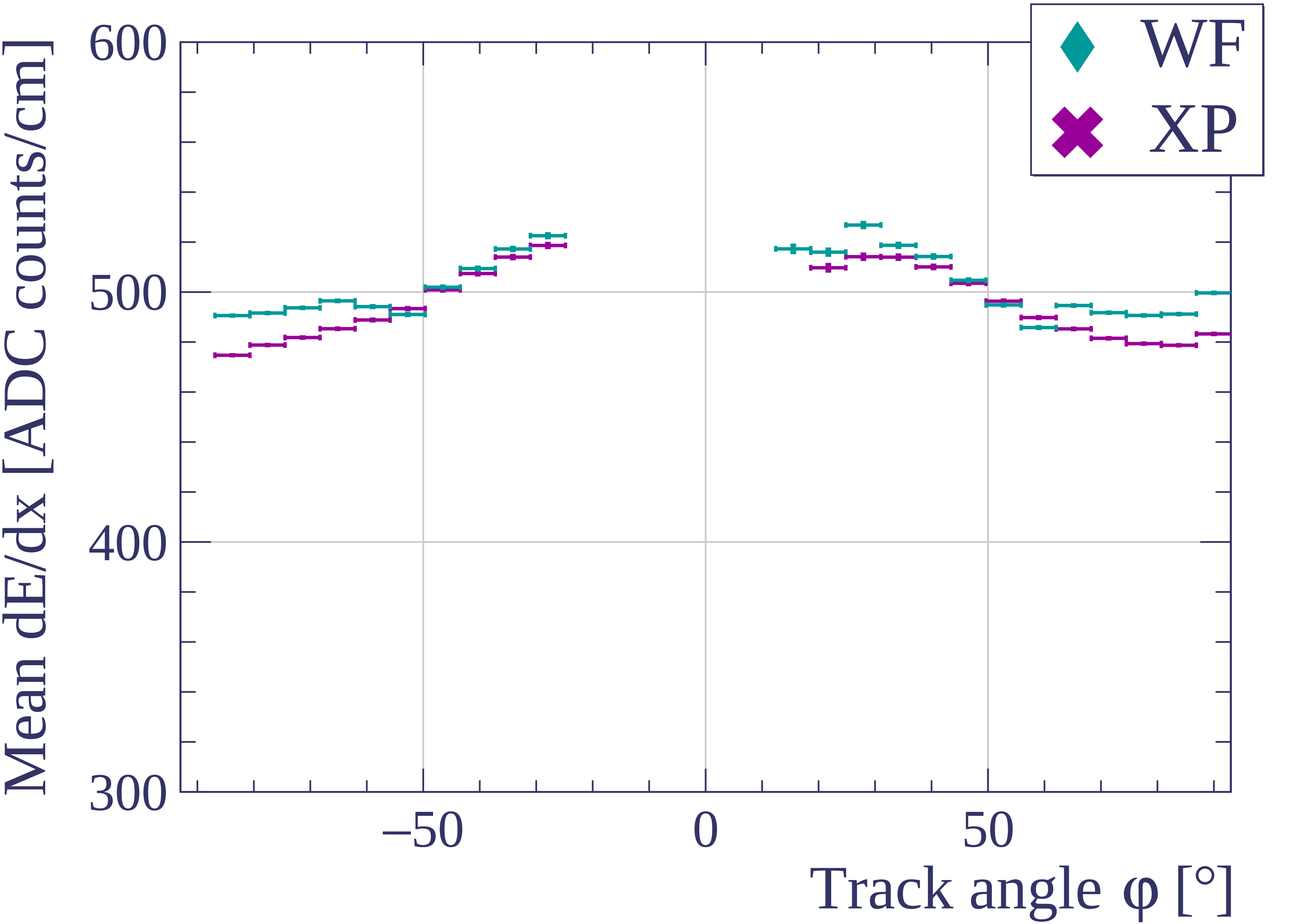
dE/dx resolution [%]
12
10
8
6
4
0
20
40
60
80
Track angle φ [°]
WF
XP



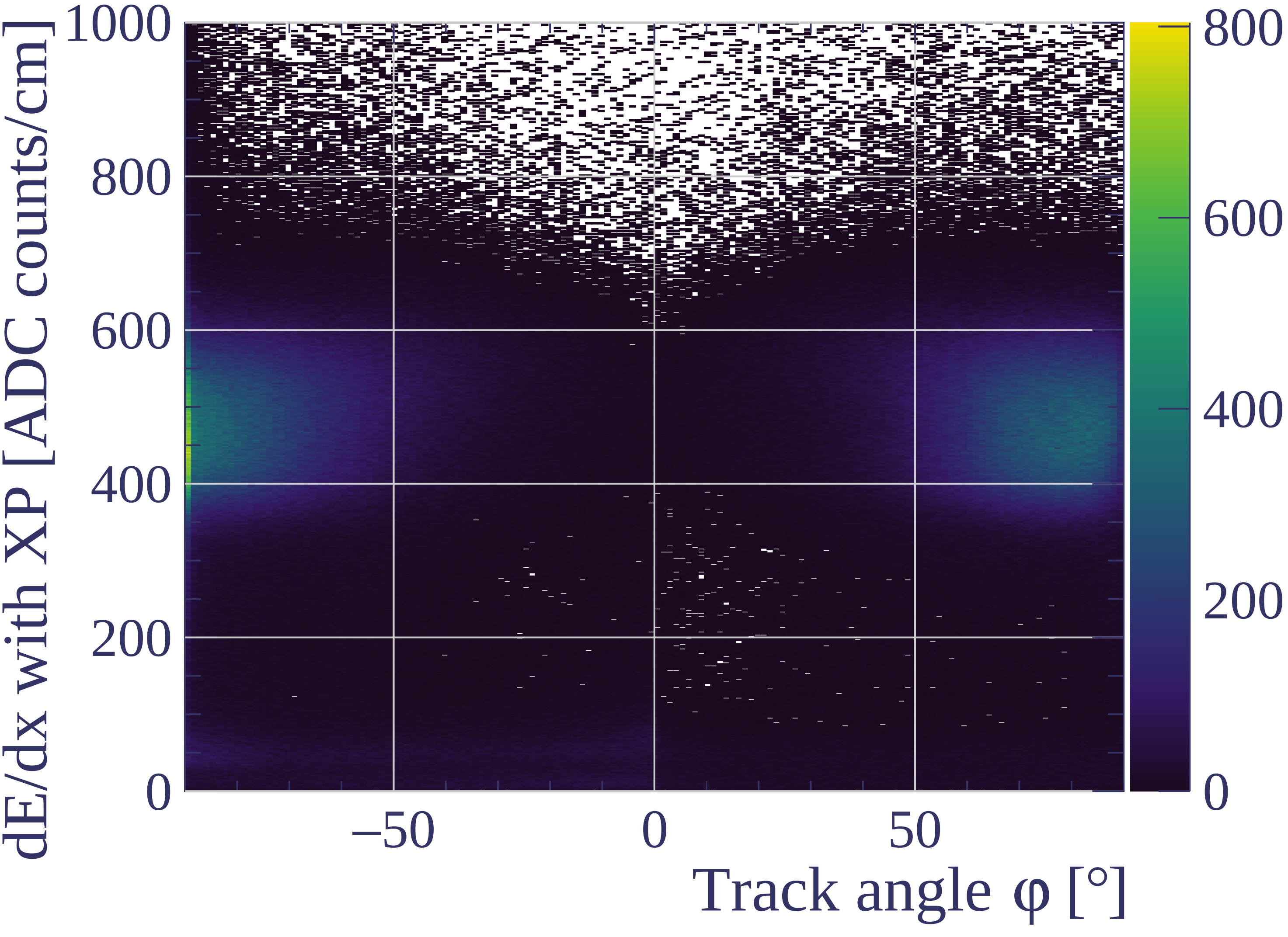
WF
XP
600
500
400
300
0
20
40
60
80
Mean dE/dx [ADC counts/cm]
Track angle φ [°]



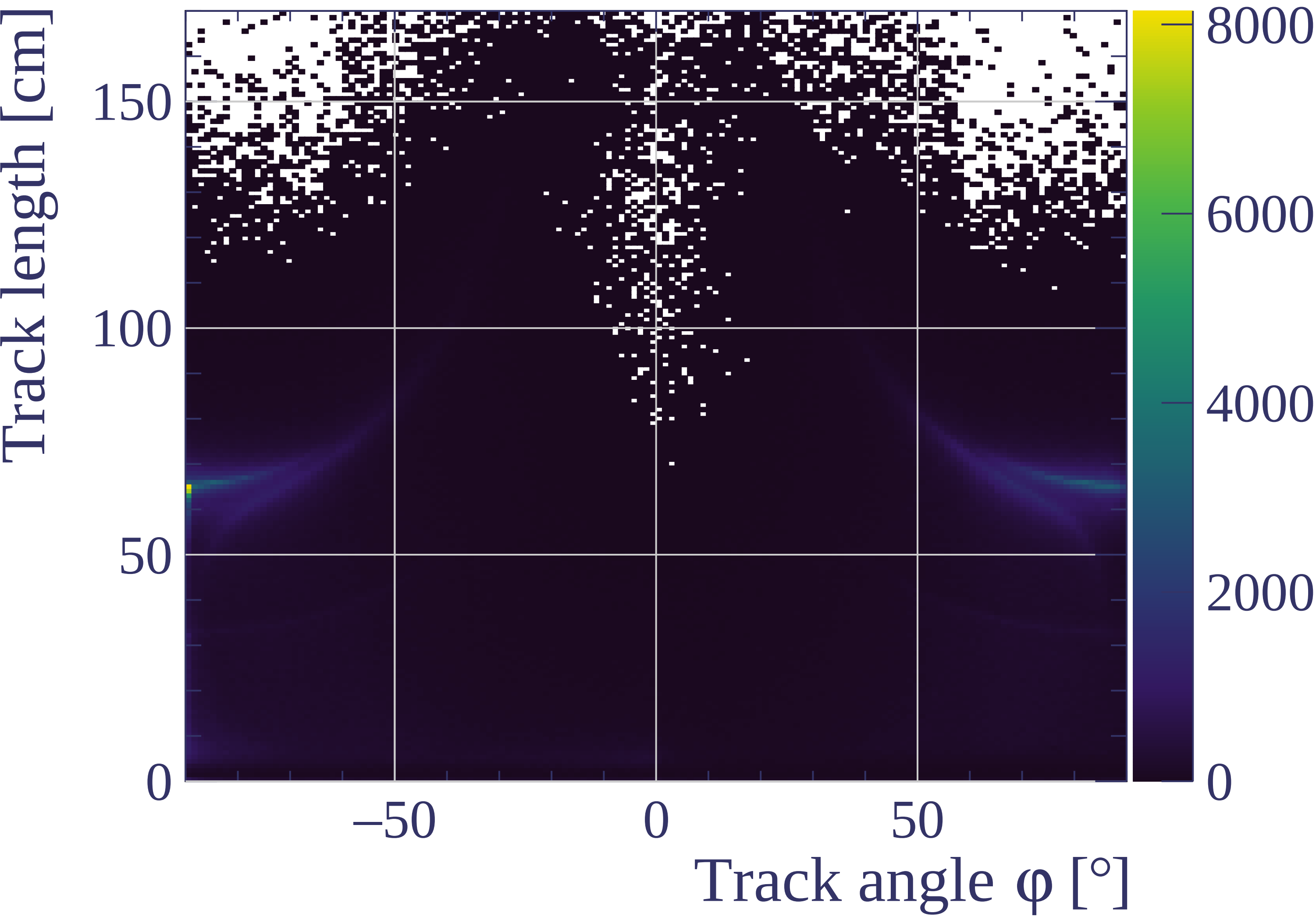
dE/dx resolution [%]
12
10
8
6
4
−50
0
50
Track angle φ [°]
WF
XP

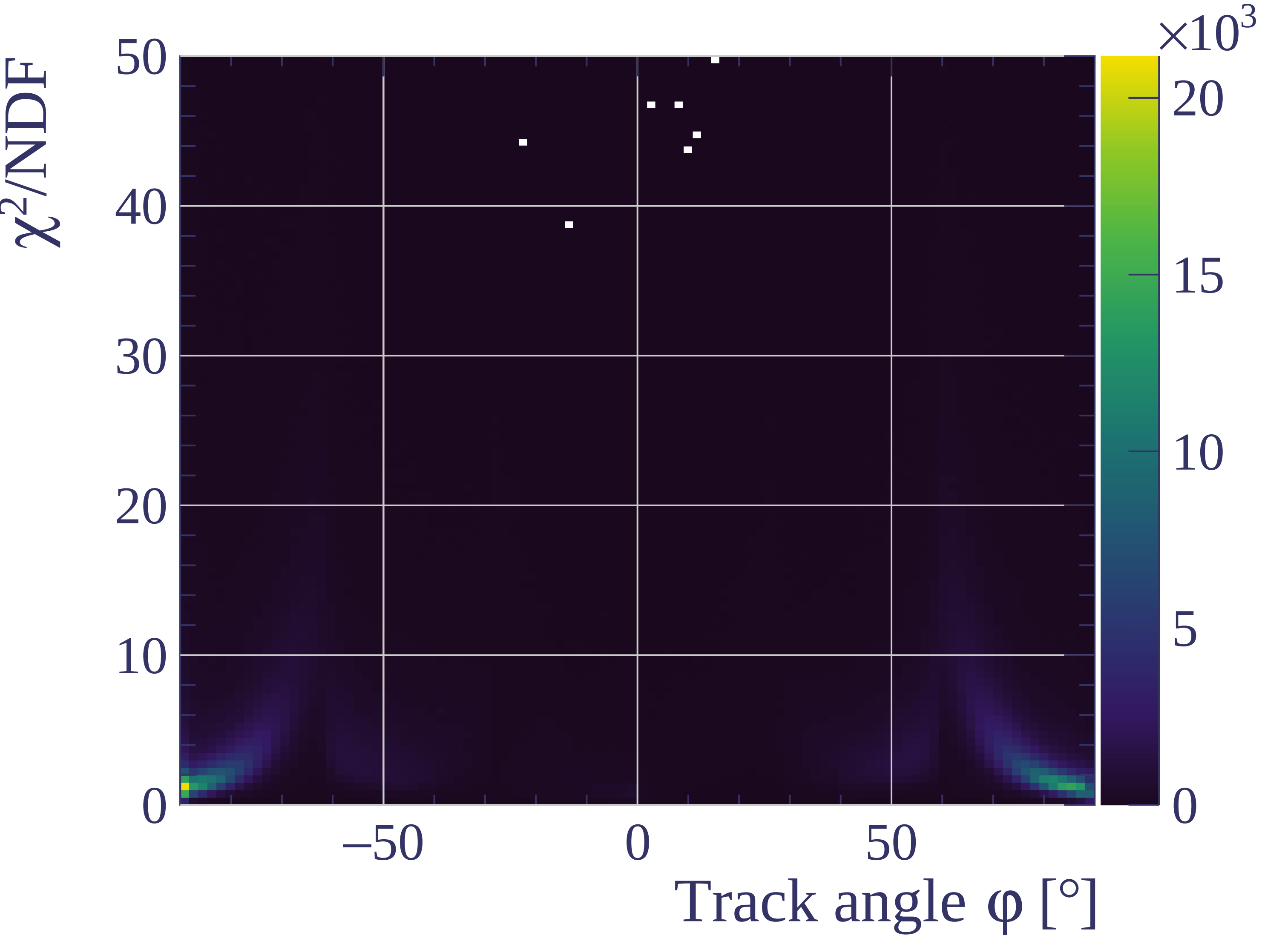


WF
XP
600
500
400
300
Mean dE/dx [ADC counts/cm]
−50
0
50
Track angle φ [°]

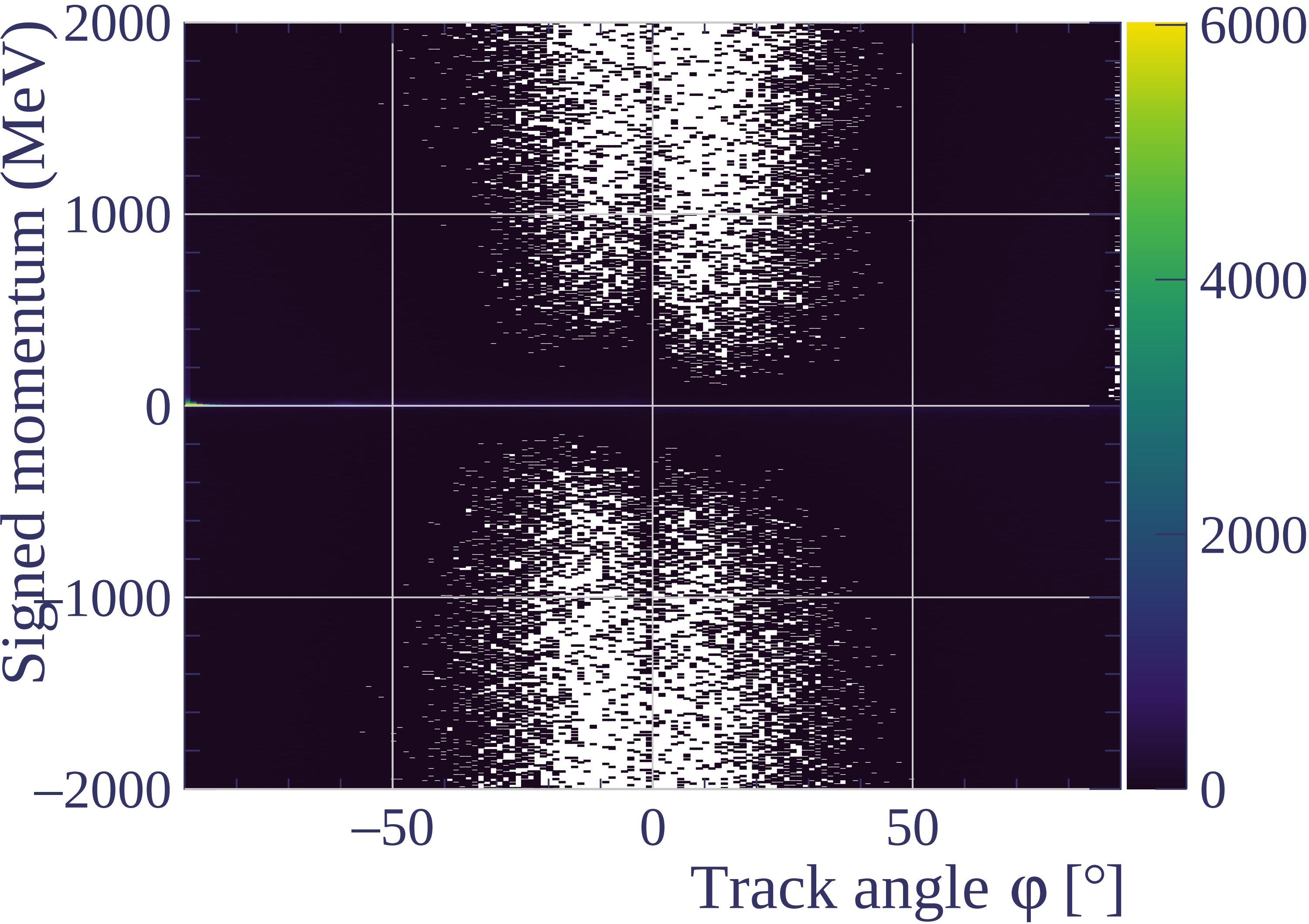


dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−50
0
50
Track angle φ [°]
800
600
400
200
0

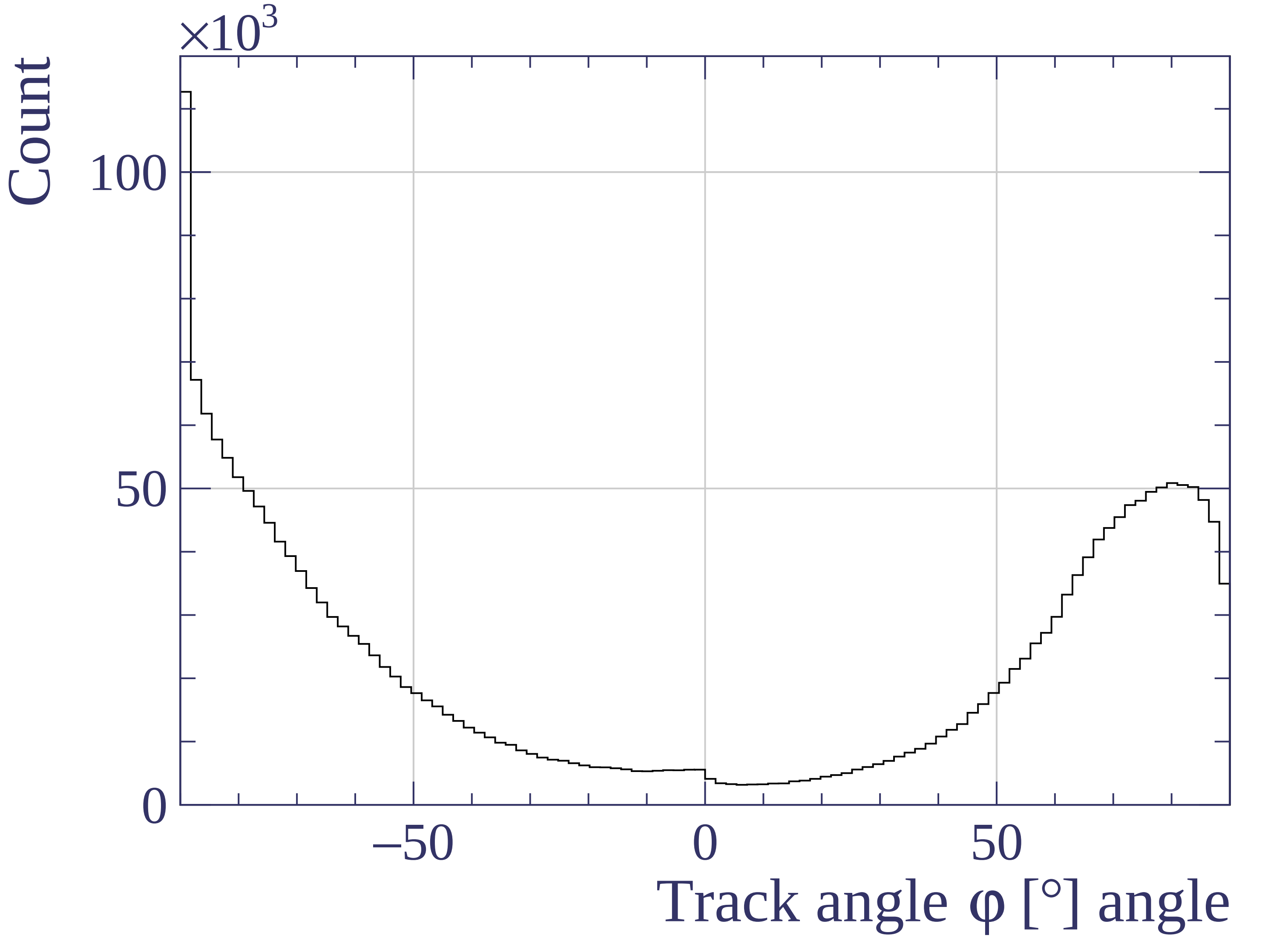

Track length [cm]
150
100
50
0
−50
0
50
Track angle φ [°]
8000
6000
4000
2000
0



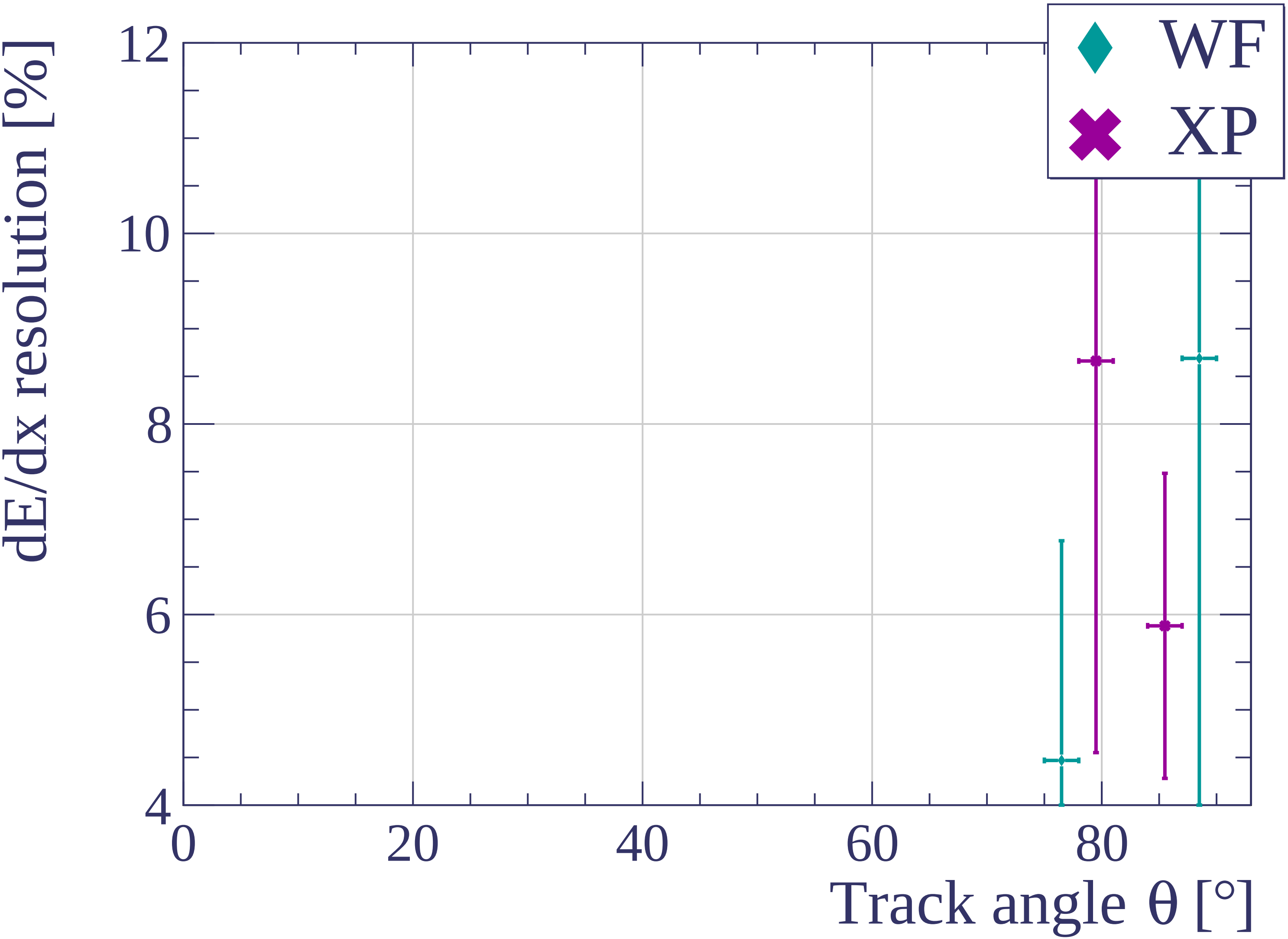
χ^2/NDF
50
40
30
20
10
0
−50
0
50
Track angle φ [°]
$\times 10^3$
20
15
10
5
0

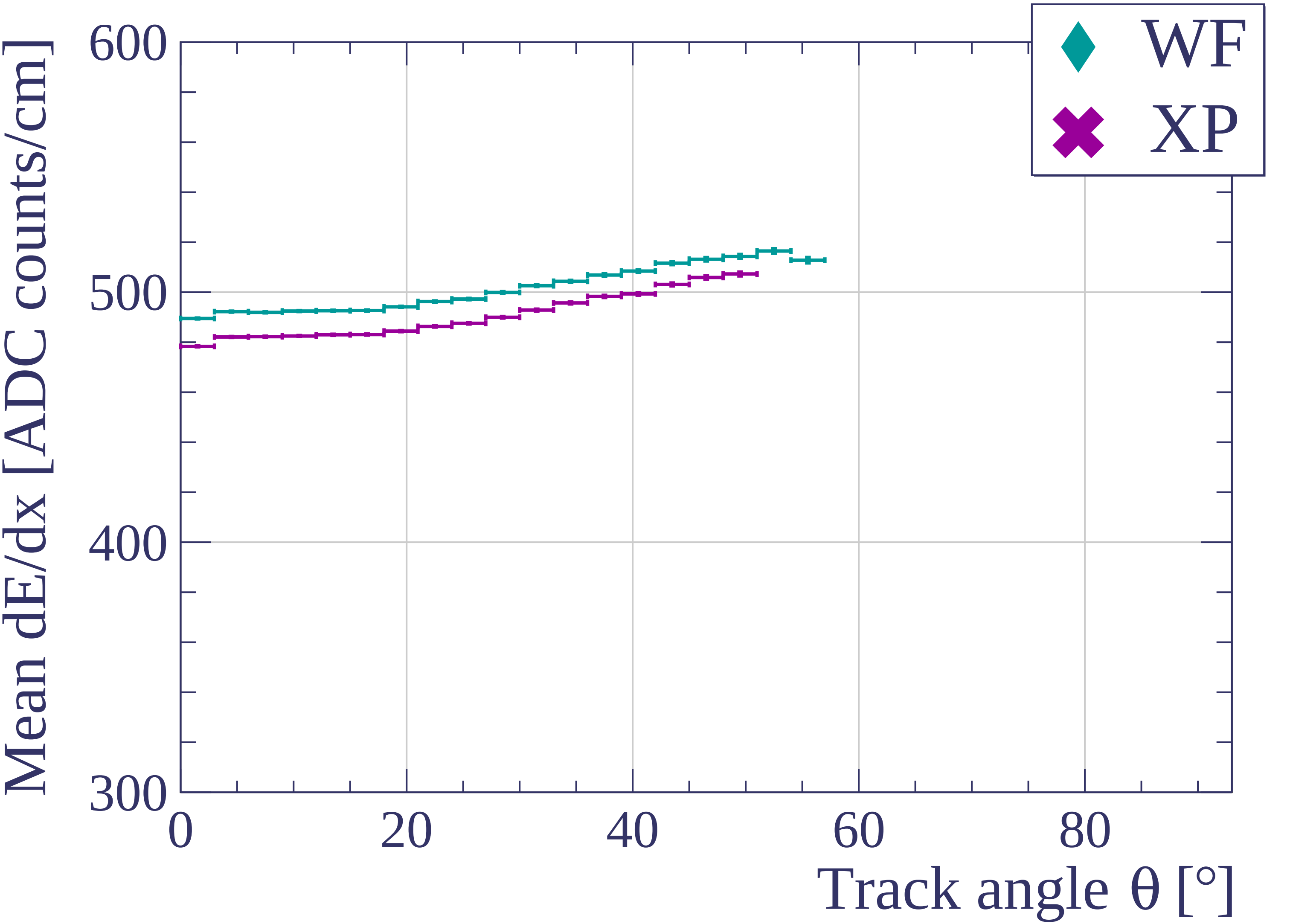


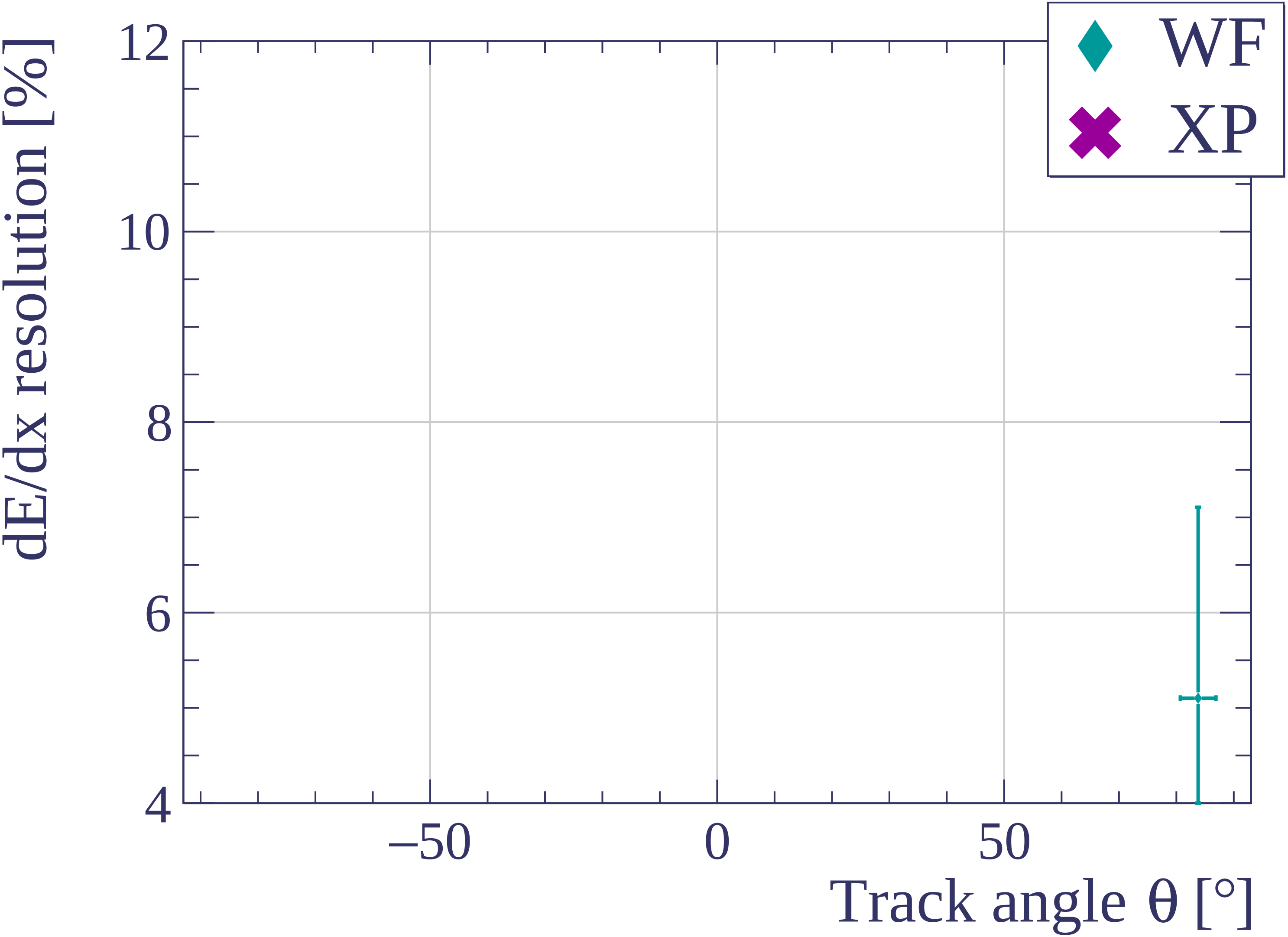
Signed momentum (MeV)
2000
1000
0
−1000
−2000
−50
0
50
Track angle φ [°]
6000
4000
2000
0



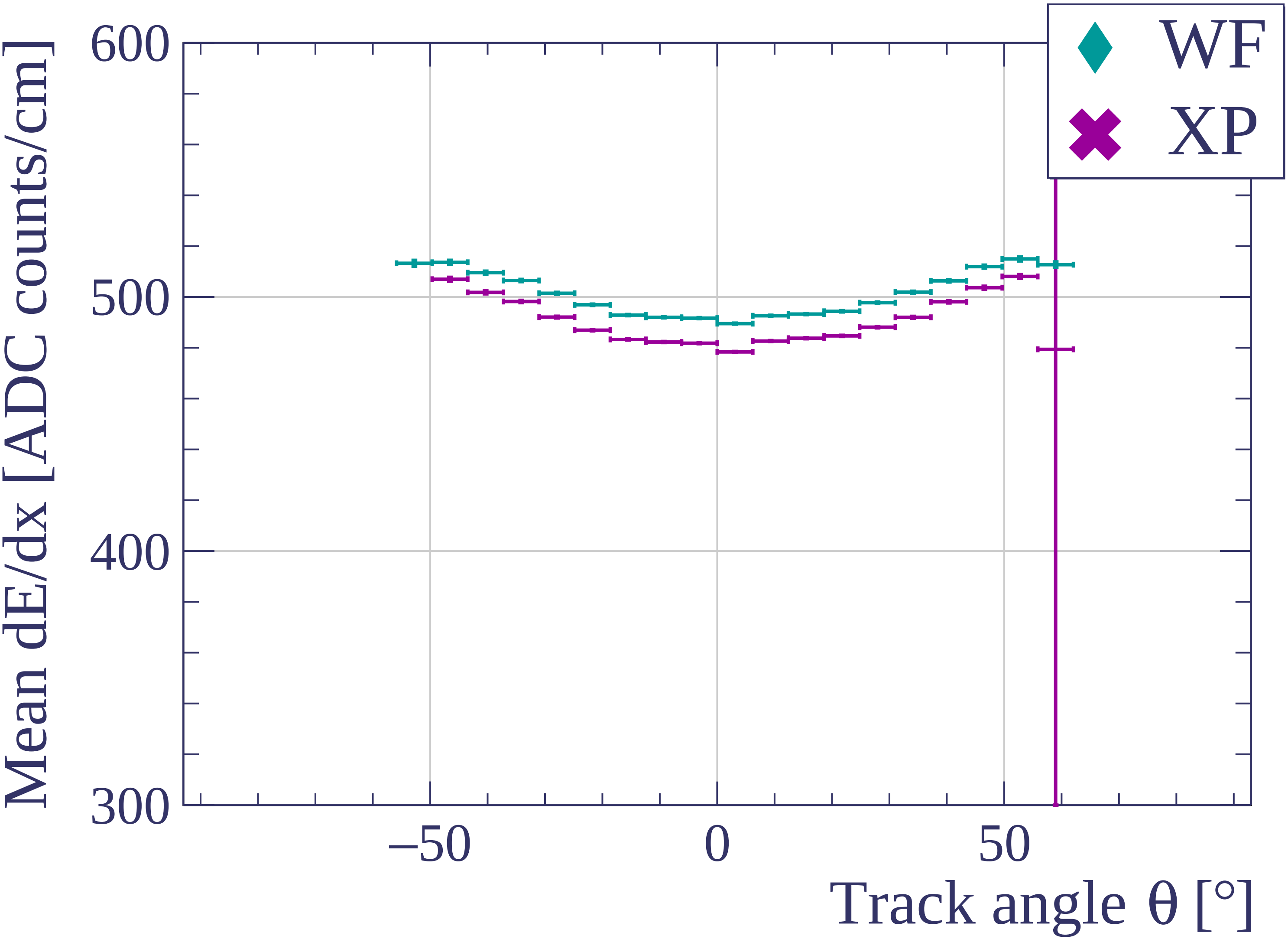
×10^3
Count
100
50
0
−50
0
50
Track angle φ [°] angle


WF
XP
dE/dx resolution [%]
Track angle θ [°]
4
6
8
10
12
0
20
40
60
80

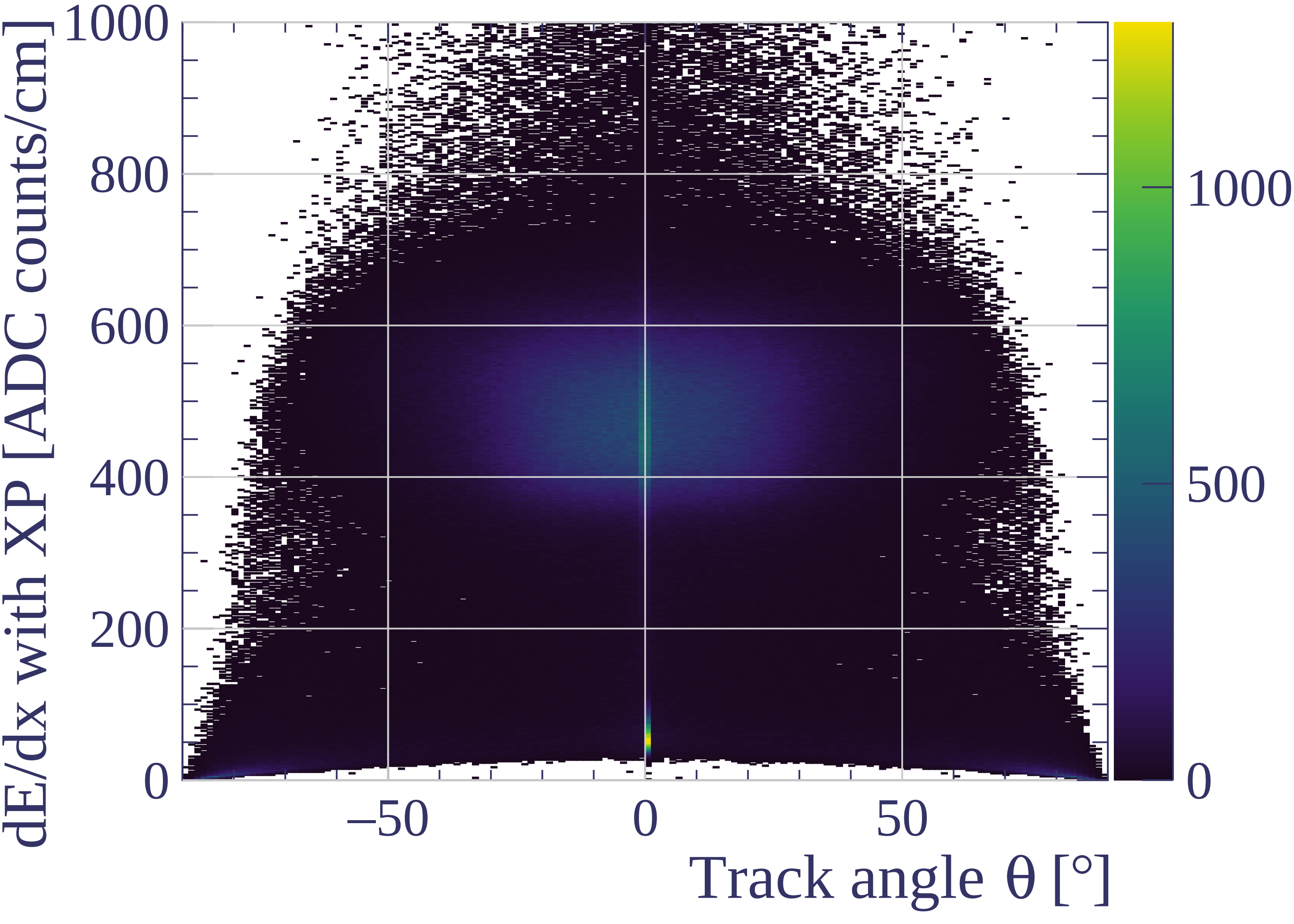

WF
XP
Mean dE/dx [ADC counts/cm]
Track angle θ [°]
300
400
500
600
0
20
40
60
80



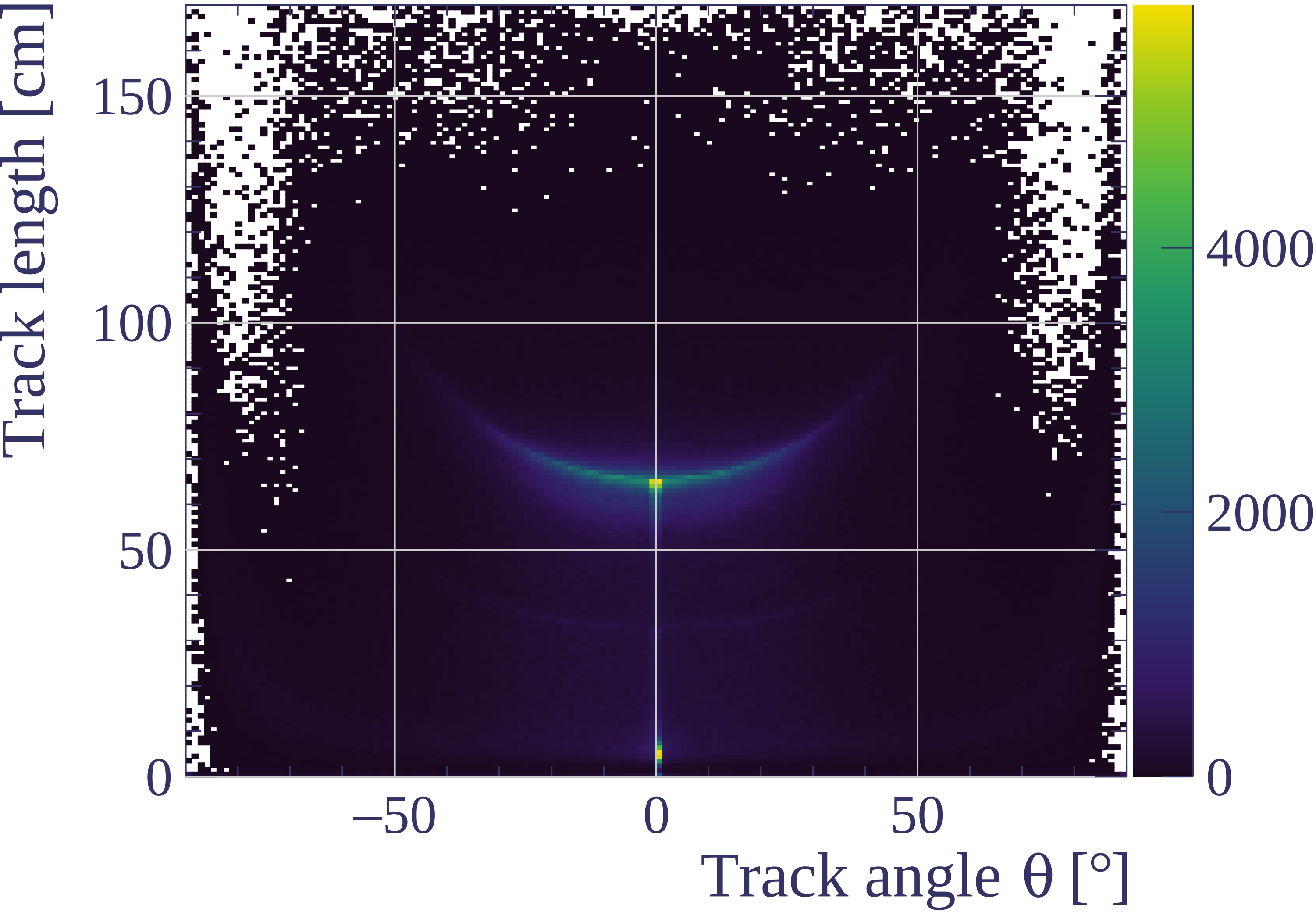
WF
XP
12
10
8
6
4
−50
0
50
dE/dx resolution [%]
Track angle θ [°]



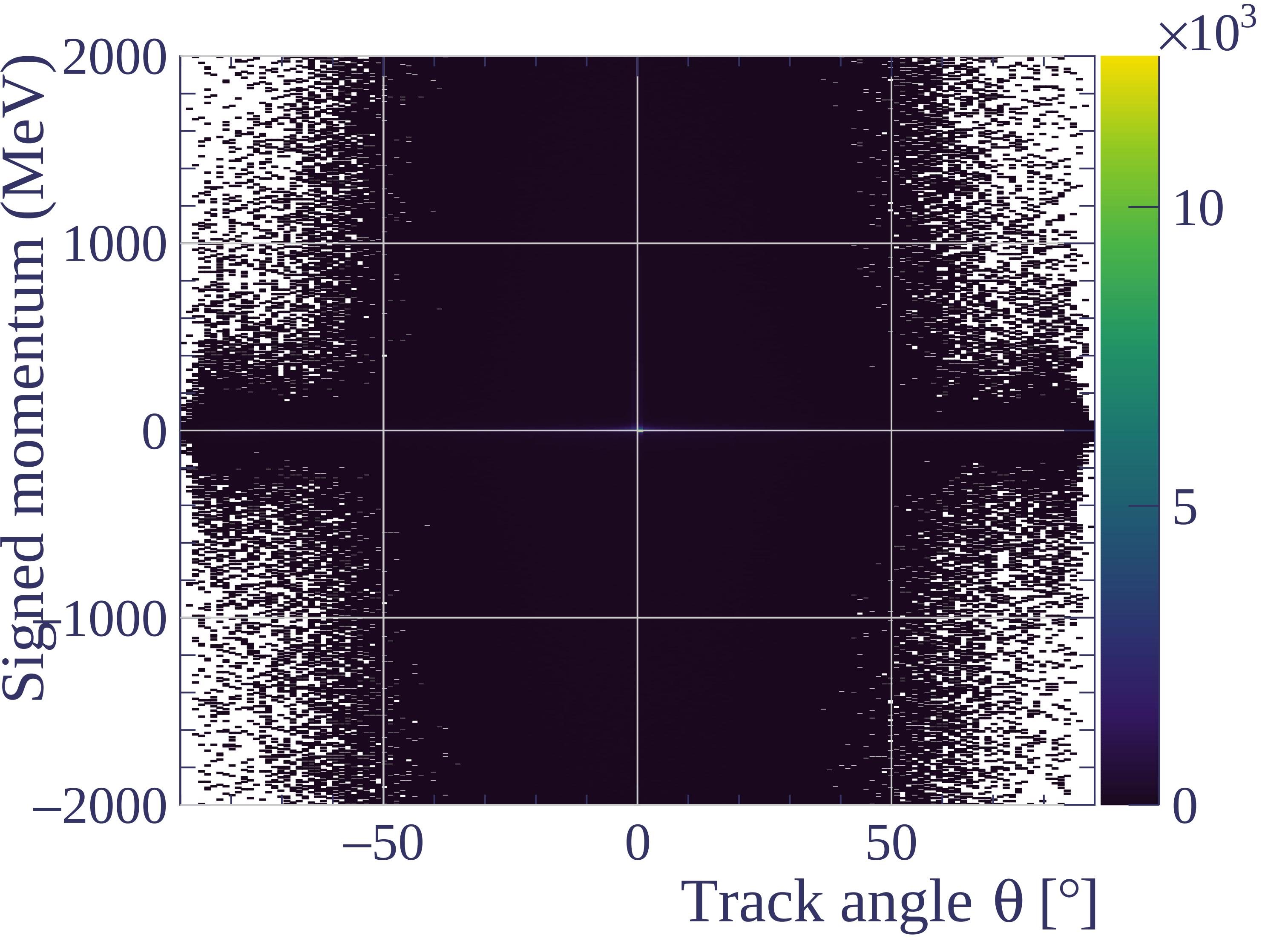
WF
XP
600
500
400
300
Mean dE/dx [ADC counts/cm]
−50
0
50
Track angle θ [°]



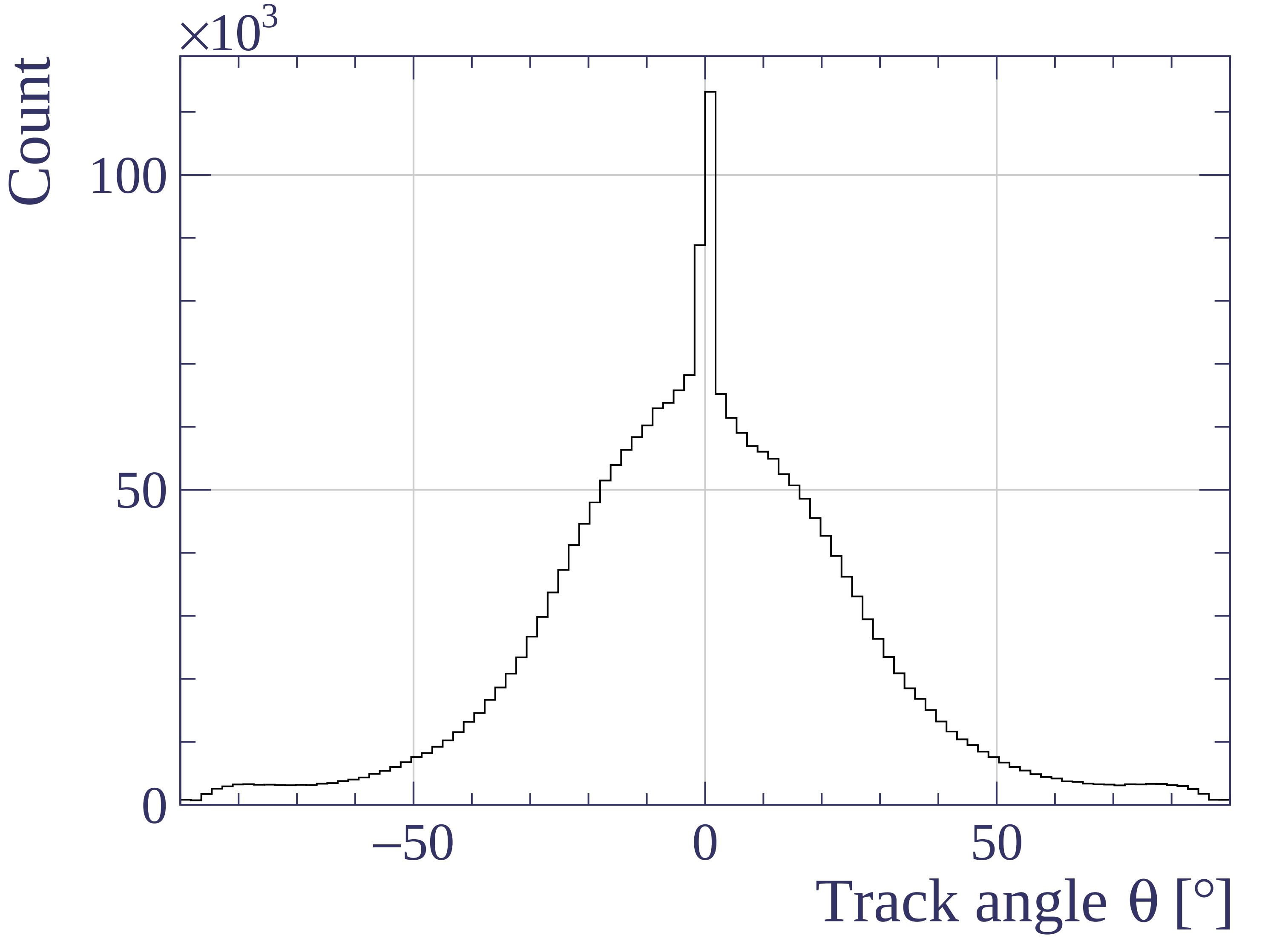
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−50
0
50
Track angle θ [°]
1000
500
0



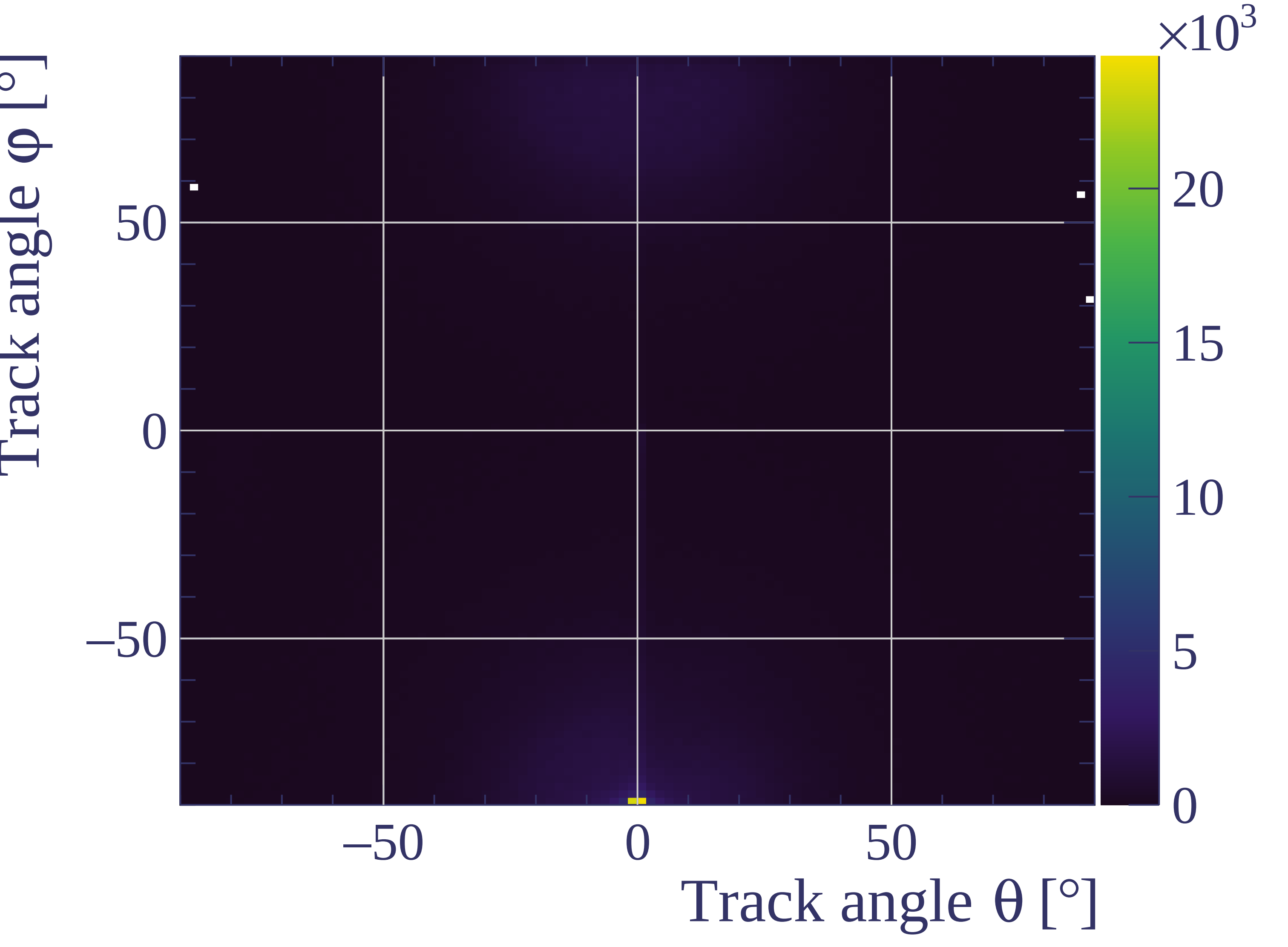
Track length [cm]
150
100
50
0
−50
0
50
Track angle θ [°]
4000
2000
0



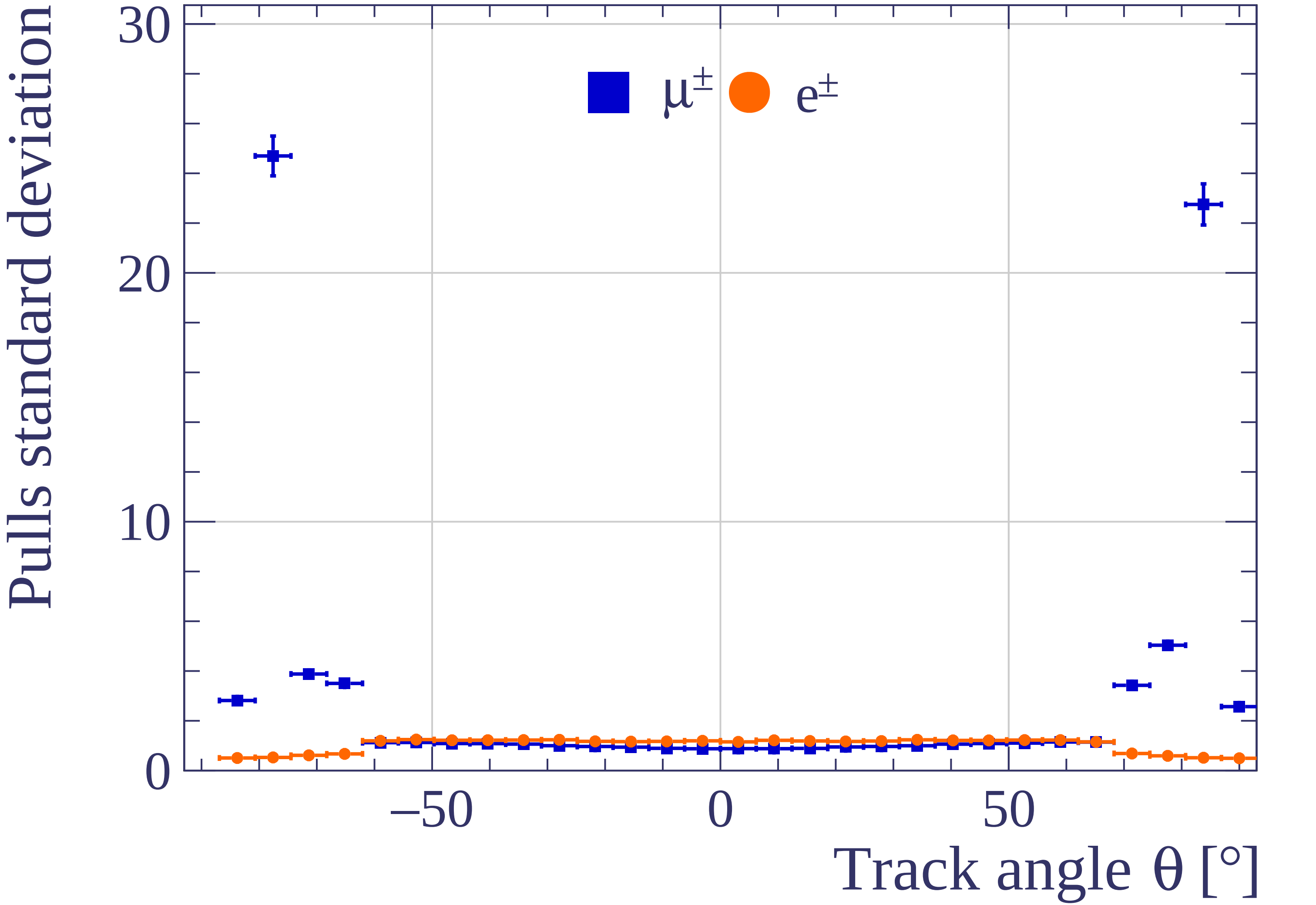
×10^3
2000
1000
0
−1000
−2000
−50
0
50
10
5
0
Signed momentum (MeV)
Track angle θ [°]



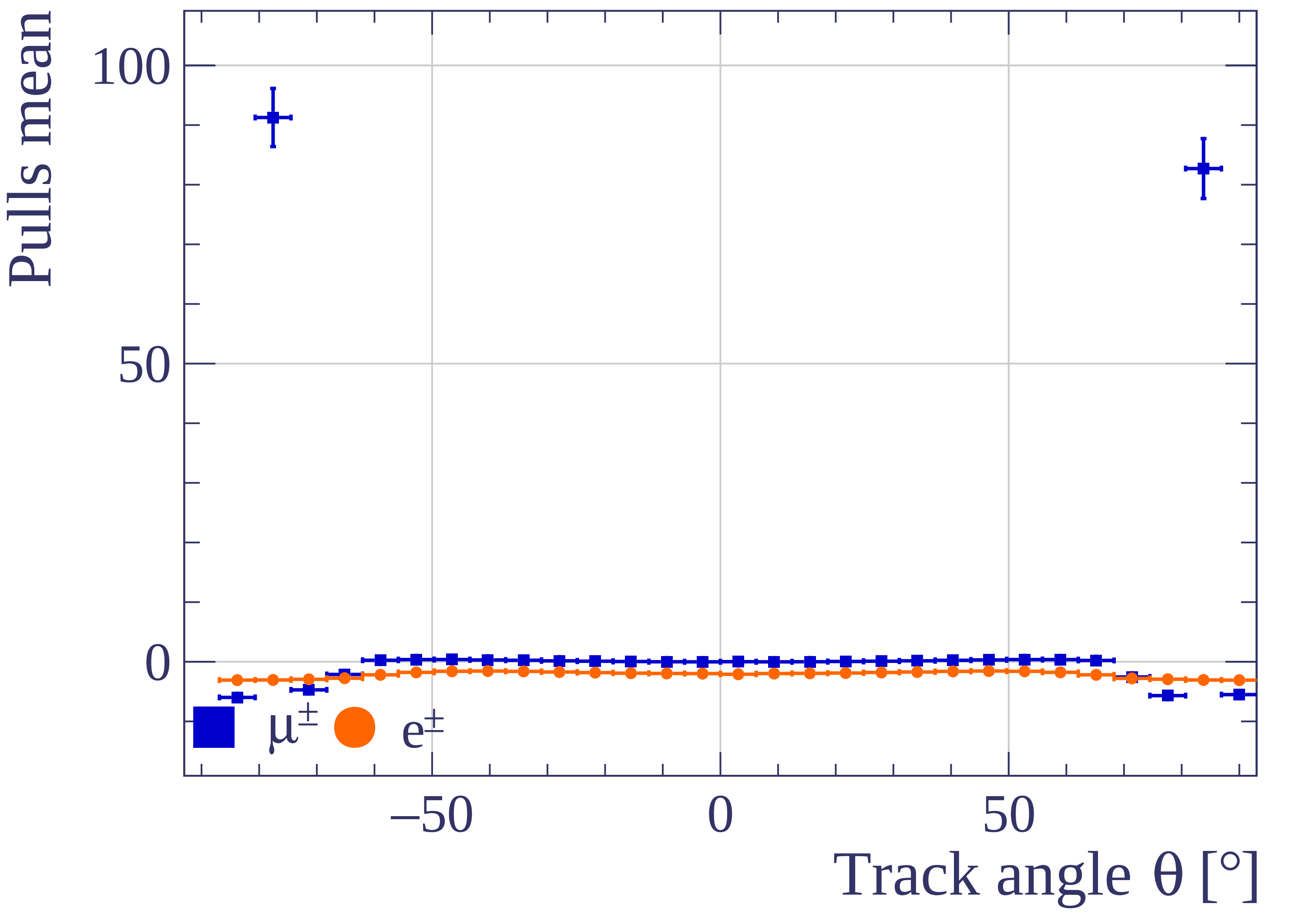
×10^3
Count
100
50
0
−50
0
50
Track angle θ [°]



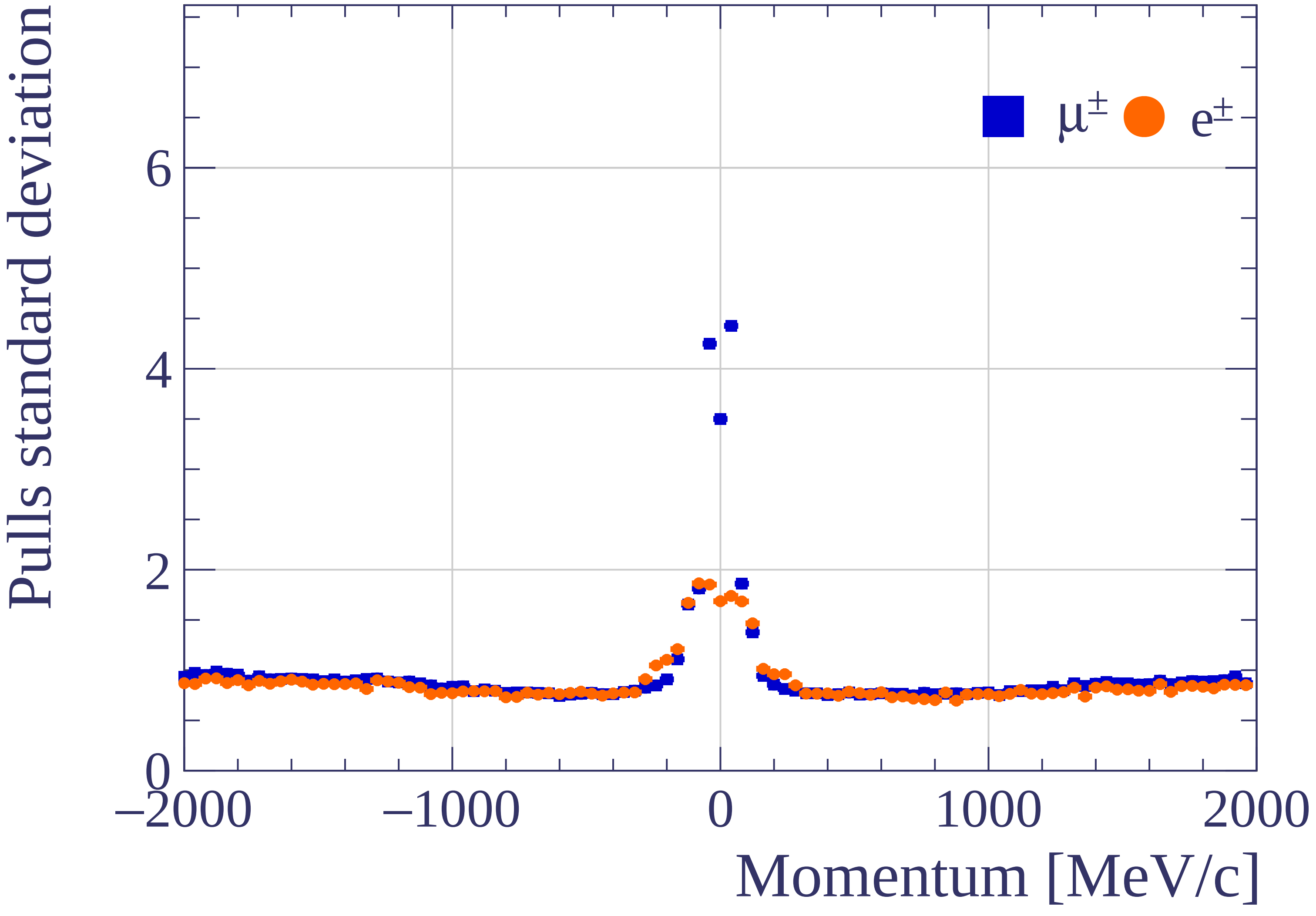
Track angle φ [°]
50
0
−50
−50
0
50
Track angle θ [°]
×10^3
20
15
10
5
0



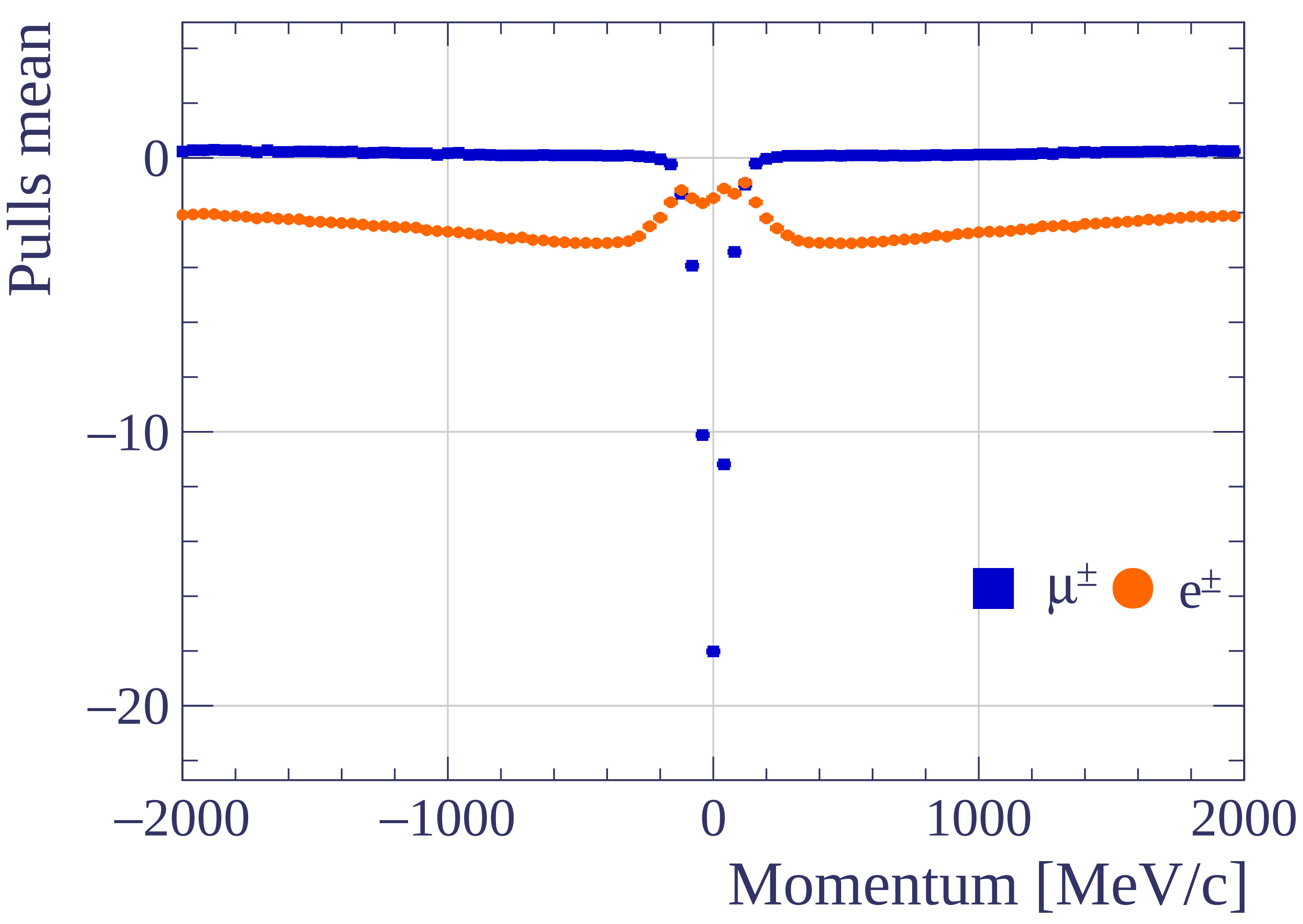
Pulls standard deviation
Track angle θ [°]
μ±
e±
0
10
20
30
−50
0
50



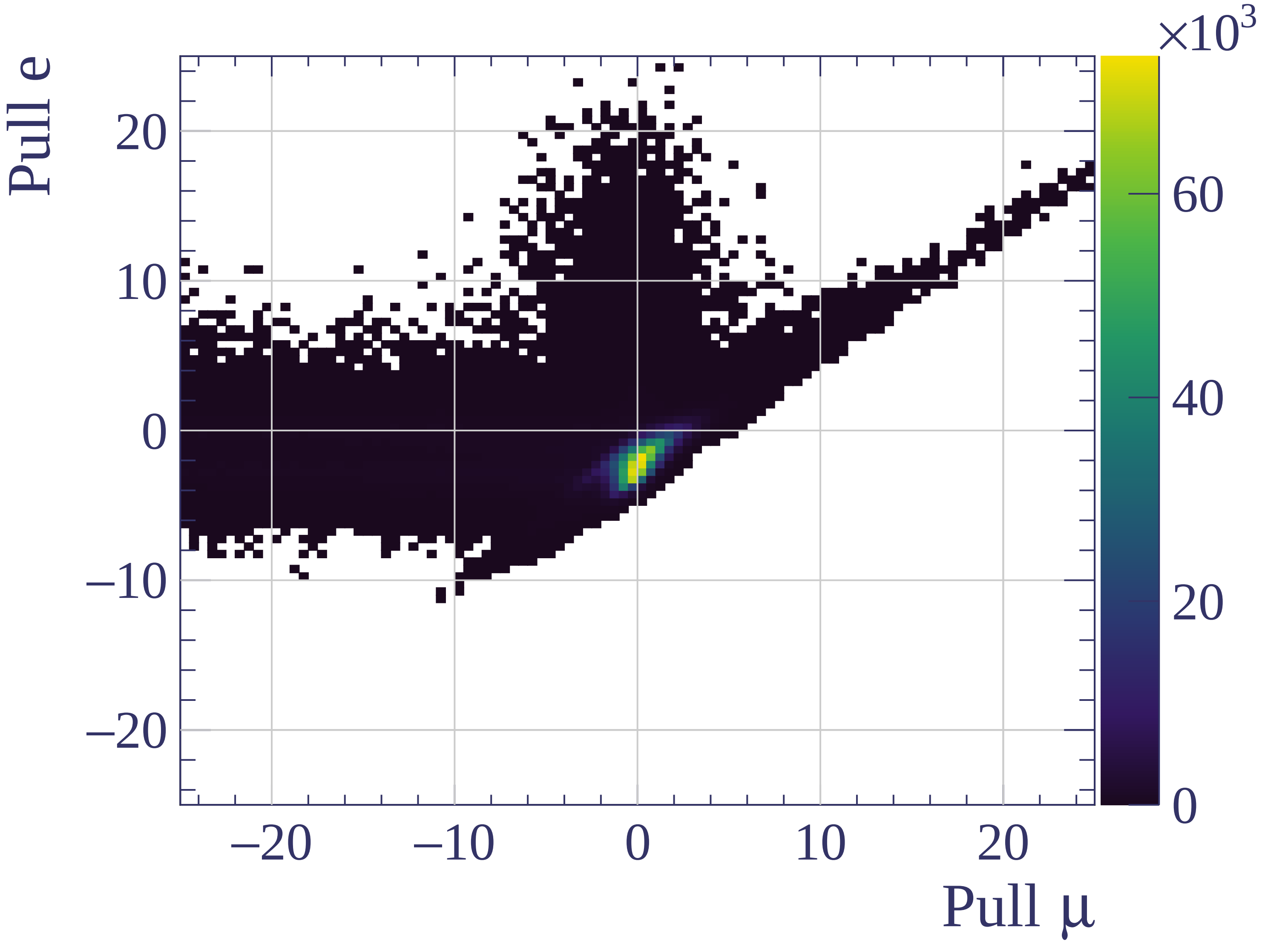
Pulls mean
100
50
0
−50
0
50
Track angle θ [°]
μ±
e±



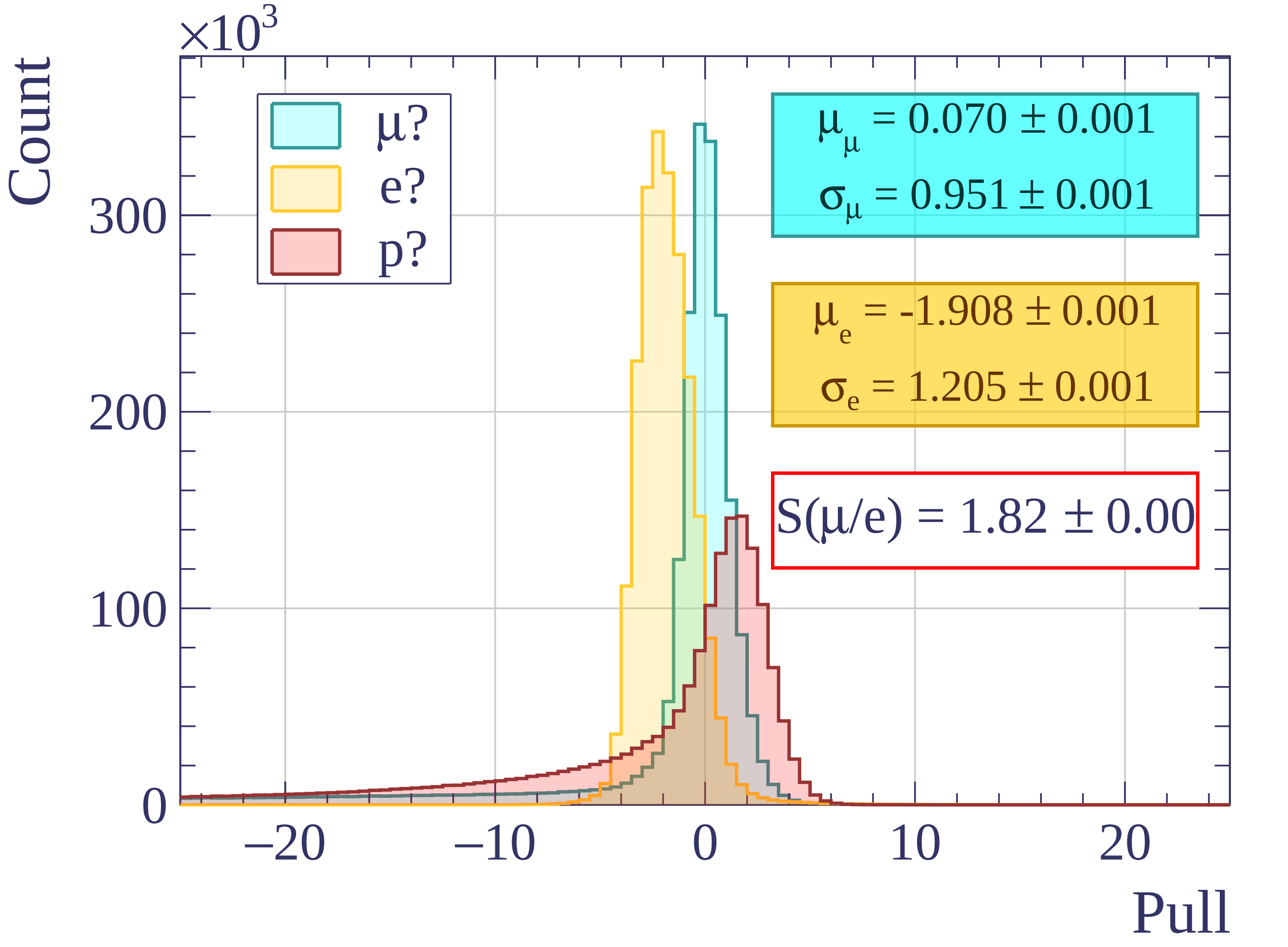
Pulls standard deviation
Momentum [MeV/c]
$\mu^{\pm}$
$e^{\pm}$
0
2
4
6
−2000
−1000
0
1000
2000



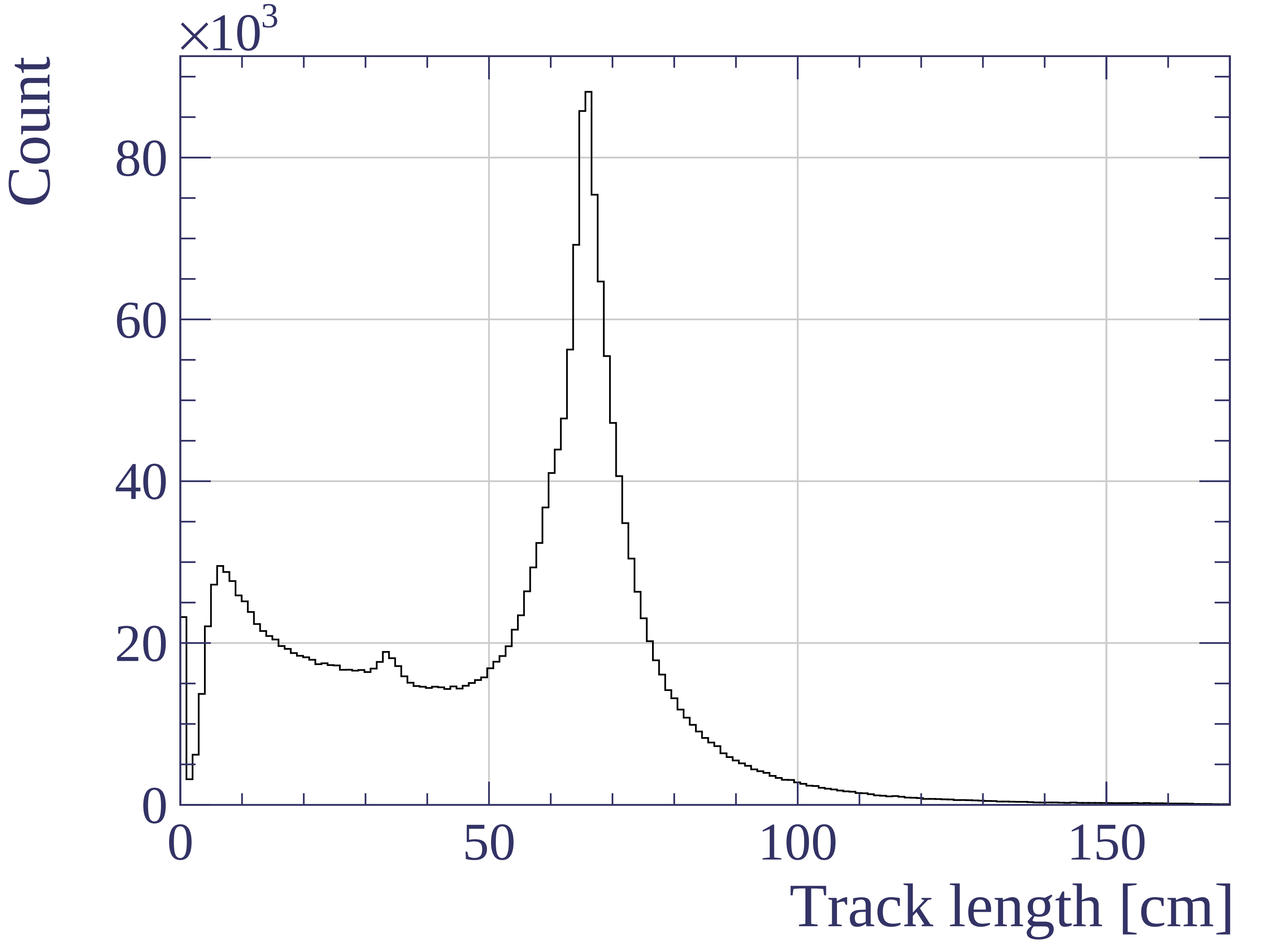
Pulls mean
0
−10
−20
−2000
−1000
0
1000
2000
Momentum [MeV/c]
$\mu^{\pm}$
$e^{\pm}$



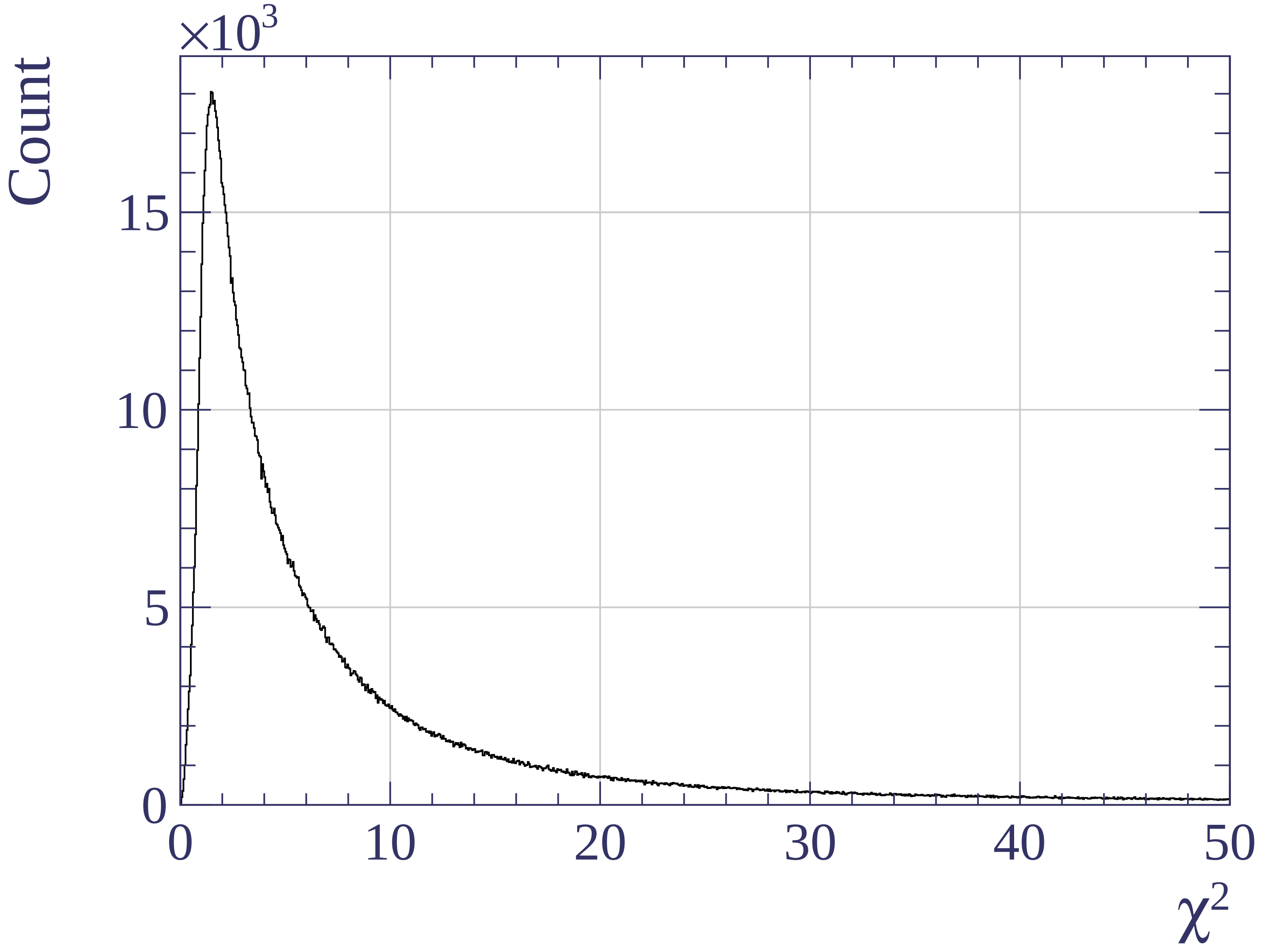
Pull e
20
10
0
−10
−20
−20
−10
0
10
20
Pull μ
×10^3
60
40
20
0



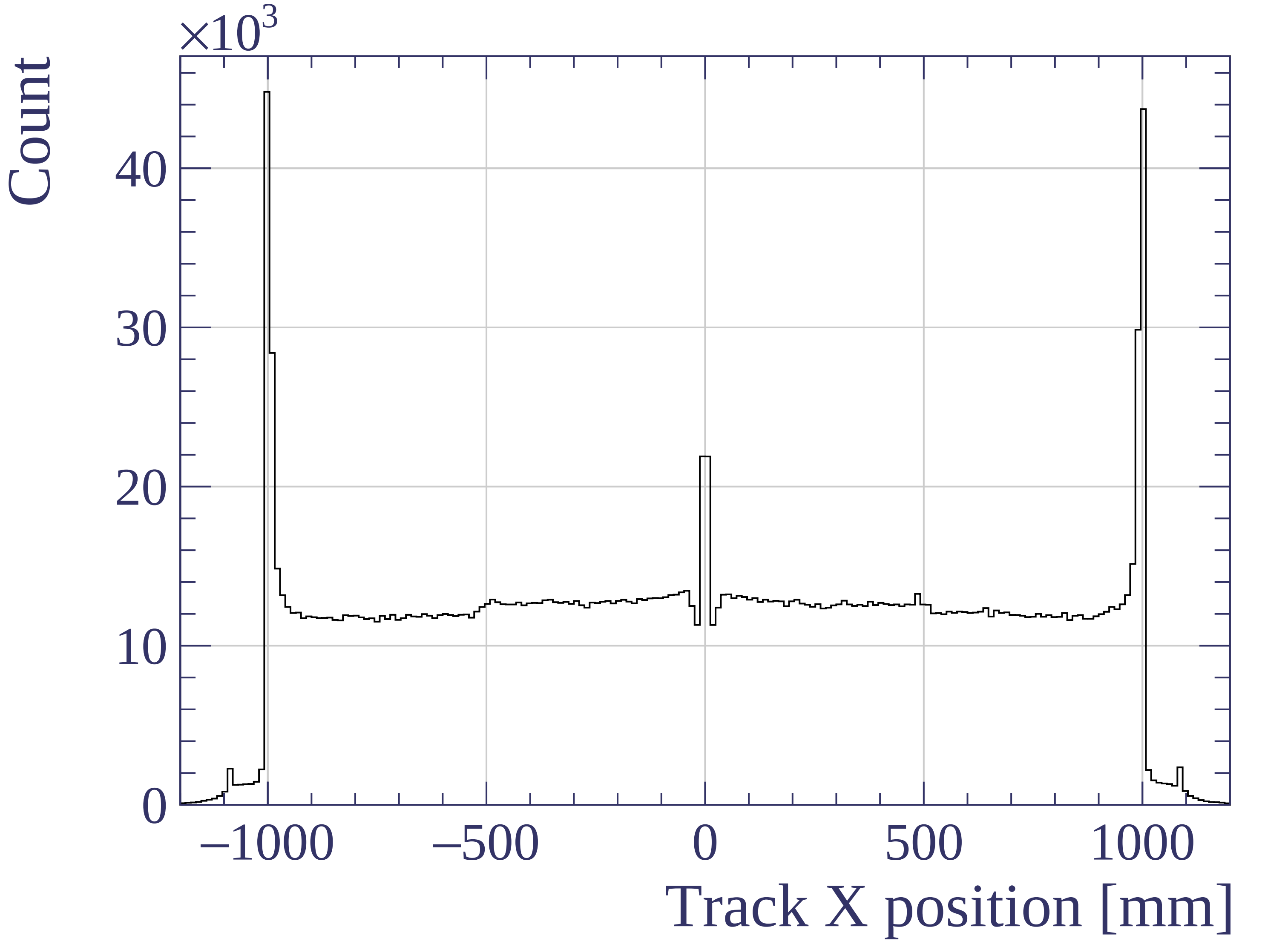
×10^3
Count
300
200
100
0
−20
−10
0
10
20
Pull
μ?
e?
p?
μ_μ = 0.070 ± 0.001
σ_μ = 0.951 ± 0.001
μ_e = -1.908 ± 0.001
σ_e = 1.205 ± 0.001
S(μ/e) = 1.82 ± 0.00



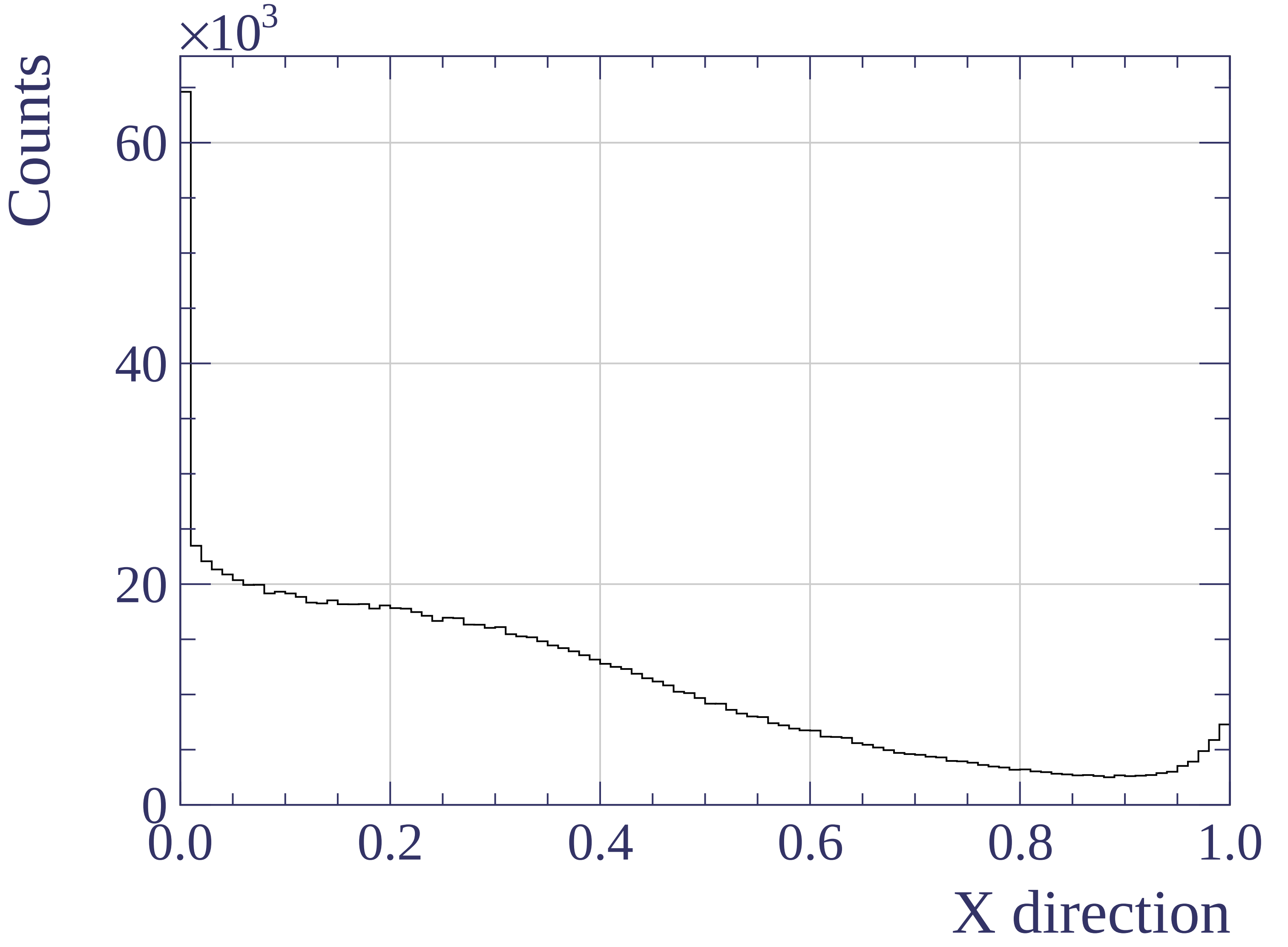
×10^3
Count
80
60
40
20
0
0
50
100
150
Track length [cm]



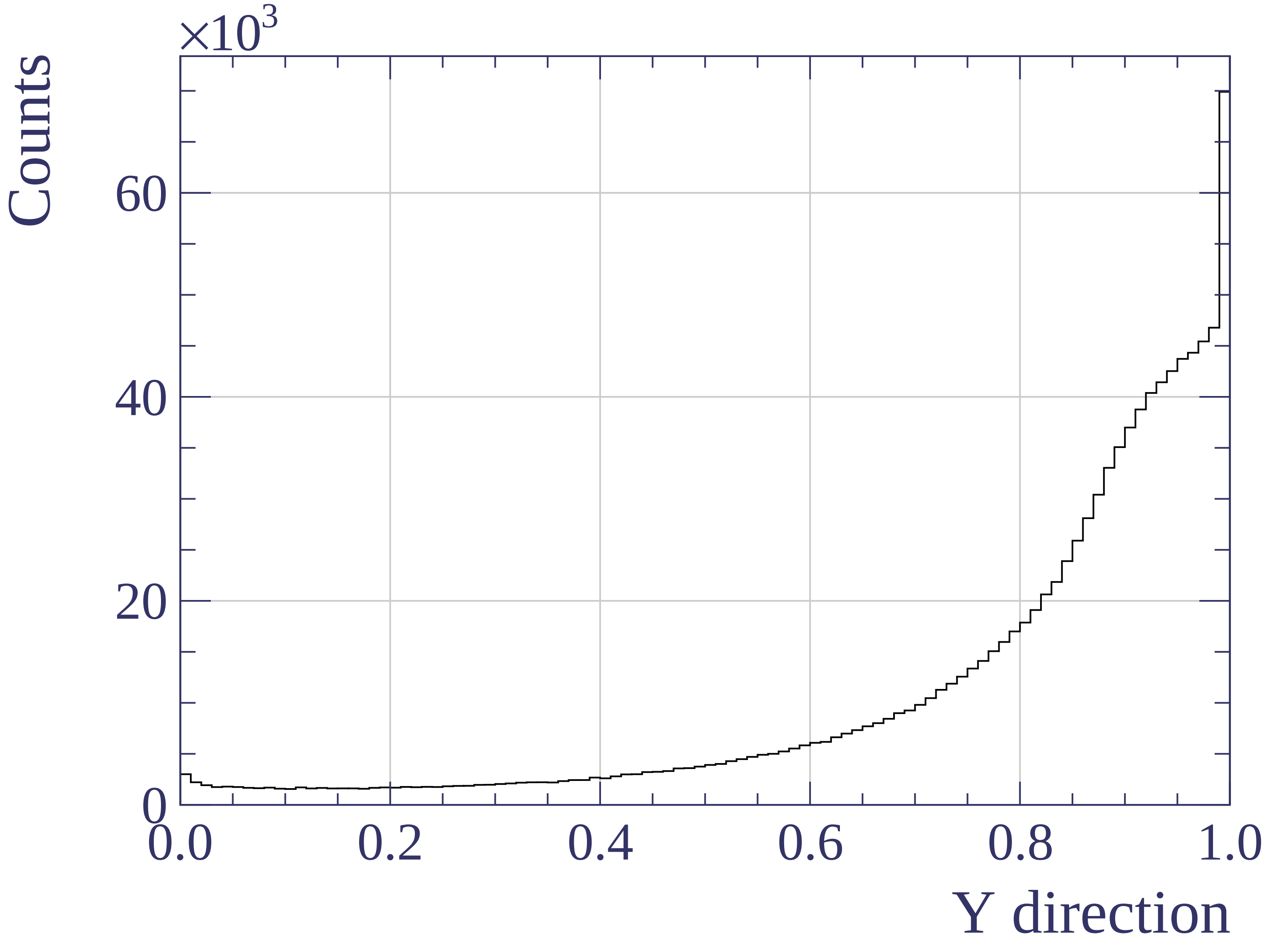
$\times 10^3$
Count
15
10
5
0
0
10
20
30
40
50
χ^2



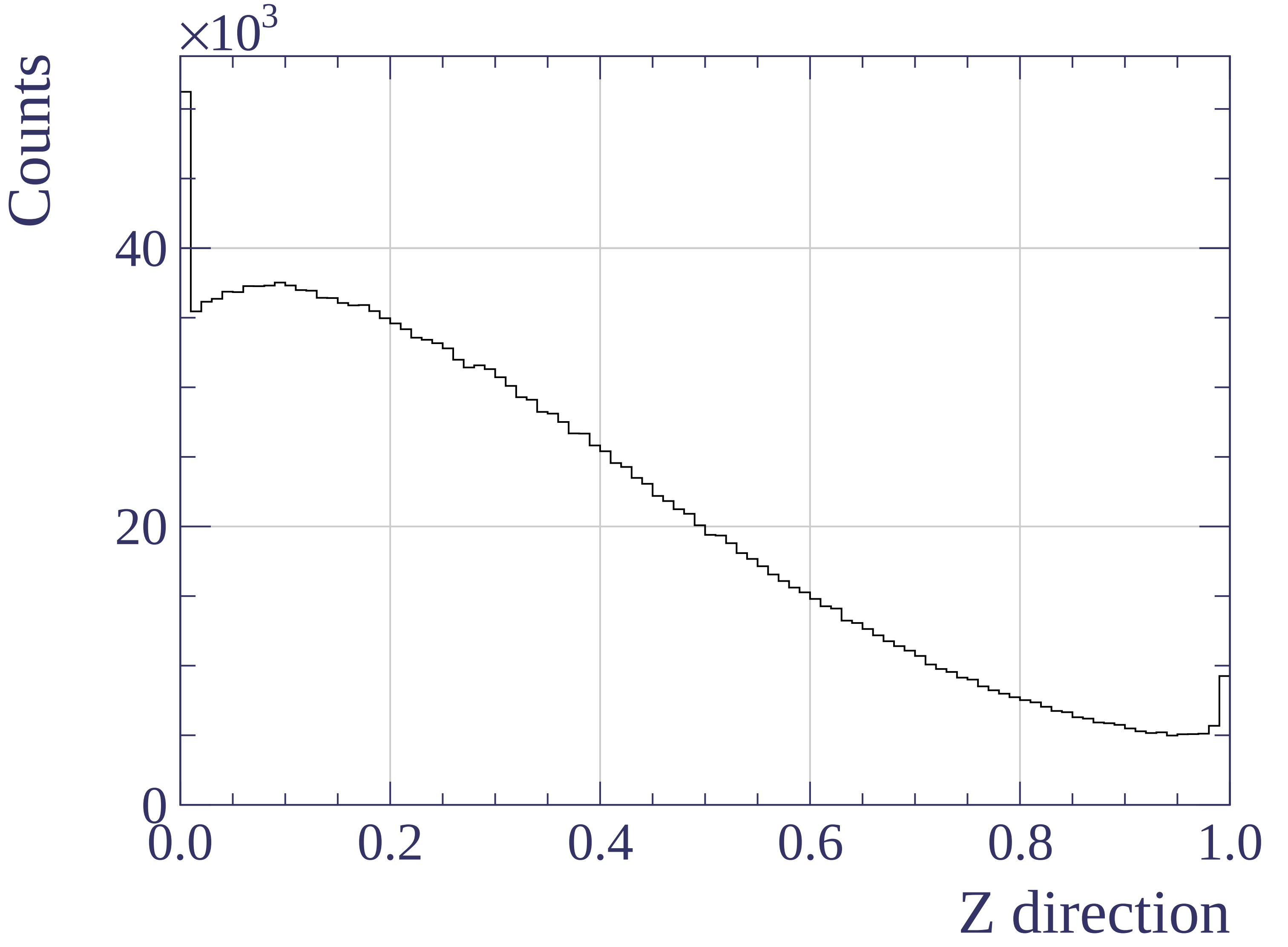
×10^3
Count
0
10
20
30
40
−1000
−500
0
500
1000
Track X position [mm]

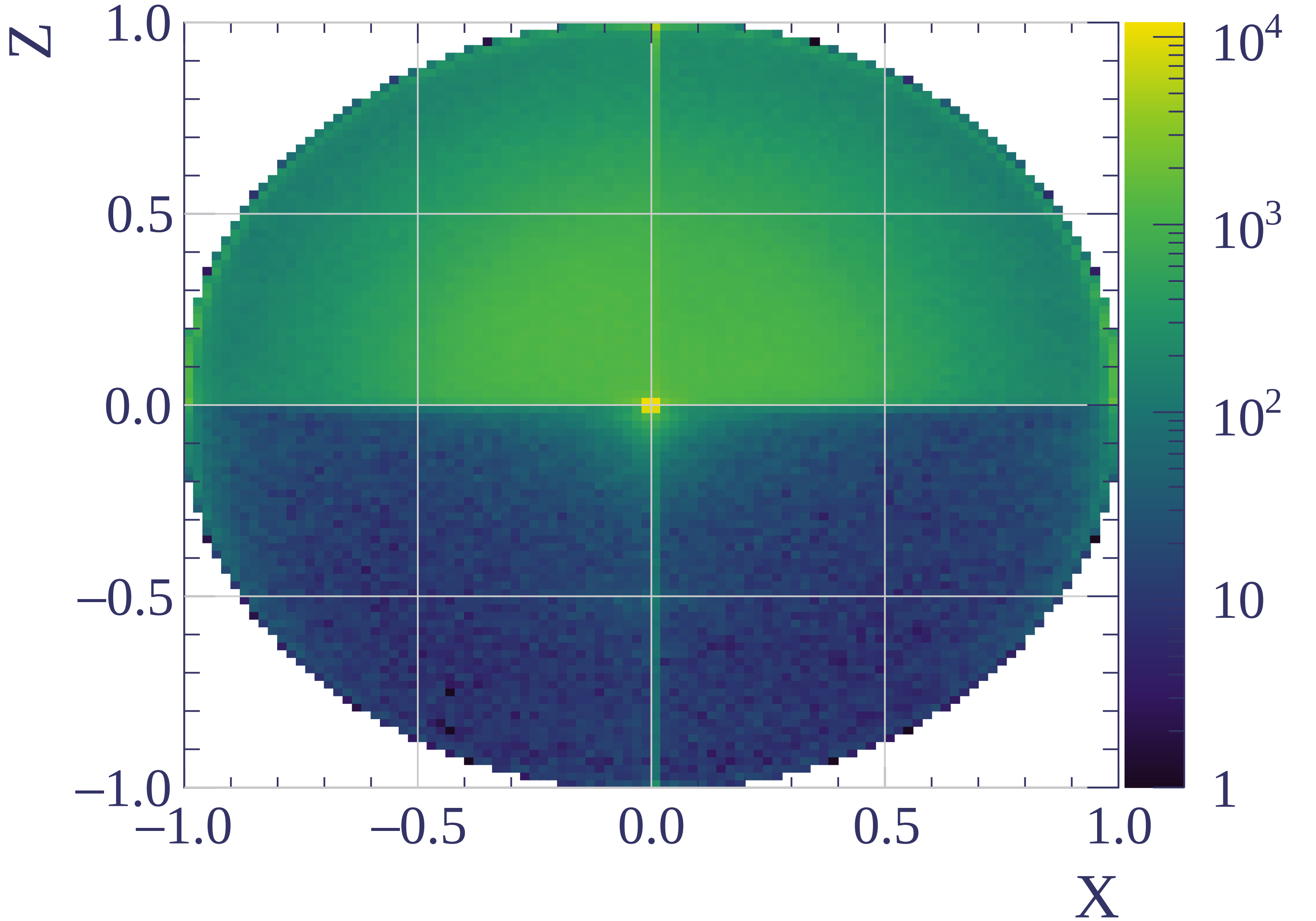


×10^3
Counts
60
40
20
0
0.0
0.2
0.4
0.6
0.8
1.0
X direction

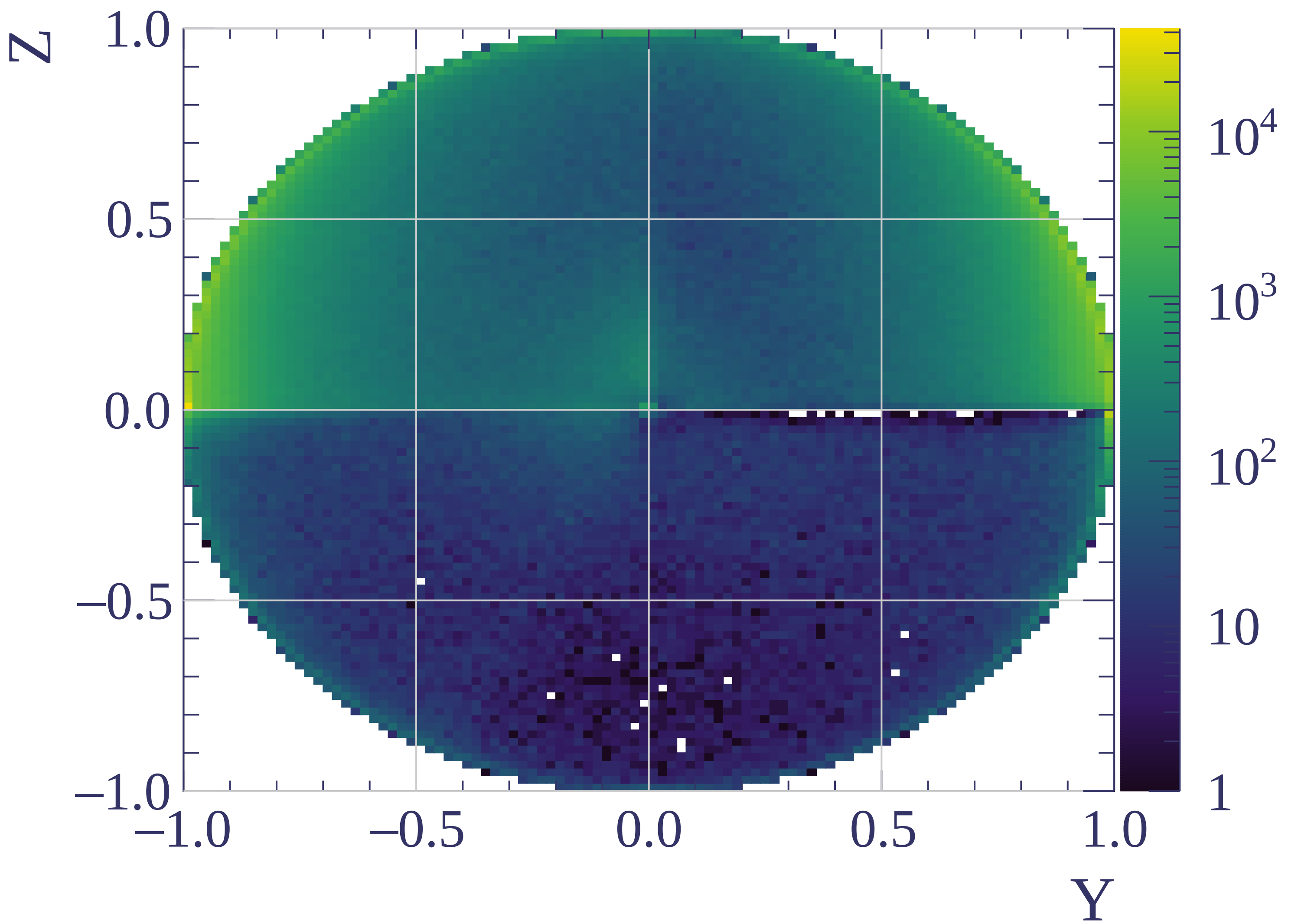


×10^3
Counts
60
40
20
0
0.0
0.2
0.4
0.6
0.8
1.0
Y direction

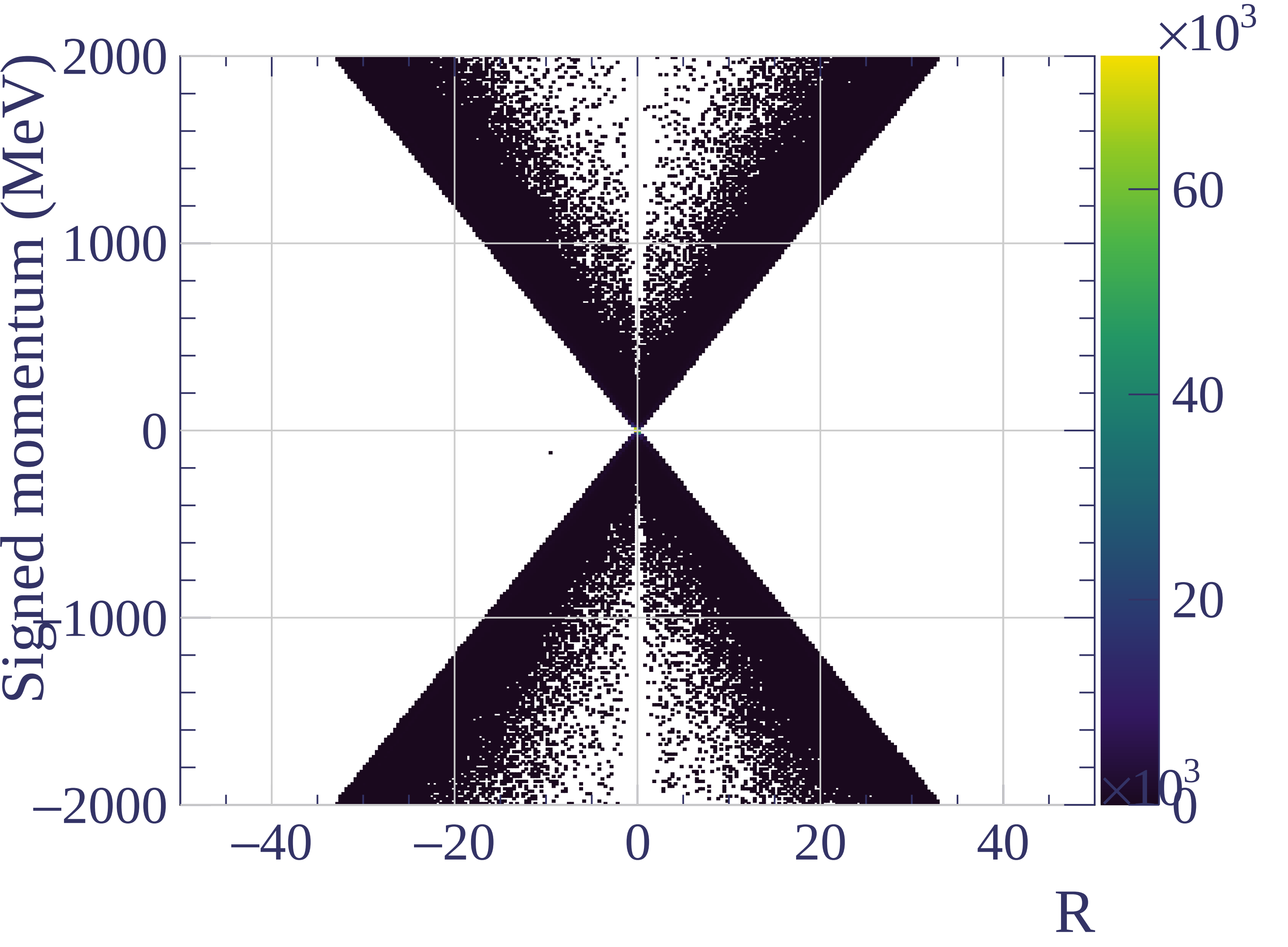

×10^3
Counts
40
20
0
0.0
0.2
0.4
0.6
0.8
1.0
Z direction



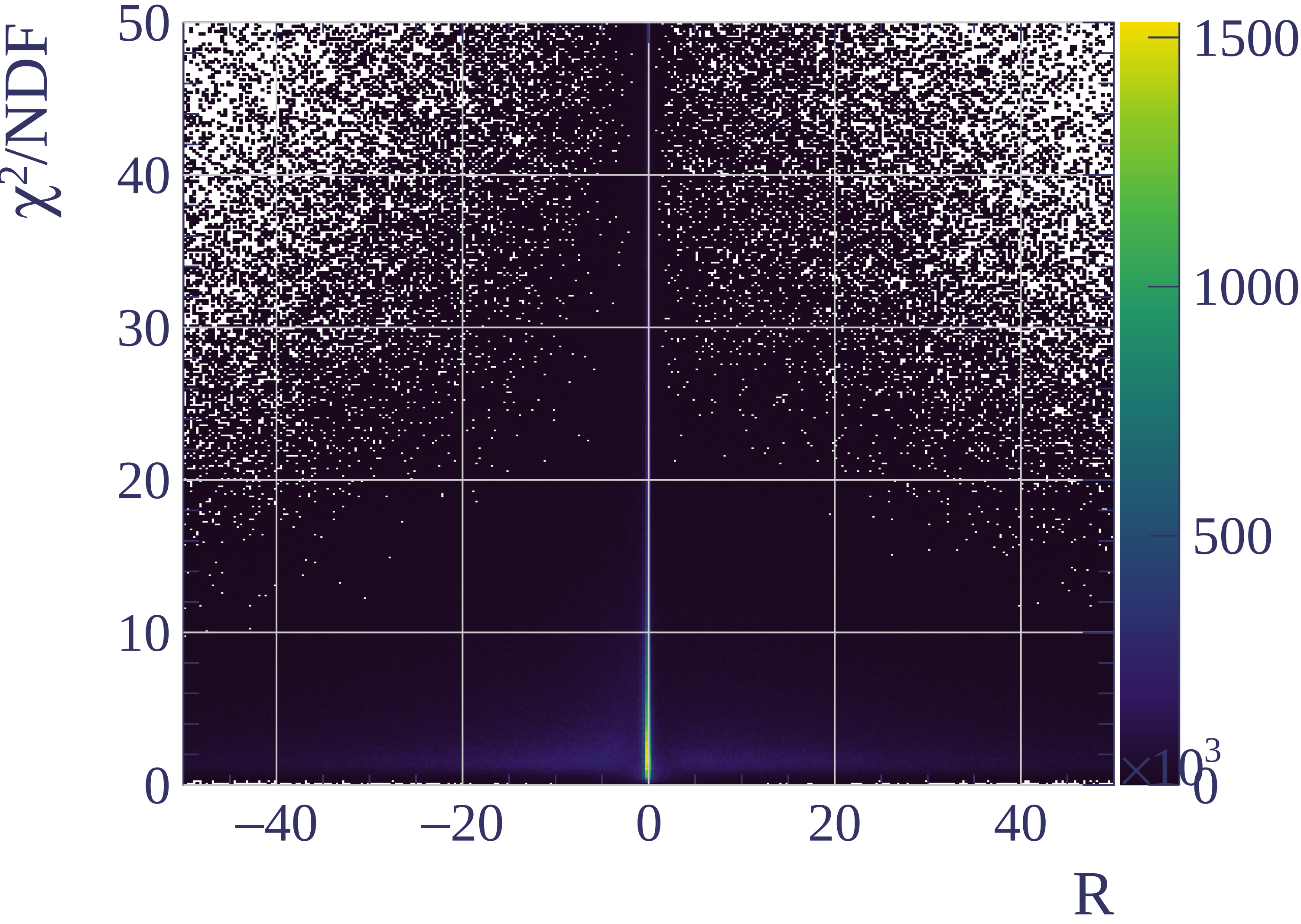
Z
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
X
10^4
10^3
10^2
10
1



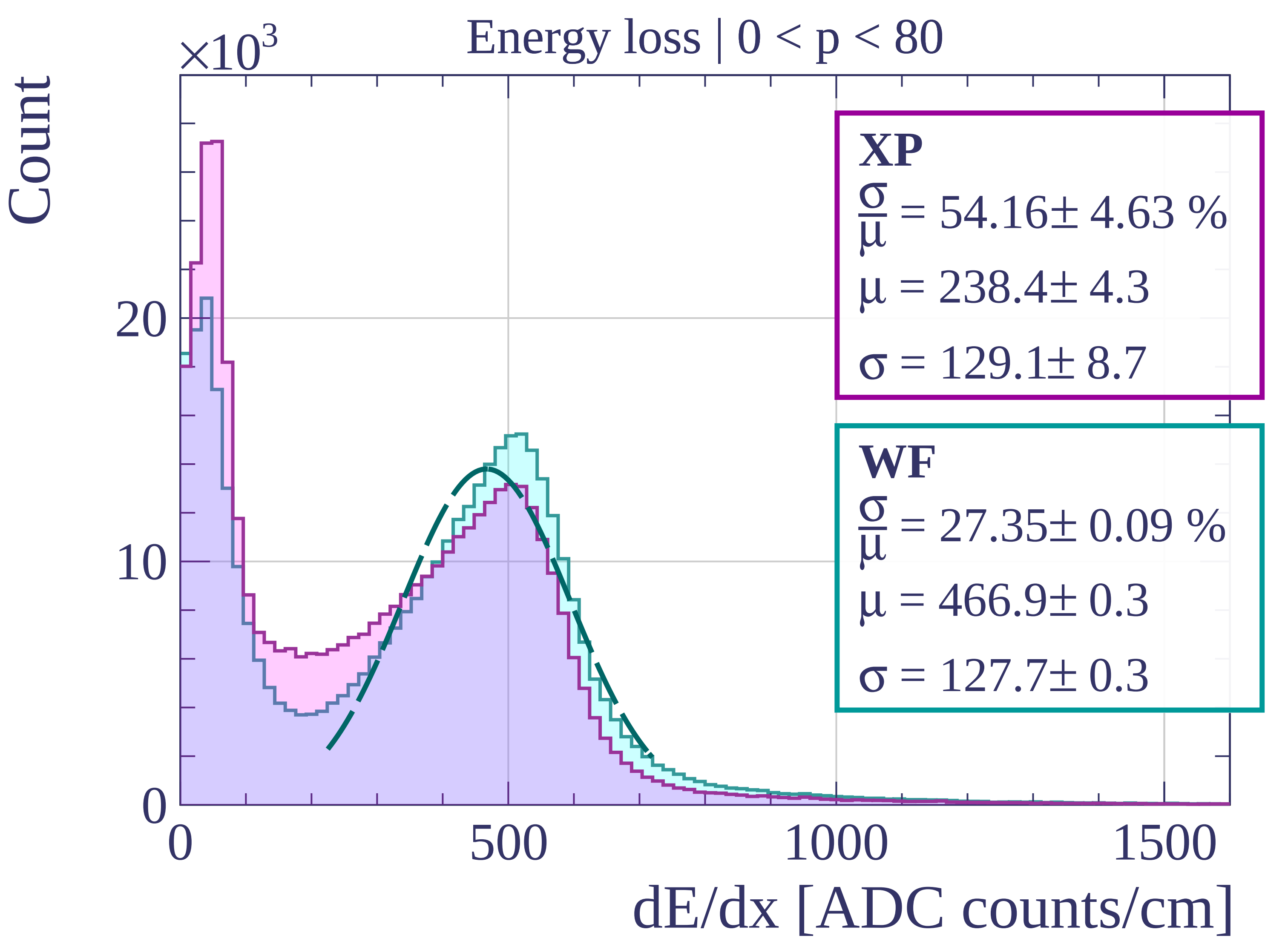
Z
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
Y
10^4
10^3
10^2
10
1



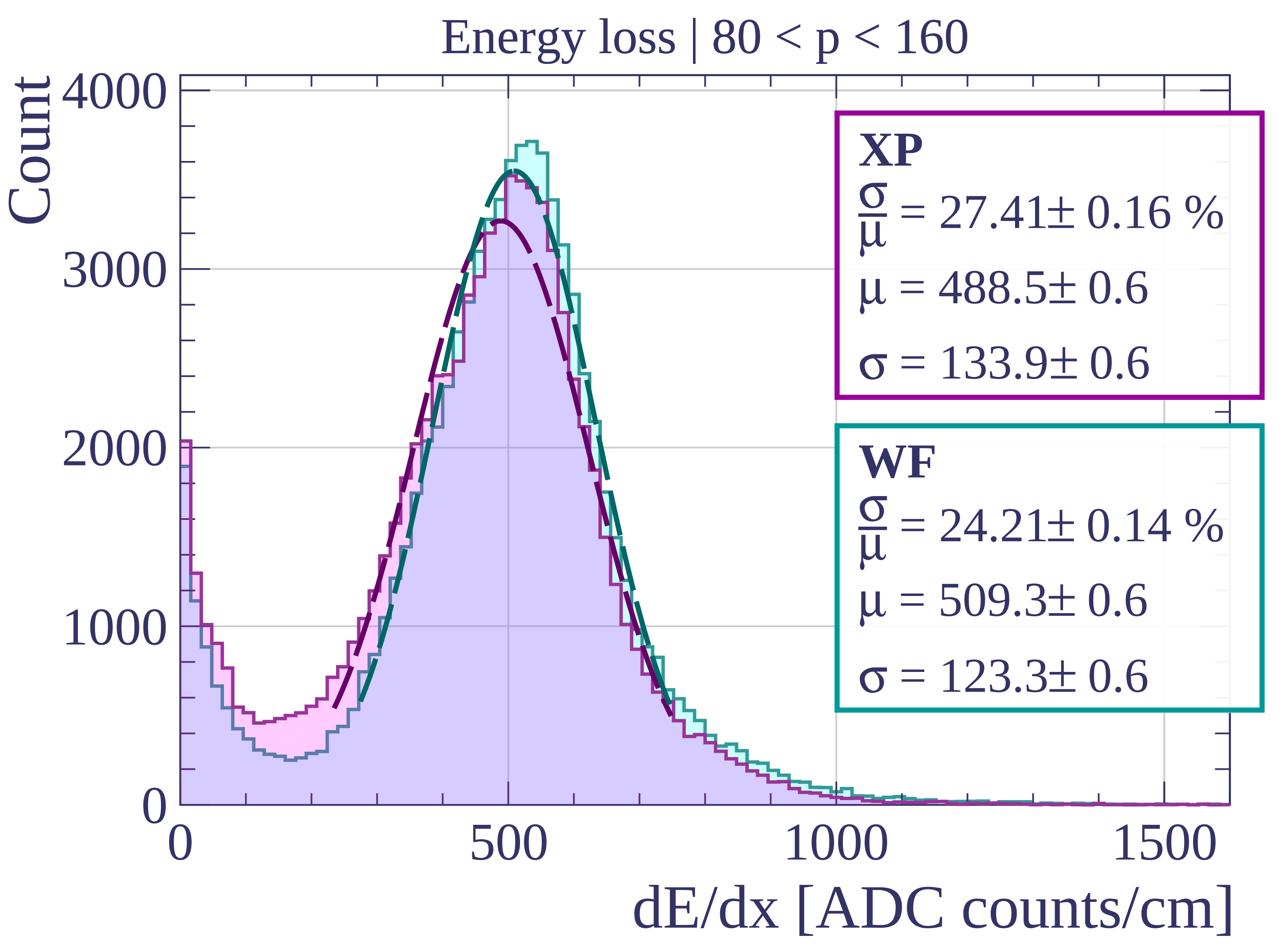
×10^3
2000
1000
0
−1000
−2000
Signed momentum (MeV)
−40
−20
0
20
40
R
60
40
20
0



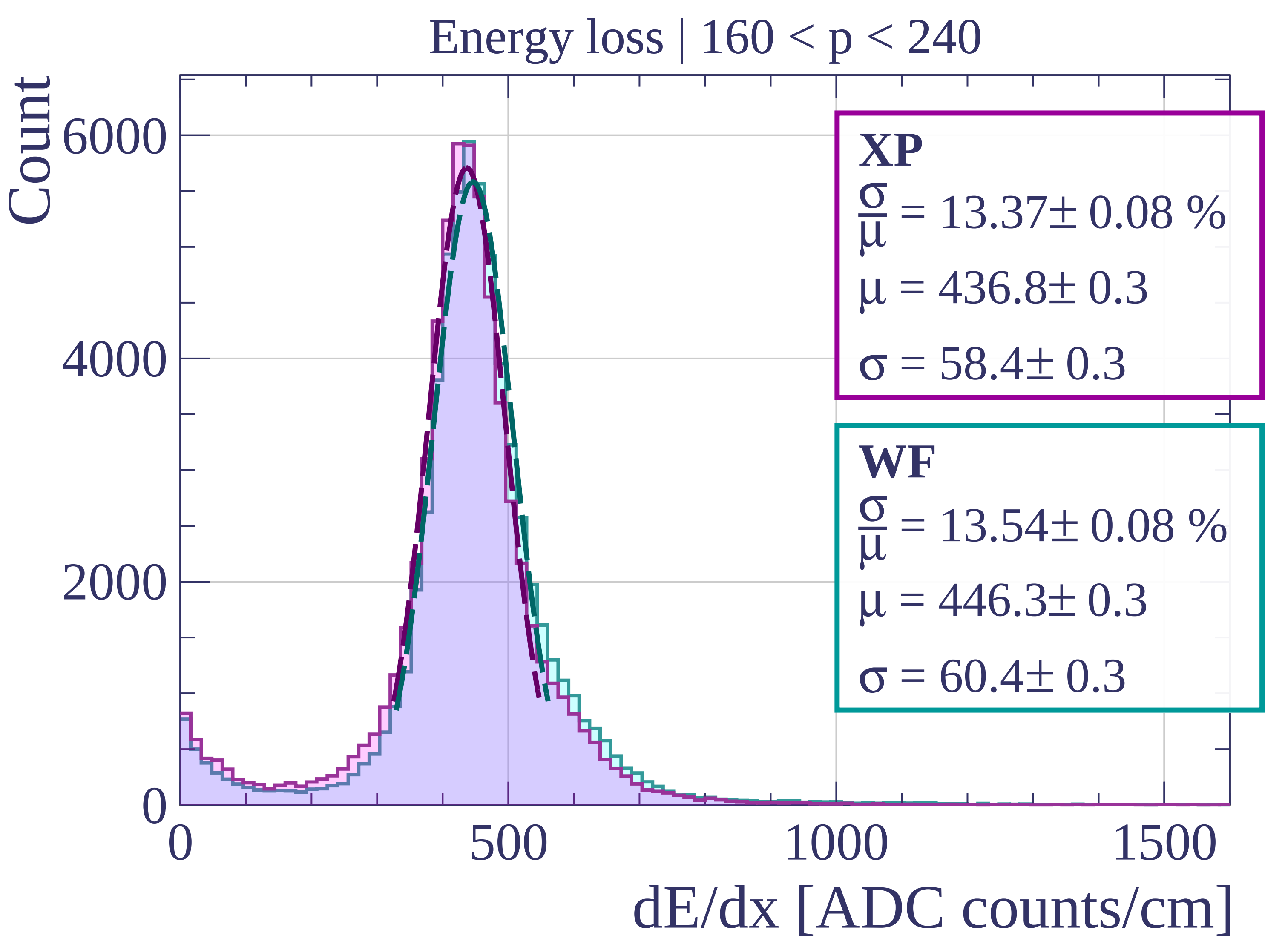
χ^2/NDF
50
40
30
20
10
0
−40
−20
0
20
40
R
1500
1000
500
0
$\times10^3$



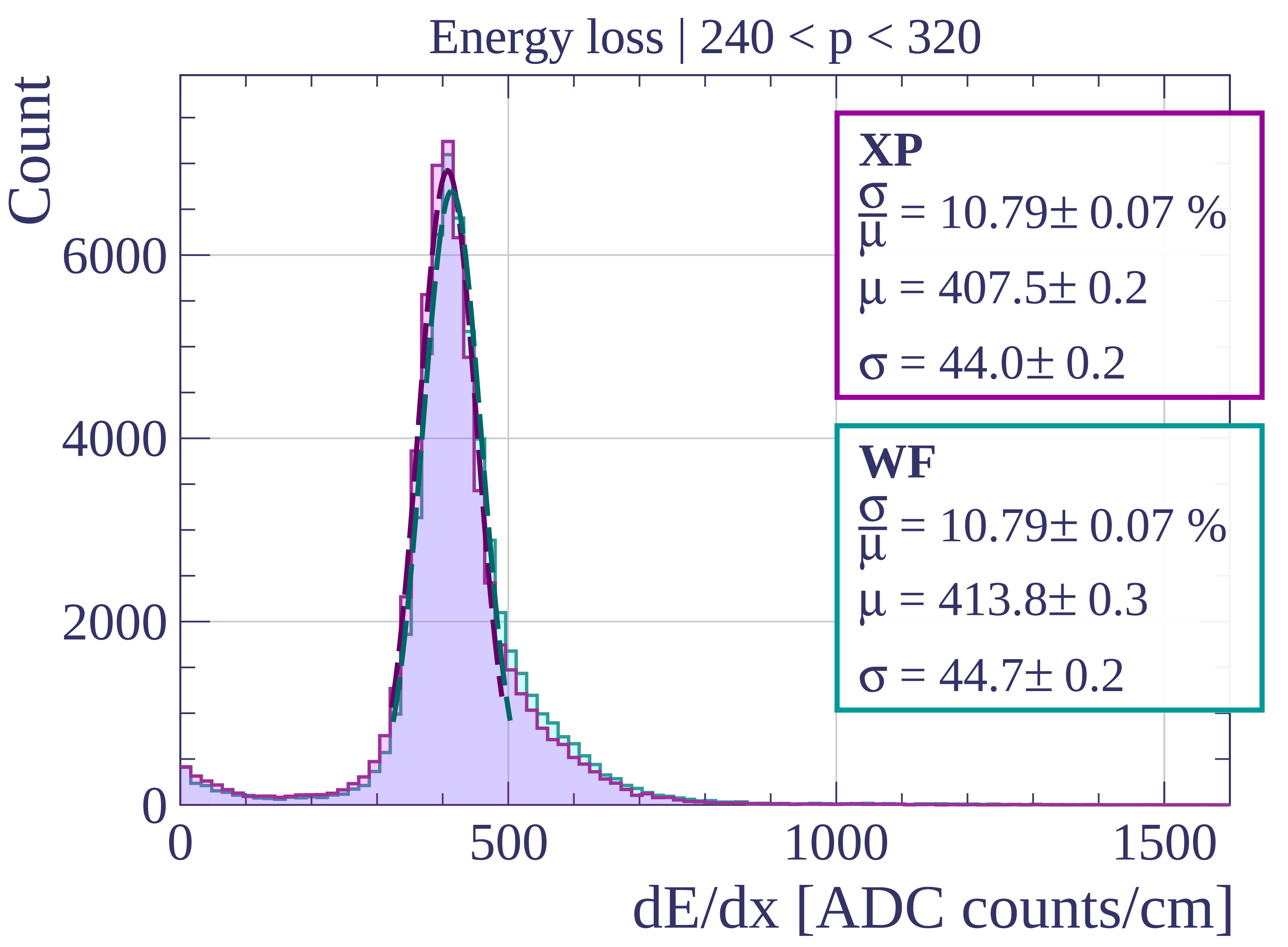
Energy loss | 0 < p < 80
×10^3
Count
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 54.16± 4.63 %
μ = 238.4± 4.3
σ = 129.1± 8.7
WF
σ/μ = 27.35± 0.09 %
μ = 466.9± 0.3
σ = 127.7± 0.3



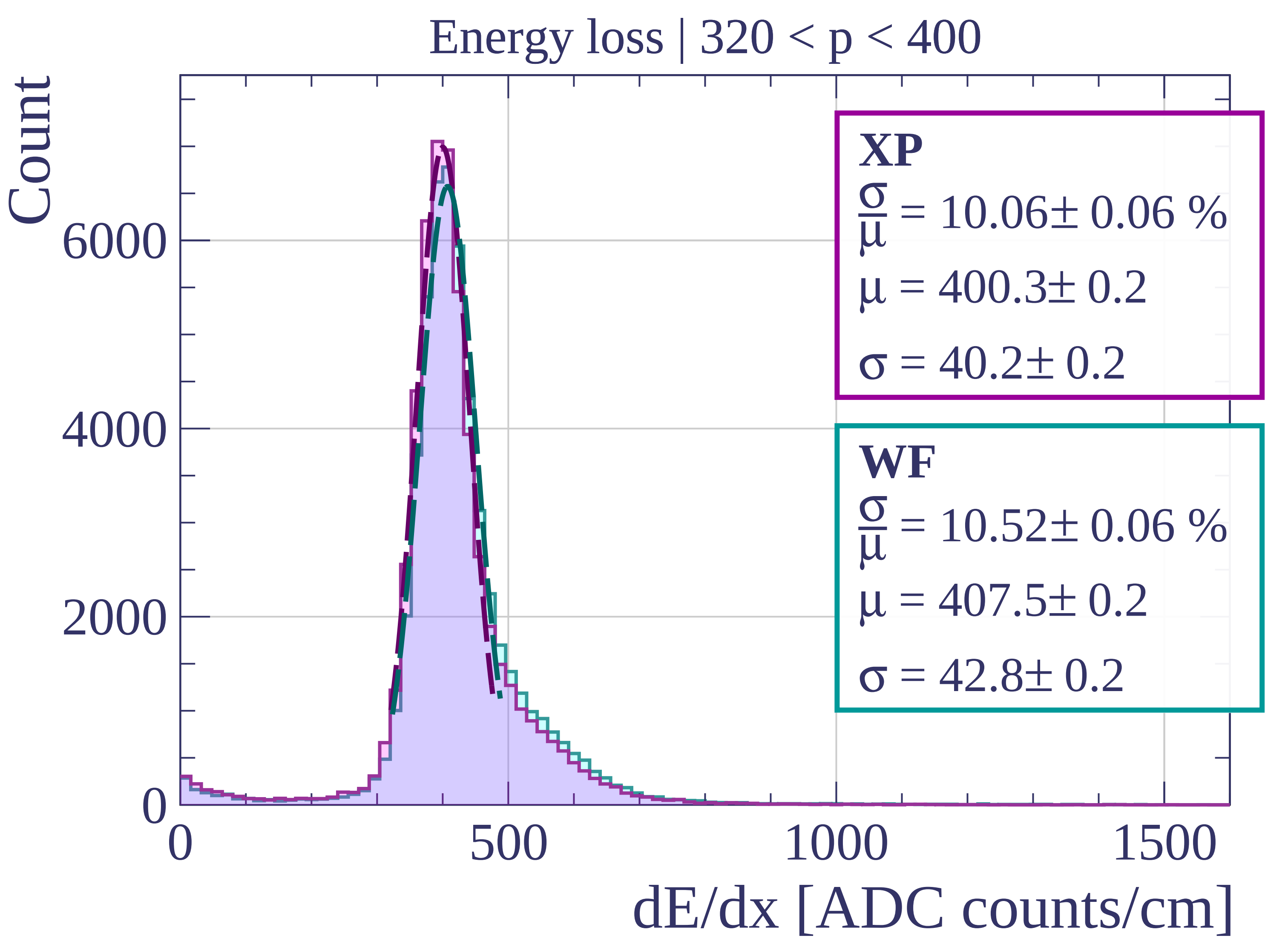
Energy loss | 80 < p < 160
Count
4000
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 27.41± 0.16 %
μ = 488.5± 0.6
σ = 133.9± 0.6
WF
σ/μ = 24.21± 0.14 %
μ = 509.3± 0.6
σ = 123.3± 0.6



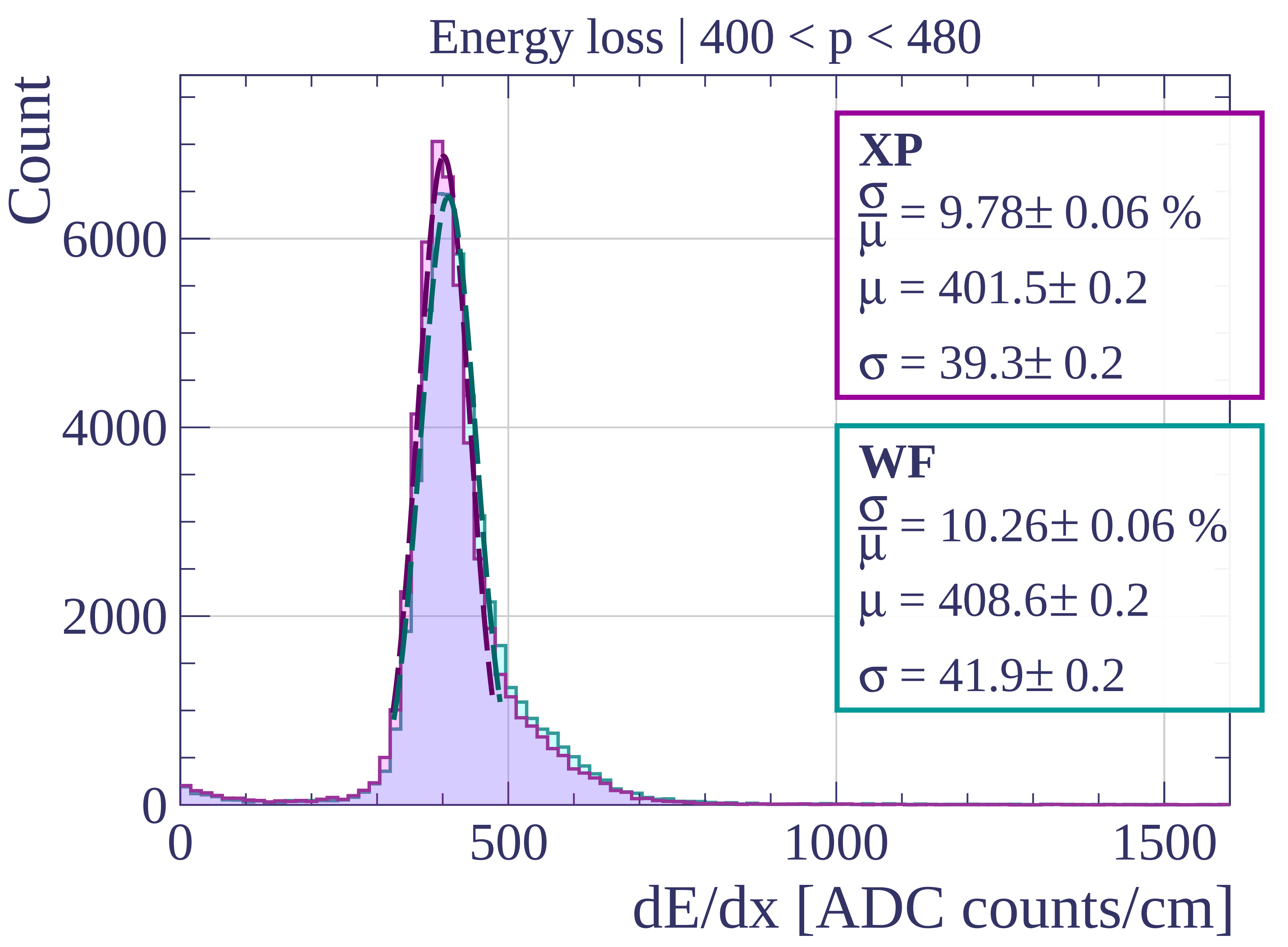
Energy loss | 160 < p < 240
Count
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 13.37± 0.08 %
μ = 436.8± 0.3
σ = 58.4± 0.3
WF
$\frac{\sigma}{\mu}$ = 13.54± 0.08 %
μ = 446.3± 0.3
σ = 60.4± 0.3



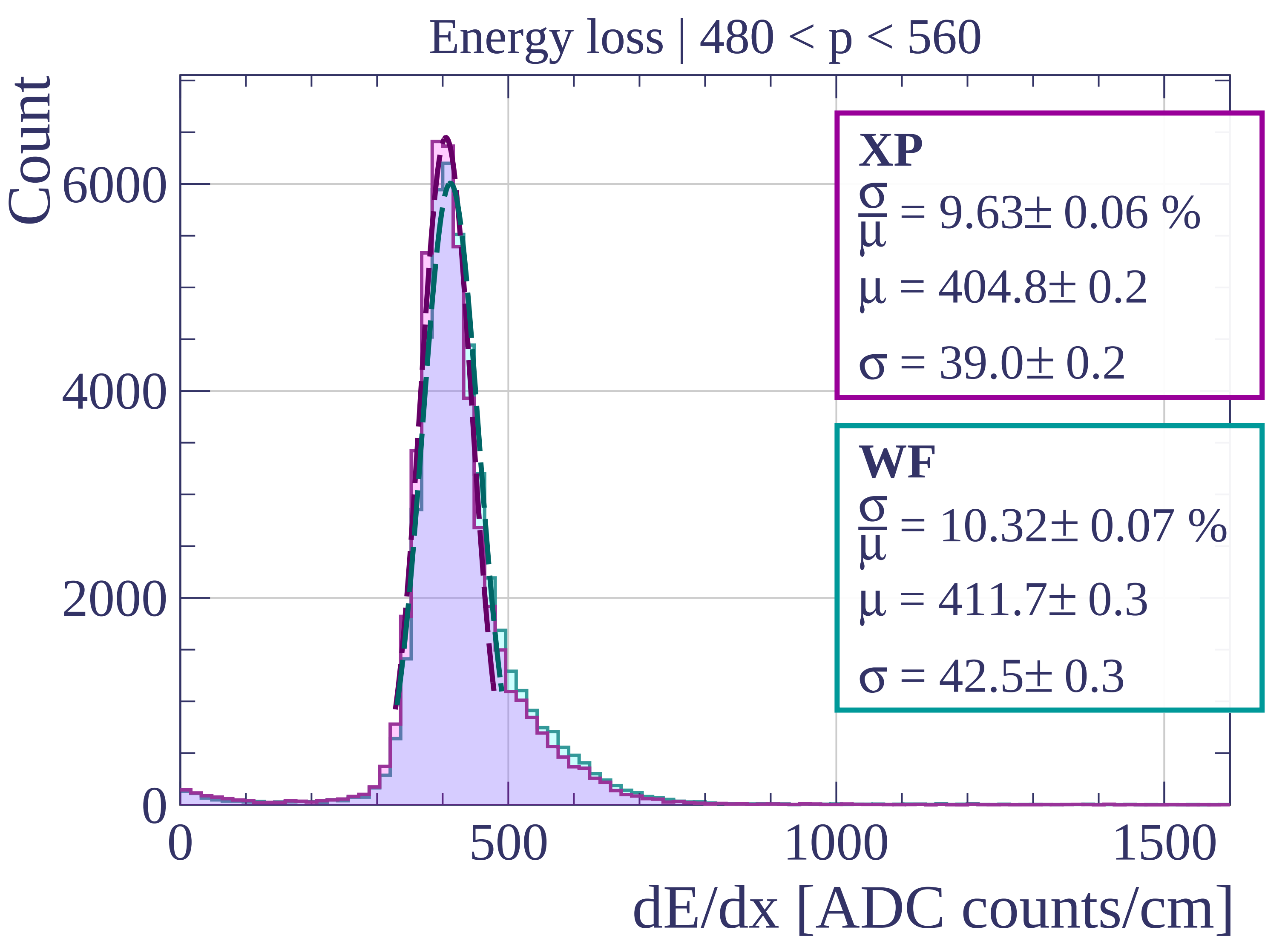
Energy loss | 240 < p < 320
Count
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 10.79± 0.07 %
μ = 407.5± 0.2
σ = 44.0± 0.2
WF
$\frac{\sigma}{\mu}$ = 10.79± 0.07 %
μ = 413.8± 0.3
σ = 44.7± 0.2

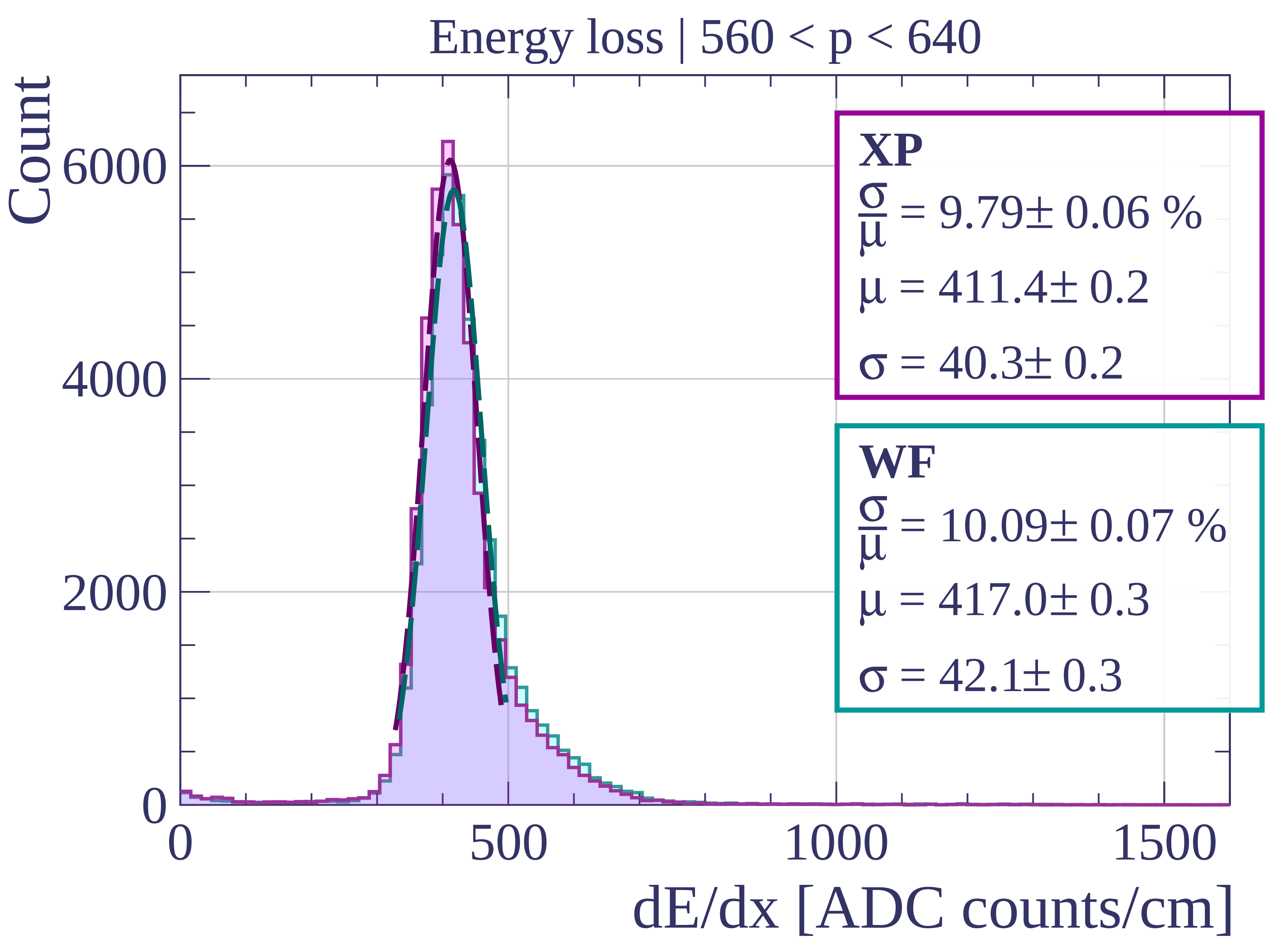


Energy loss | 320 < p < 400
Count
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu} = 10.06 \pm 0.06$ %
$\mu = 400.3 \pm 0.2$
$\sigma = 40.2 \pm 0.2$
WF
$\frac{\sigma}{\mu} = 10.52 \pm 0.06$ %
$\mu = 407.5 \pm 0.2$
$\sigma = 42.8 \pm 0.2$

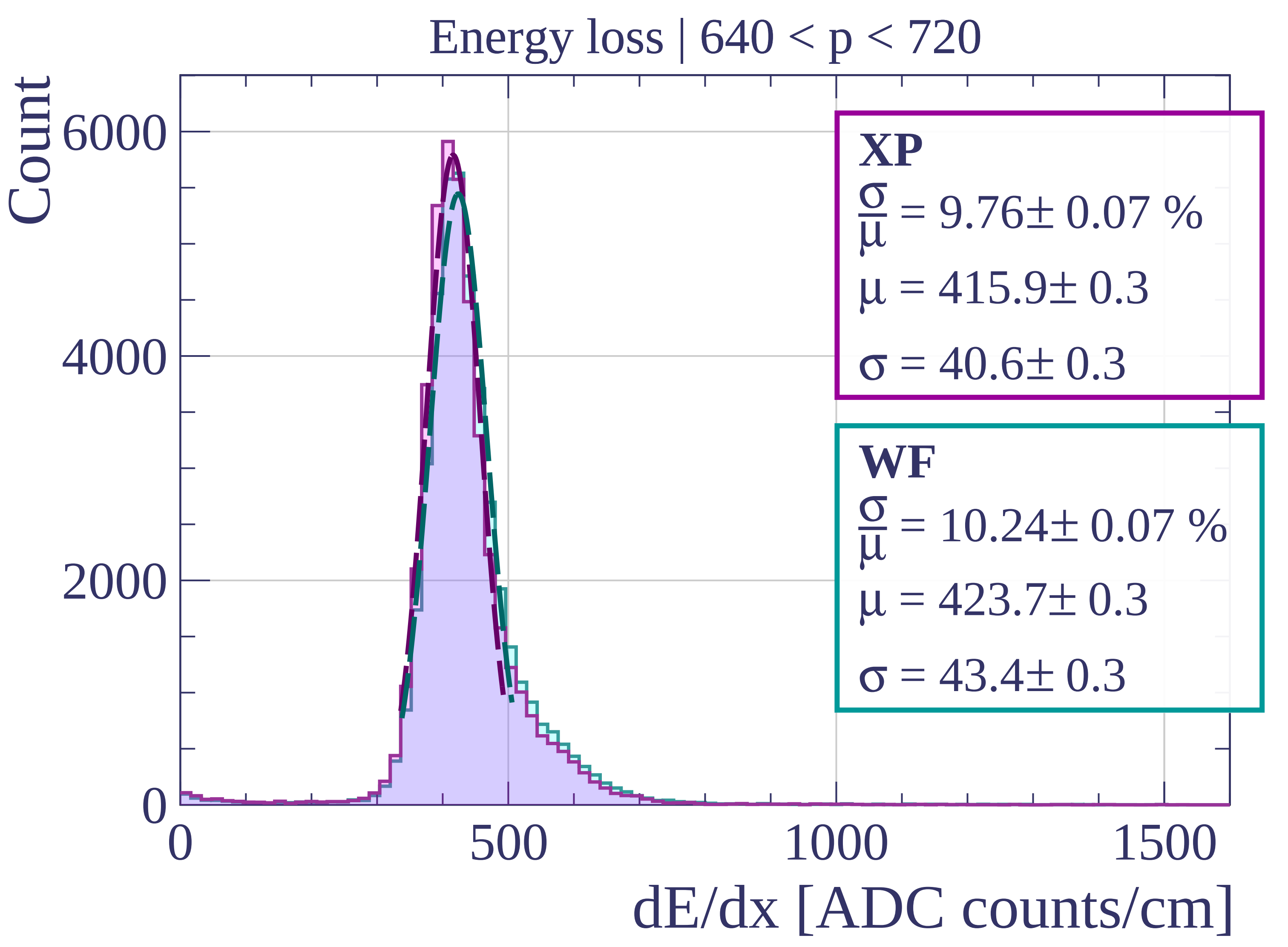


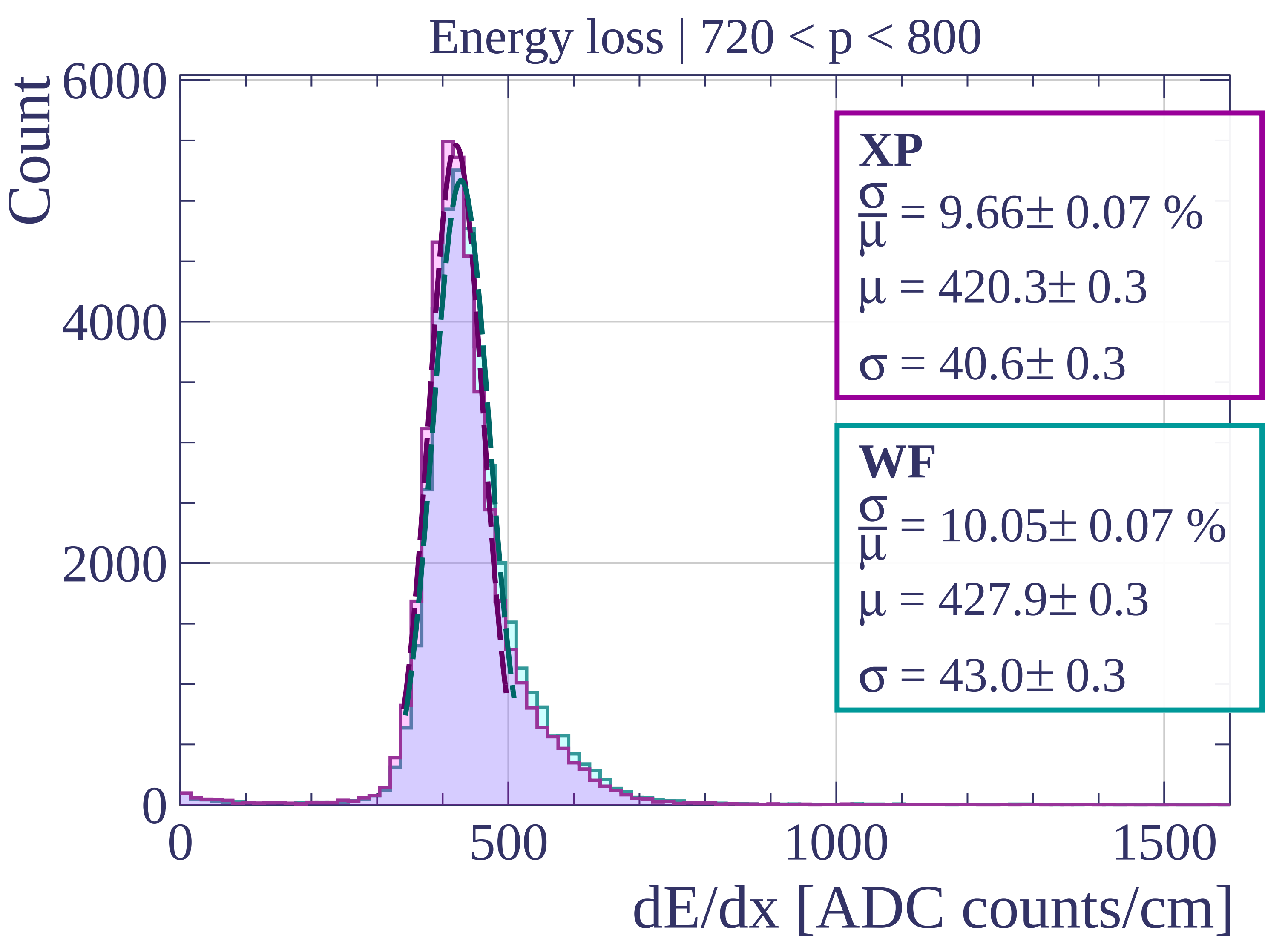
Energy loss | 400 < p < 480
Count
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 9.78± 0.06 %
μ = 401.5± 0.2
σ = 39.3± 0.2
WF
$\frac{\sigma}{\mu}$ = 10.26± 0.06 %
μ = 408.6± 0.2
σ = 41.9± 0.2


Energy loss | 480 < p < 560
Count
dE/dx [ADC counts/cm]
0
2000
4000
6000
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = 9.63± 0.06 %
μ = 404.8± 0.2
σ = 39.0± 0.2
WF
$\frac{\sigma}{\mu}$ = 10.32± 0.07 %
μ = 411.7± 0.3
σ = 42.5± 0.3

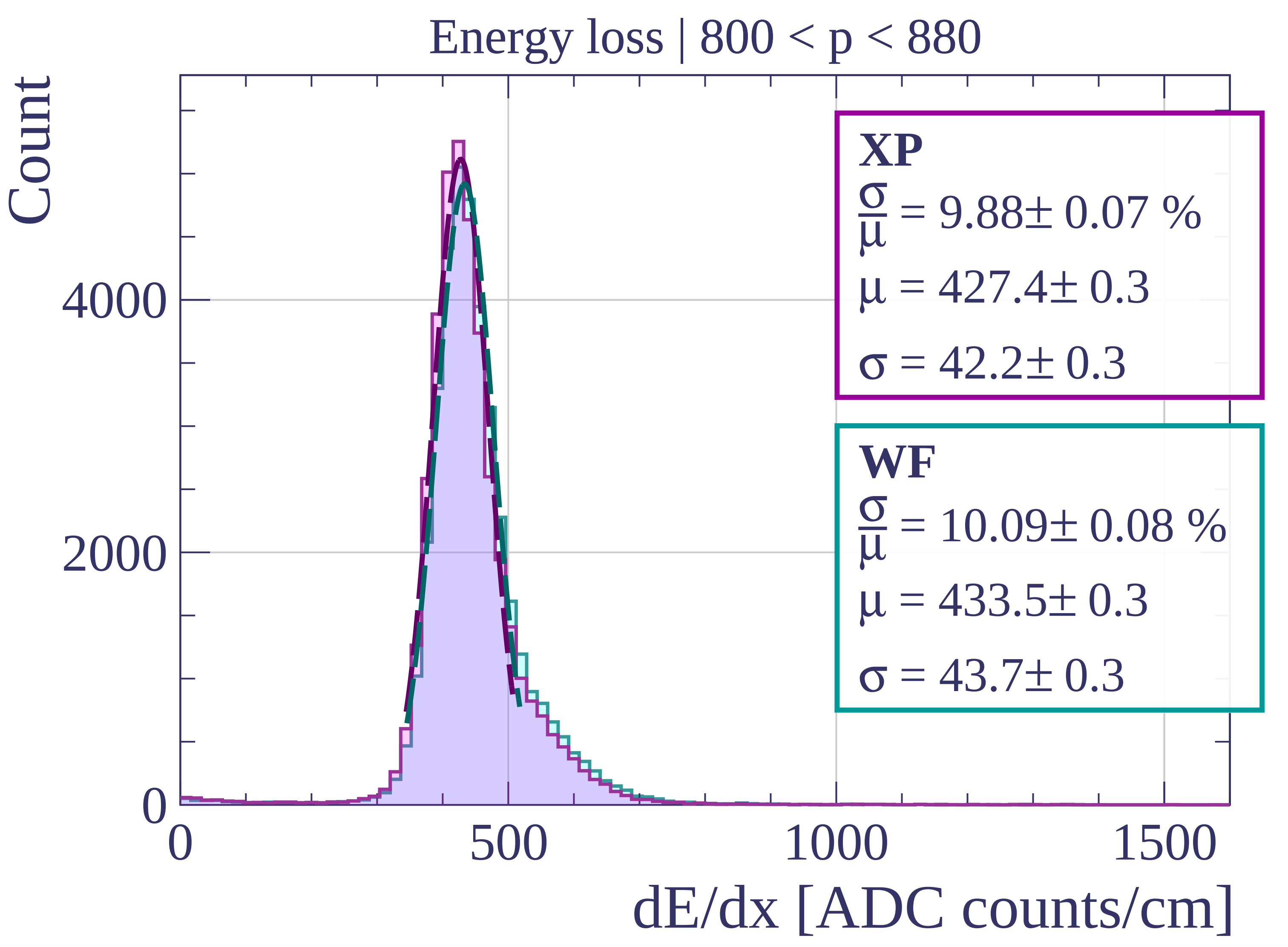


Energy loss | 560 < p < 640
Count
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 9.79± 0.06 %
μ = 411.4± 0.2
σ = 40.3± 0.2
WF
σ/μ = 10.09± 0.07 %
μ = 417.0± 0.3
σ = 42.1± 0.3

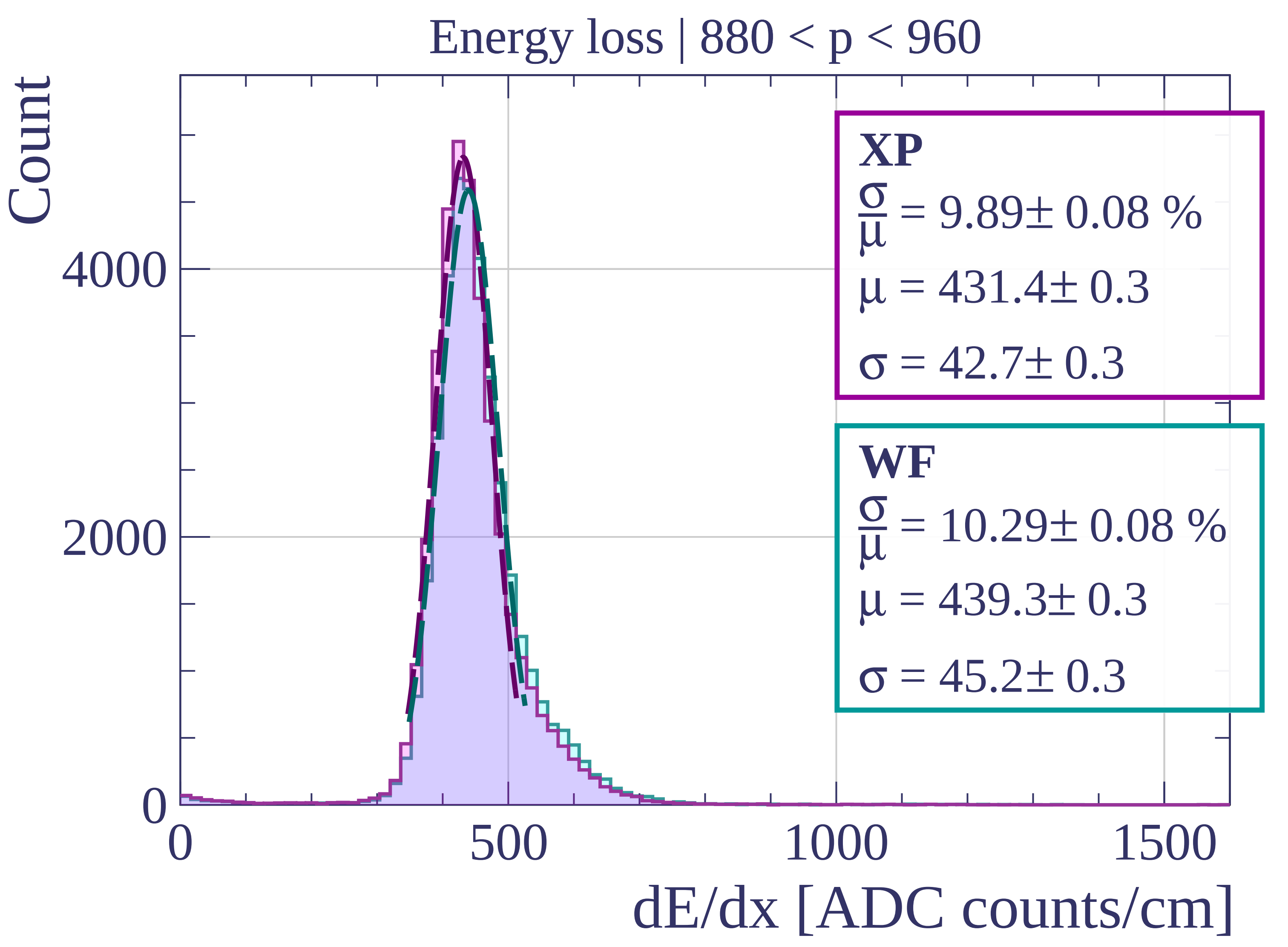

Energy loss | 640 < p < 720
Count
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 9.76± 0.07 %
μ = 415.9± 0.3
σ = 40.6± 0.3
WF
σ/μ = 10.24± 0.07 %
μ = 423.7± 0.3
σ = 43.4± 0.3

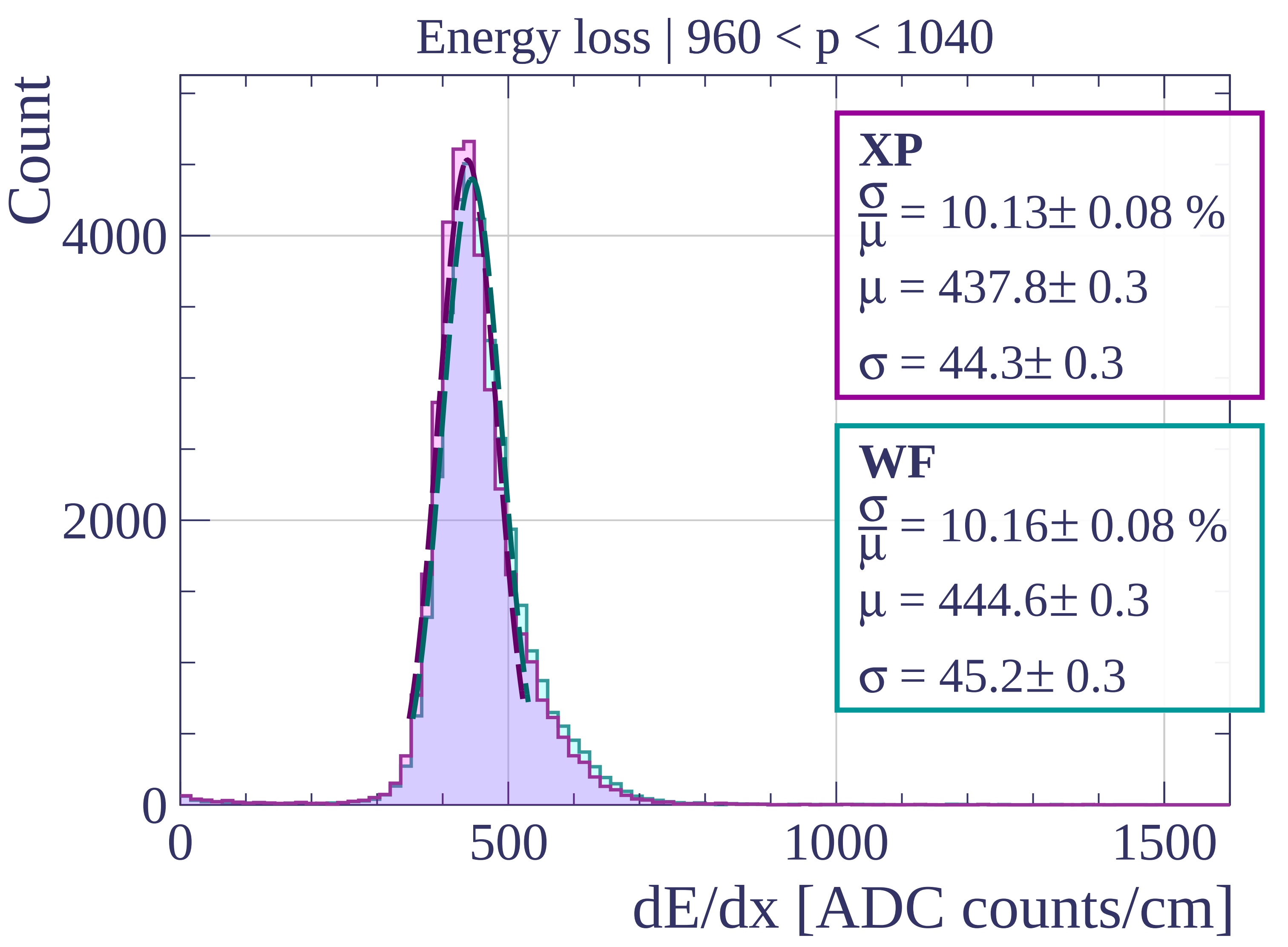


Energy loss | 720 < p < 800
Count
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 9.66± 0.07 %
μ = 420.3± 0.3
σ = 40.6± 0.3
WF
σ/μ = 10.05± 0.07 %
μ = 427.9± 0.3
σ = 43.0± 0.3

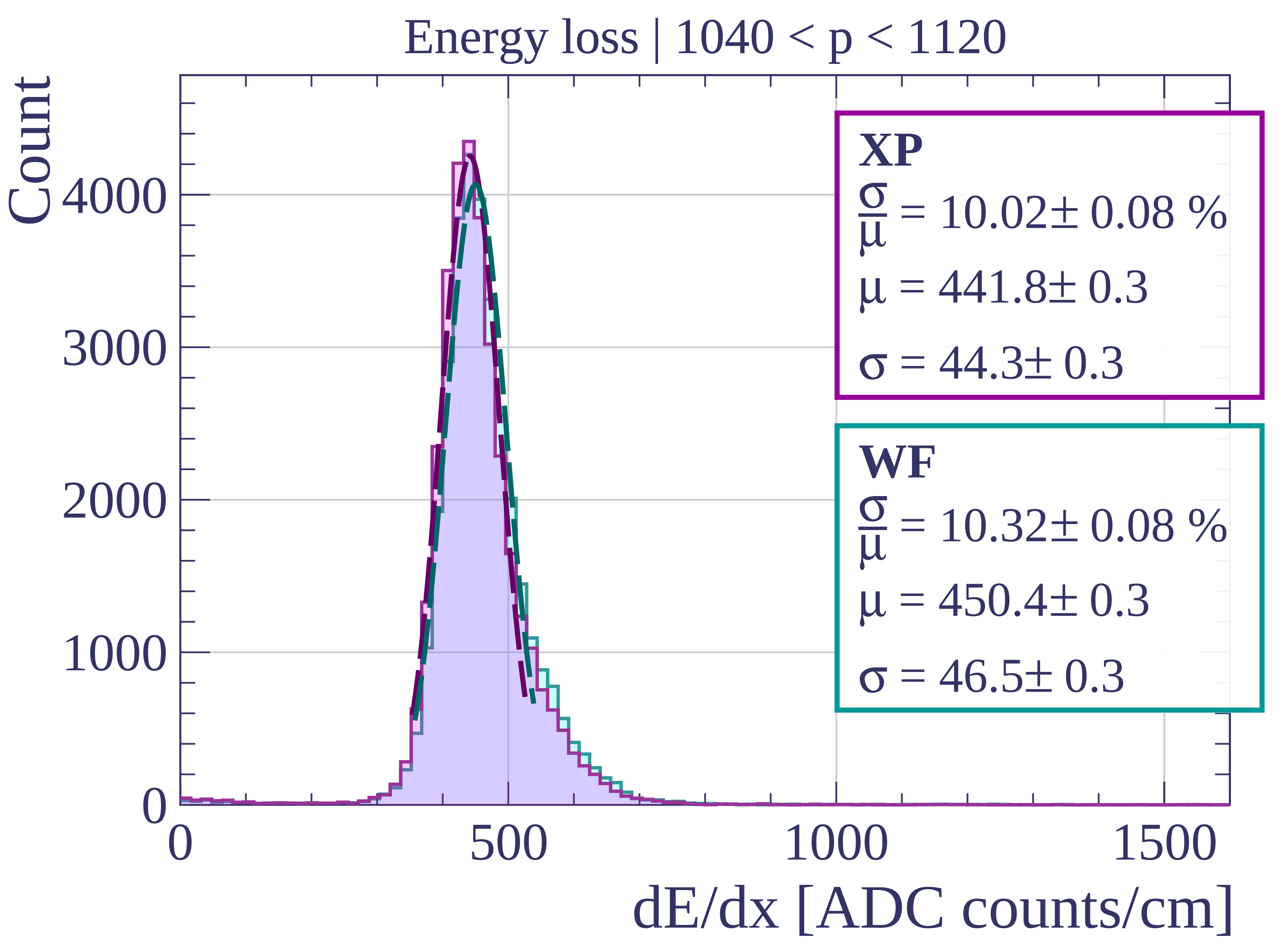


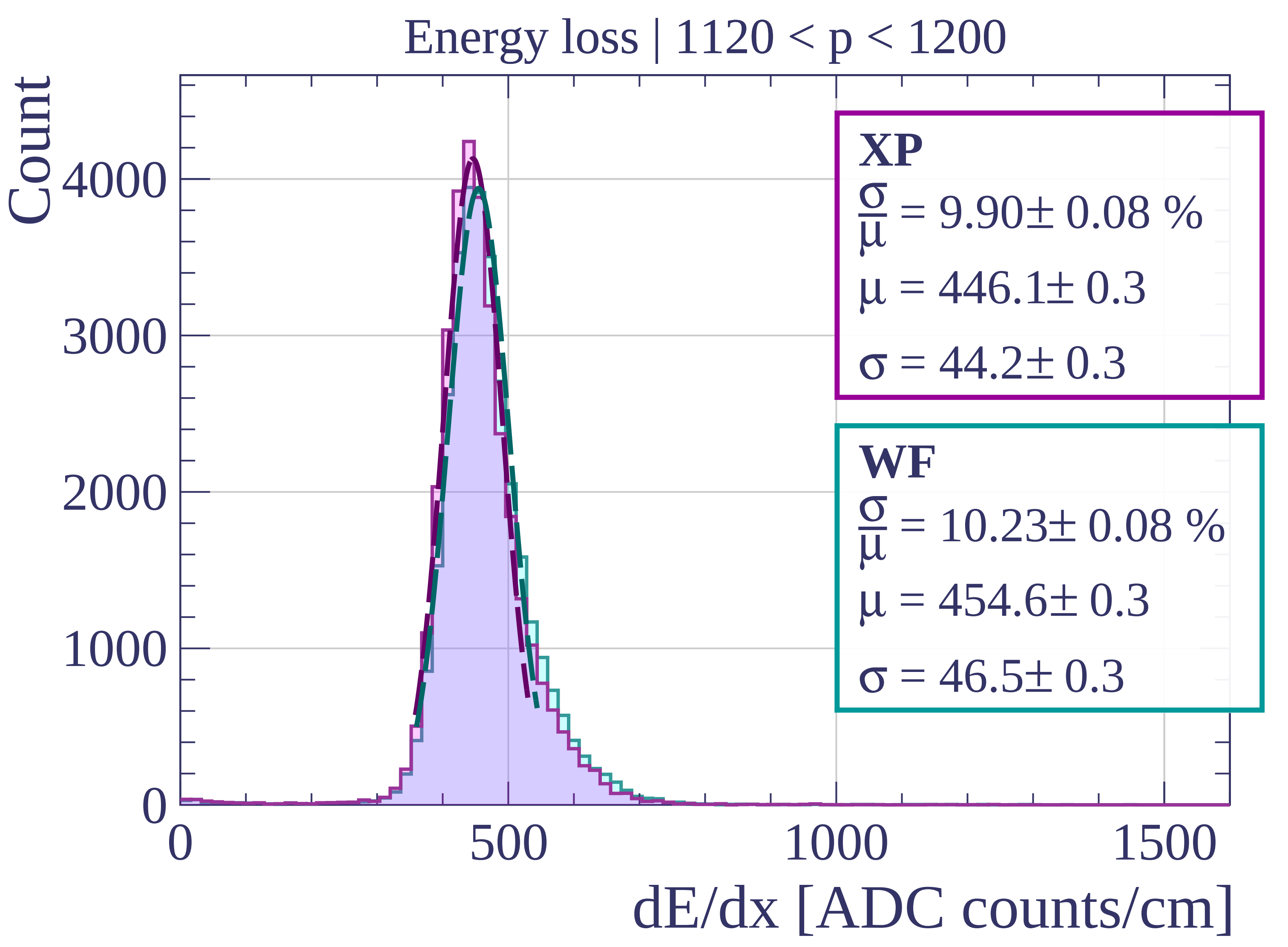
Energy loss | 800 < p < 880
Count
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 9.88± 0.07 %
μ = 427.4± 0.3
σ = 42.2± 0.3
WF
$\frac{\sigma}{\mu}$ = 10.09± 0.08 %
μ = 433.5± 0.3
σ = 43.7± 0.3


Energy loss | 880 < p < 960
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
2000
4000
XP
σ/μ = 9.89± 0.08 %
μ = 431.4± 0.3
σ = 42.7± 0.3
WF
σ/μ = 10.29± 0.08 %
μ = 439.3± 0.3
σ = 45.2± 0.3

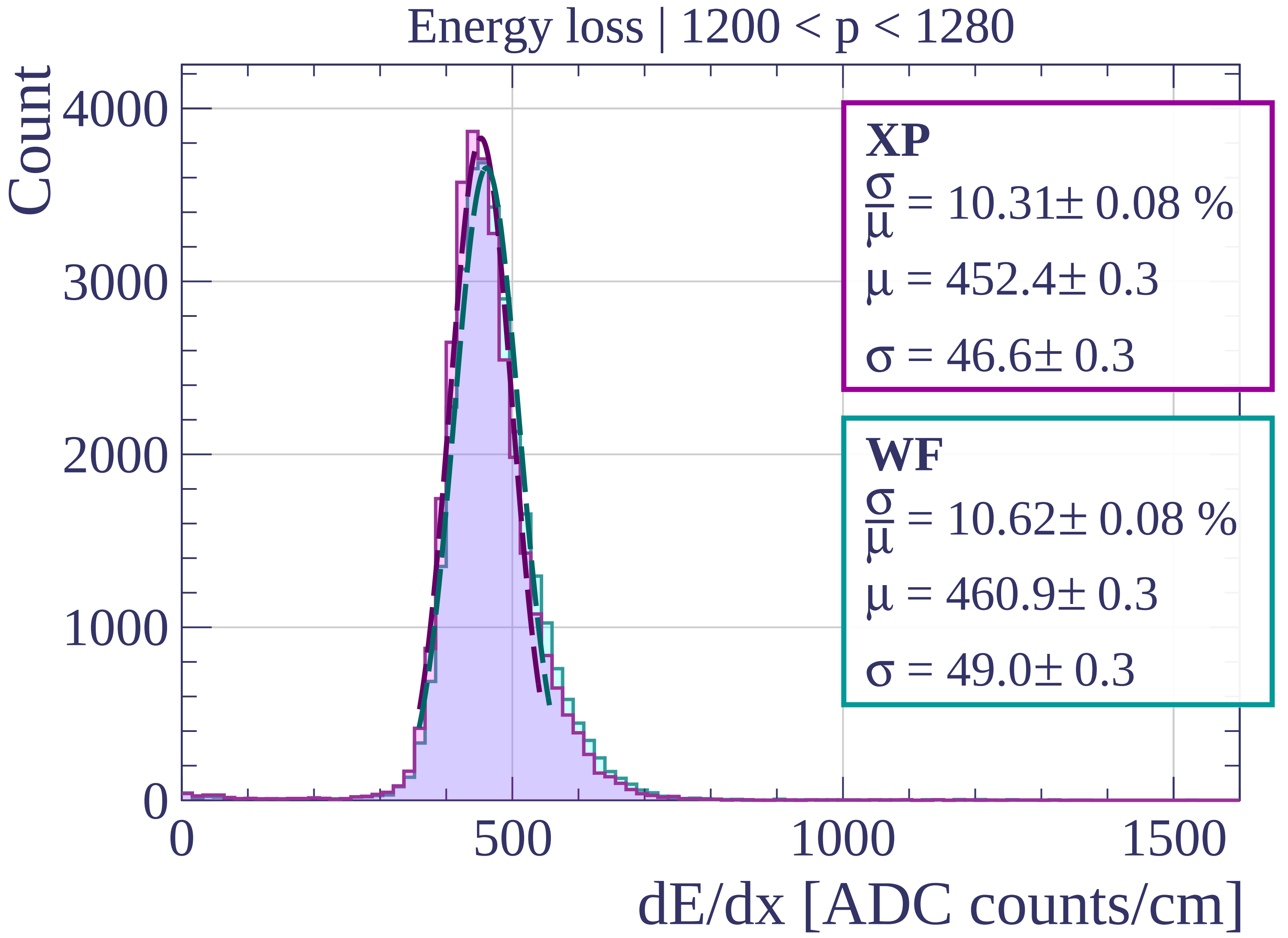


Energy loss | 960 < p < 1040
Count
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 10.13± 0.08 %
μ = 437.8± 0.3
σ = 44.3± 0.3
WF
σ/μ = 10.16± 0.08 %
μ = 444.6± 0.3
σ = 45.2± 0.3

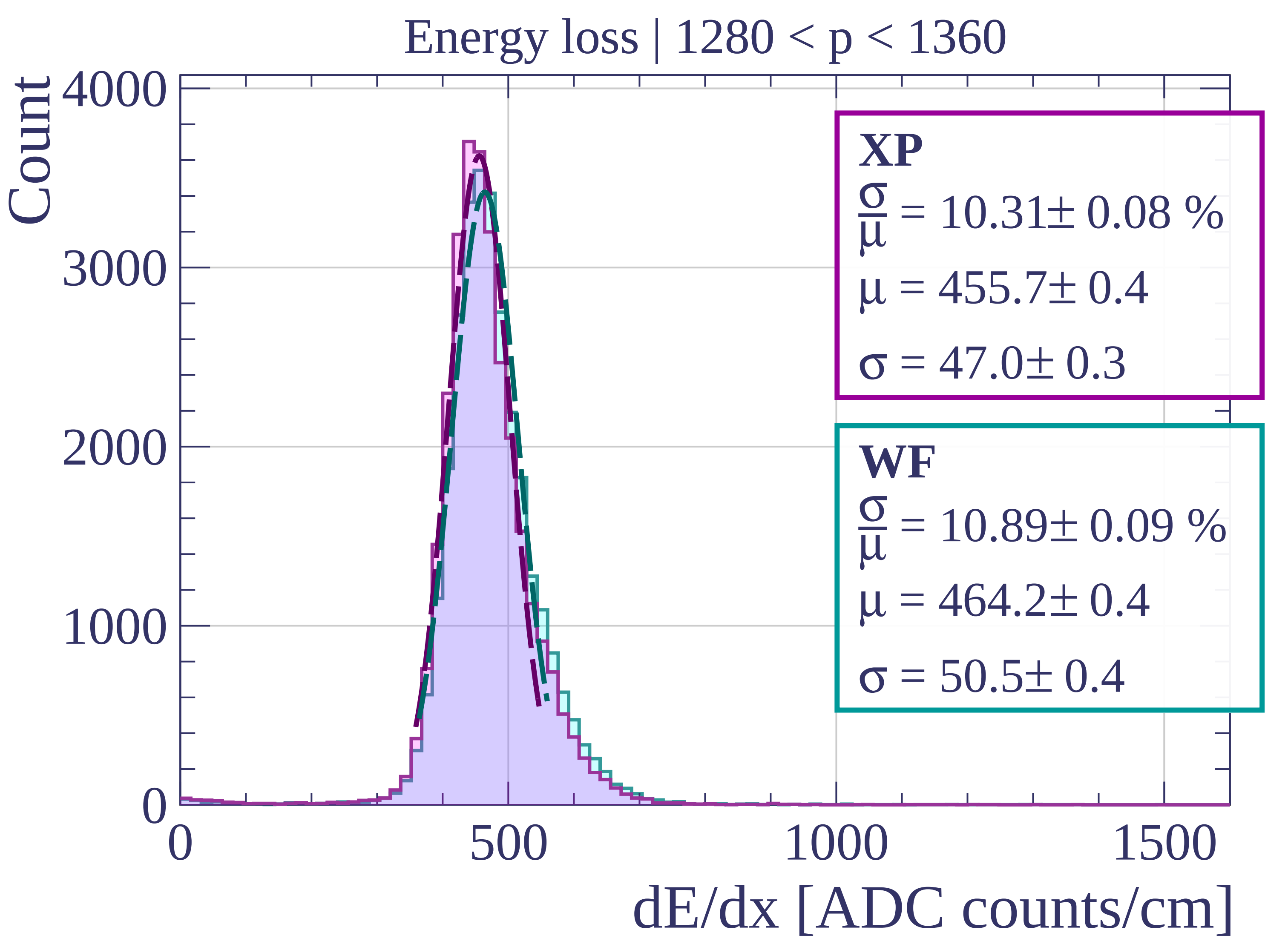

Energy loss | 1040 < p < 1120
Count
dE/dx [ADC counts/cm]
0
1000
2000
3000
4000
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = 10.02± 0.08 %
μ = 441.8± 0.3
σ = 44.3± 0.3
WF
$\frac{\sigma}{\mu}$ = 10.32± 0.08 %
μ = 450.4± 0.3
σ = 46.5± 0.3



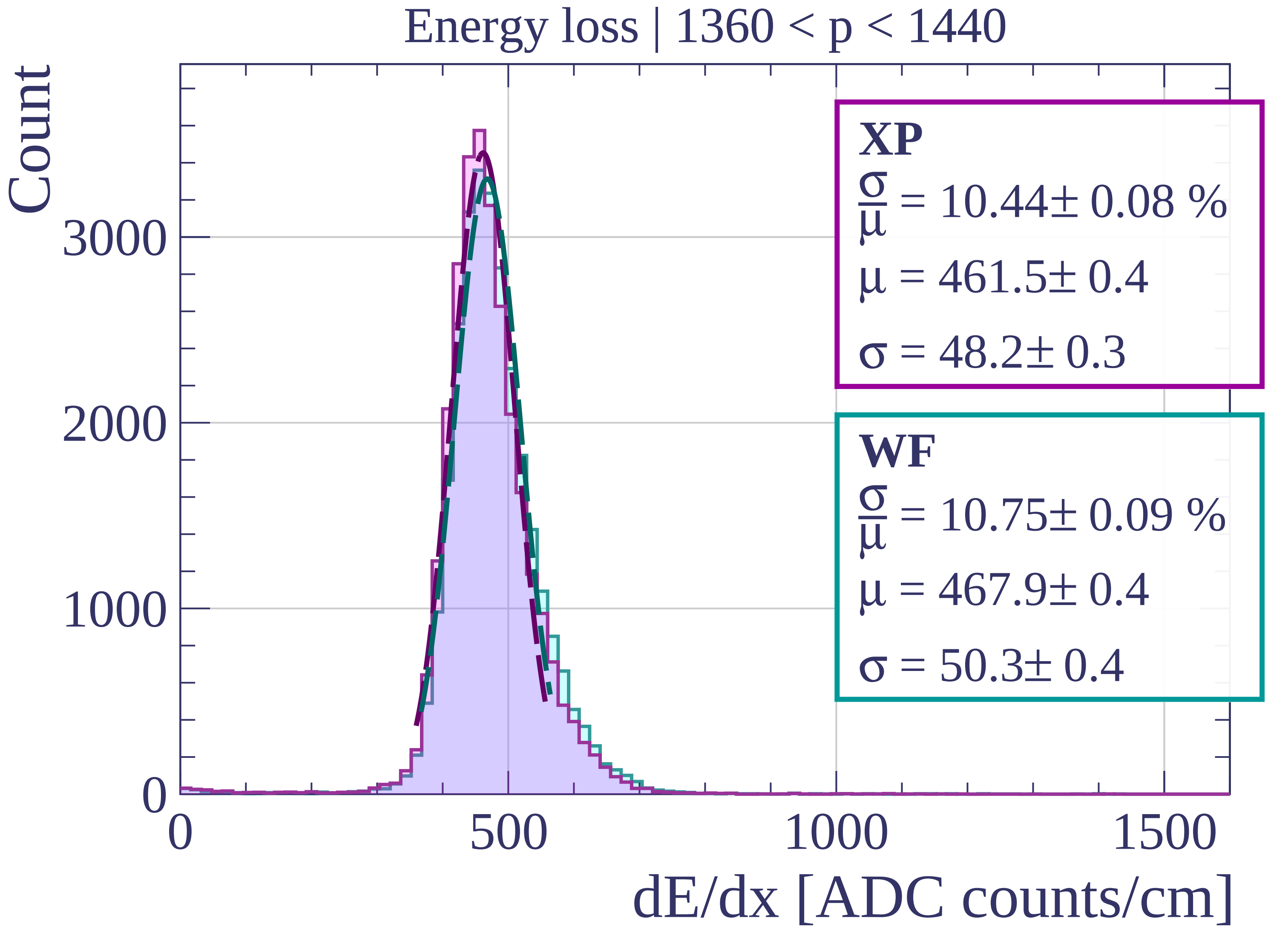
Energy loss | 1120 < p < 1200
Count
4000
3000
2000
1000
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 9.90± 0.08 %
μ = 446.1± 0.3
σ = 44.2± 0.3
WF
$\frac{\sigma}{\mu}$ = 10.23± 0.08 %
μ = 454.6± 0.3
σ = 46.5± 0.3



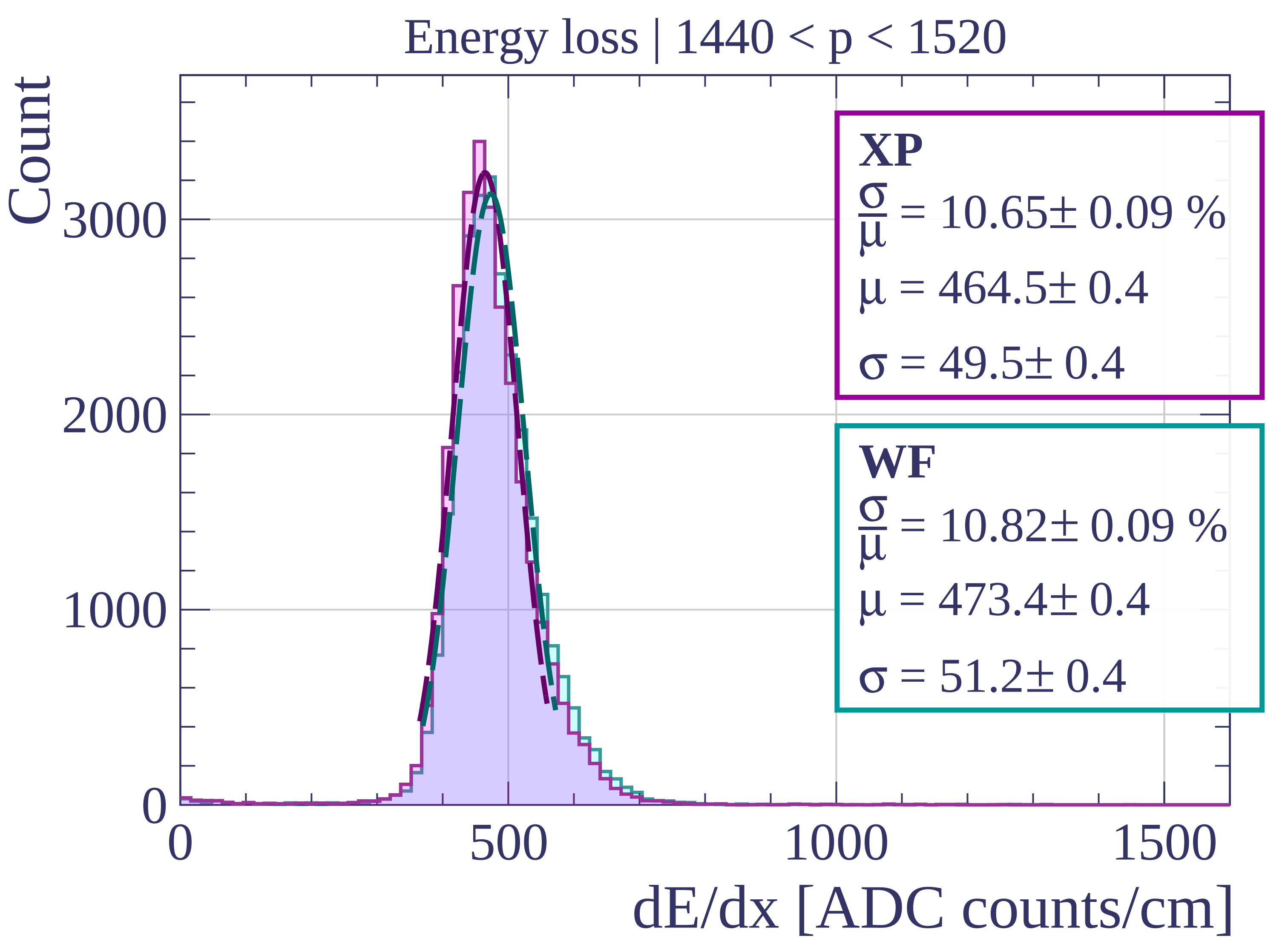
Energy loss | 1200 < p < 1280
Count
4000
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 10.31± 0.08 %
μ = 452.4± 0.3
σ = 46.6± 0.3
WF
σ/μ = 10.62± 0.08 %
μ = 460.9± 0.3
σ = 49.0± 0.3

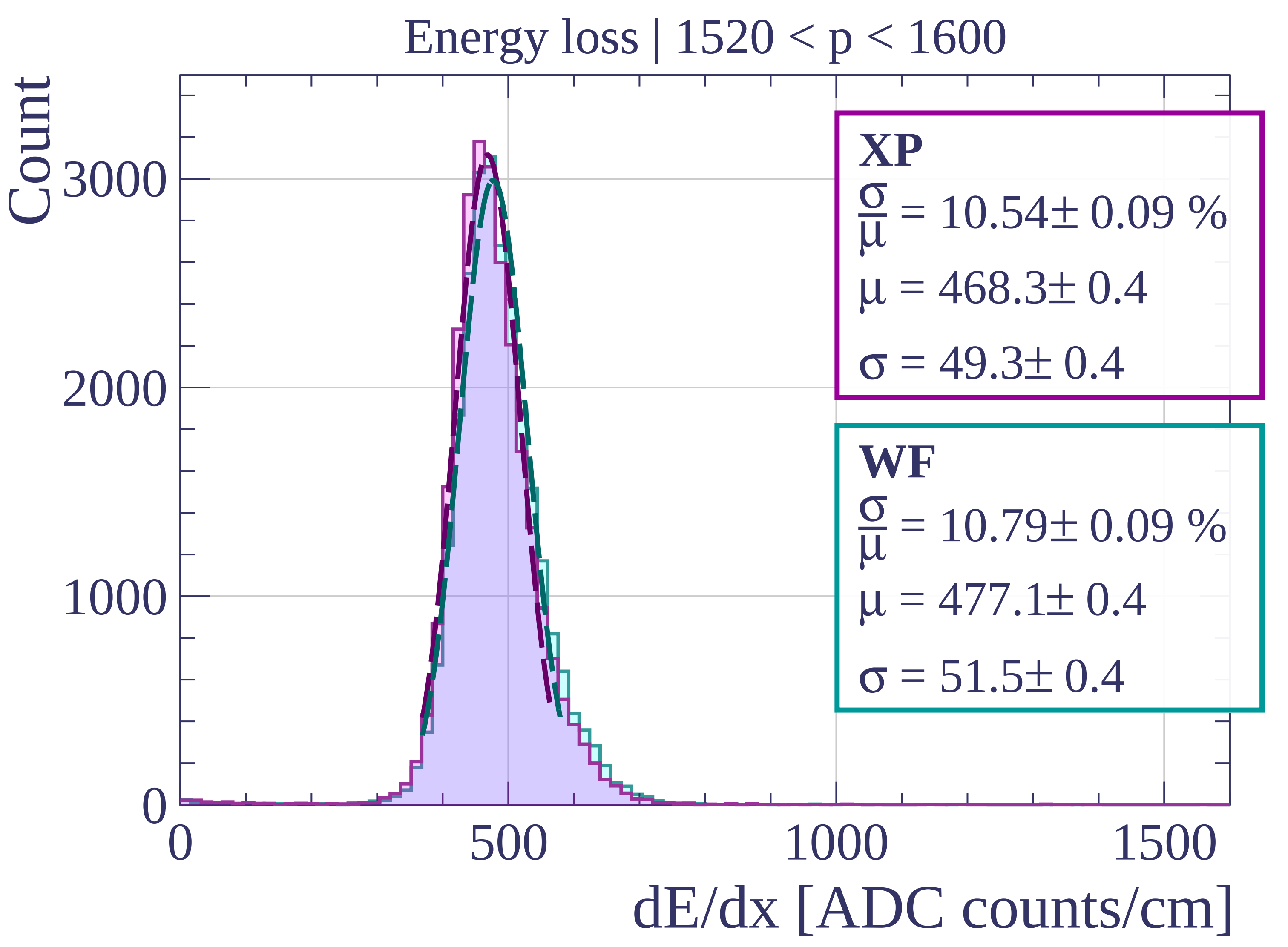


Energy loss | 1280 < p < 1360
Count
4000
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 10.31± 0.08 %
μ = 455.7± 0.4
σ = 47.0± 0.3
WF
σ/μ = 10.89± 0.09 %
μ = 464.2± 0.4
σ = 50.5± 0.4

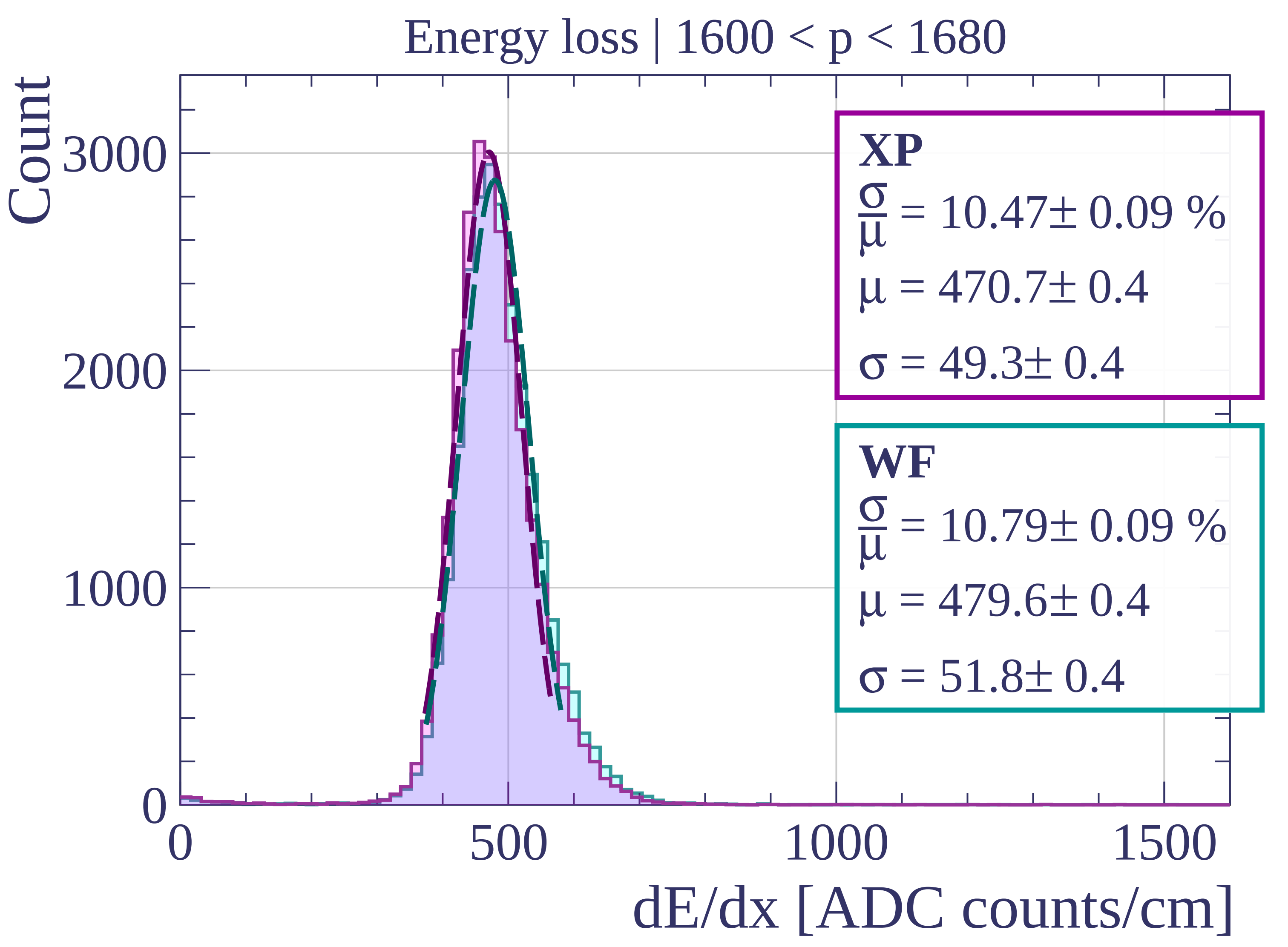


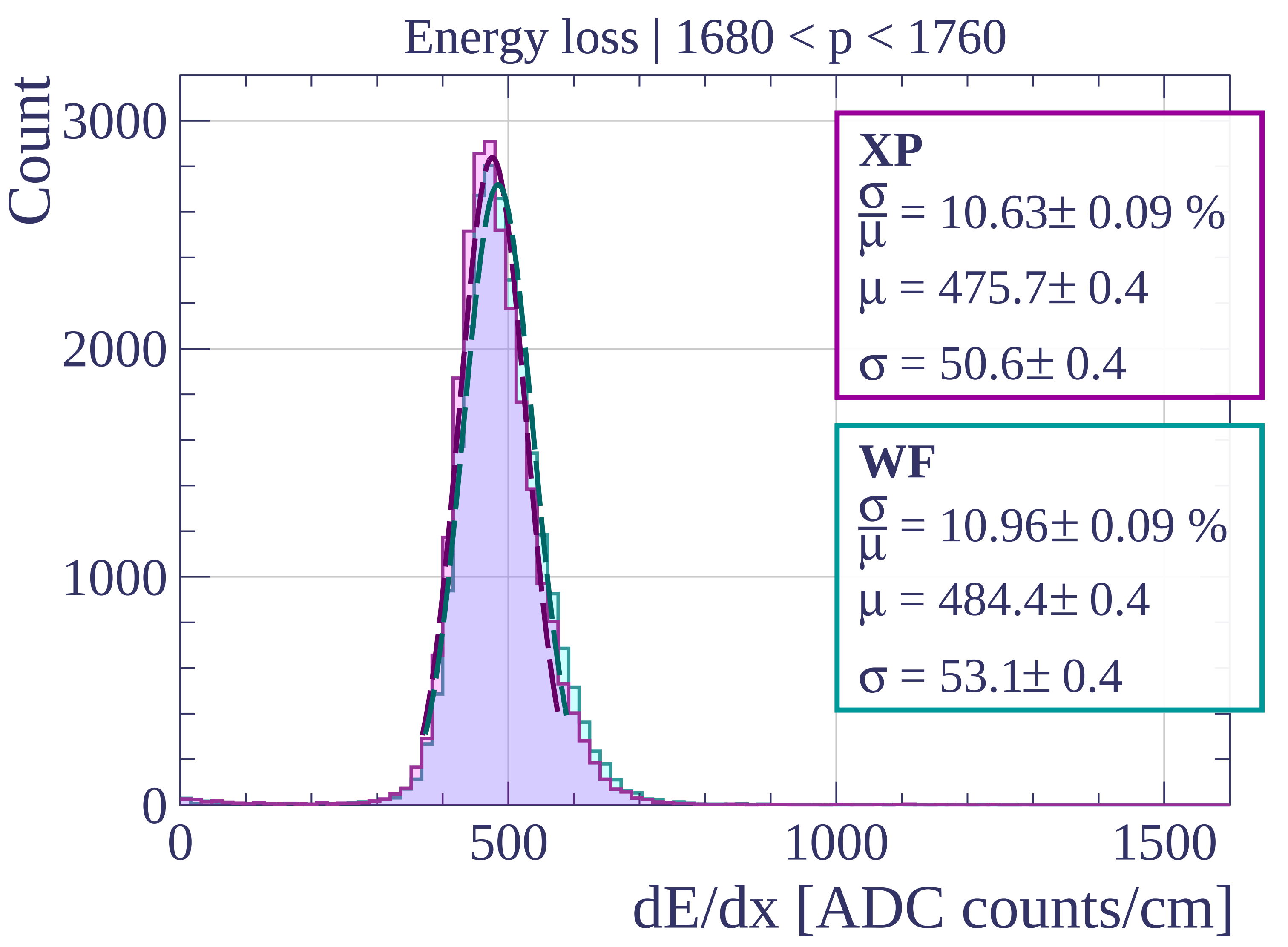
Energy loss | 1360 < p < 1440
Count
3000
2000
1000
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 10.44± 0.08 %
μ = 461.5± 0.4
σ = 48.2± 0.3
WF
$\frac{\sigma}{\mu}$ = 10.75± 0.09 %
μ = 467.9± 0.4
σ = 50.3± 0.4


Energy loss | 1440 < p < 1520
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 10.65± 0.09 %
μ = 464.5± 0.4
σ = 49.5± 0.4
WF
$\frac{\sigma}{\mu}$ = 10.82± 0.09 %
μ = 473.4± 0.4
σ = 51.2± 0.4

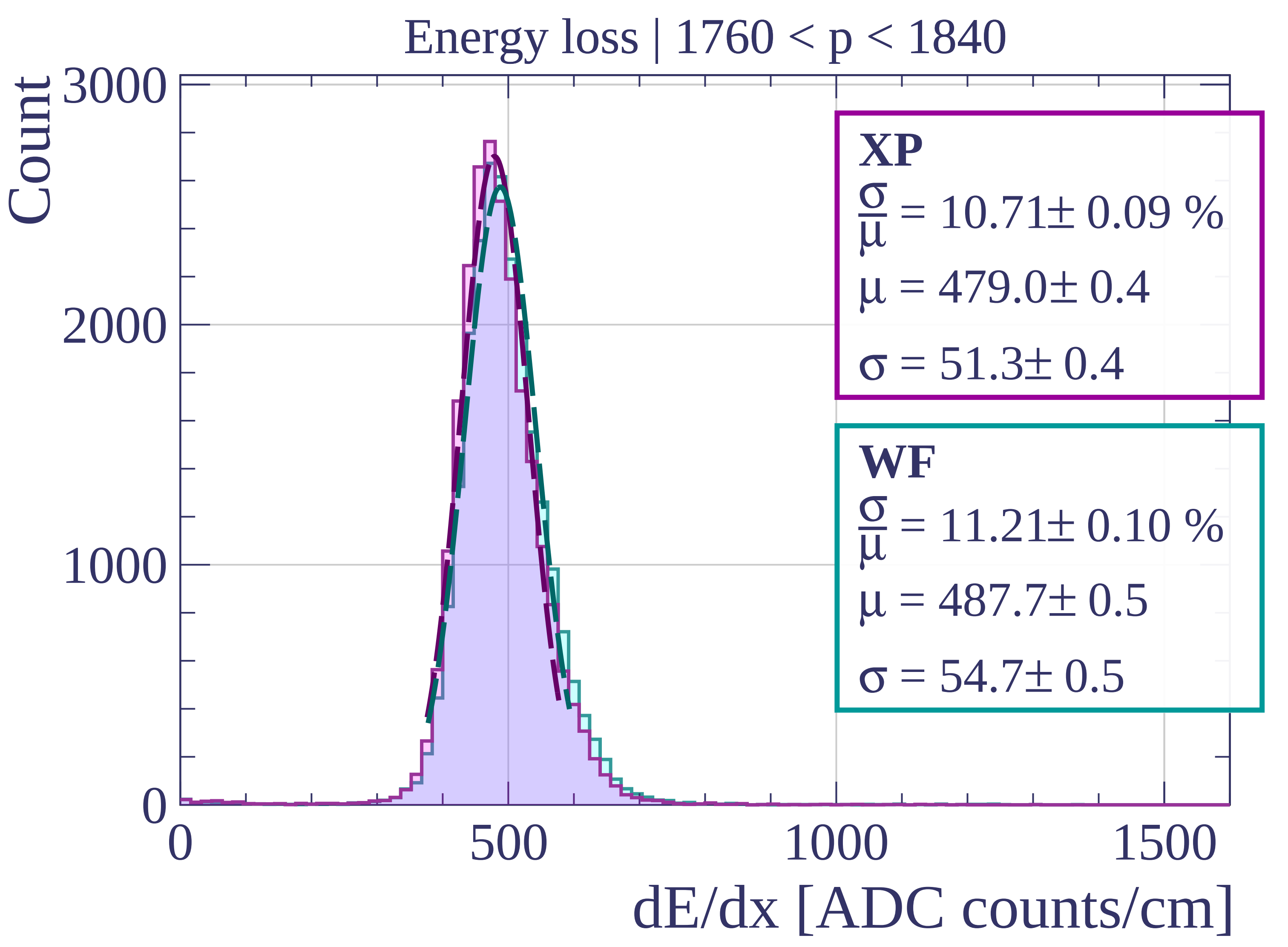


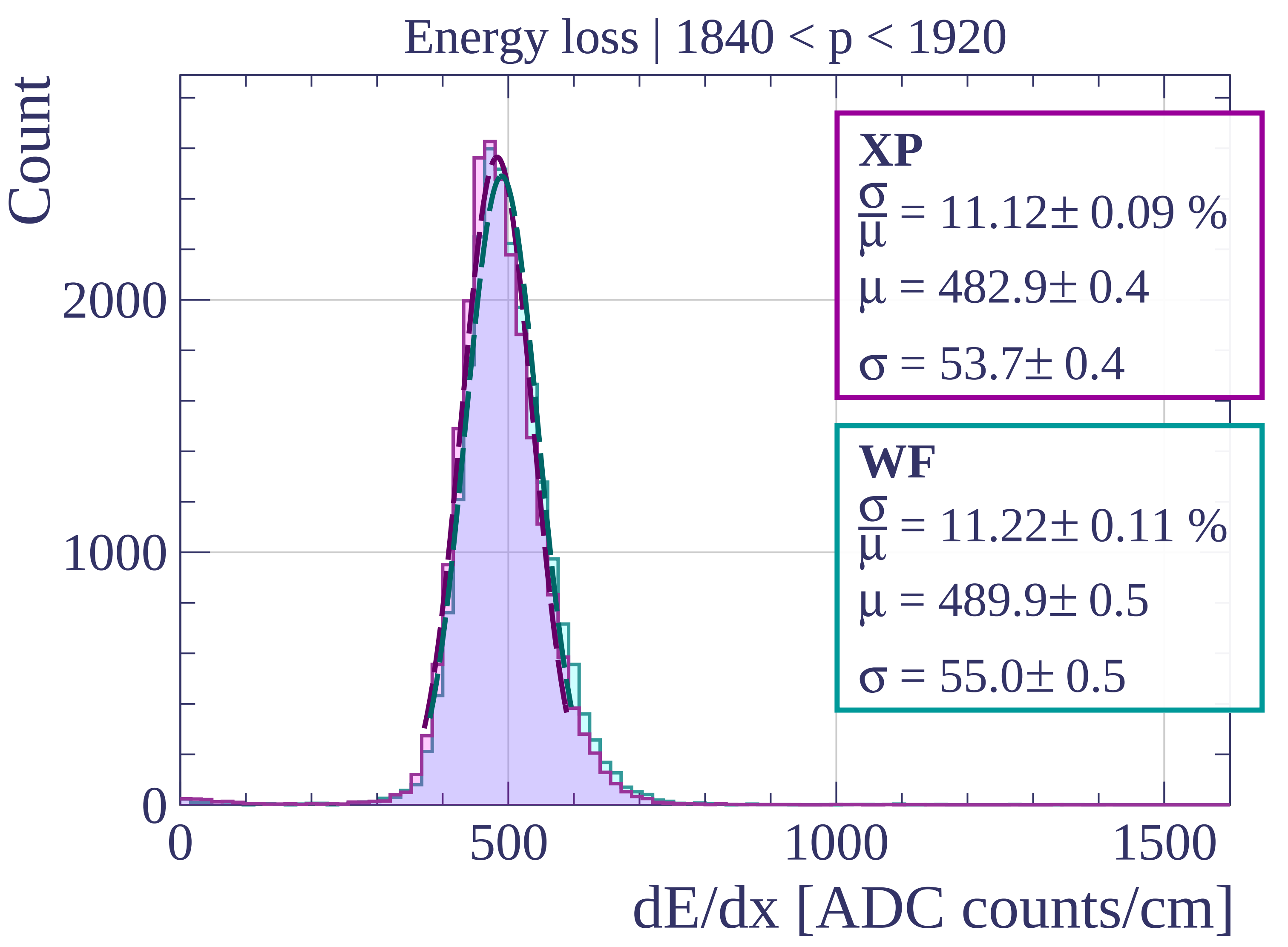
Energy loss | 1520 < p < 1600
Count
dE/dx [ADC counts/cm]
0
1000
2000
3000
500
1500
XP
σ/μ = 10.54± 0.09 %
μ = 468.3± 0.4
σ = 49.3± 0.4
WF
σ/μ = 10.79± 0.09 %
μ = 477.1± 0.4
σ = 51.5± 0.4


Energy loss | 1600 < p < 1680
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 10.47± 0.09 %
μ = 470.7± 0.4
σ = 49.3± 0.4
WF
$\frac{\sigma}{\mu}$ = 10.79± 0.09 %
μ = 479.6± 0.4
σ = 51.8± 0.4

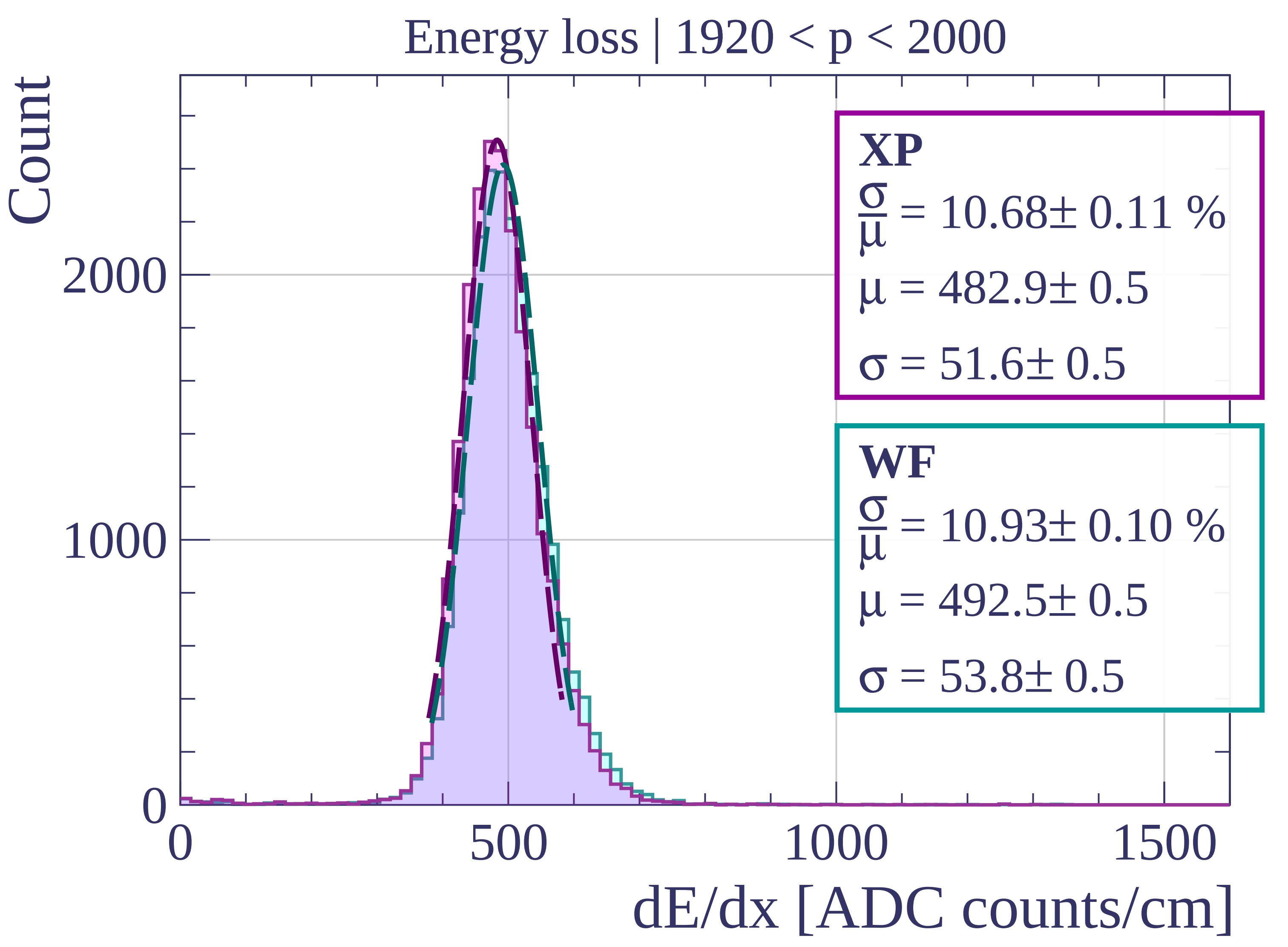


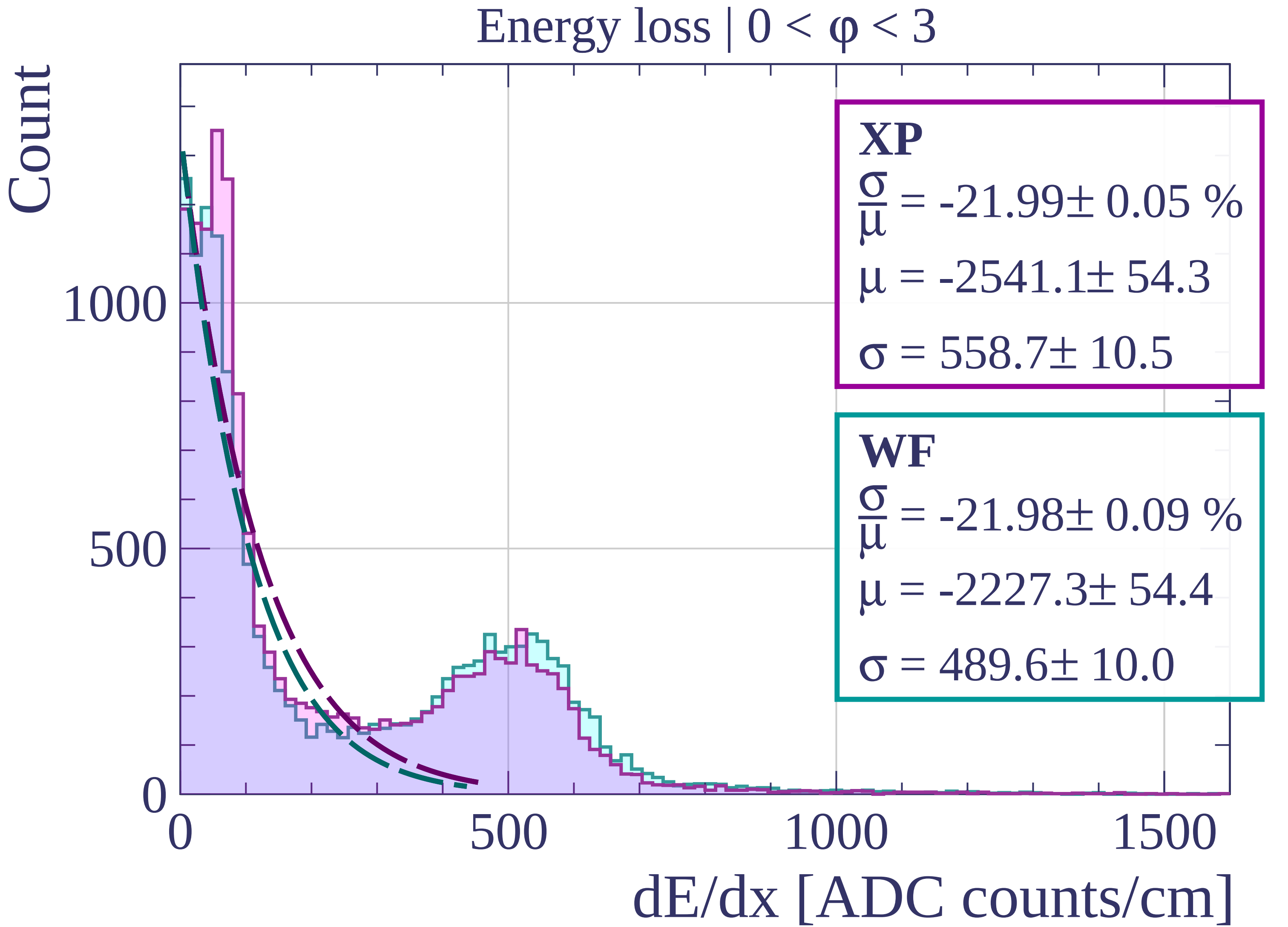
Energy loss | 1680 < p < 1760
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 10.63± 0.09 %
μ = 475.7± 0.4
σ = 50.6± 0.4
WF
σ/μ = 10.96± 0.09 %
μ = 484.4± 0.4
σ = 53.1± 0.4

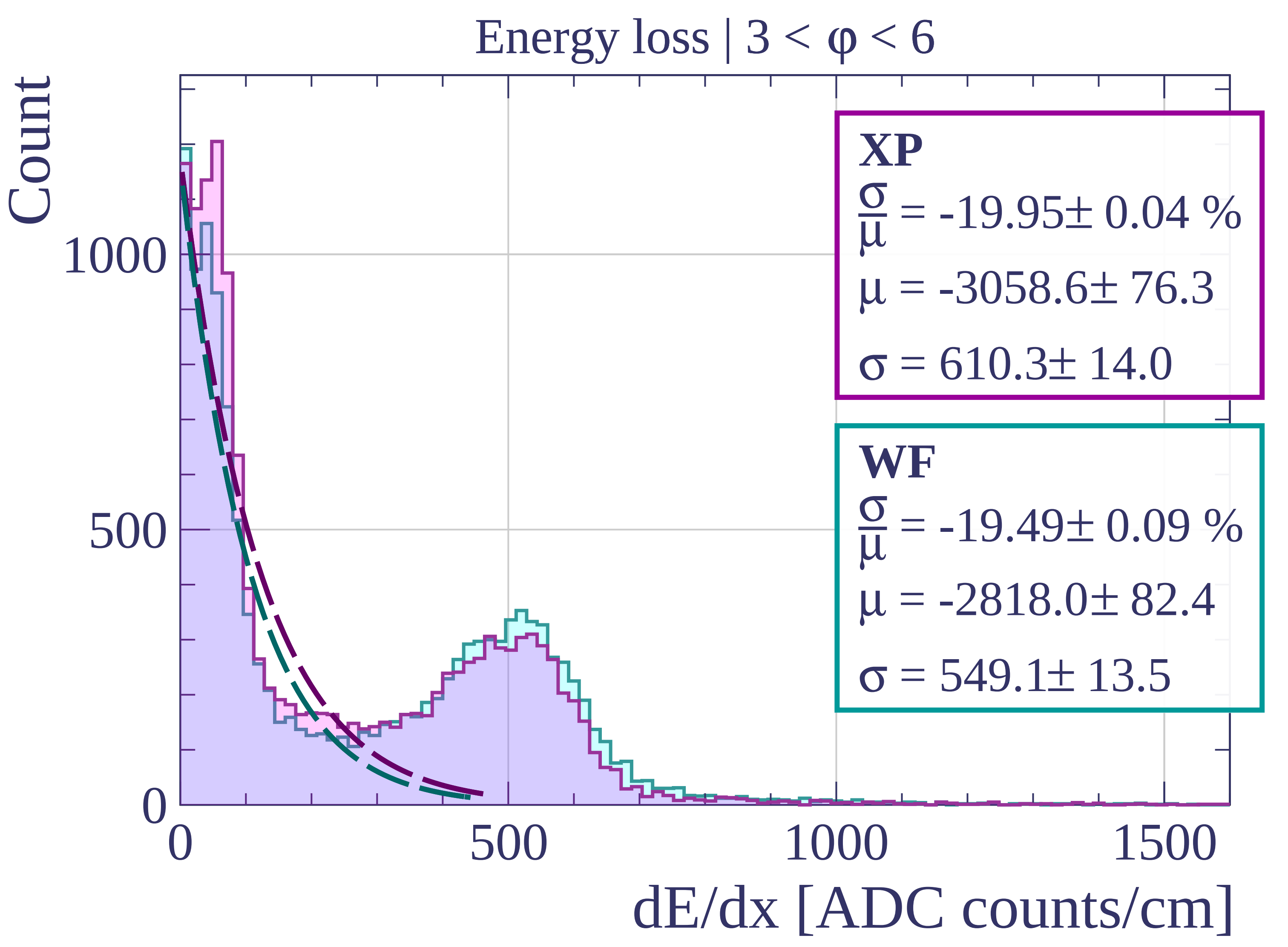

Energy loss | 1760 < p < 1840
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 10.71± 0.09 %
μ = 479.0± 0.4
σ = 51.3± 0.4
WF
σ/μ = 11.21± 0.10 %
μ = 487.7± 0.5
σ = 54.7± 0.5

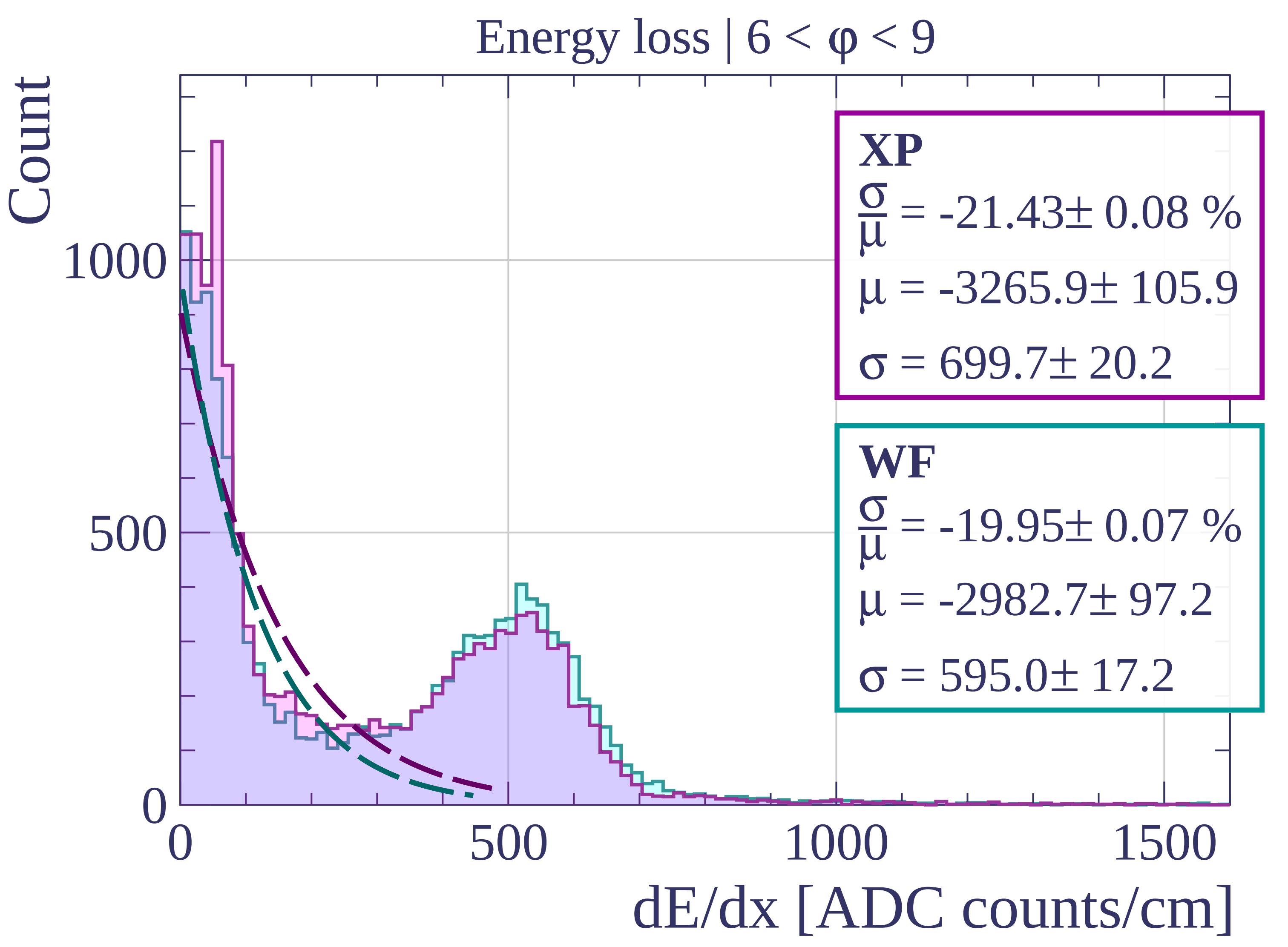


Energy loss | 1840 < p < 1920
Count
0
1000
2000
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 11.12± 0.09 %
μ = 482.9± 0.4
σ = 53.7± 0.4
WF
σ/μ = 11.22± 0.11 %
μ = 489.9± 0.5
σ = 55.0± 0.5

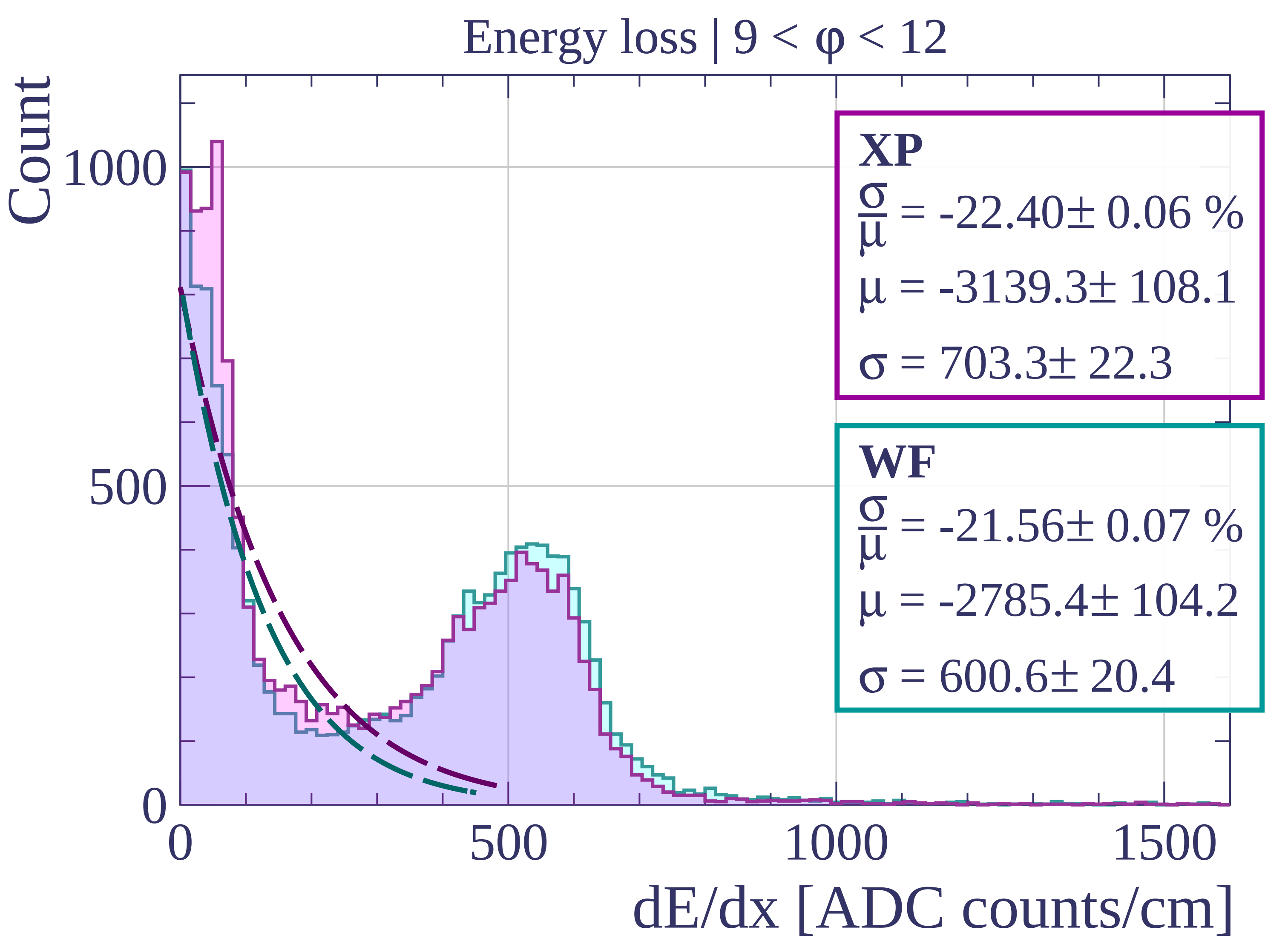

Energy loss | 1920 < p < 2000
Count
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 10.68± 0.11 %
μ = 482.9± 0.5
σ = 51.6± 0.5
WF
$\frac{\sigma}{\mu}$ = 10.93± 0.10 %
μ = 492.5± 0.5
σ = 53.8± 0.5
0
1000
2000
0
500
1000
1500

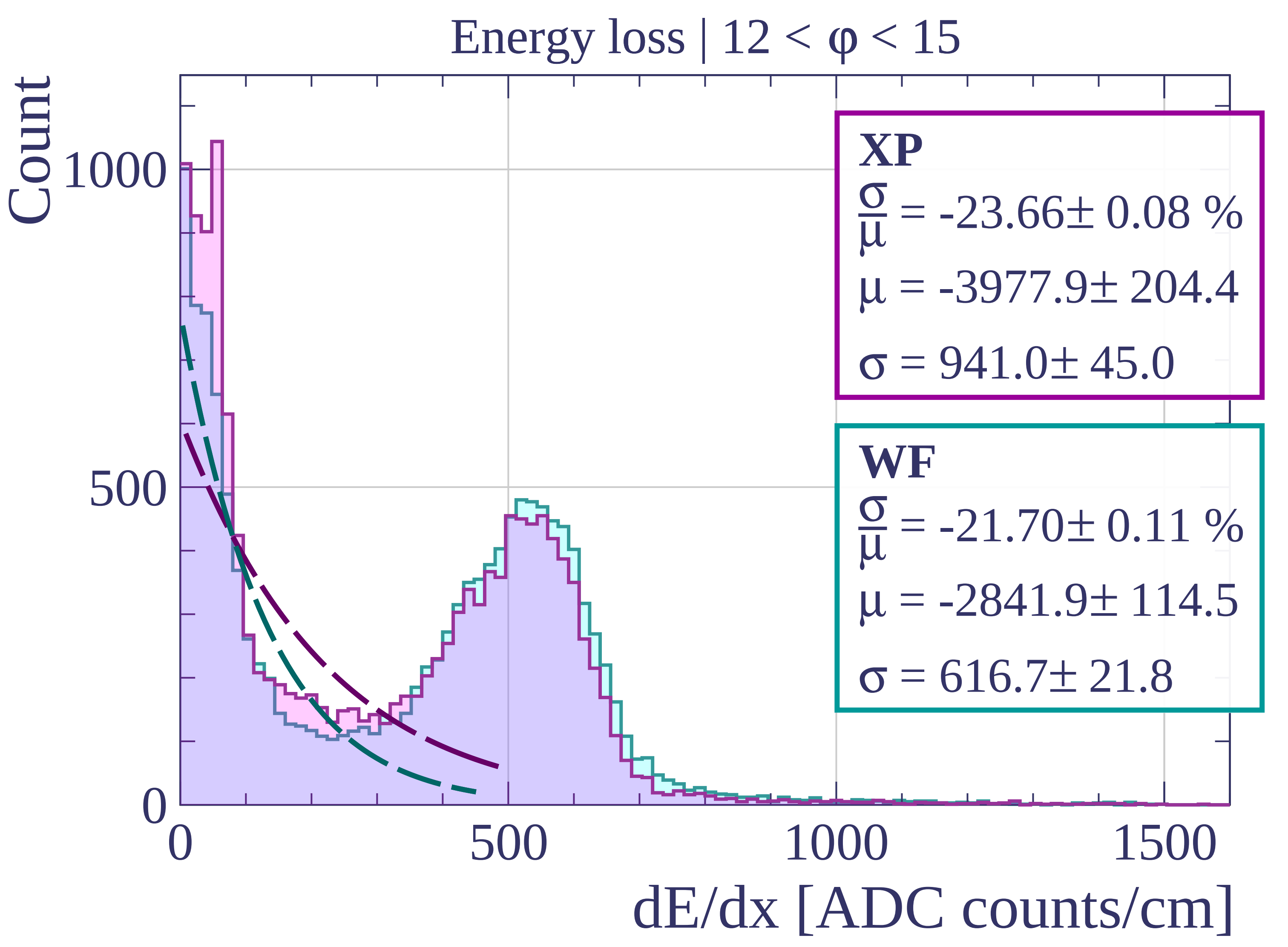

Energy loss | 0 < φ < 3
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = -21.99± 0.05 %
μ = -2541.1± 54.3
σ = 558.7± 10.5
WF
σ/μ = -21.98± 0.09 %
μ = -2227.3± 54.4
σ = 489.6± 10.0


Energy loss | 3 < φ < 6
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
σ/μ = -19.95± 0.04 %
μ = -3058.6± 76.3
σ = 610.3± 14.0
WF
σ/μ = -19.49± 0.09 %
μ = -2818.0± 82.4
σ = 549.1± 13.5

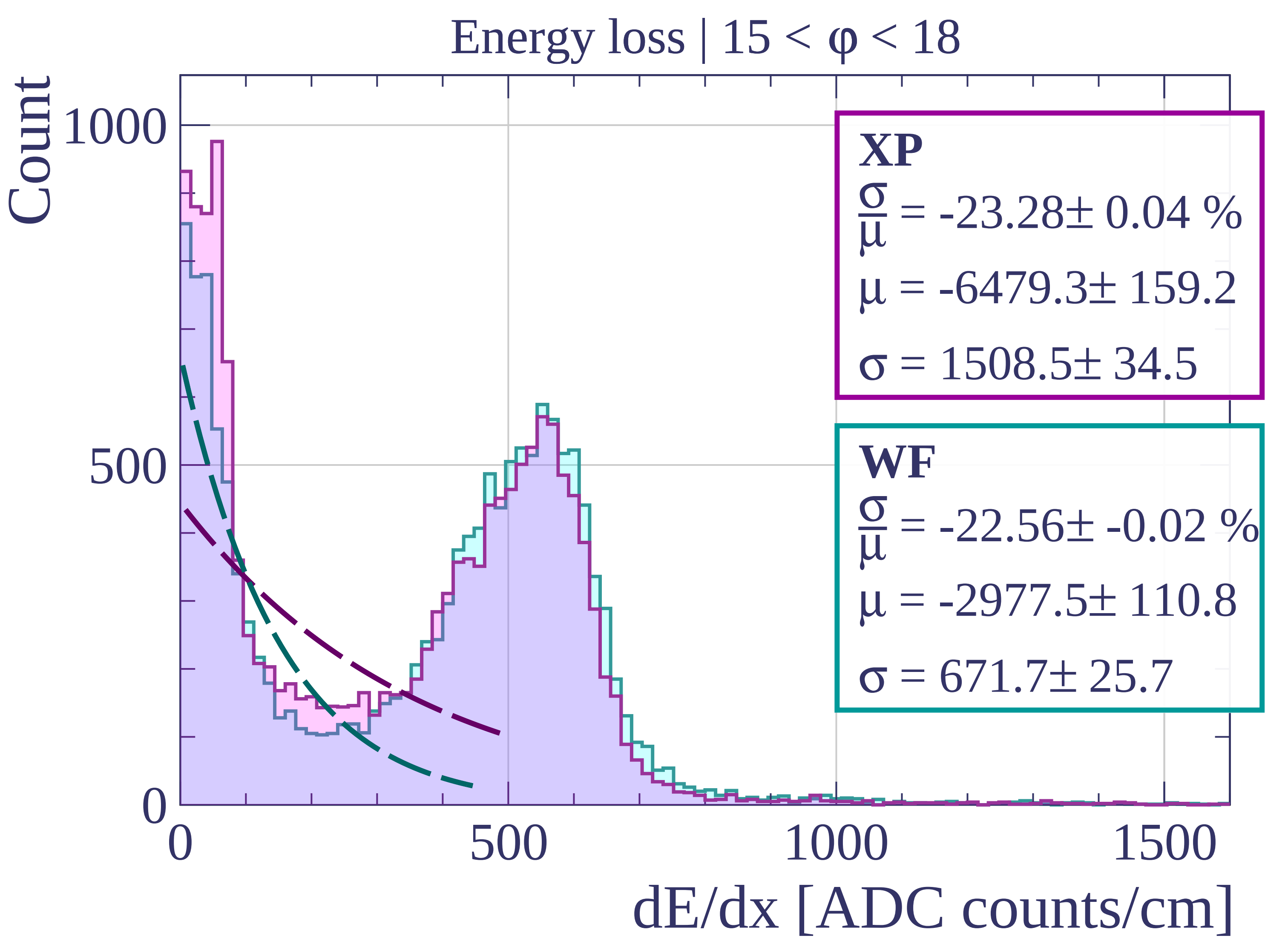


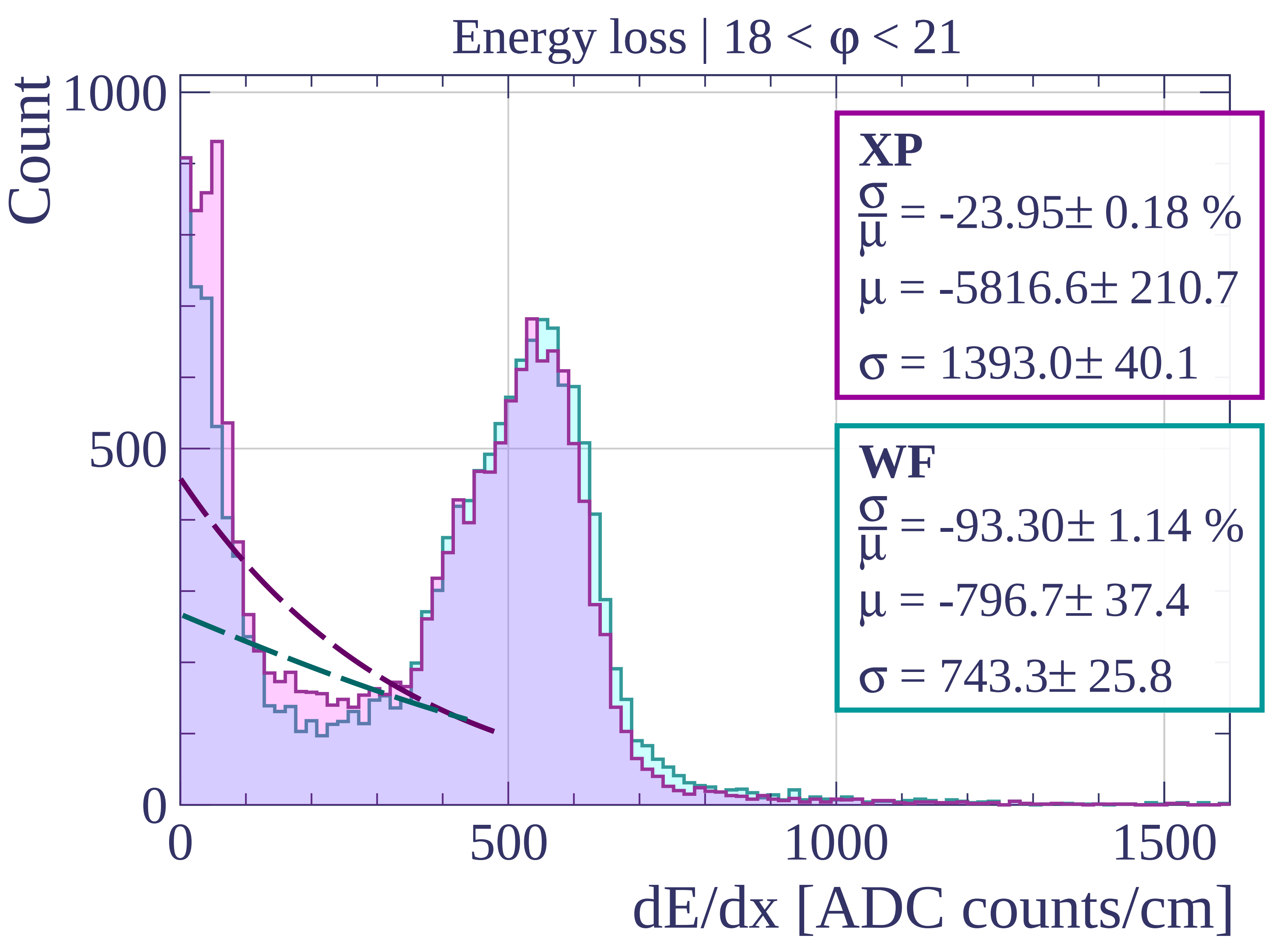
Energy loss | 6 < φ < 9
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
σ/μ = -21.43± 0.08 %
μ = -3265.9± 105.9
σ = 699.7± 20.2
WF
σ/μ = -19.95± 0.07 %
μ = -2982.7± 97.2
σ = 595.0± 17.2


Energy loss | 9 < φ < 12
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
σ/μ = -22.40± 0.06 %
μ = -3139.3± 108.1
σ = 703.3± 22.3
WF
σ/μ = -21.56± 0.07 %
μ = -2785.4± 104.2
σ = 600.6± 20.4

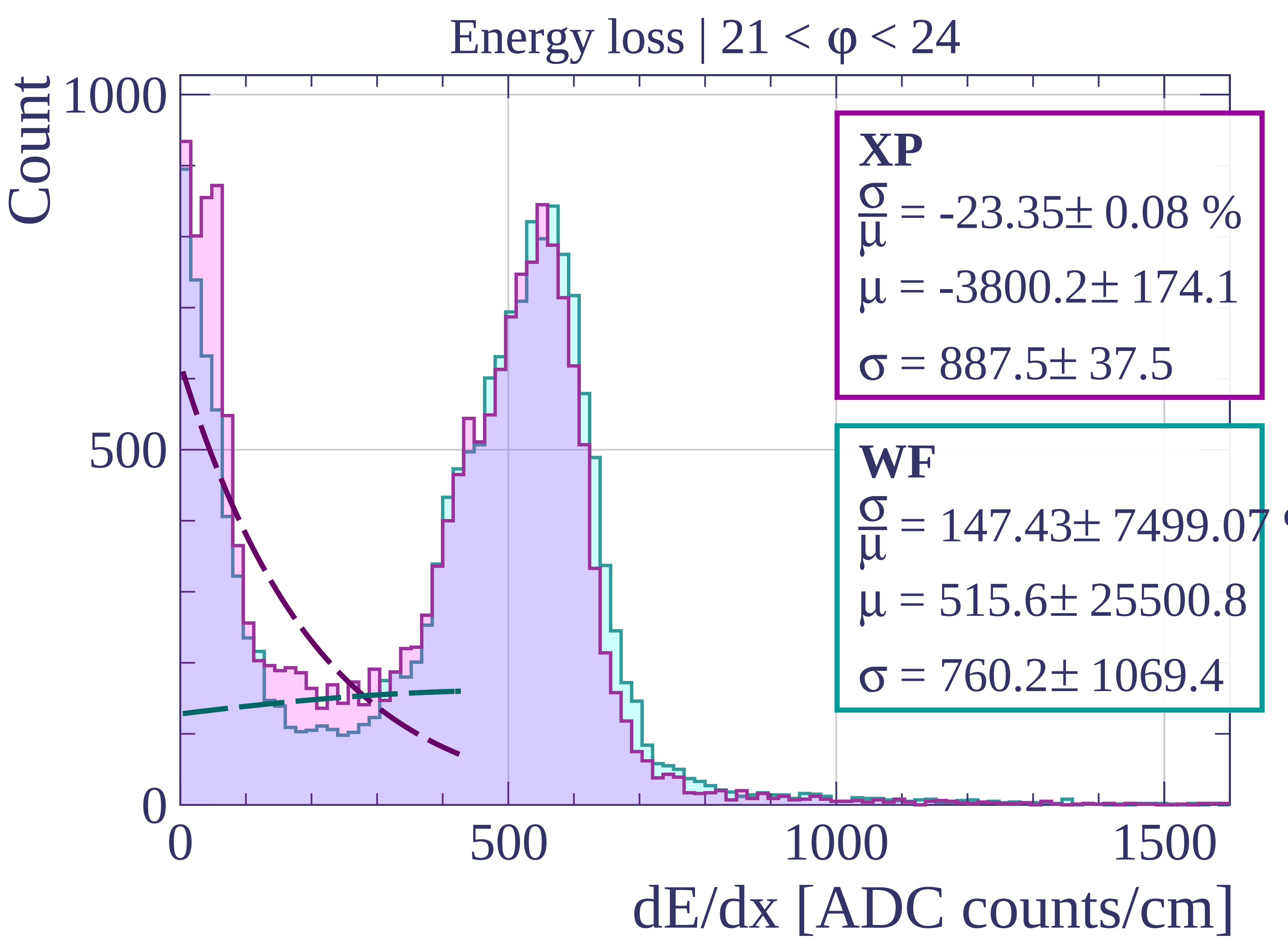


Energy loss | 12 < φ < 15
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = -23.66± 0.08 %
μ = -3977.9± 204.4
σ = 941.0± 45.0
WF
σ/μ = -21.70± 0.11 %
μ = -2841.9± 114.5
σ = 616.7± 21.8

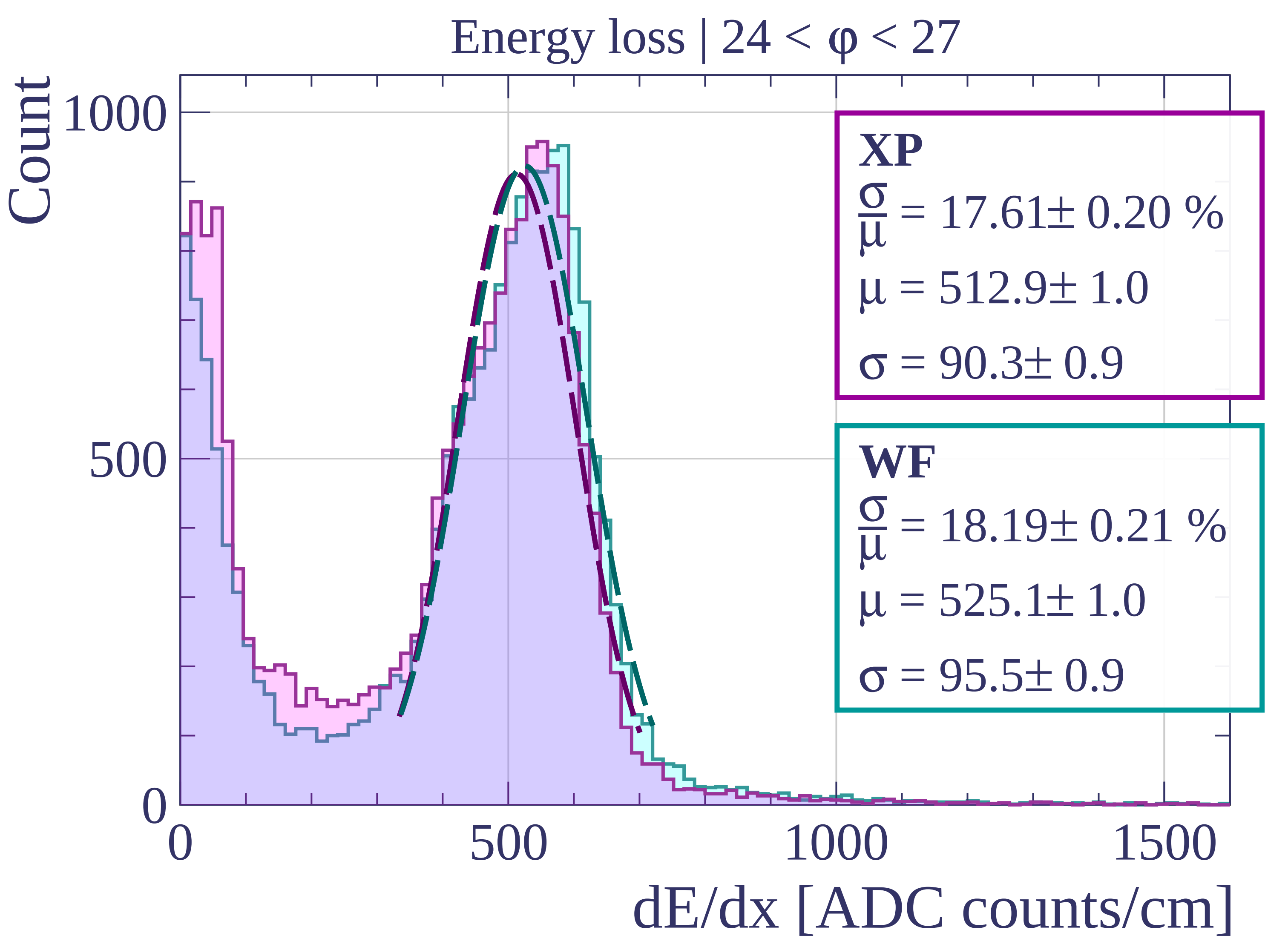

Energy loss | 15 < φ < 18
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = -23.28± 0.04 %
μ = -6479.3± 159.2
σ = 1508.5± 34.5
WF
σ/μ = -22.56± -0.02 %
μ = -2977.5± 110.8
σ = 671.7± 25.7

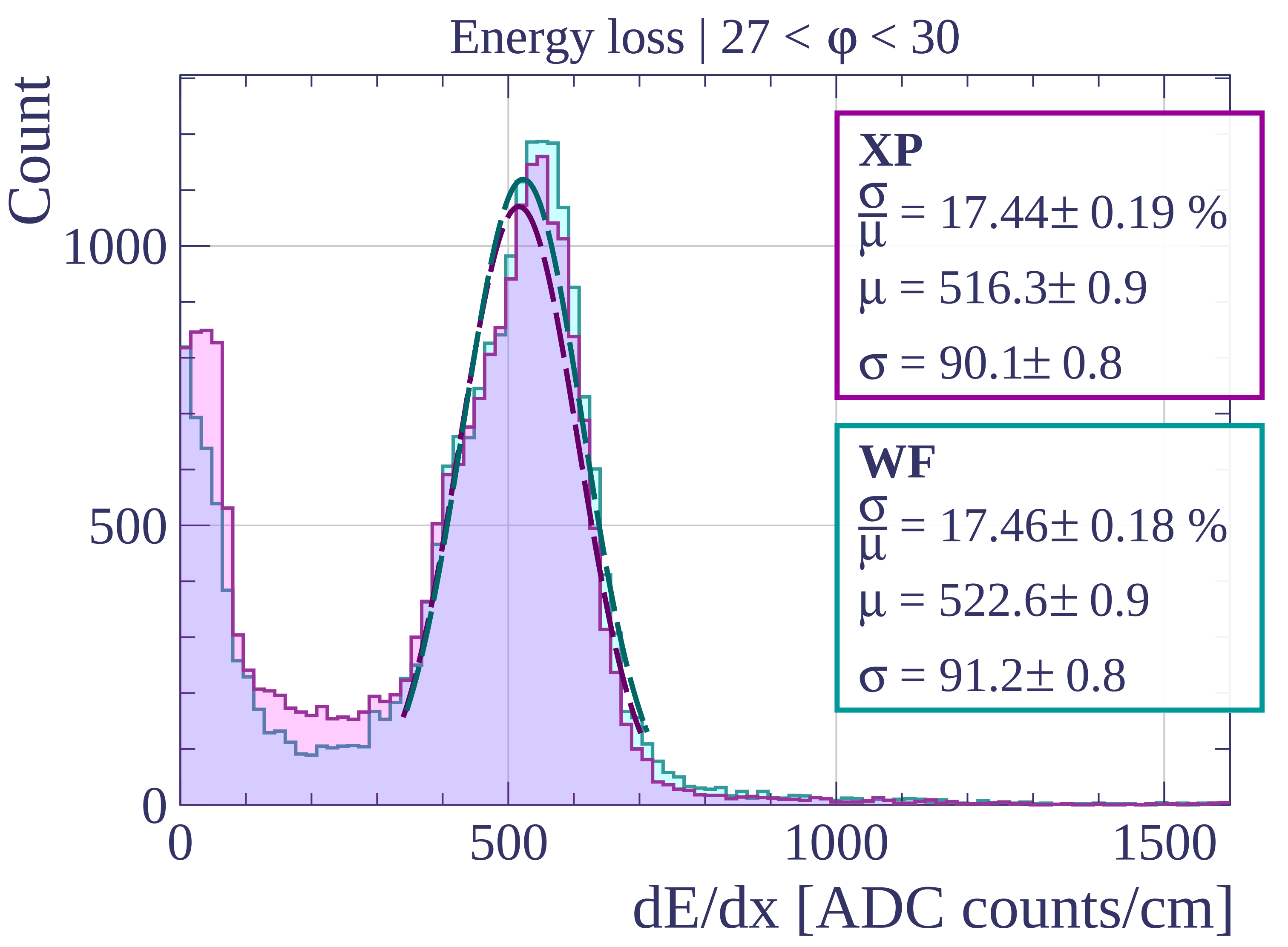


Energy loss | 18 < φ < 21
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = -23.95± 0.18 %
μ = -5816.6± 210.7
σ = 1393.0± 40.1
WF
σ/μ = -93.30± 1.14 %
μ = -796.7± 37.4
σ = 743.3± 25.8

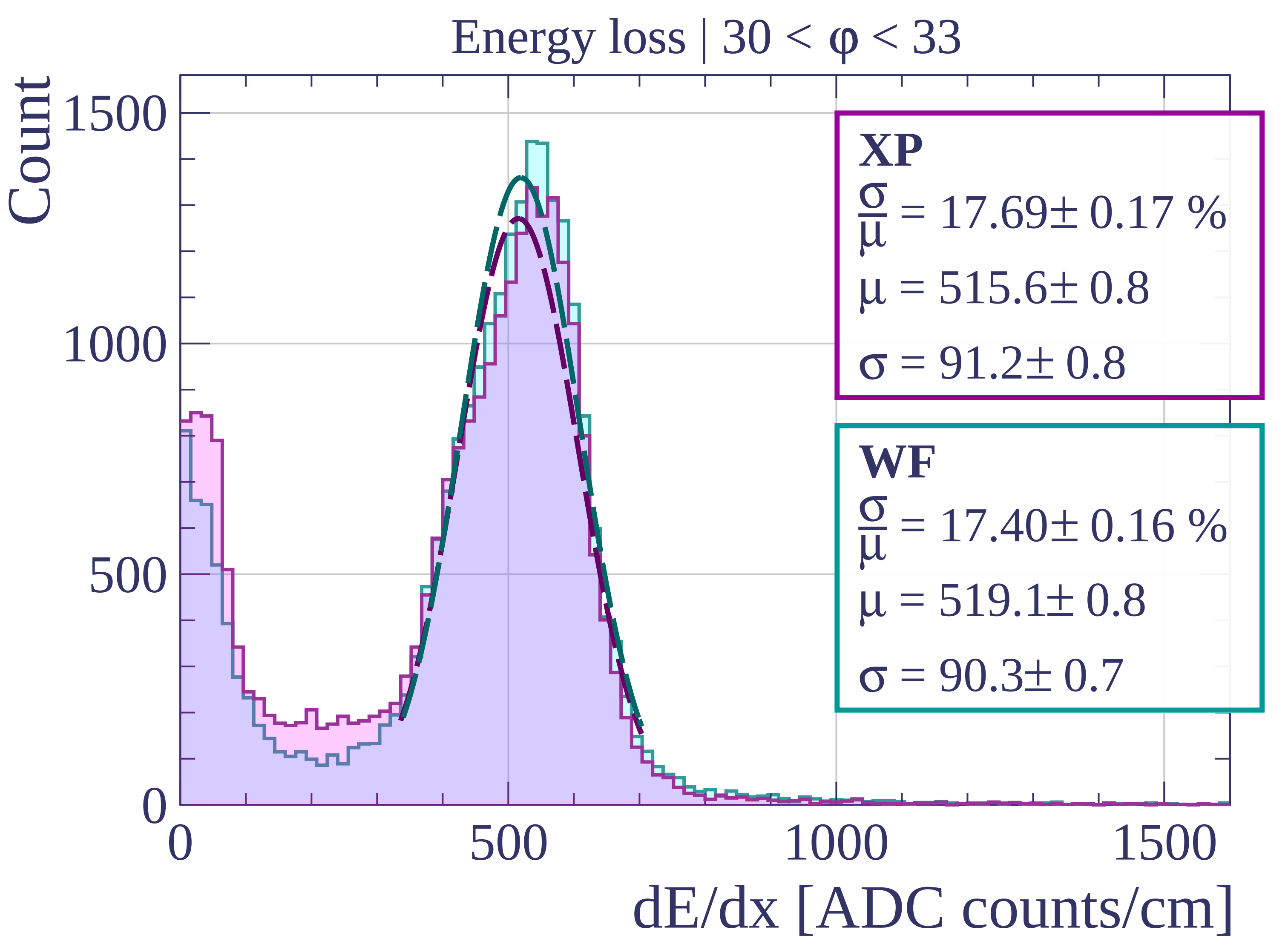


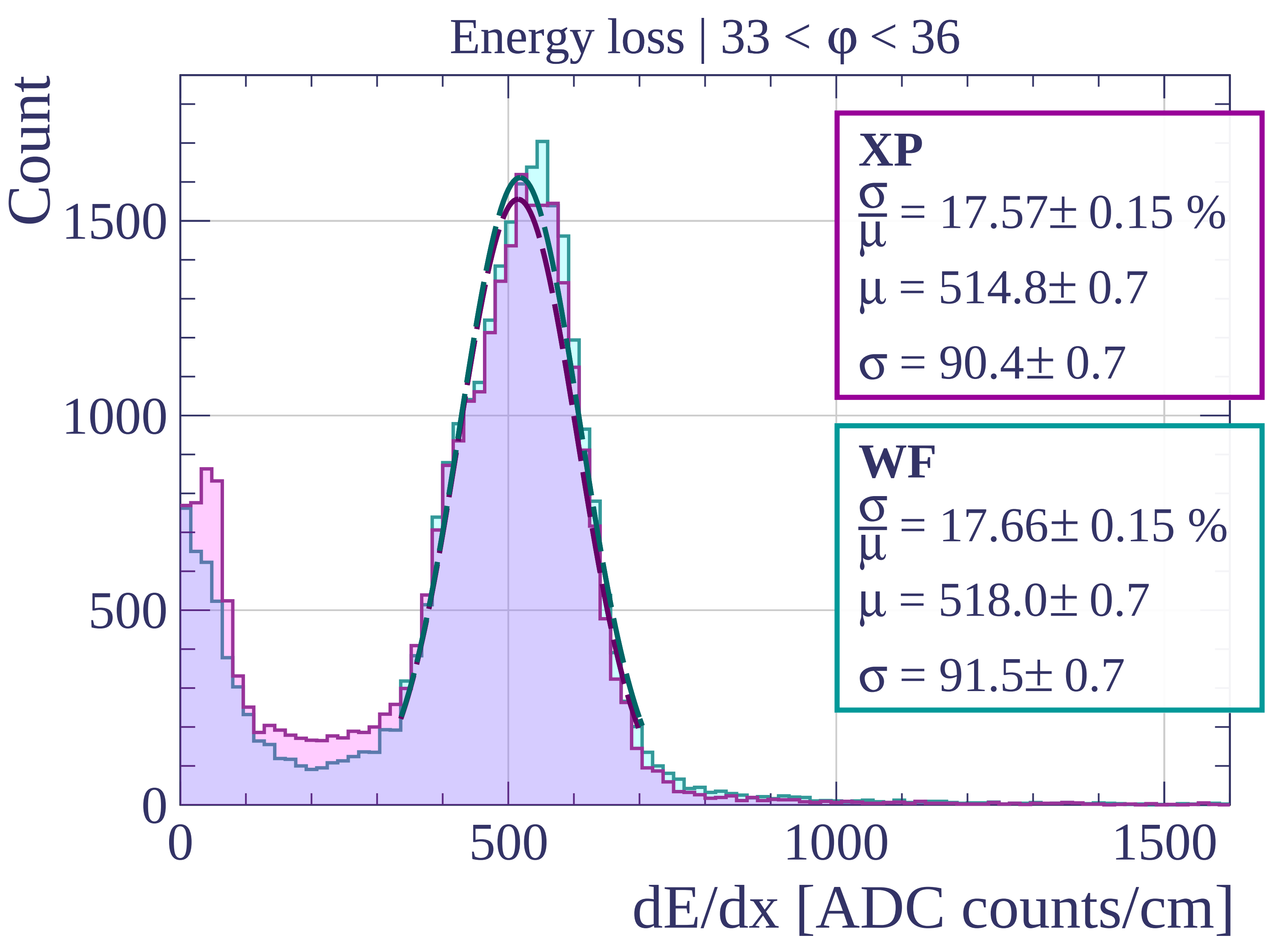
Energy loss | 21 < φ < 24
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = -23.35± 0.08 %
μ = -3800.2± 174.1
σ = 887.5± 37.5
WF
σ/μ = 147.43± 7499.07
μ = 515.6± 25500.8
σ = 760.2± 1069.4

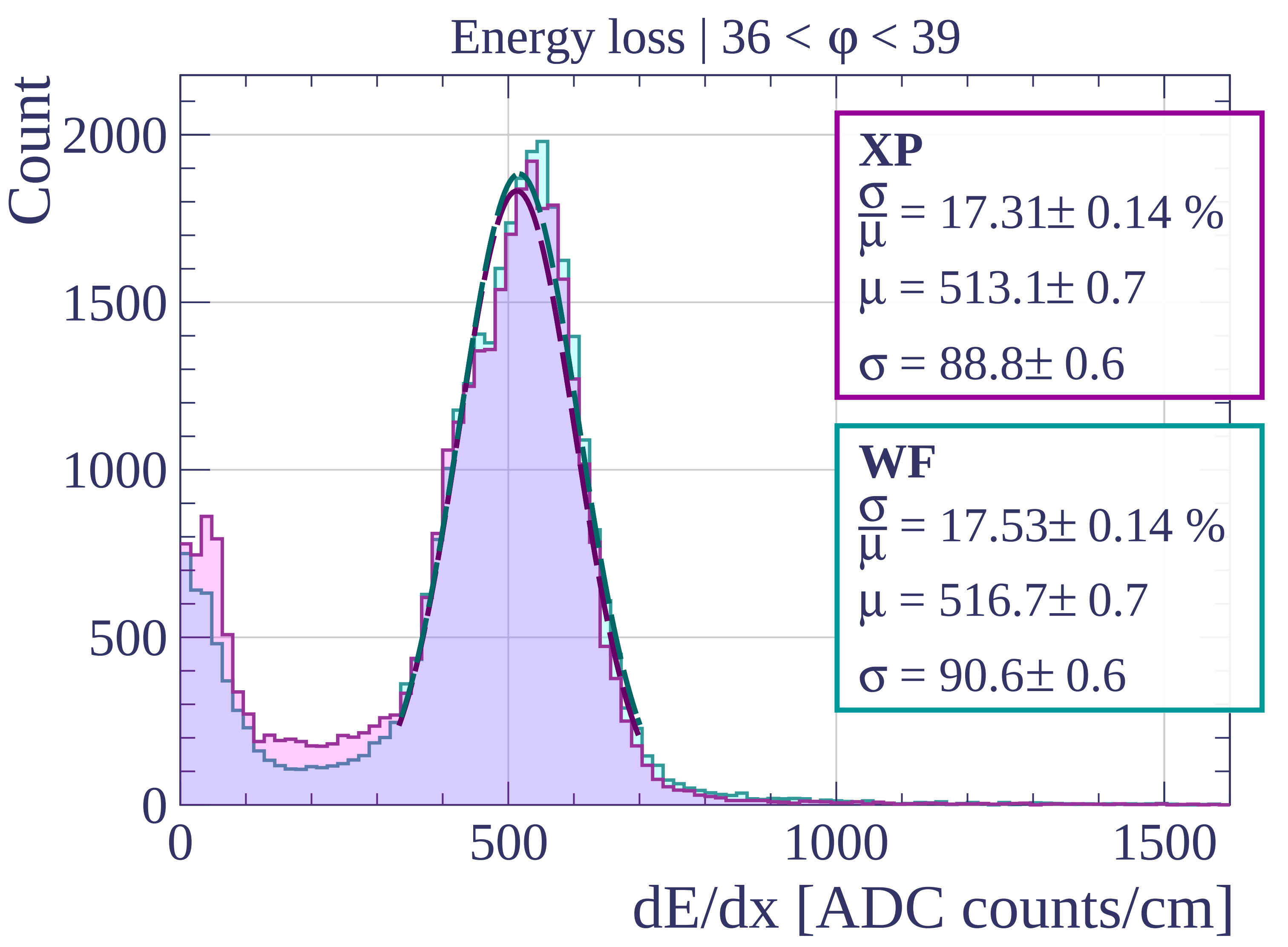

Energy loss | 24 < φ < 27
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 17.61± 0.20 %
μ = 512.9± 1.0
σ = 90.3± 0.9
WF
σ/μ = 18.19± 0.21 %
μ = 525.1± 1.0
σ = 95.5± 0.9



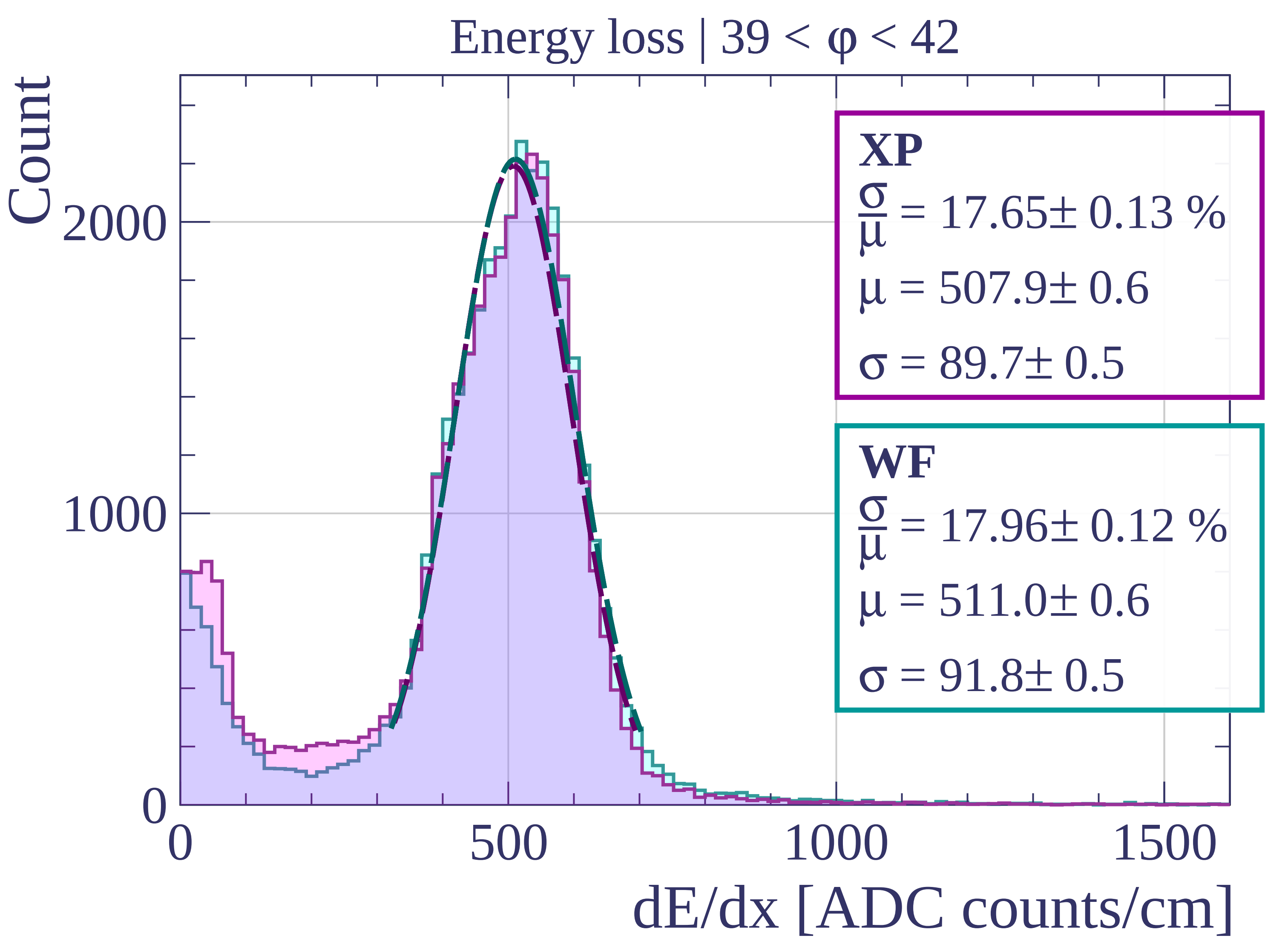
Energy loss | 27 < φ < 30
Count
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 17.44± 0.19 %
μ = 516.3± 0.9
σ = 90.1± 0.8
WF
$\frac{\sigma}{\mu}$ = 17.46± 0.18 %
μ = 522.6± 0.9
σ = 91.2± 0.8


Energy loss | 30 < φ < 33
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 17.69± 0.17 %
μ = 515.6± 0.8
σ = 91.2± 0.8
WF
σ/μ = 17.40± 0.16 %
μ = 519.1± 0.8
σ = 90.3± 0.7

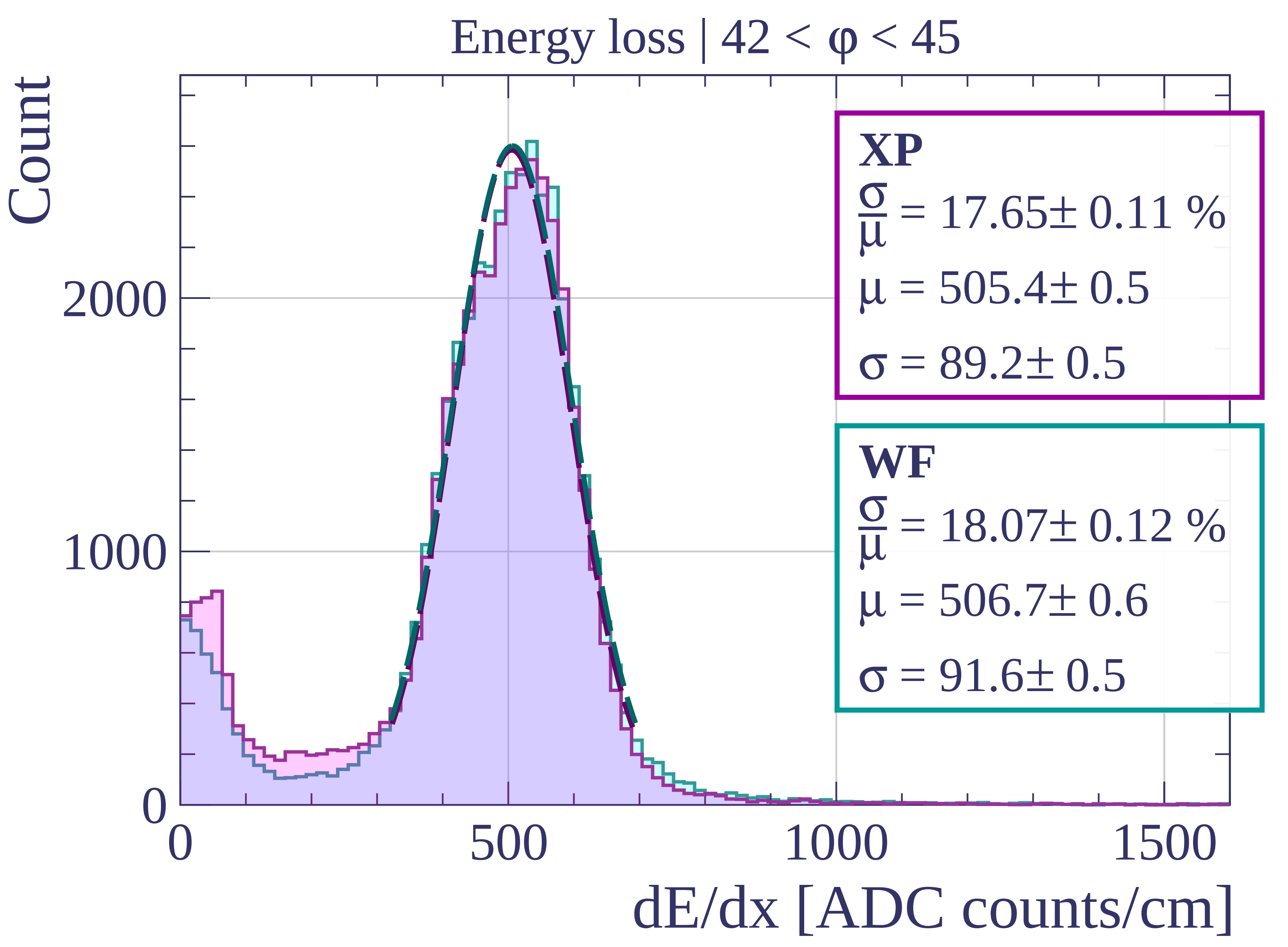

Energy loss | 33 < φ < 36
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
σ/μ = 17.57± 0.15 %
μ = 514.8± 0.7
σ = 90.4± 0.7
WF
σ/μ = 17.66± 0.15 %
μ = 518.0± 0.7
σ = 91.5± 0.7

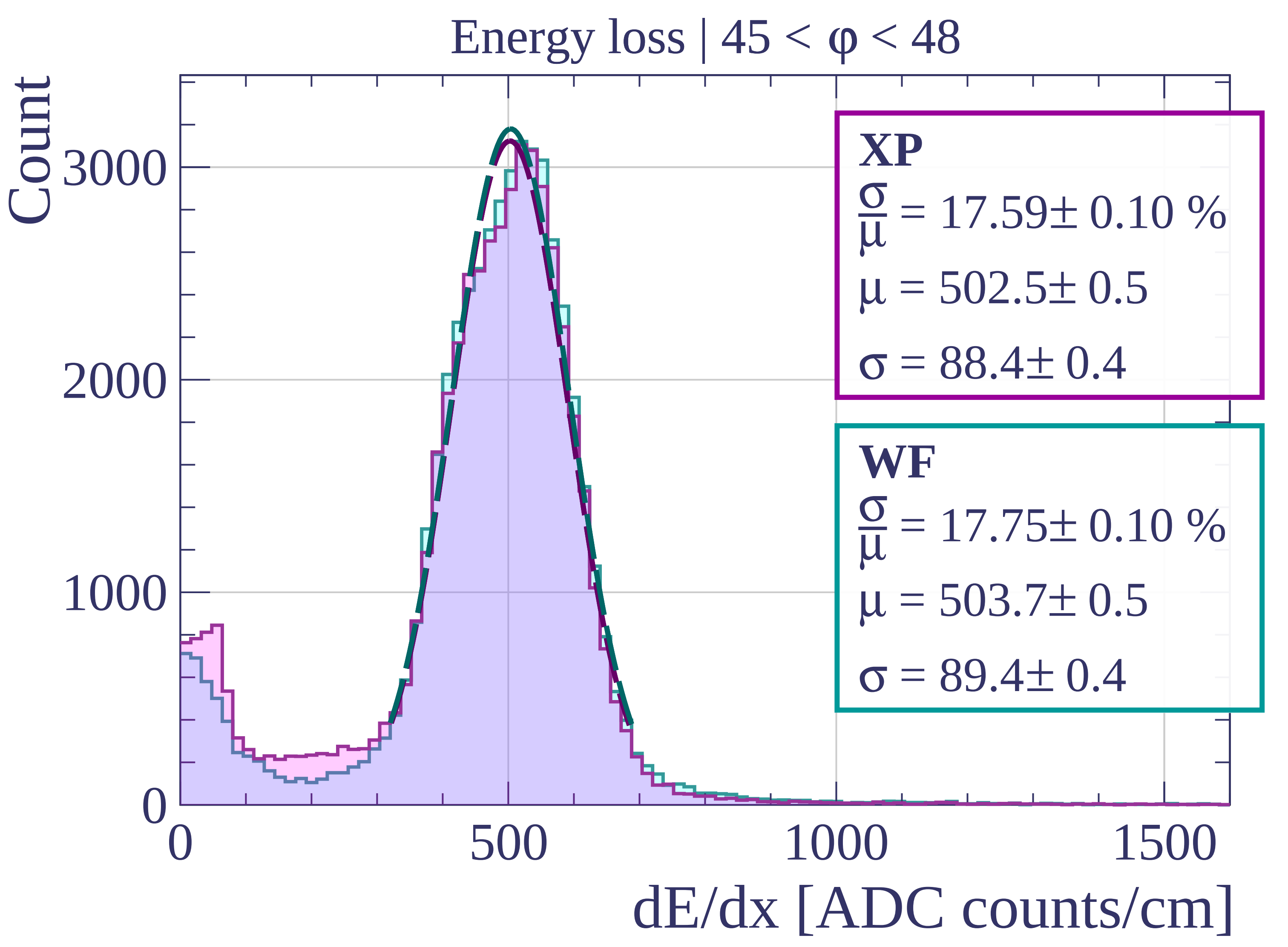


Energy loss | 36 < φ < 39
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
2000
0
500
1000
1500
XP
σ/μ = 17.31± 0.14 %
μ = 513.1± 0.7
σ = 88.8± 0.6
WF
σ/μ = 17.53± 0.14 %
μ = 516.7± 0.7
σ = 90.6± 0.6

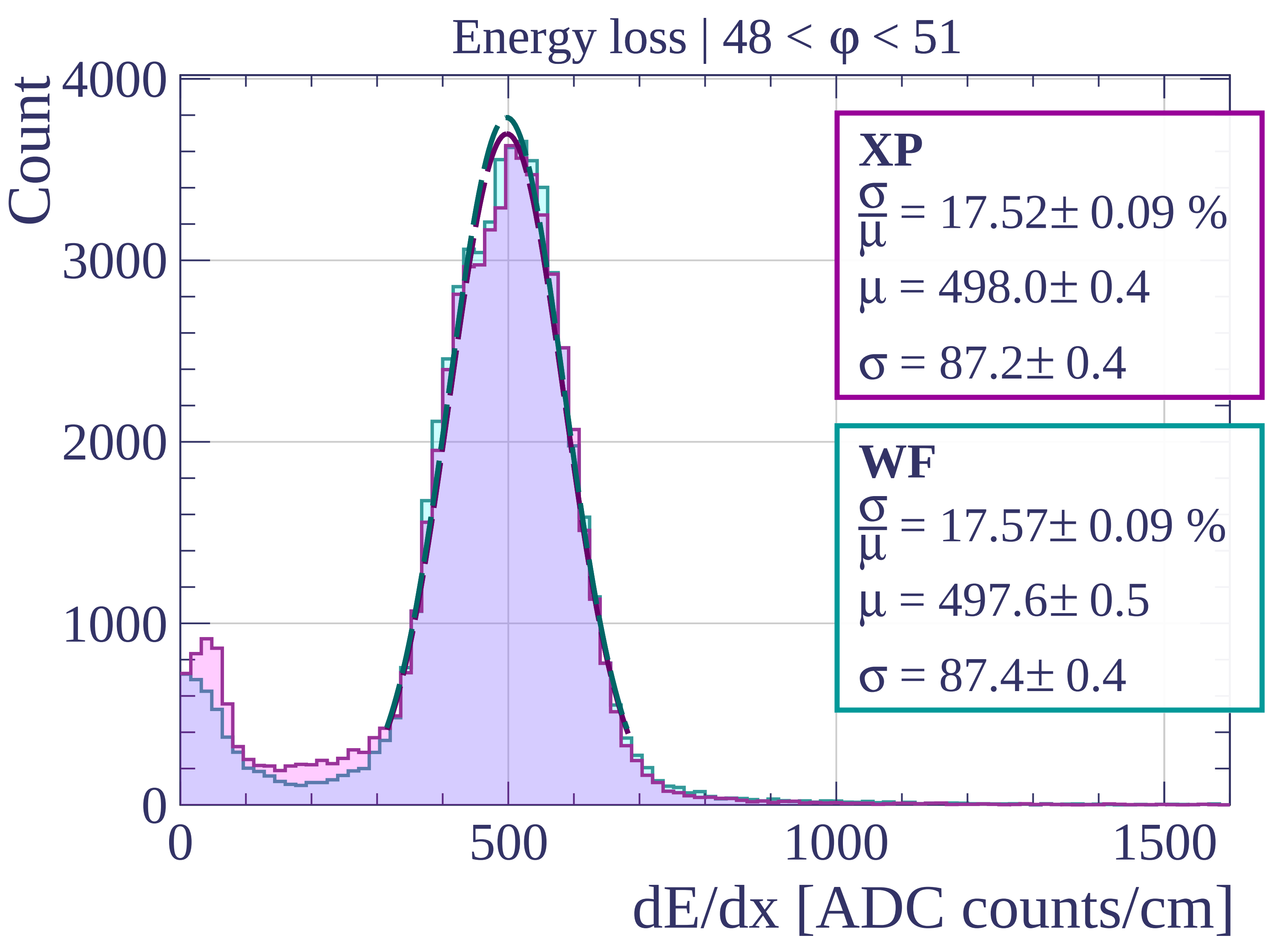


Energy loss | 39 < φ < 42
Count
2000
1000
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 17.65± 0.13 %
μ = 507.9± 0.6
σ = 89.7± 0.5
WF
σ/μ = 17.96± 0.12 %
μ = 511.0± 0.6
σ = 91.8± 0.5

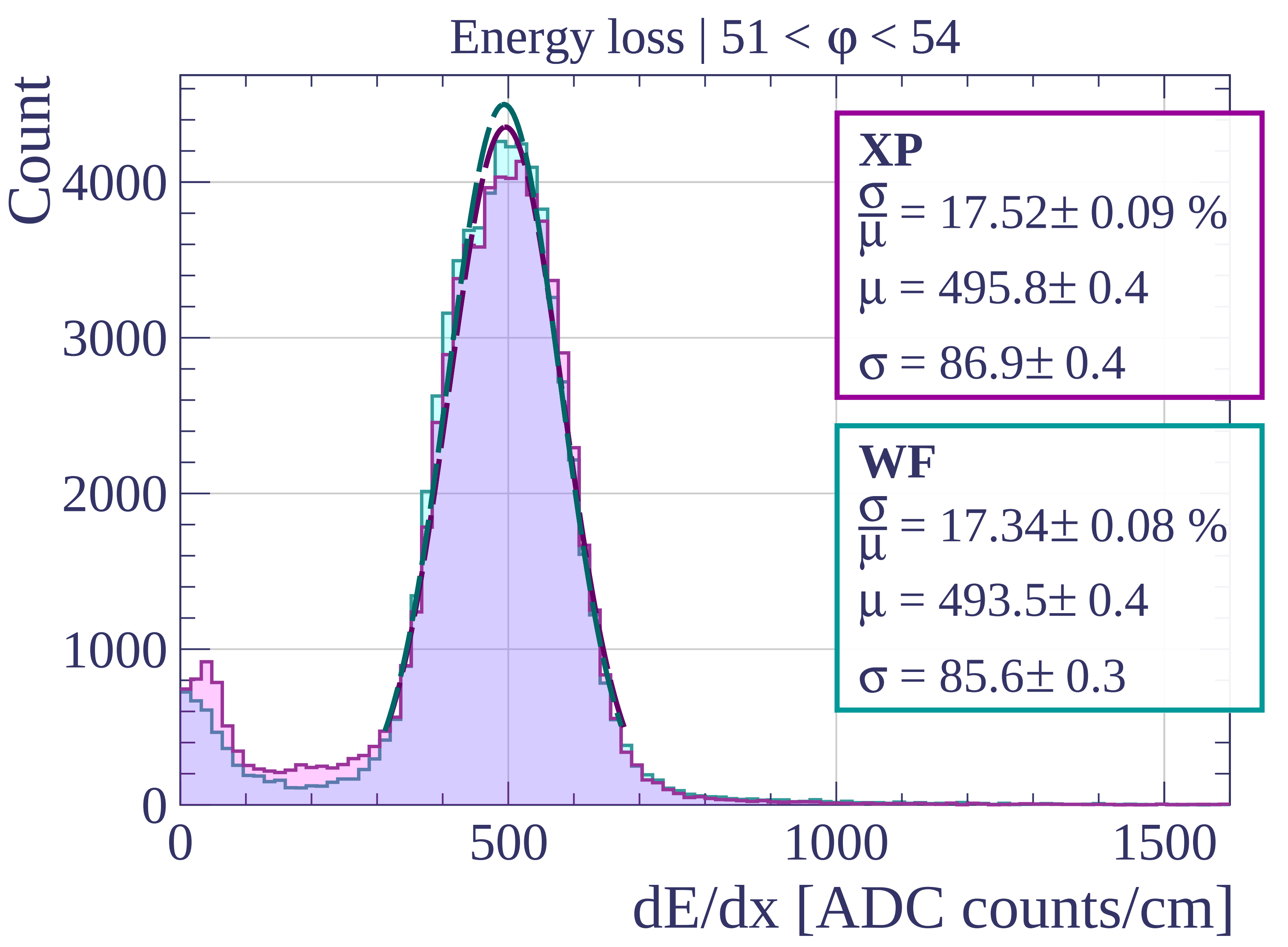

Energy loss | 42 < φ < 45
Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 17.65± 0.11 %
μ = 505.4± 0.5
σ = 89.2± 0.5
WF
σ/μ = 18.07± 0.12 %
μ = 506.7± 0.6
σ = 91.6± 0.5



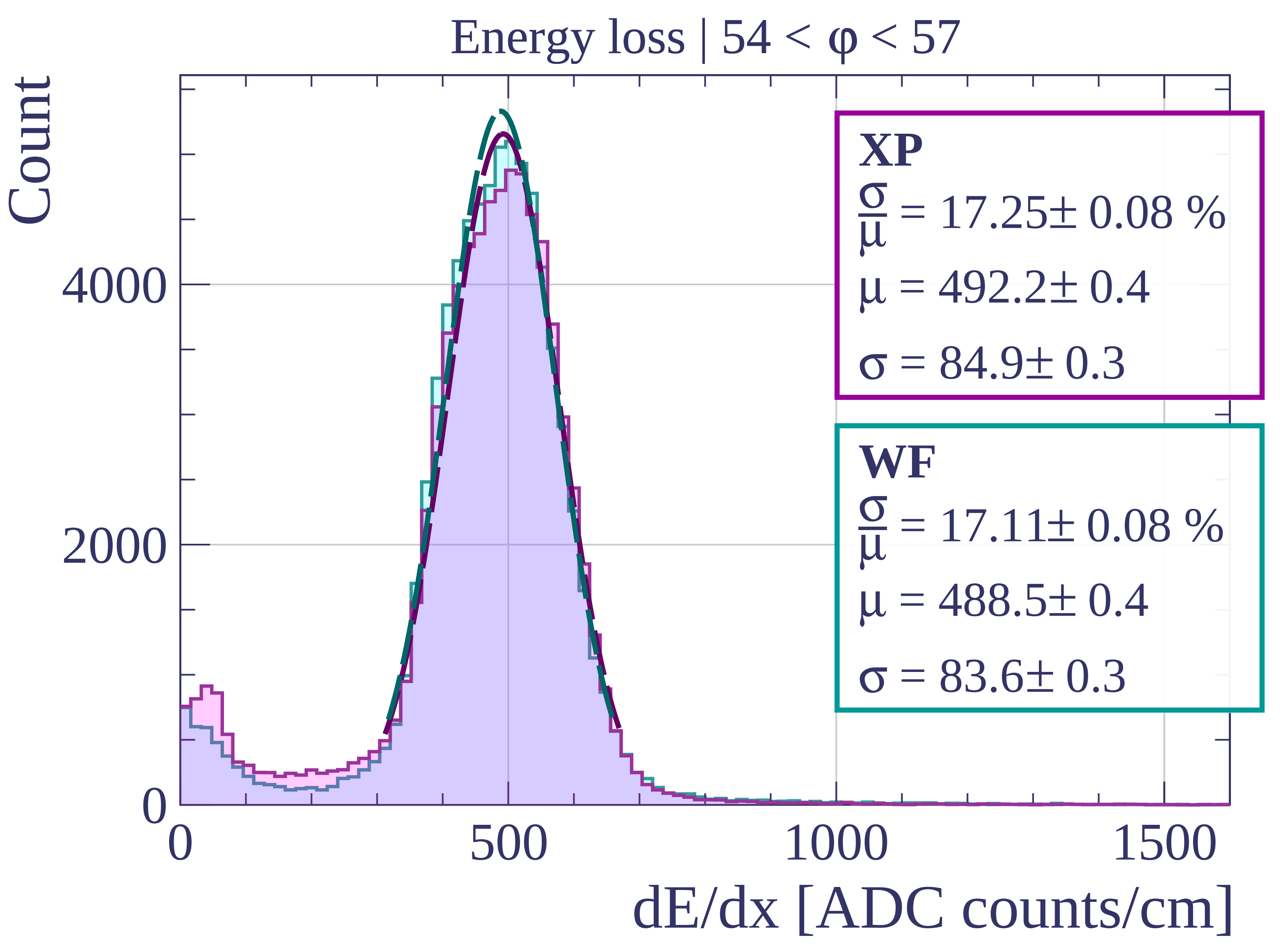
Energy loss | 45 < φ < 48
Count
dE/dx [ADC counts/cm]
0
1000
2000
3000
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = 17.59± 0.10 %
μ = 502.5± 0.5
σ = 88.4± 0.4
WF
$\frac{\sigma}{\mu}$ = 17.75± 0.10 %
μ = 503.7± 0.5
σ = 89.4± 0.4



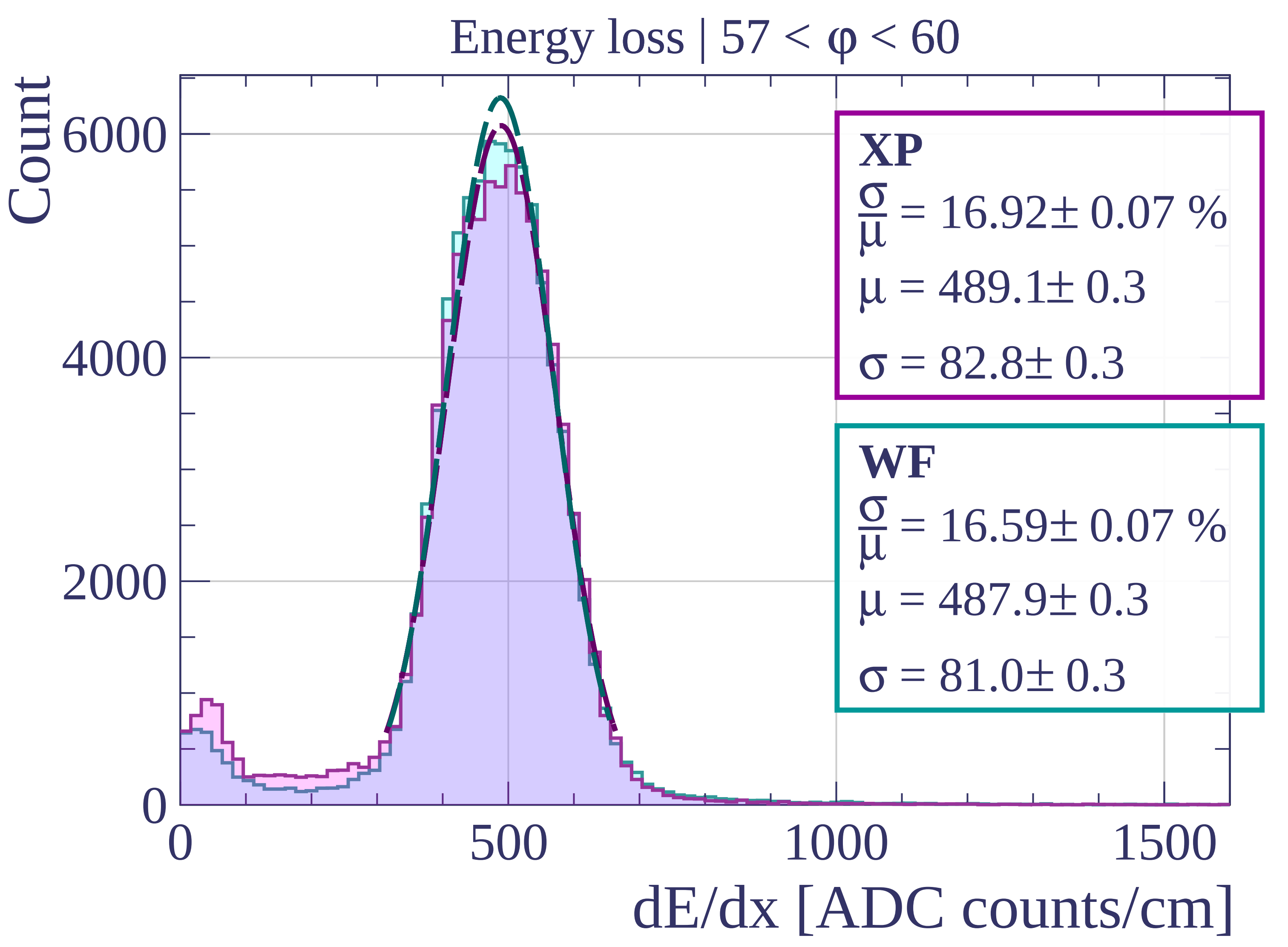
Energy loss | 48 < φ < 51
Count
4000
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 17.52± 0.09 %
μ = 498.0± 0.4
σ = 87.2± 0.4
WF
σ/μ = 17.57± 0.09 %
μ = 497.6± 0.5
σ = 87.4± 0.4



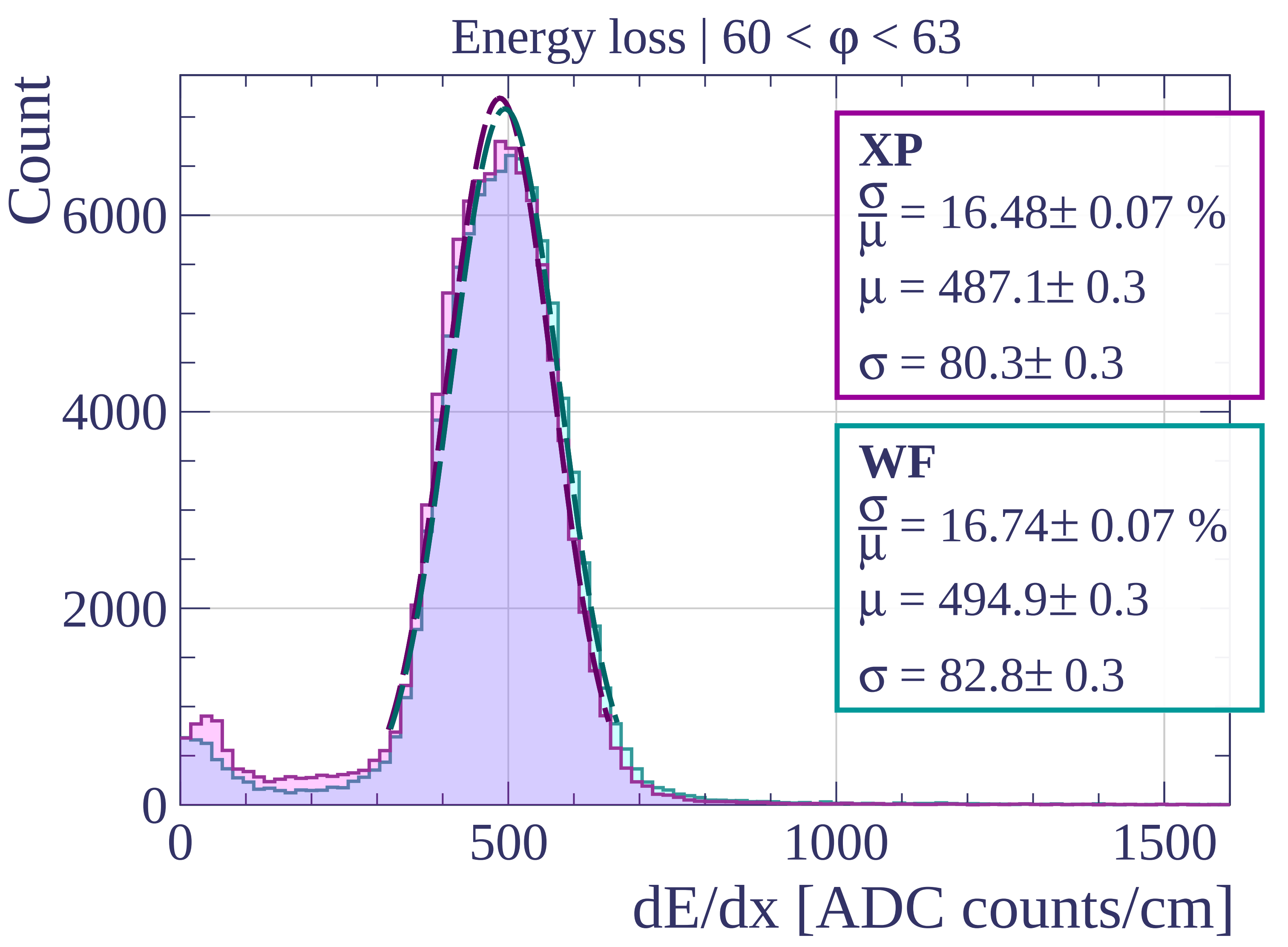
Energy loss | 51 < φ < 54
Count
4000
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 17.52± 0.09 %
μ = 495.8± 0.4
σ = 86.9± 0.4
WF
σ/μ = 17.34± 0.08 %
μ = 493.5± 0.4
σ = 85.6± 0.3



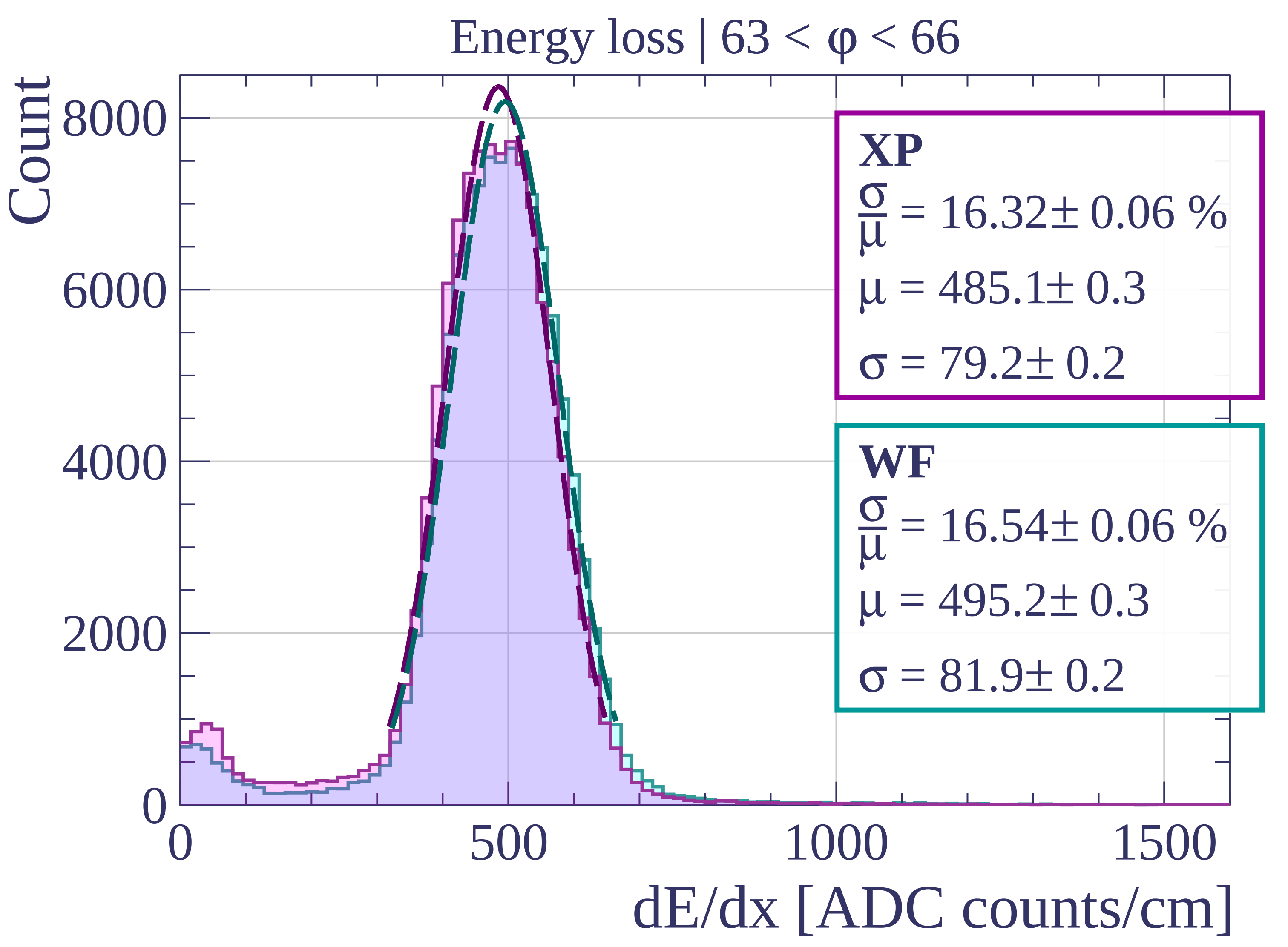
Energy loss | 54 < φ < 57
Count
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 17.25± 0.08 %
μ = 492.2± 0.4
σ = 84.9± 0.3
WF
σ/μ = 17.11± 0.08 %
μ = 488.5± 0.4
σ = 83.6± 0.3



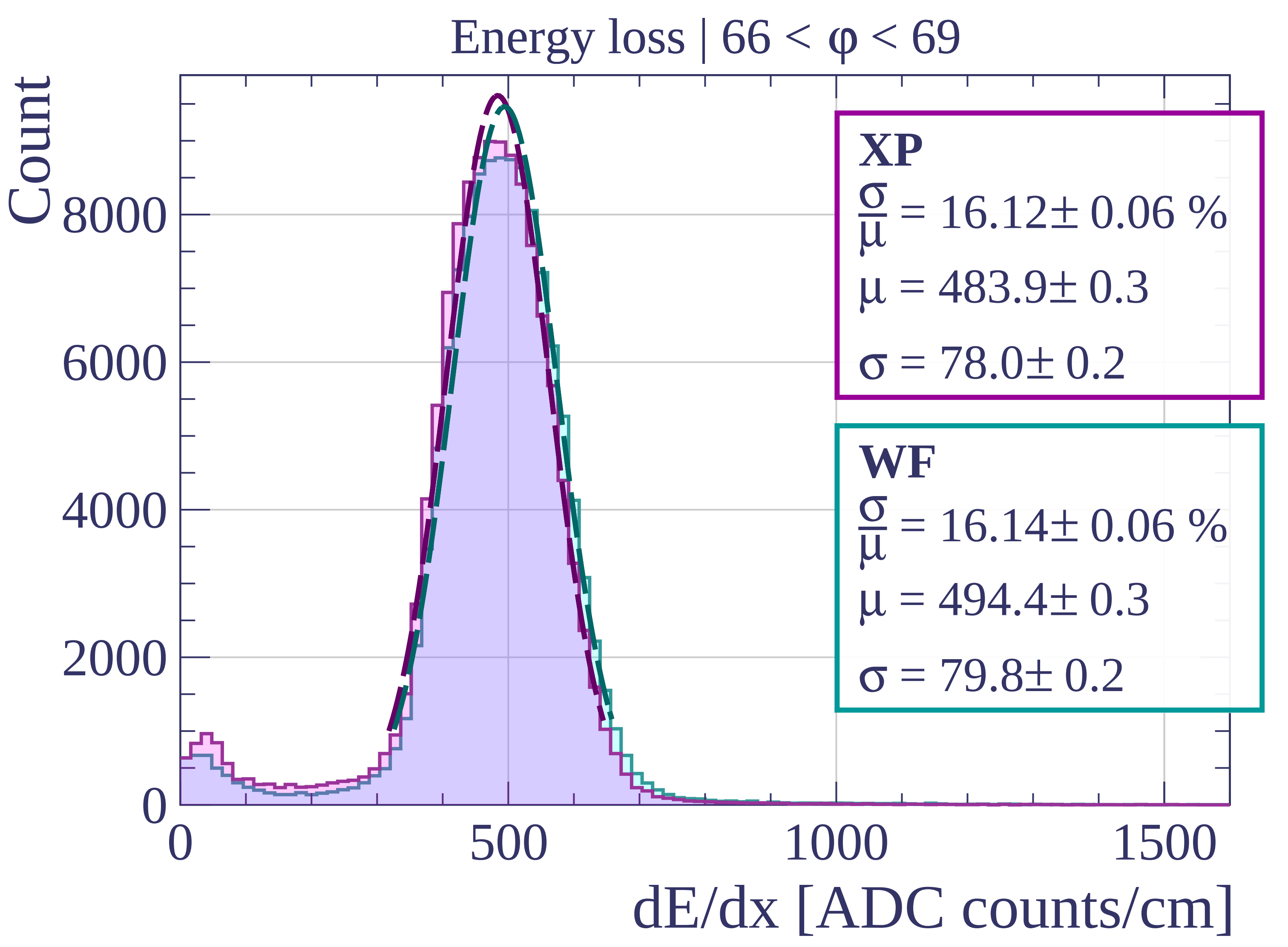
Energy loss | 57 < φ < 60
Count
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 16.92± 0.07 %
μ = 489.1± 0.3
σ = 82.8± 0.3
WF
σ/μ = 16.59± 0.07 %
μ = 487.9± 0.3
σ = 81.0± 0.3



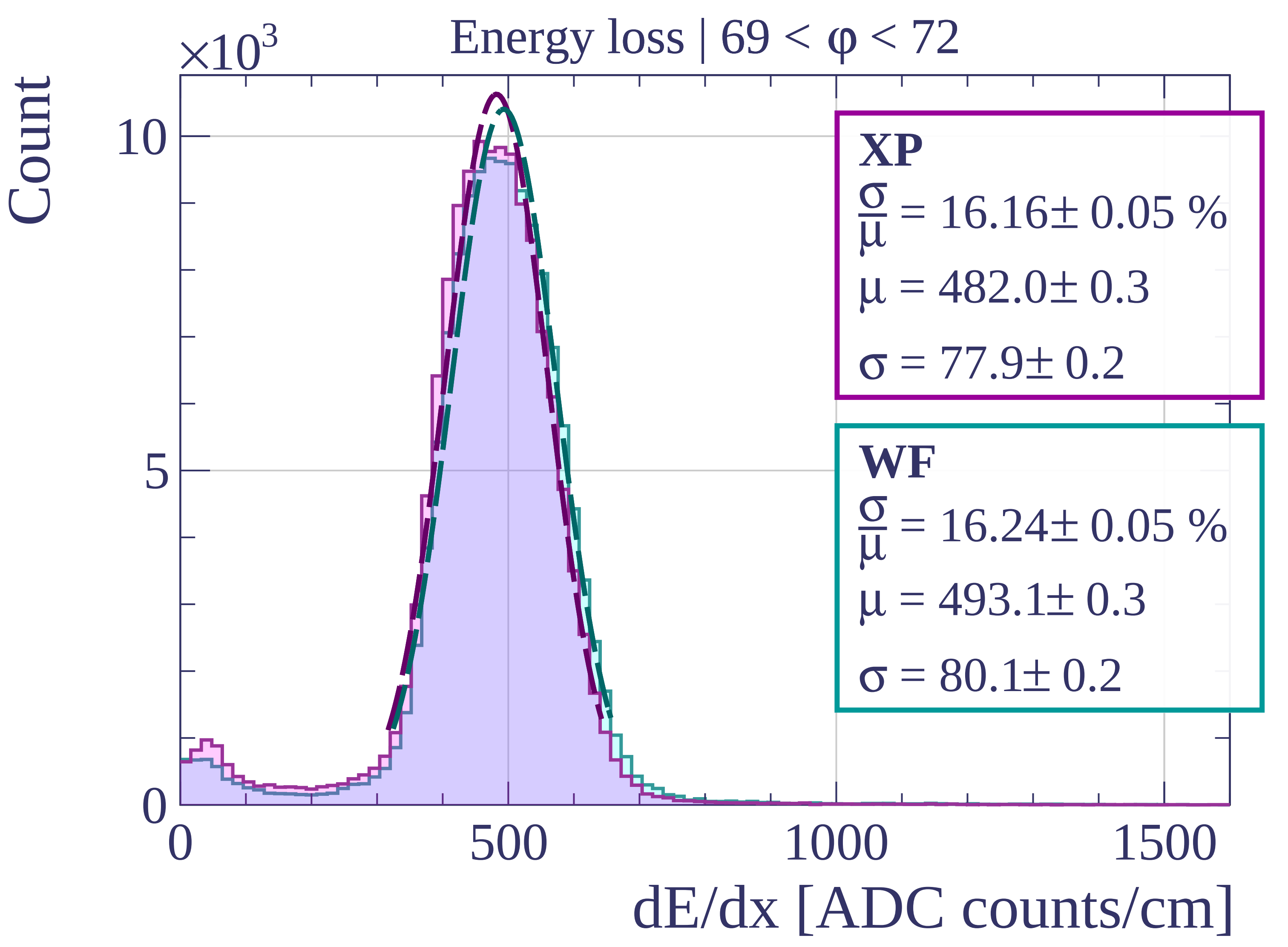
Energy loss | 60 < φ < 63
Count
0
2000
4000
6000
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 16.48± 0.07 %
μ = 487.1± 0.3
σ = 80.3± 0.3
WF
σ/μ = 16.74± 0.07 %
μ = 494.9± 0.3
σ = 82.8± 0.3



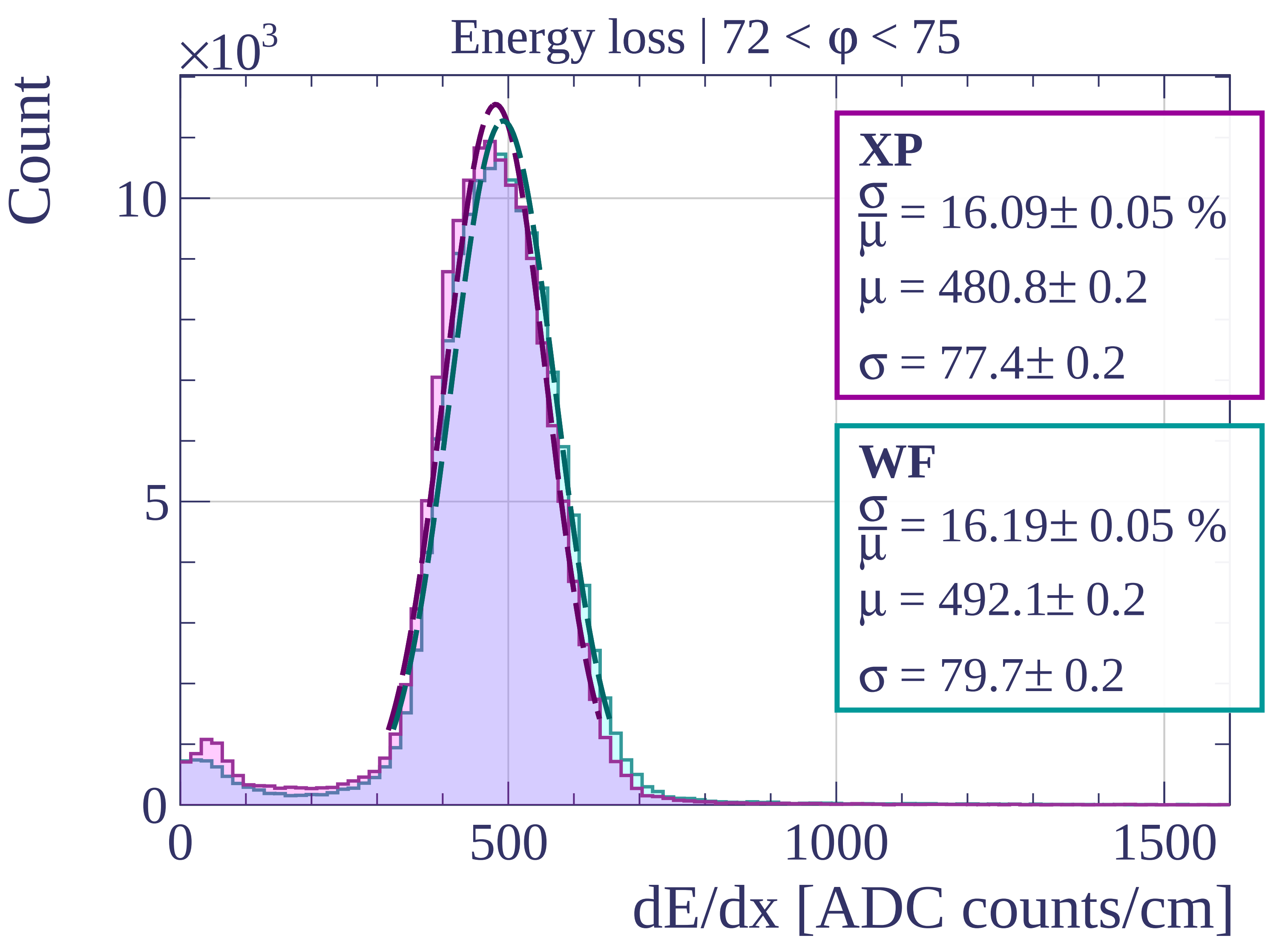
Energy loss | 63 < φ < 66
Count
8000
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 16.32± 0.06 %
μ = 485.1± 0.3
σ = 79.2± 0.2
WF
σ/μ = 16.54± 0.06 %
μ = 495.2± 0.3
σ = 81.9± 0.2



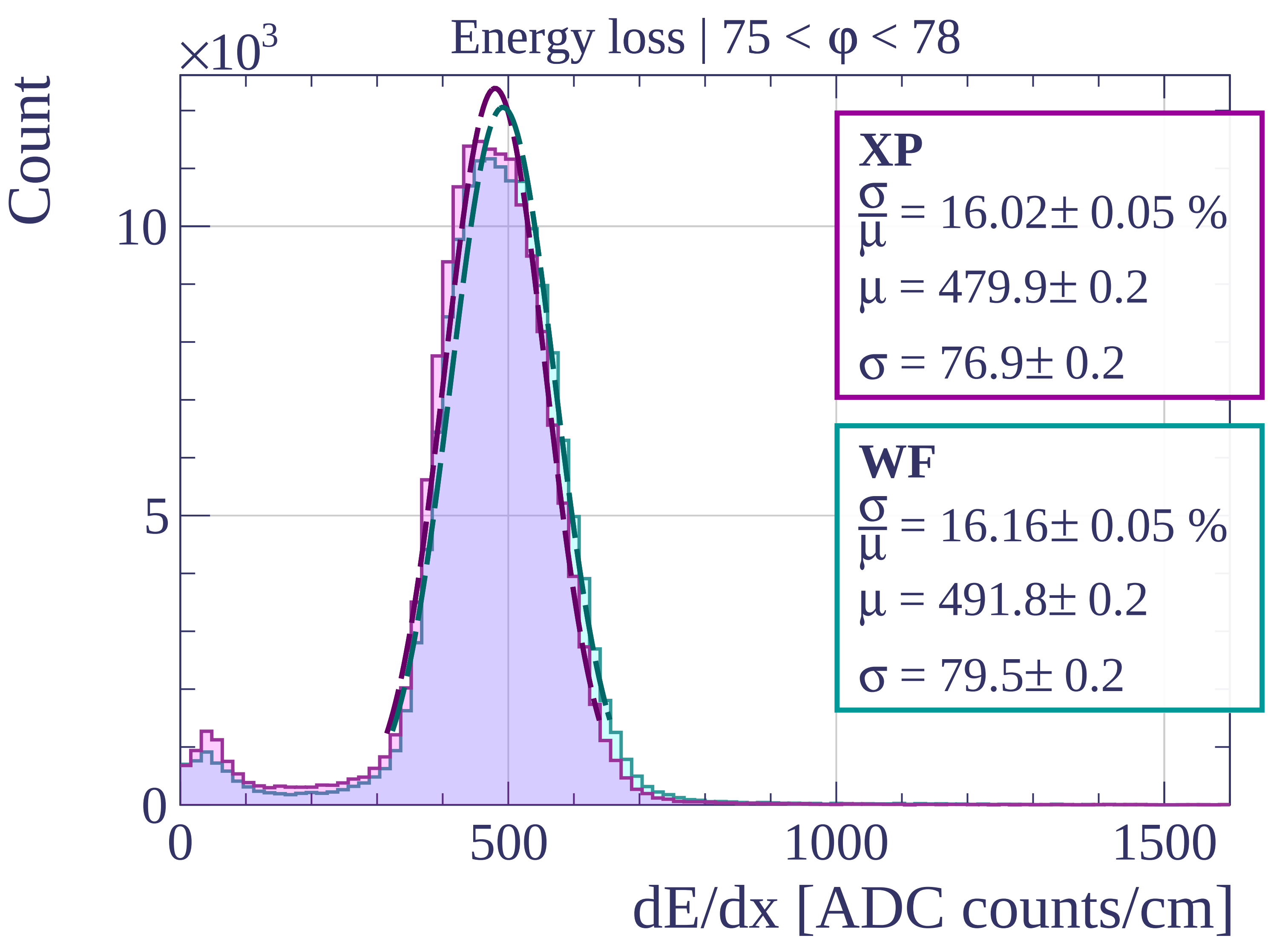
Energy loss | 66 < φ < 69
Count
dE/dx [ADC counts/cm]
0
2000
4000
6000
8000
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = 16.12± 0.06 %
μ = 483.9± 0.3
σ = 78.0± 0.2
WF
$\frac{\sigma}{\mu}$ = 16.14± 0.06 %
μ = 494.4± 0.3
σ = 79.8± 0.2



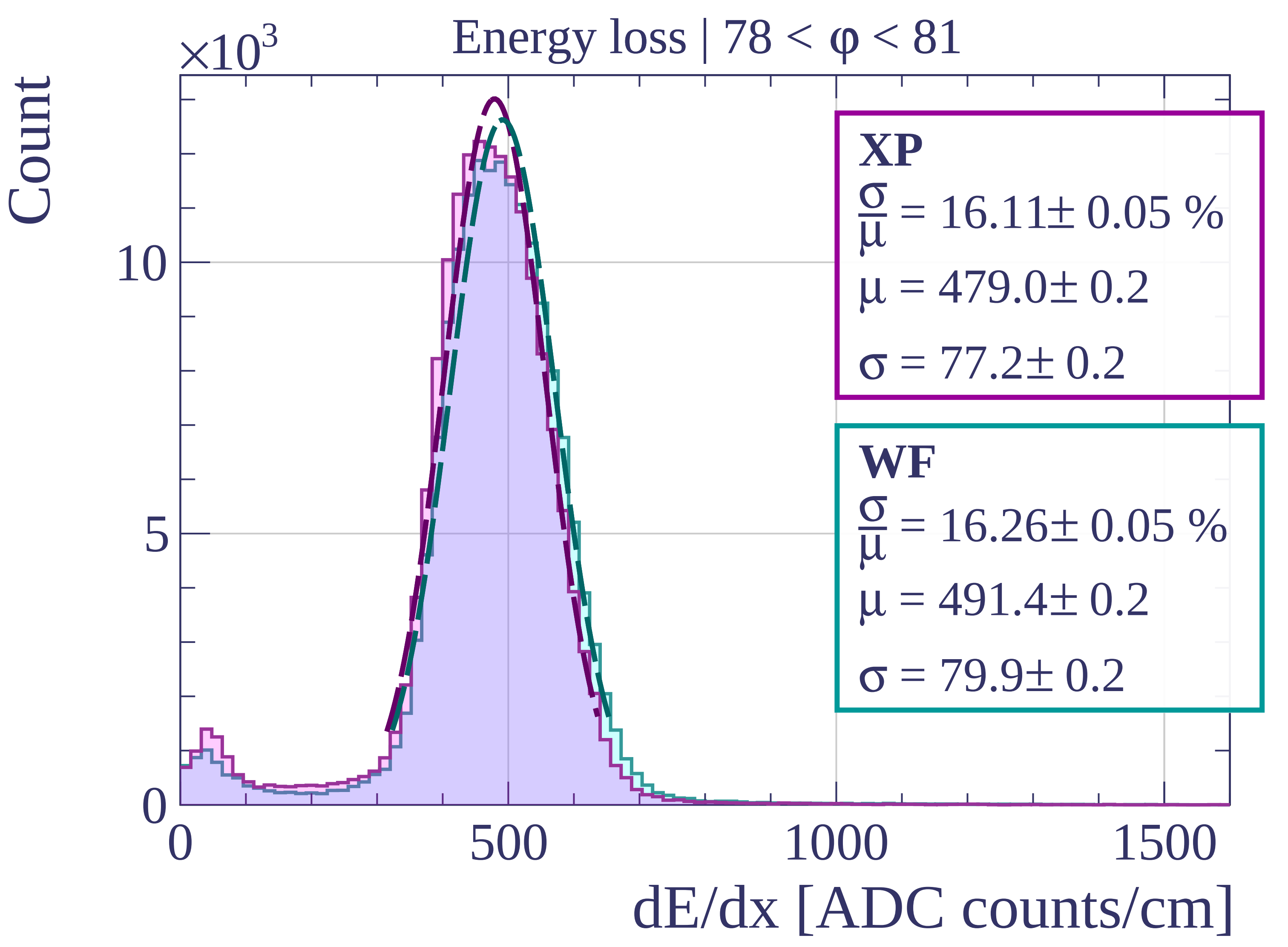
Energy loss | 69 < φ < 72
×10³
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 16.16± 0.05 %
μ = 482.0± 0.3
σ = 77.9± 0.2
WF
σ/μ = 16.24± 0.05 %
μ = 493.1± 0.3
σ = 80.1± 0.2

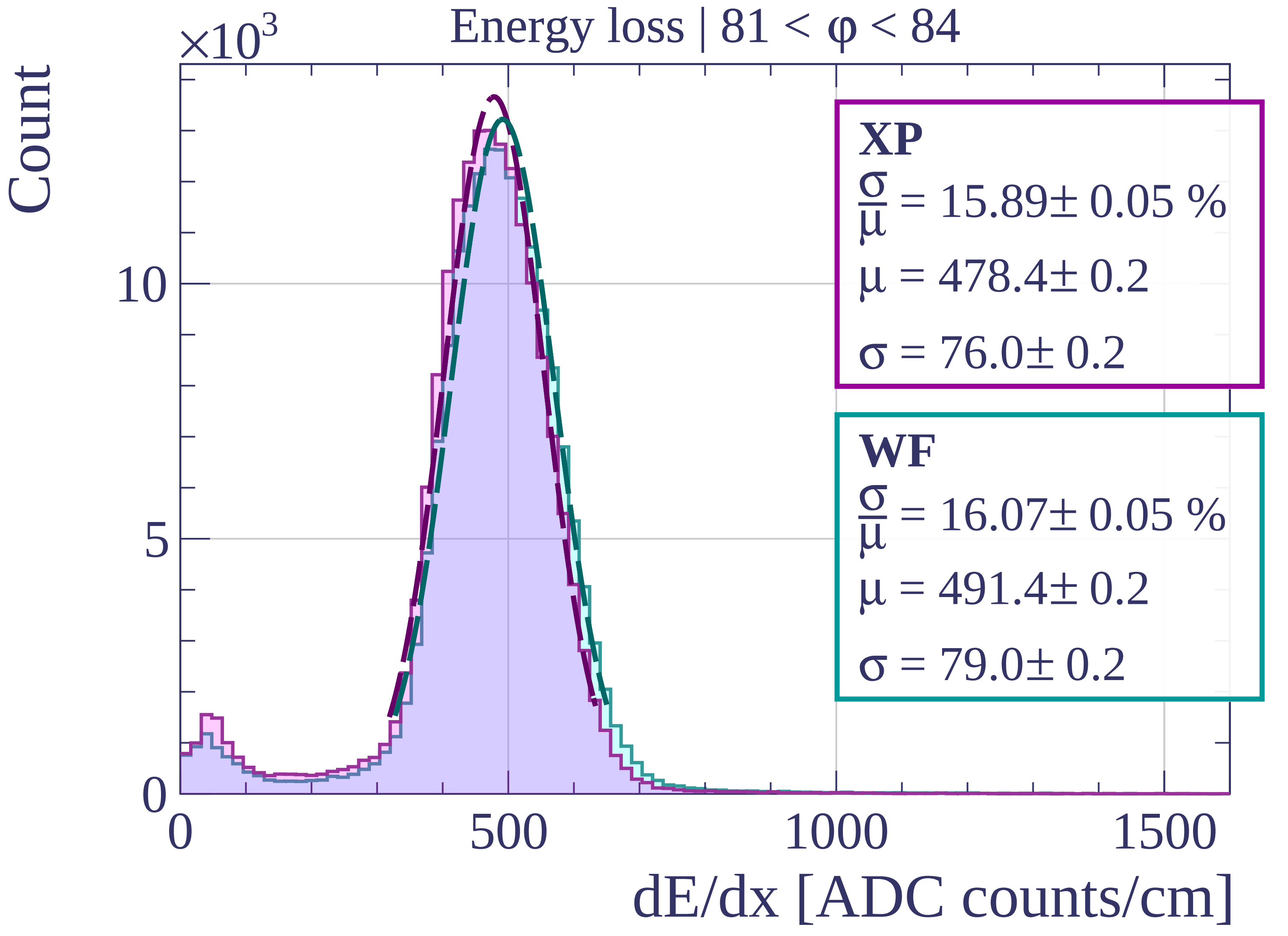


Energy loss | 72 < φ < 75
×10^3
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 16.09± 0.05 %
μ = 480.8± 0.2
σ = 77.4± 0.2
WF
σ/μ = 16.19± 0.05 %
μ = 492.1± 0.2
σ = 79.7± 0.2

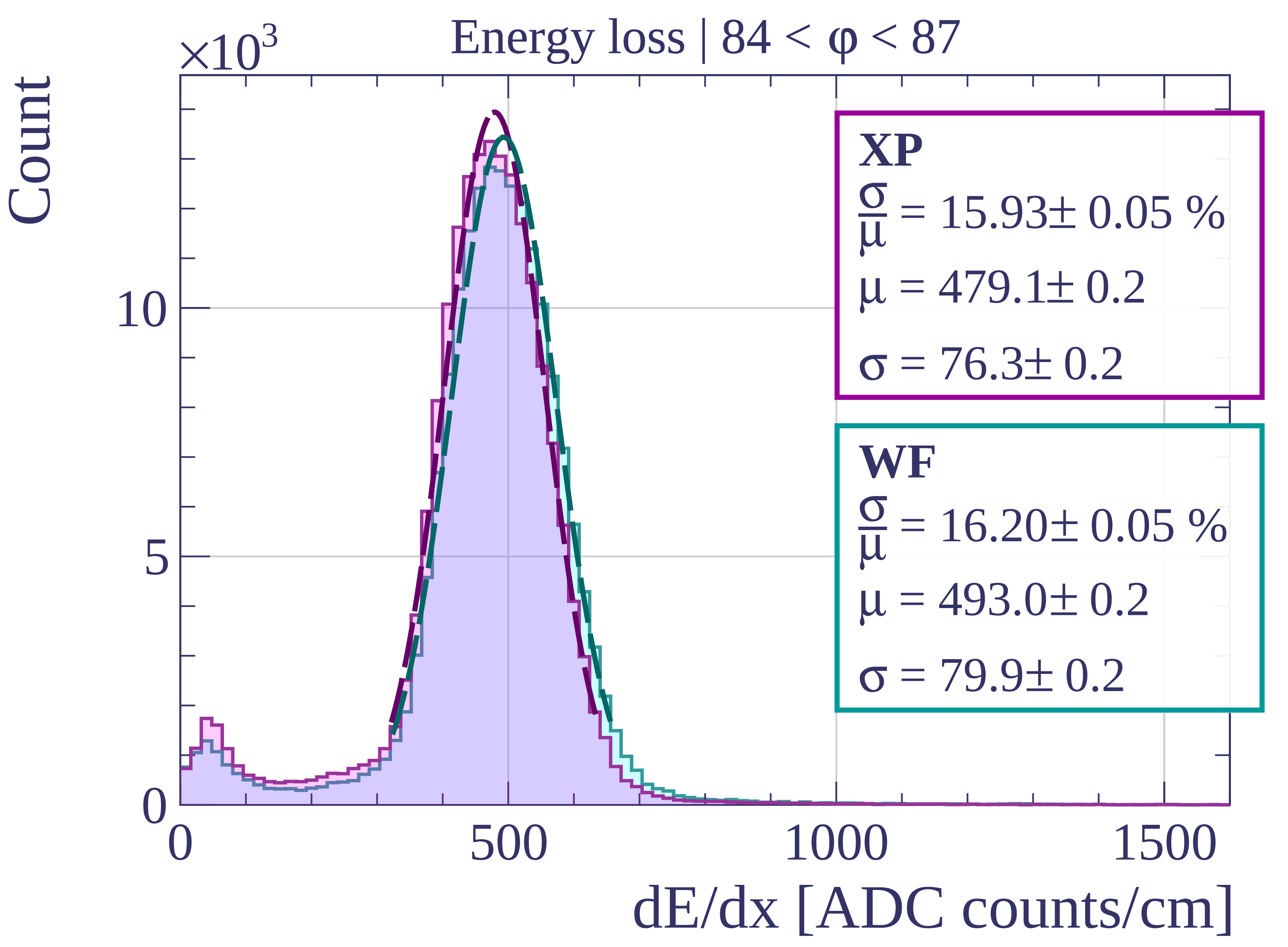


Energy loss | 75 < φ < 78
×10^3
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 16.02± 0.05 %
μ = 479.9± 0.2
σ = 76.9± 0.2
WF
σ/μ = 16.16± 0.05 %
μ = 491.8± 0.2
σ = 79.5± 0.2

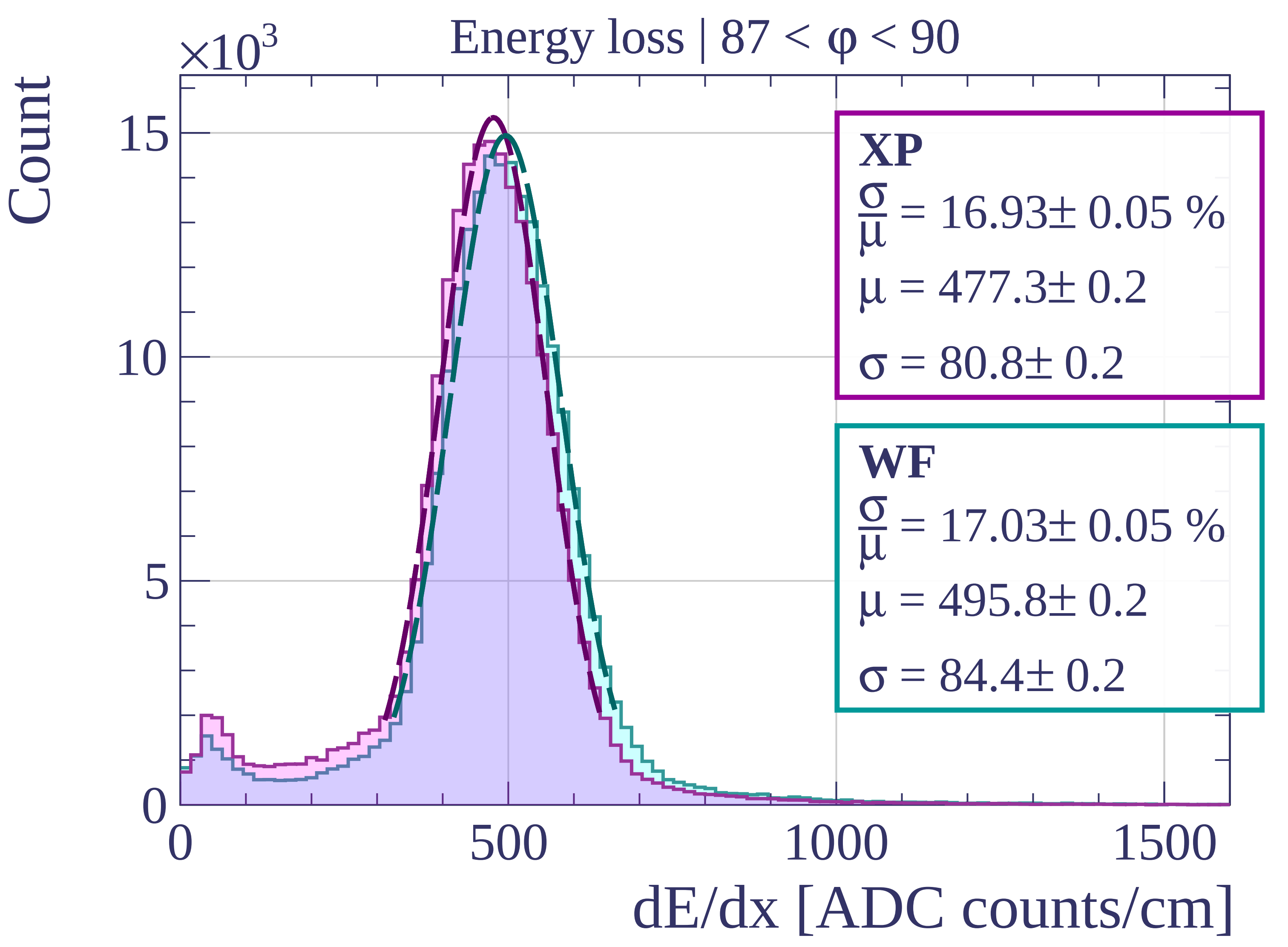

Energy loss | 78 < φ < 81
×10^3
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 16.11± 0.05 %
μ = 479.0± 0.2
σ = 77.2± 0.2
WF
σ/μ = 16.26± 0.05 %
μ = 491.4± 0.2
σ = 79.9± 0.2

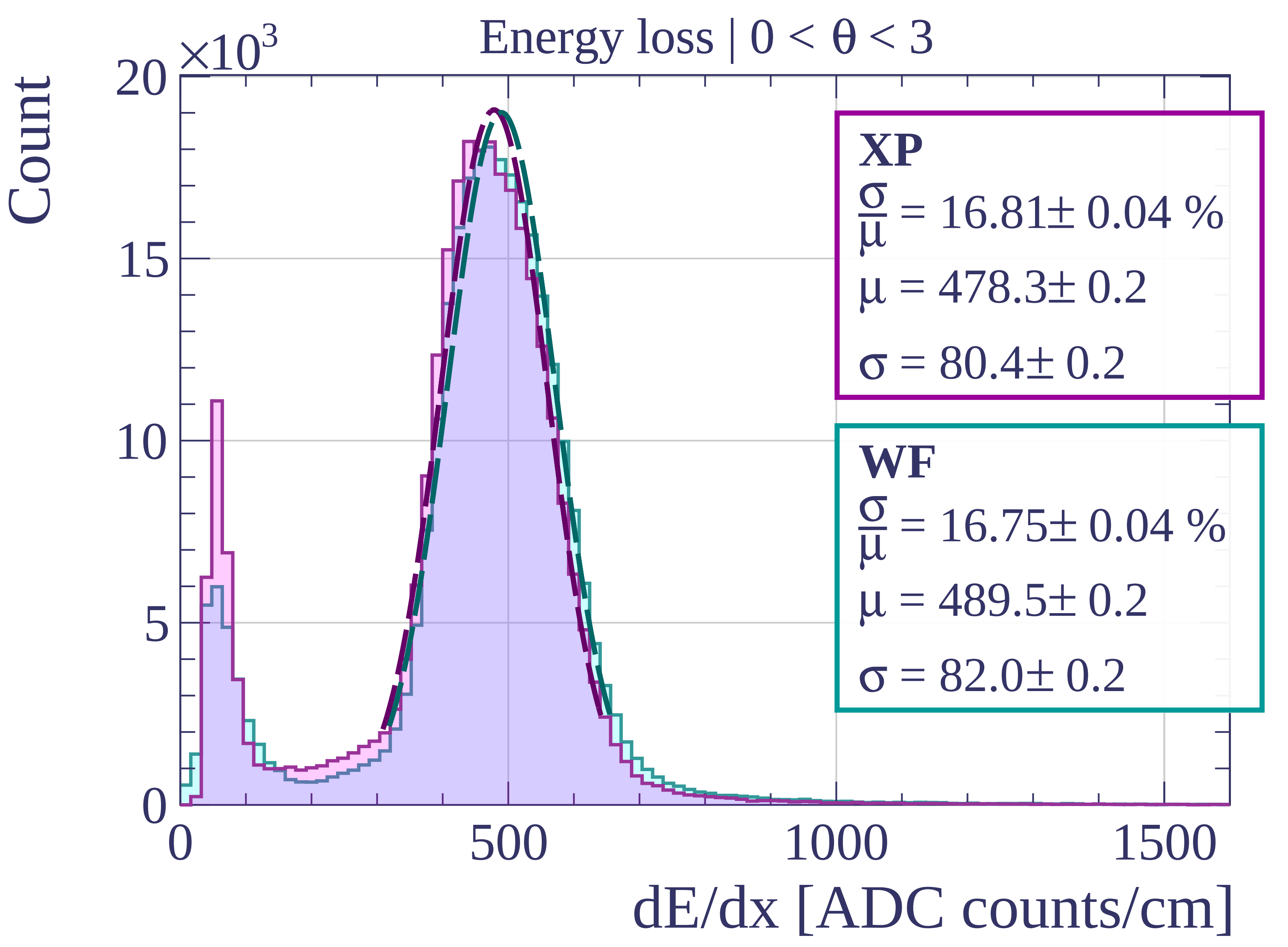


Energy loss | 81 < φ < 84
×10^3
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 15.89± 0.05 %
μ = 478.4± 0.2
σ = 76.0± 0.2
WF
σ/μ = 16.07± 0.05 %
μ = 491.4± 0.2
σ = 79.0± 0.2

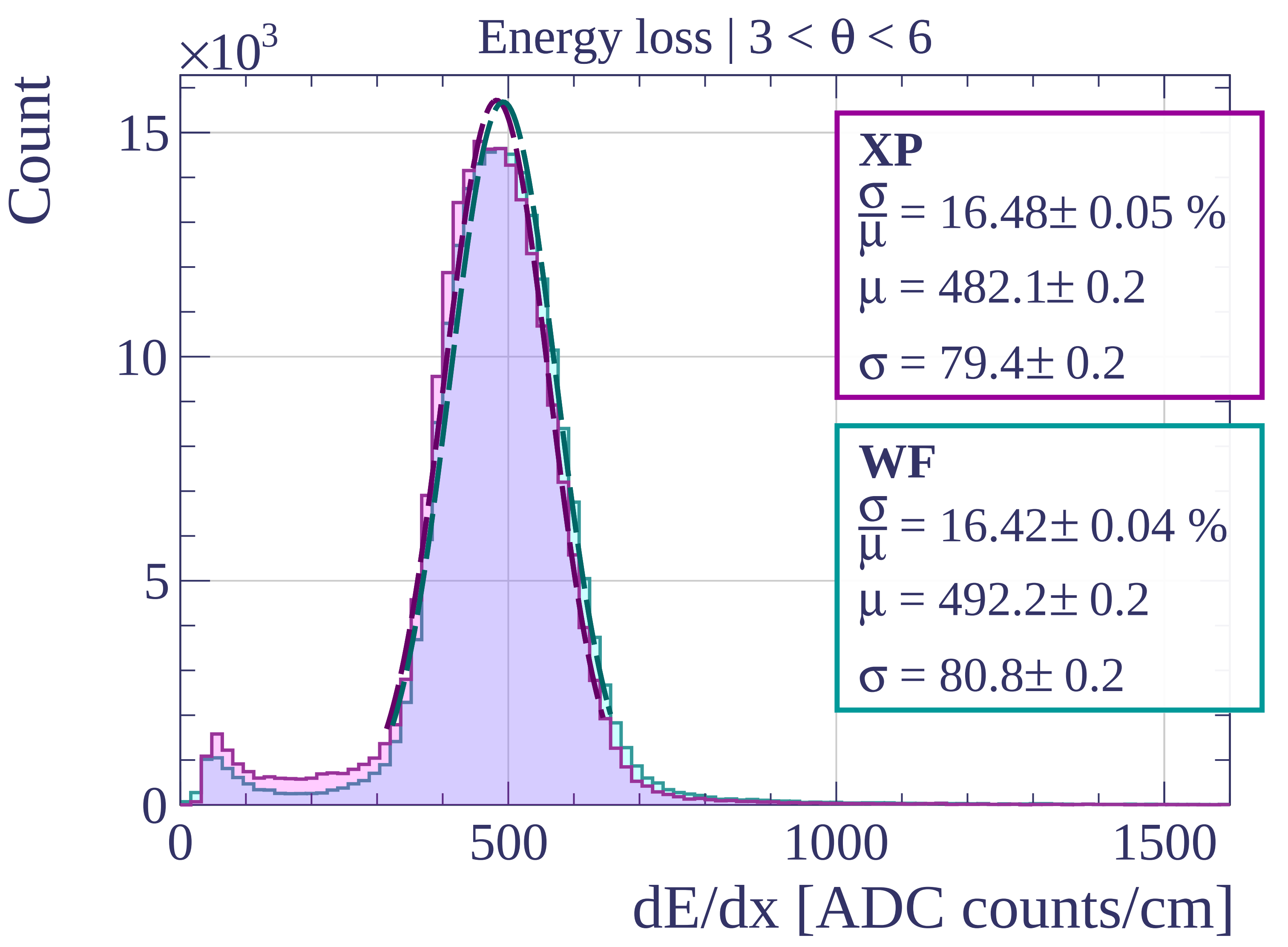


Energy loss | 84 < φ < 87
×10^3
Count
10
5
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 15.93± 0.05 %
μ = 479.1± 0.2
σ = 76.3± 0.2
WF
σ/μ = 16.20± 0.05 %
μ = 493.0± 0.2
σ = 79.9± 0.2

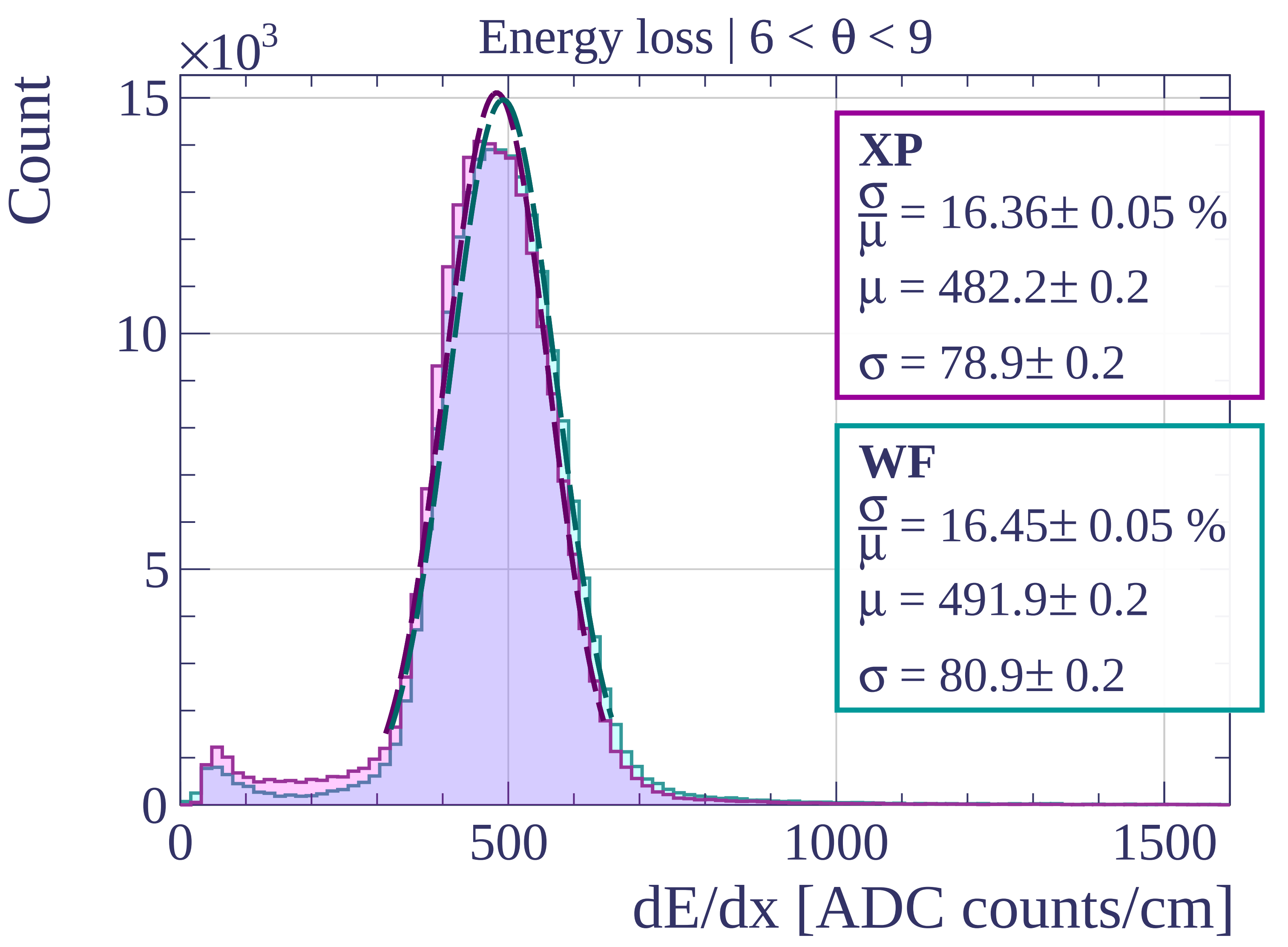

Energy loss | 87 < φ < 90
×10^3
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 16.93± 0.05 %
μ = 477.3± 0.2
σ = 80.8± 0.2
WF
σ/μ = 17.03± 0.05 %
μ = 495.8± 0.2
σ = 84.4± 0.2



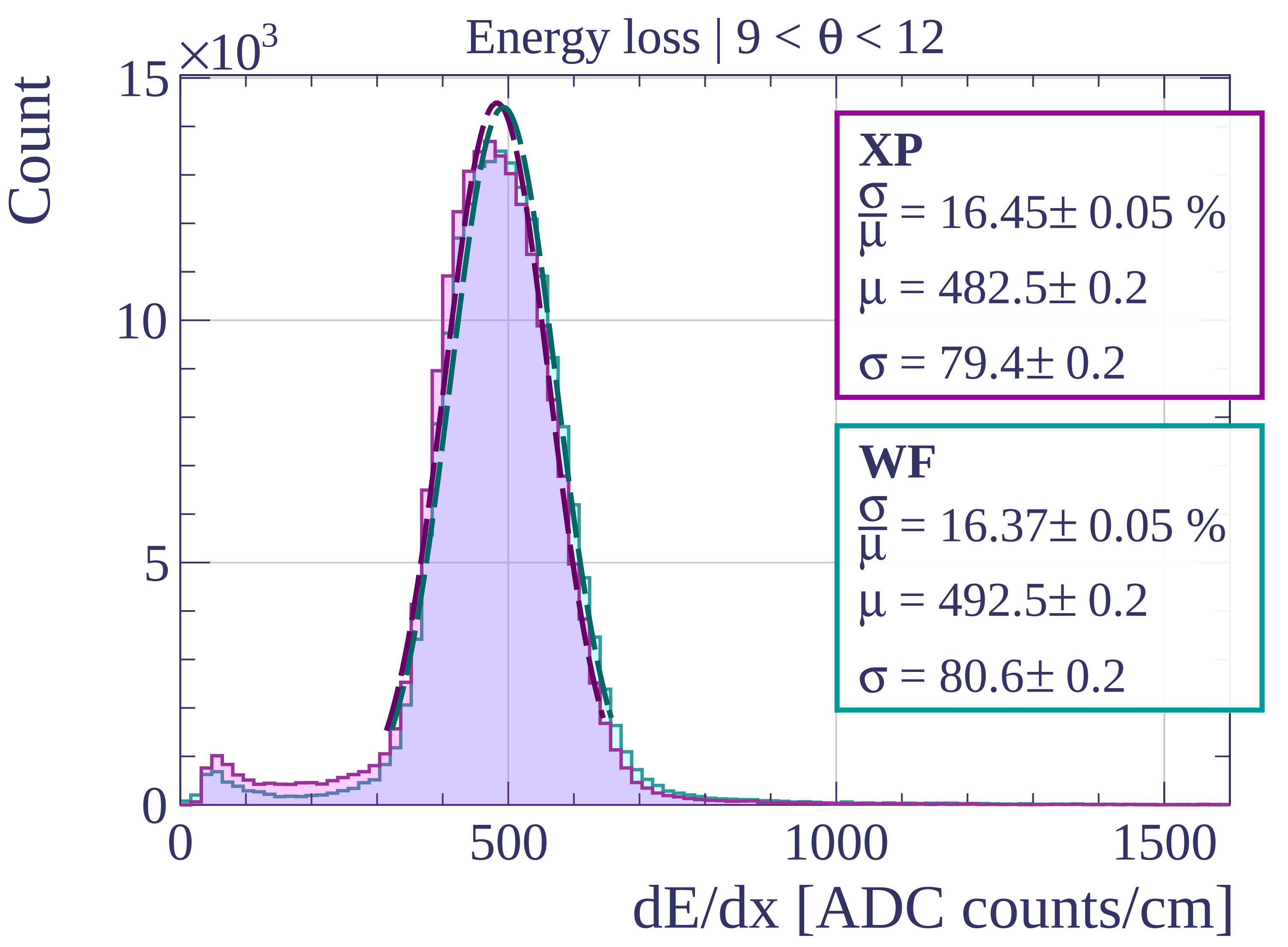
Energy loss | 0 < θ < 3
×10³
Count
20
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 16.81± 0.04 %
μ = 478.3± 0.2
σ = 80.4± 0.2
WF
σ/μ = 16.75± 0.04 %
μ = 489.5± 0.2
σ = 82.0± 0.2

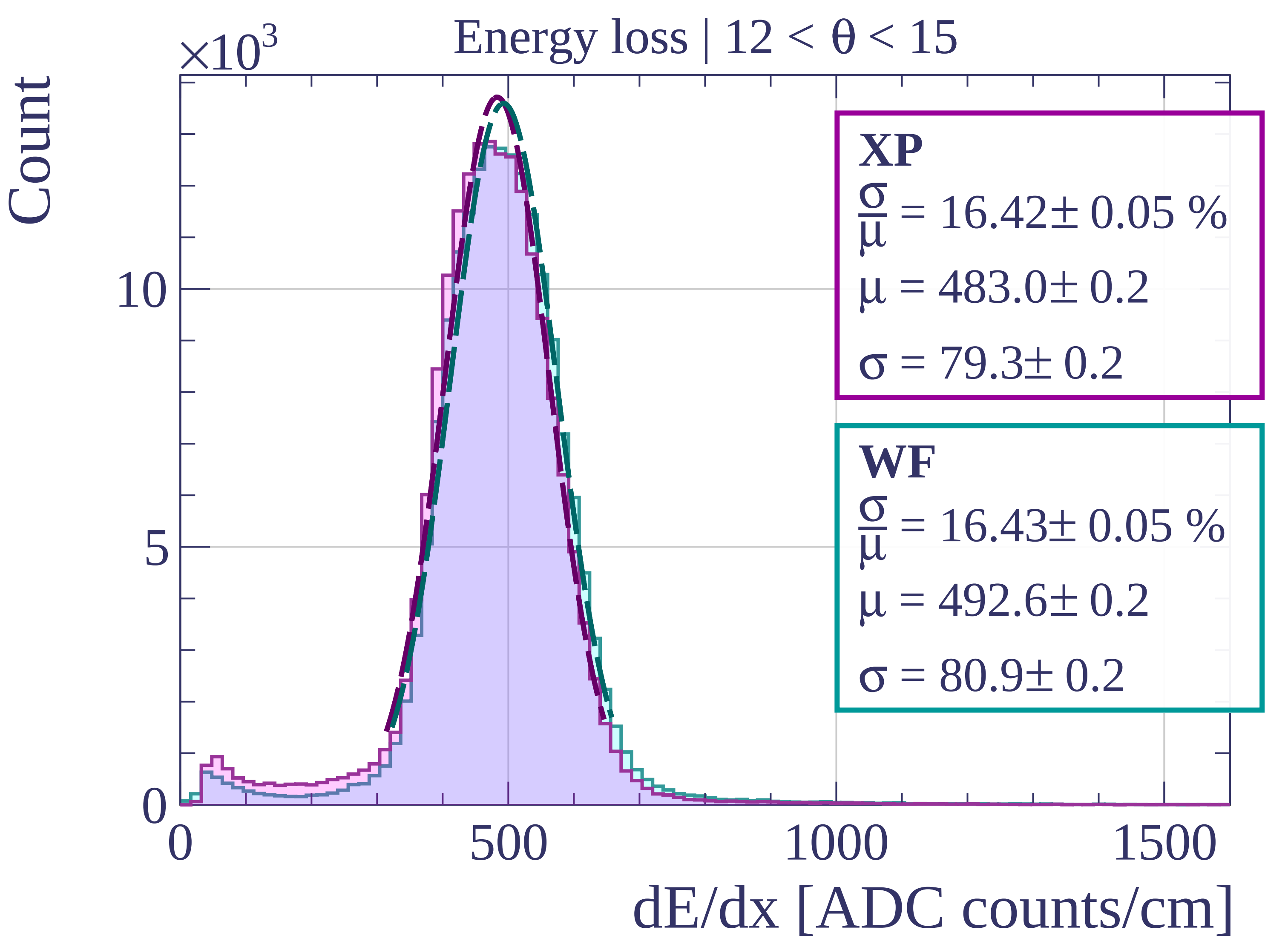


Energy loss | 3 < θ < 6
×10^3
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 16.48± 0.05 %
μ = 482.1± 0.2
σ = 79.4± 0.2
WF
σ/μ = 16.42± 0.04 %
μ = 492.2± 0.2
σ = 80.8± 0.2

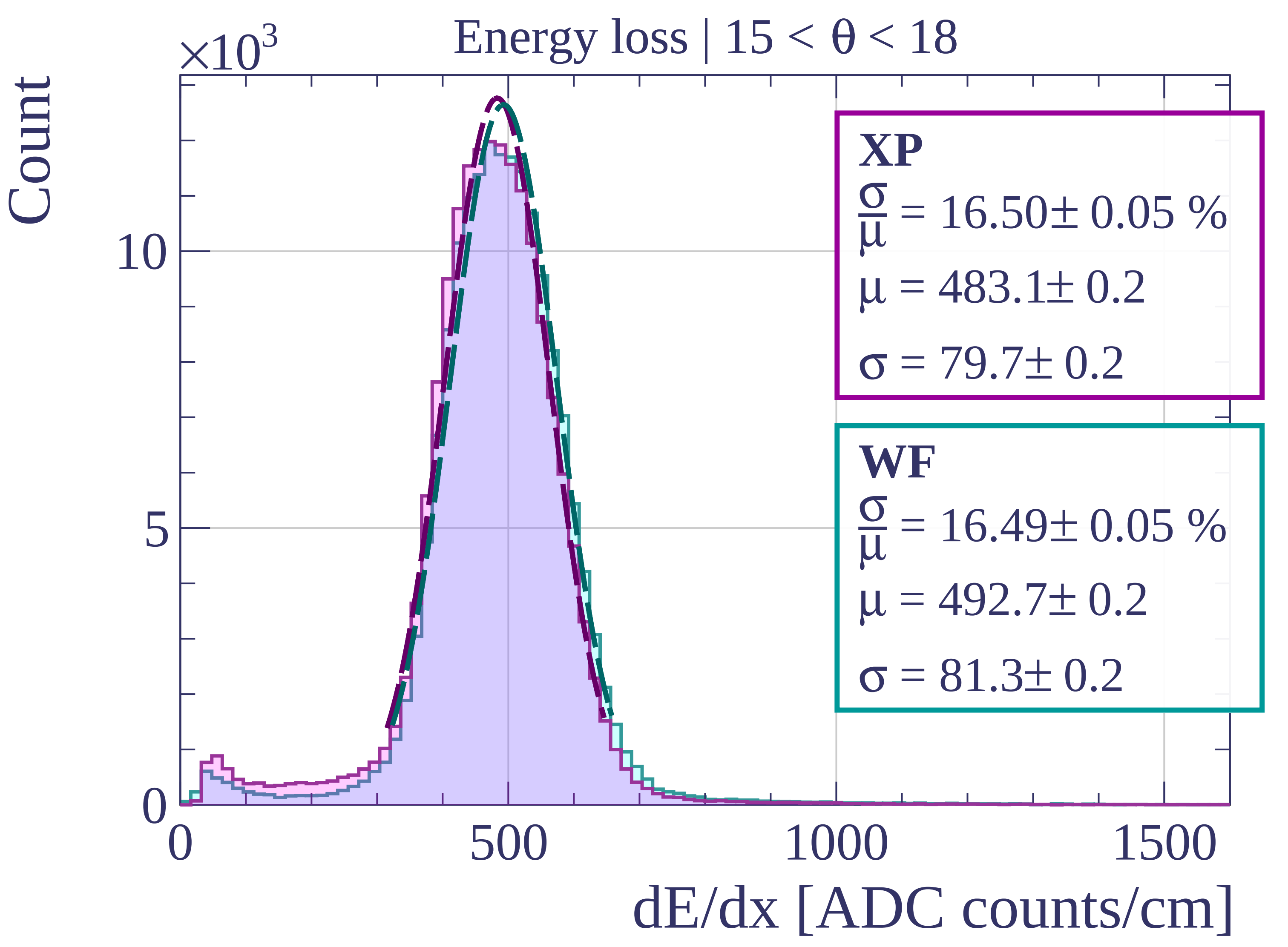


Energy loss | 6 < θ < 9
×10^3
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 16.36± 0.05 %
μ = 482.2± 0.2
σ = 78.9± 0.2
WF
σ/μ = 16.45± 0.05 %
μ = 491.9± 0.2
σ = 80.9± 0.2

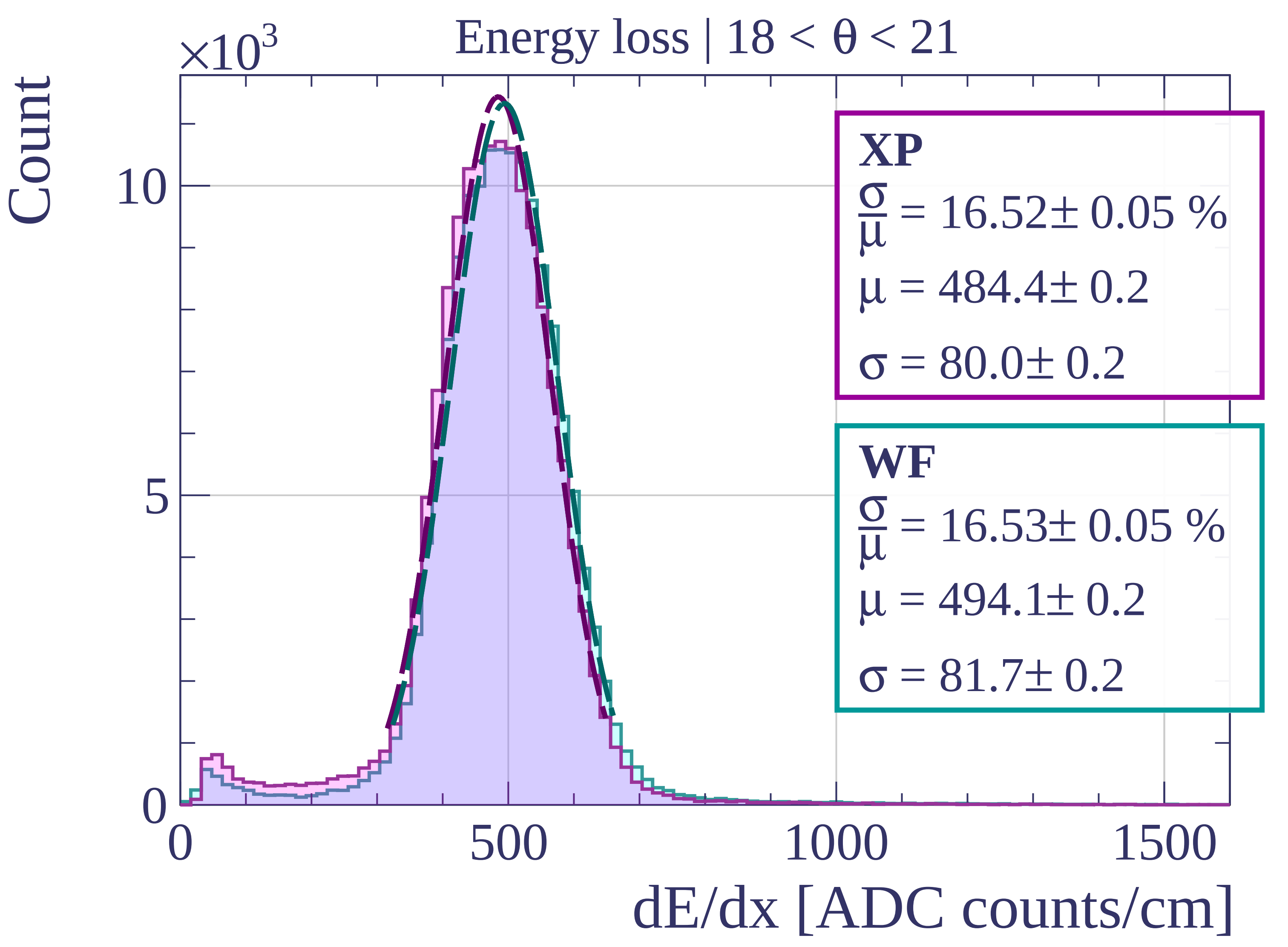

Energy loss | 9 < θ < 12
×10^3
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 16.45± 0.05 %
μ = 482.5± 0.2
σ = 79.4± 0.2
WF
σ/μ = 16.37± 0.05 %
μ = 492.5± 0.2
σ = 80.6± 0.2



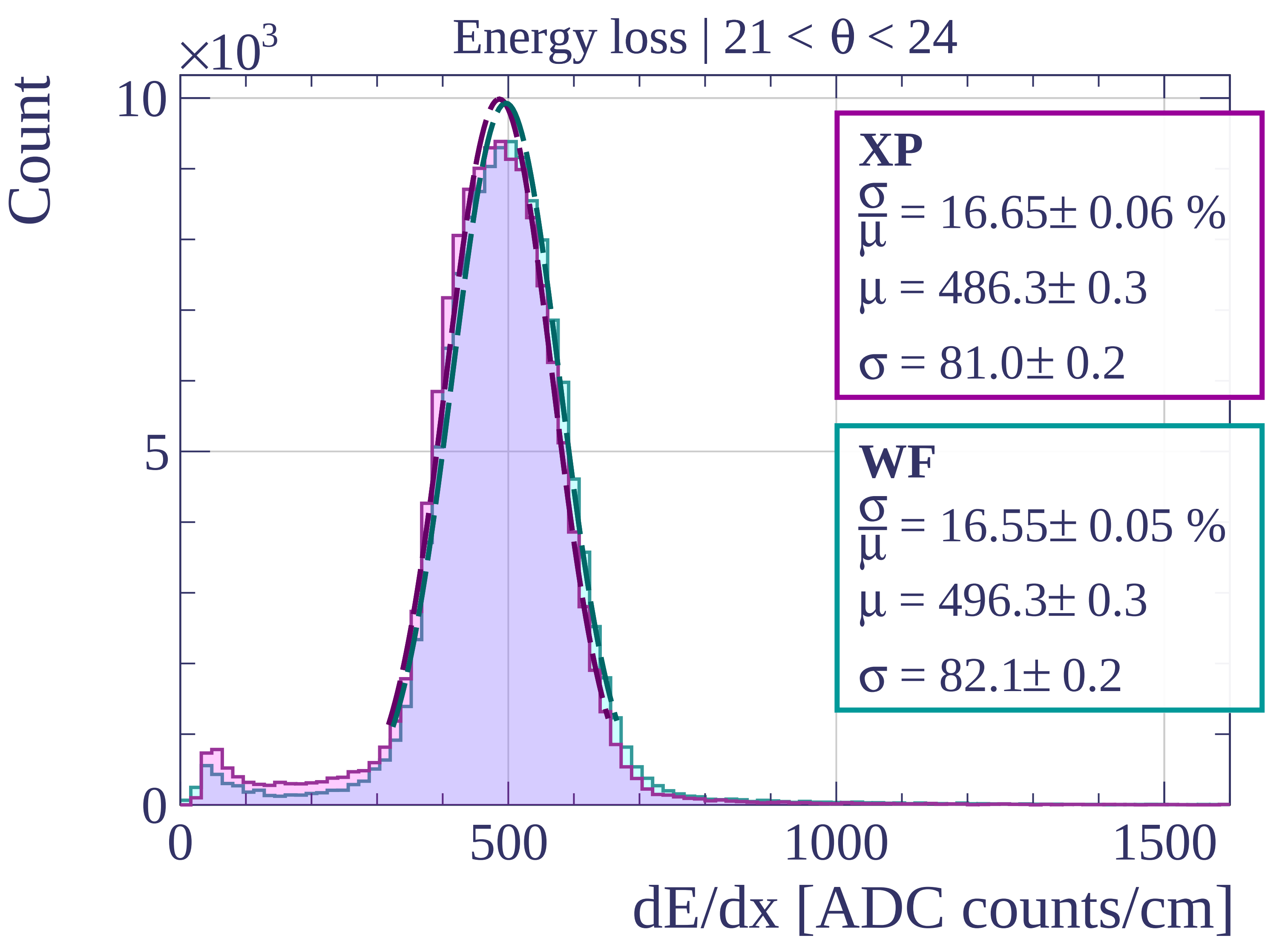
Energy loss | 12 < θ < 15
×10³
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 16.42± 0.05 %
μ = 483.0± 0.2
σ = 79.3± 0.2
WF
σ/μ = 16.43± 0.05 %
μ = 492.6± 0.2
σ = 80.9± 0.2

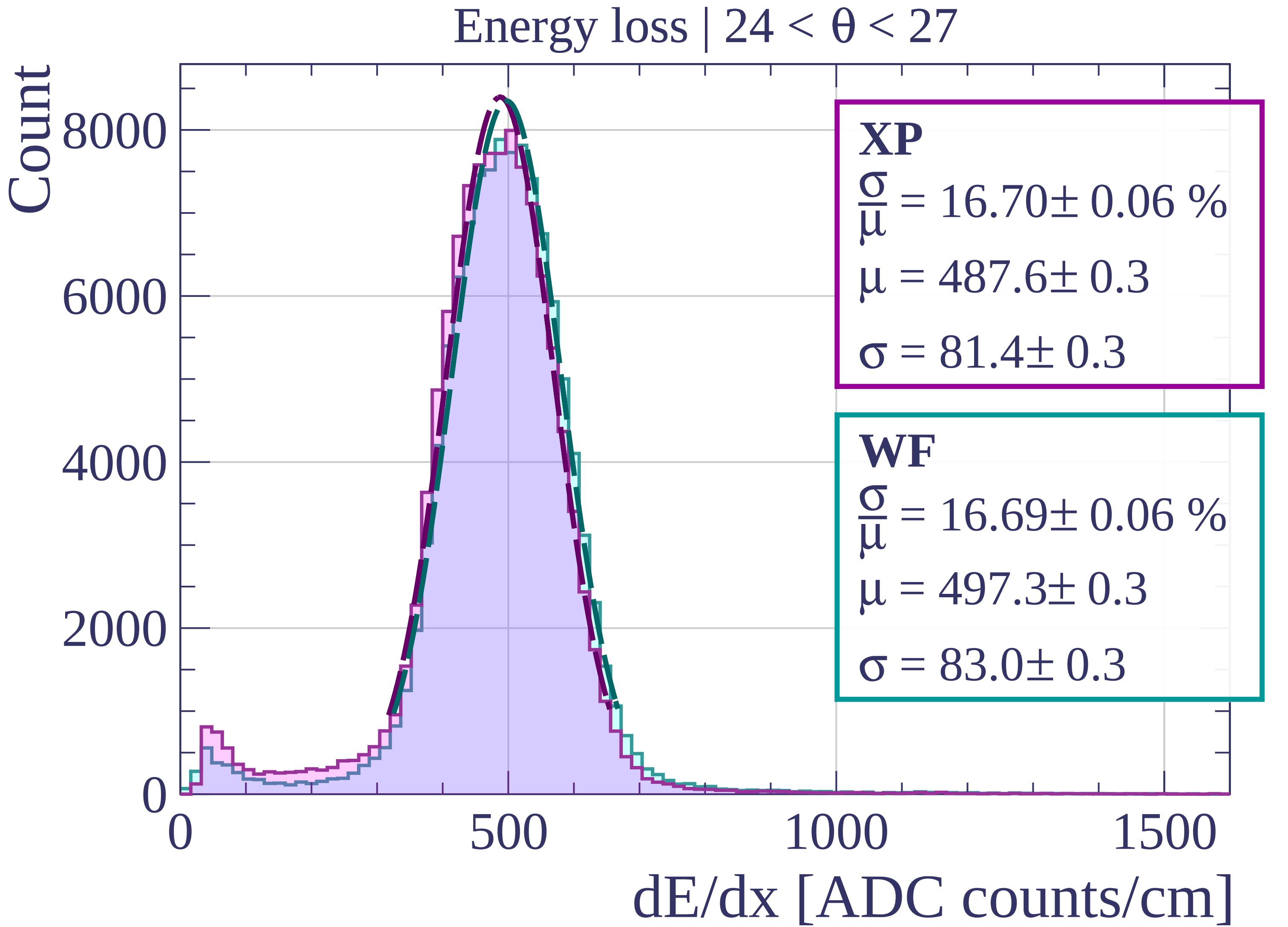


Energy loss | 15 < θ < 18
×10^3
Count
dE/dx [ADC counts/cm]
0
5
10
0
500
1000
1500
XP
σ/μ = 16.50± 0.05 %
μ = 483.1± 0.2
σ = 79.7± 0.2
WF
σ/μ = 16.49± 0.05 %
μ = 492.7± 0.2
σ = 81.3± 0.2

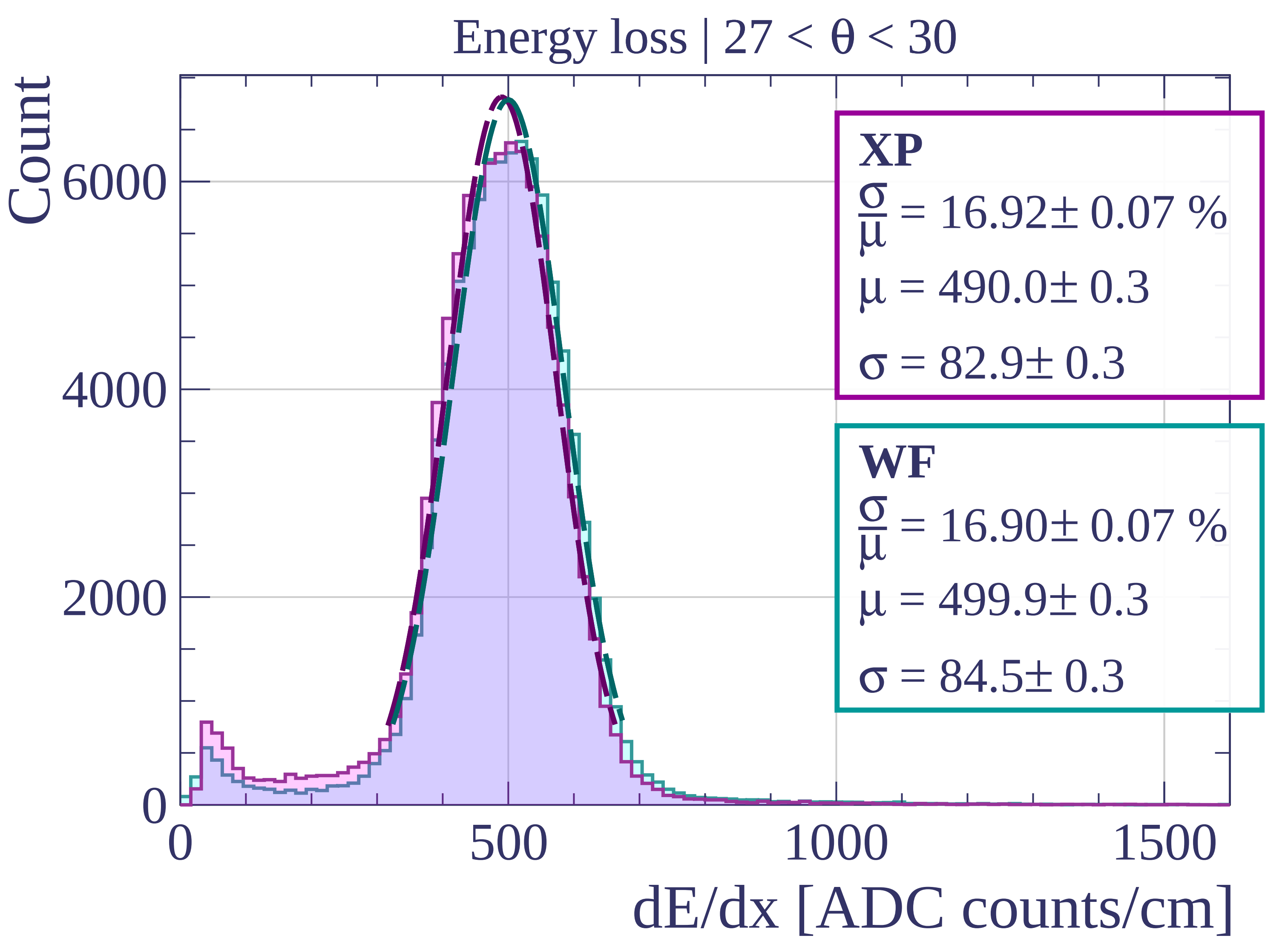


Energy loss | 18 < θ < 21
×10^3
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 16.52± 0.05 %
μ = 484.4± 0.2
σ = 80.0± 0.2
WF
σ/μ = 16.53± 0.05 %
μ = 494.1± 0.2
σ = 81.7± 0.2

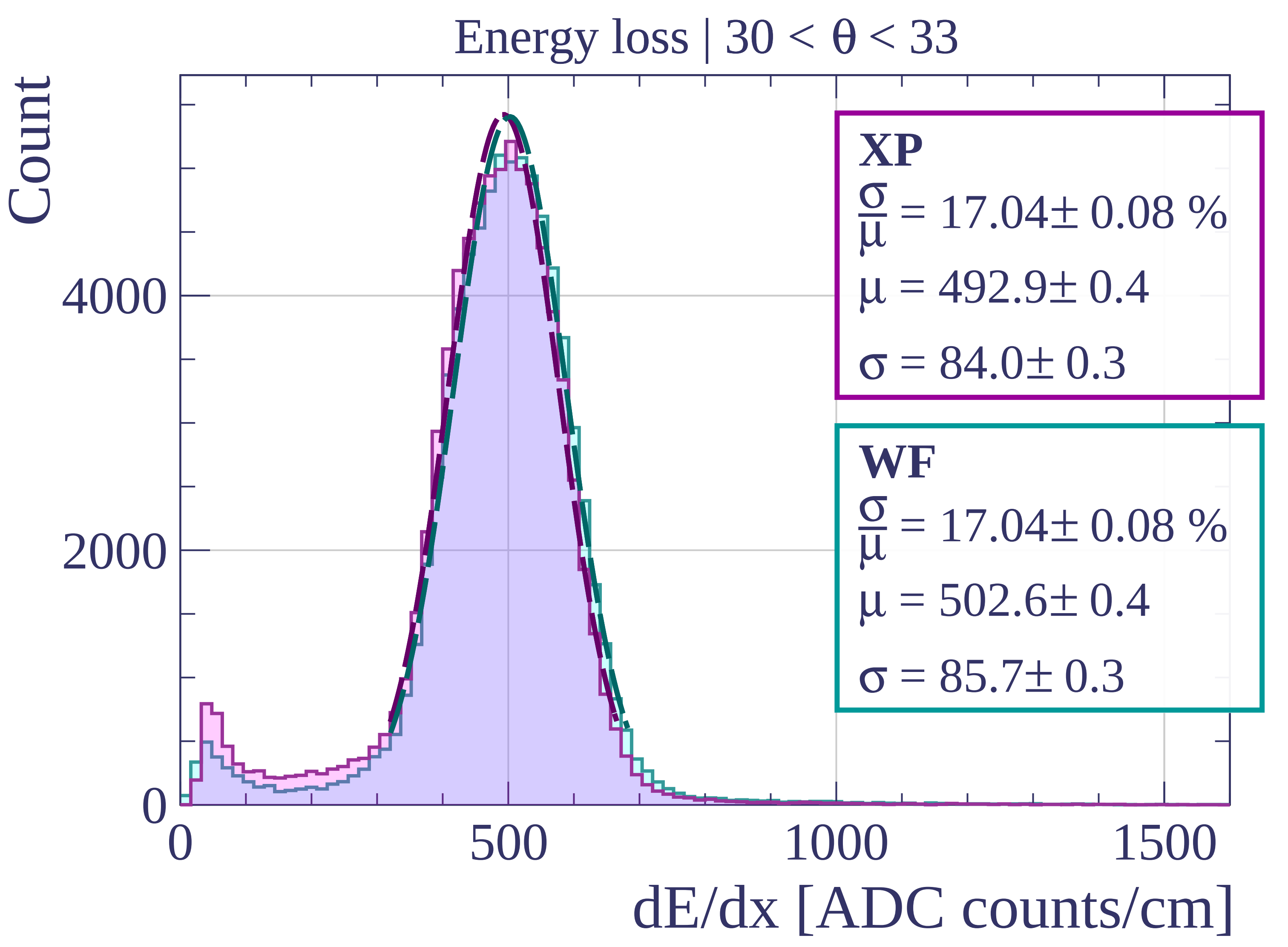

Energy loss | 21 < θ < 24
×10^3
Count
dE/dx [ADC counts/cm]
0
5
10
0
500
1000
1500
XP
σ/μ = 16.65± 0.06 %
μ = 486.3± 0.3
σ = 81.0± 0.2
WF
σ/μ = 16.55± 0.05 %
μ = 496.3± 0.3
σ = 82.1± 0.2

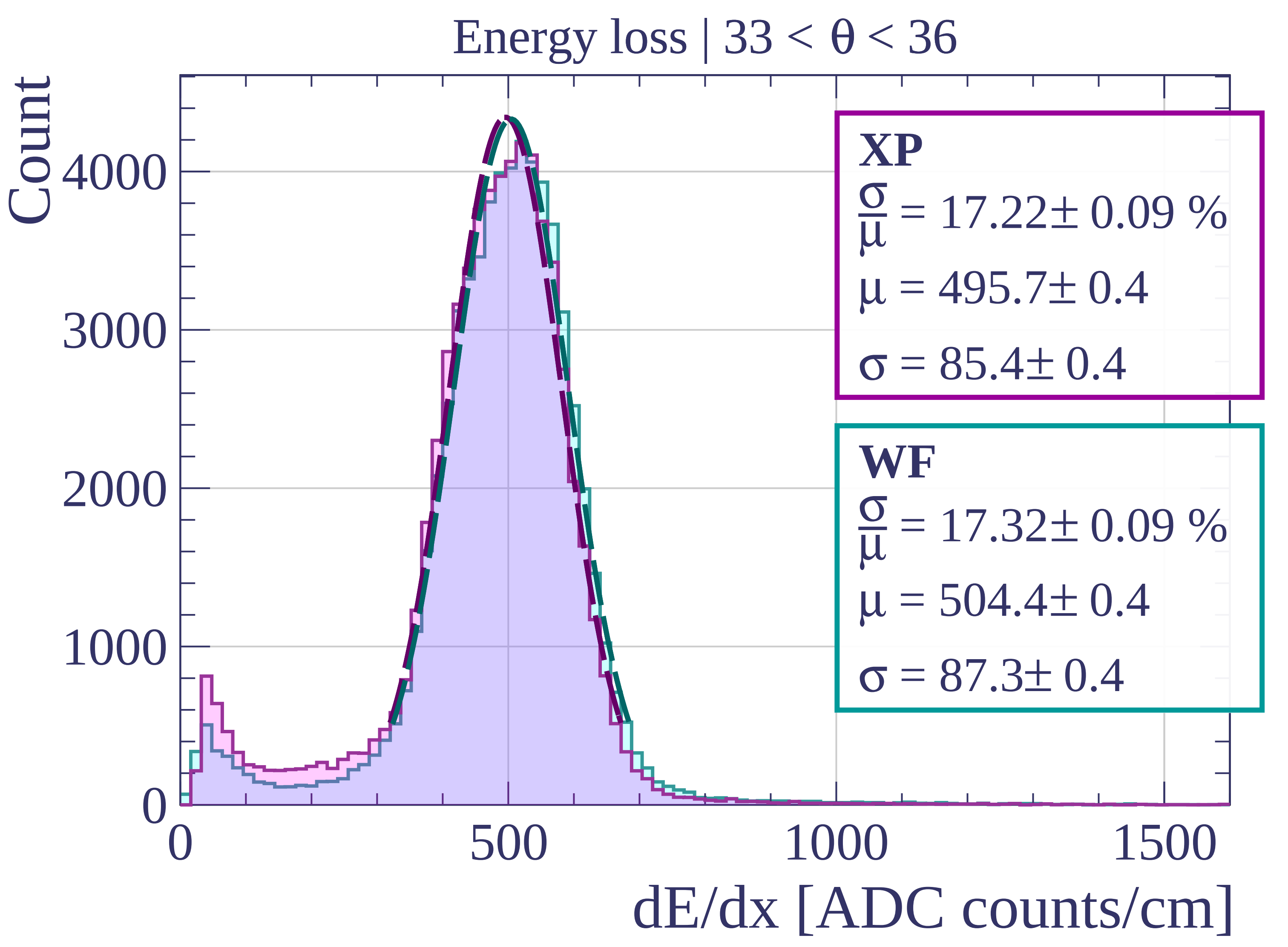


Energy loss | 24 < θ < 27
Count
8000
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 16.70± 0.06 %
μ = 487.6± 0.3
σ = 81.4± 0.3
WF
σ/μ = 16.69± 0.06 %
μ = 497.3± 0.3
σ = 83.0± 0.3

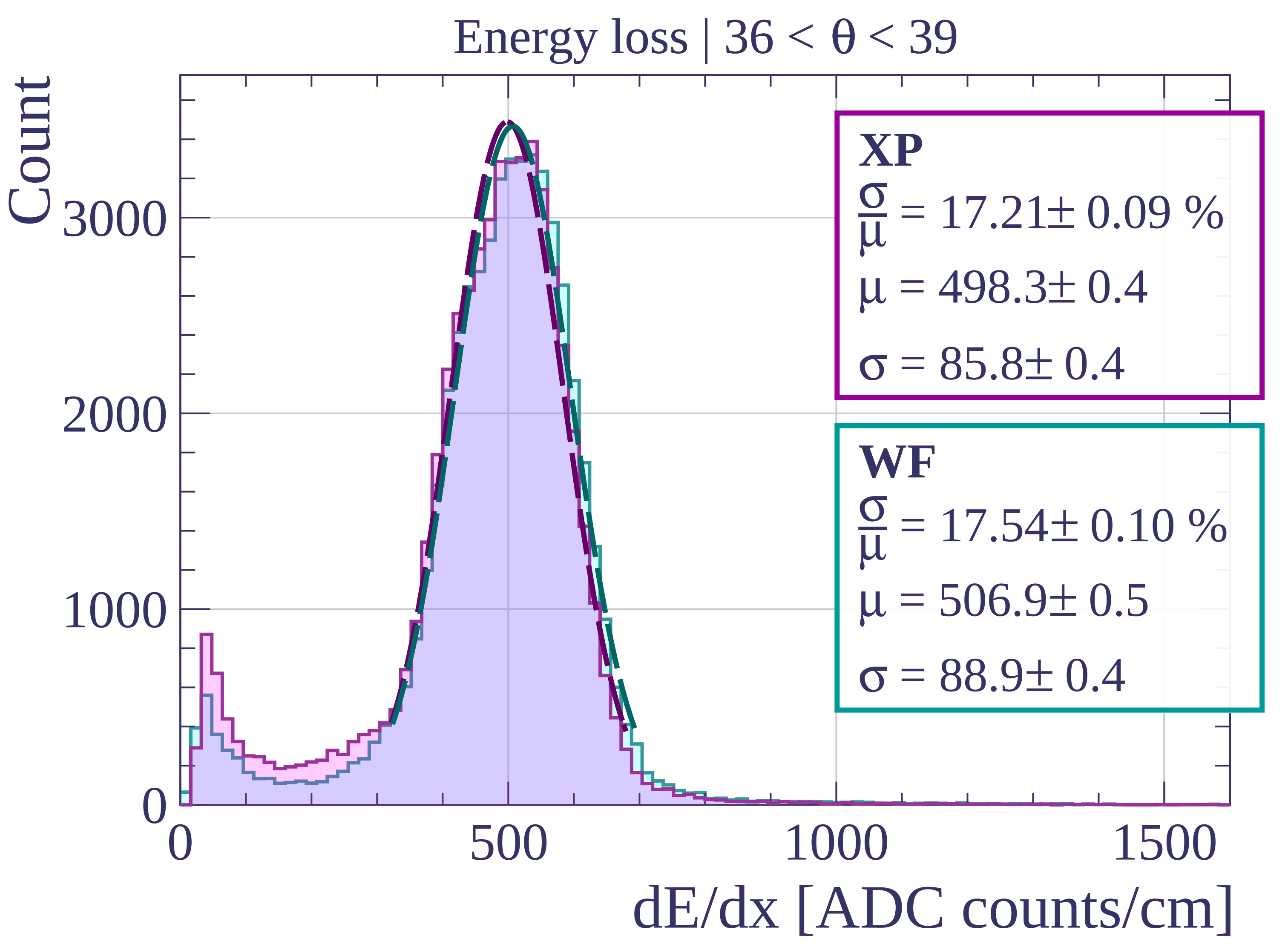


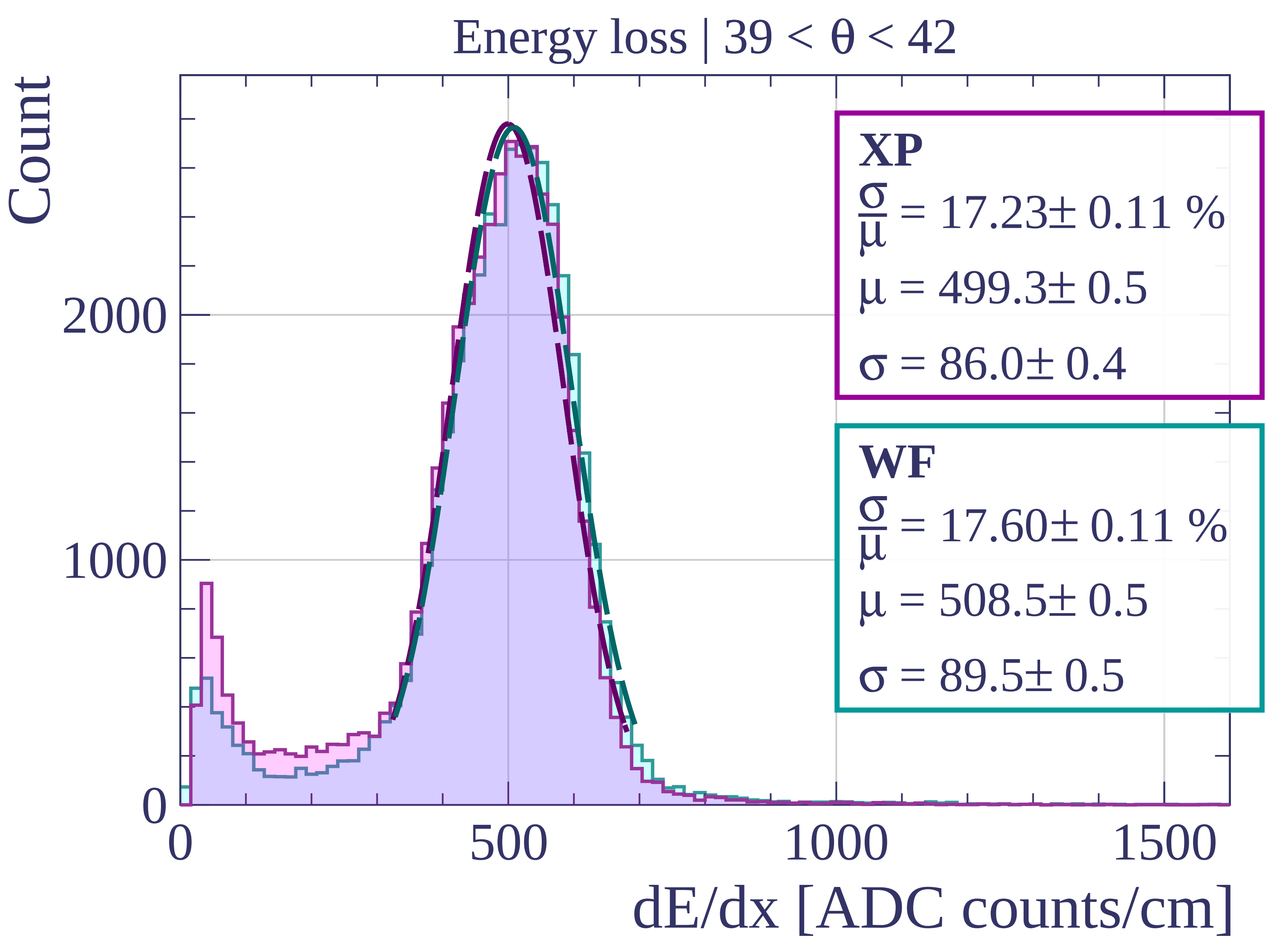
Energy loss | 27 < θ < 30
Count
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 16.92± 0.07 %
μ = 490.0± 0.3
σ = 82.9± 0.3
WF
$\frac{\sigma}{\mu}$ = 16.90± 0.07 %
μ = 499.9± 0.3
σ = 84.5± 0.3


Energy loss | 30 < θ < 33
Count
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 17.04± 0.08 %
μ = 492.9± 0.4
σ = 84.0± 0.3
WF
σ/μ = 17.04± 0.08 %
μ = 502.6± 0.4
σ = 85.7± 0.3

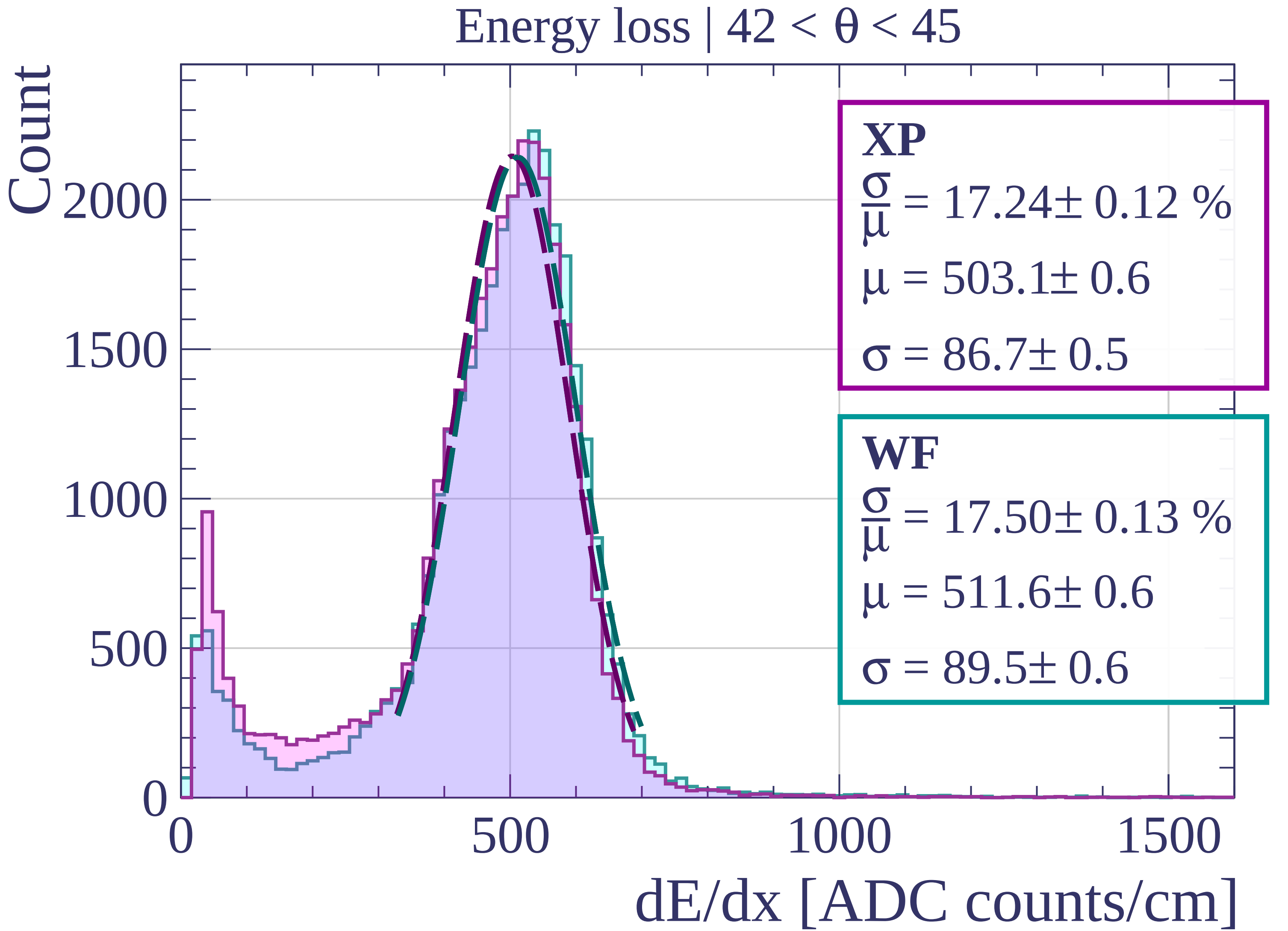


Energy loss | 33 < θ < 36
Count
4000
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 17.22± 0.09 %
μ = 495.7± 0.4
σ = 85.4± 0.4
WF
σ/μ = 17.32± 0.09 %
μ = 504.4± 0.4
σ = 87.3± 0.4

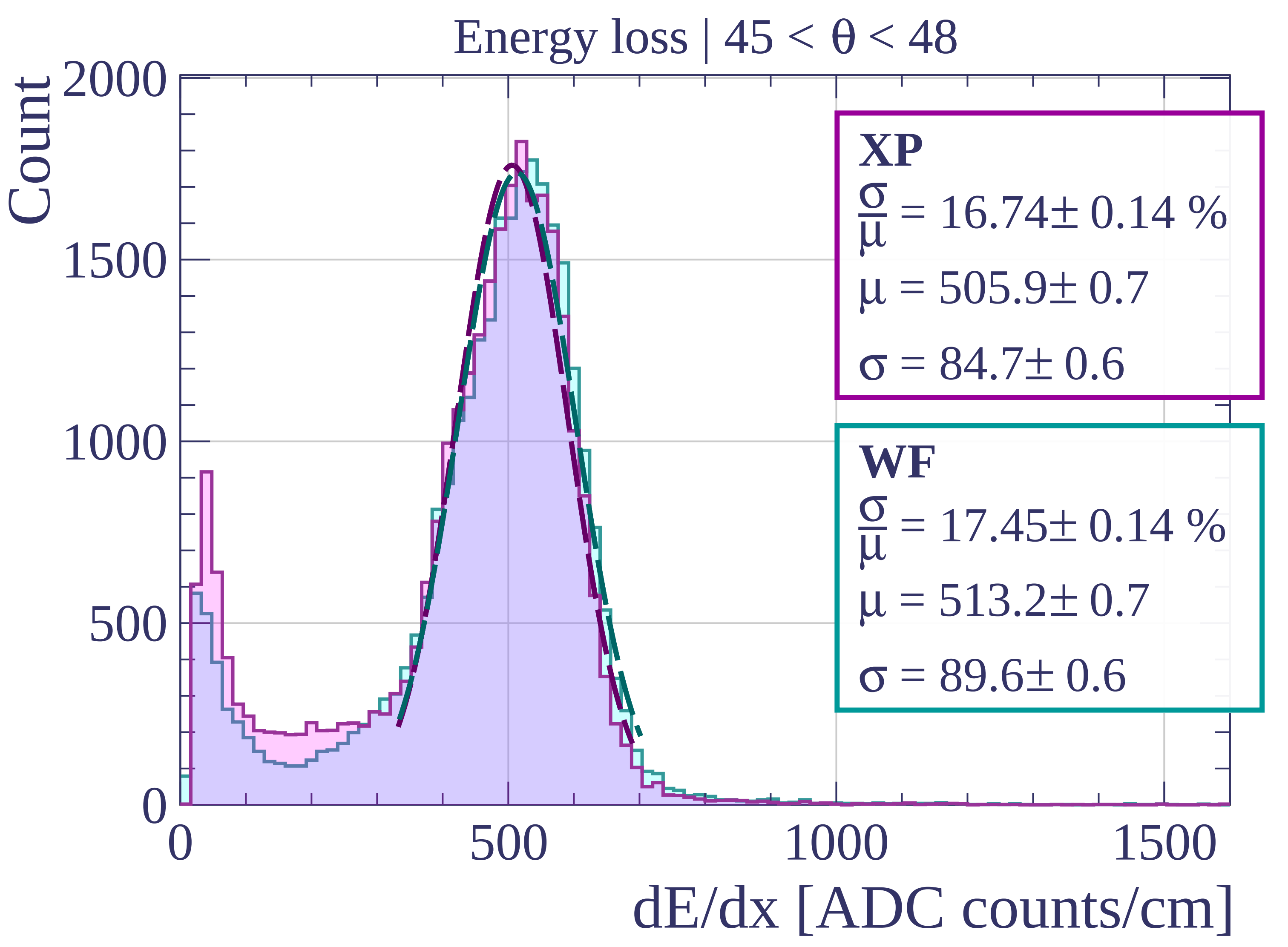

Energy loss | 36 < θ < 39
Count
3000
2000
1000
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 17.21± 0.09 %
μ = 498.3± 0.4
σ = 85.8± 0.4
WF
σ/μ = 17.54± 0.10 %
μ = 506.9± 0.5
σ = 88.9± 0.4



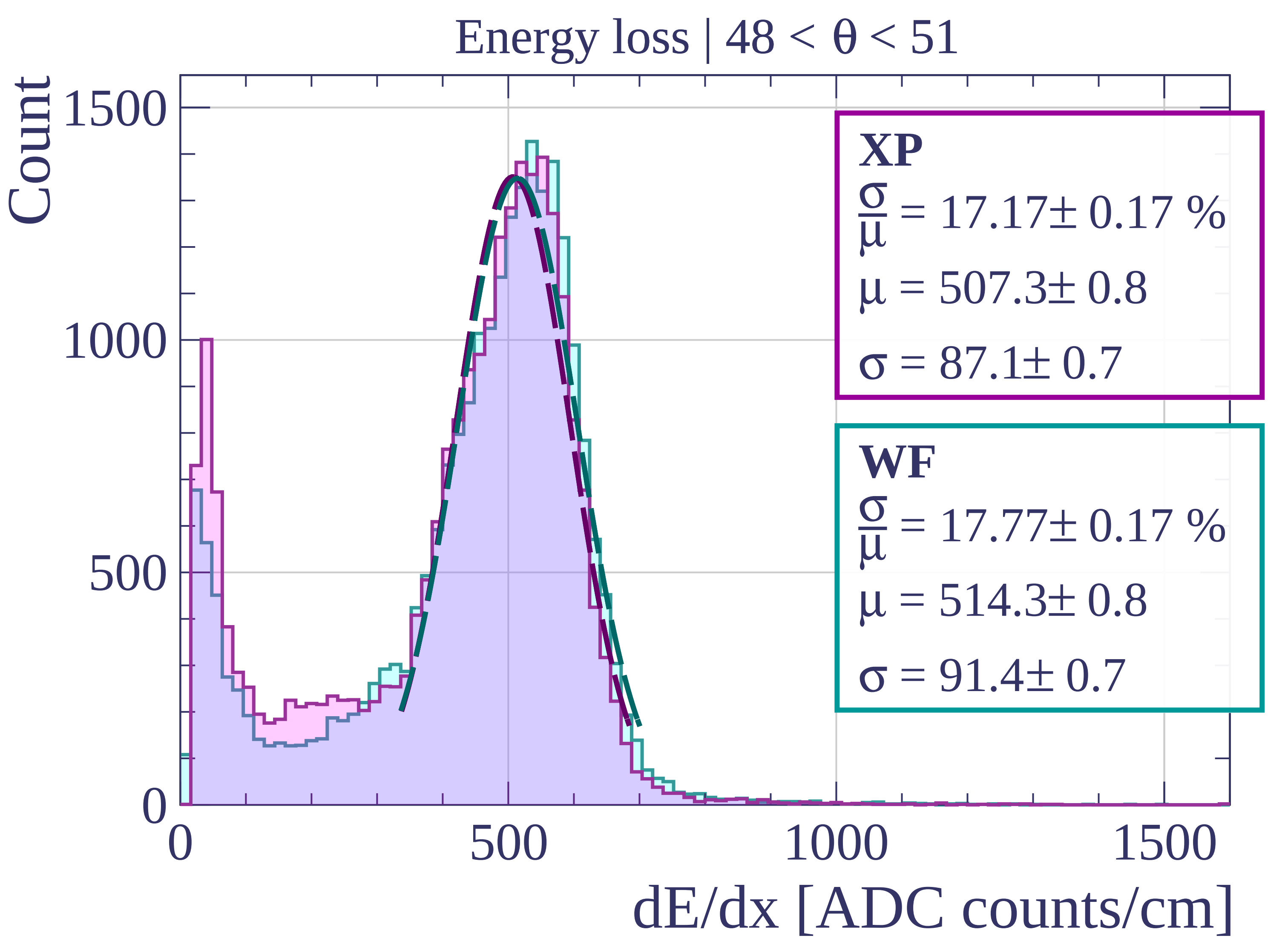
Energy loss | 39 < θ < 42
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
1000
2000
XP
σ/μ = 17.23± 0.11 %
μ = 499.3± 0.5
σ = 86.0± 0.4
WF
σ/μ = 17.60± 0.11 %
μ = 508.5± 0.5
σ = 89.5± 0.5

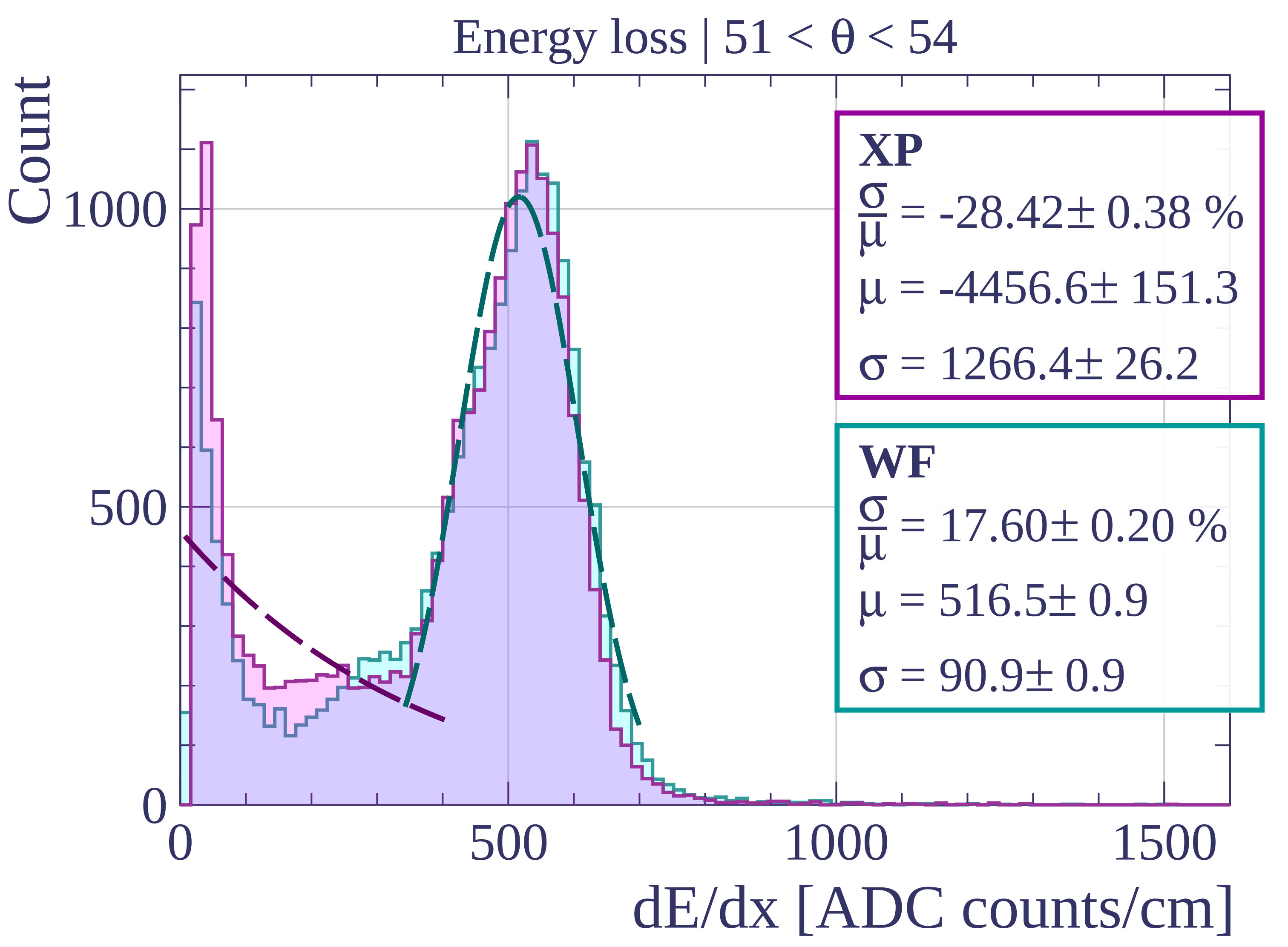


Energy loss | 42 < θ < 45
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
2000
0
500
1000
1500
XP
σ/μ = 17.24± 0.12 %
μ = 503.1± 0.6
σ = 86.7± 0.5
WF
σ/μ = 17.50± 0.13 %
μ = 511.6± 0.6
σ = 89.5± 0.6

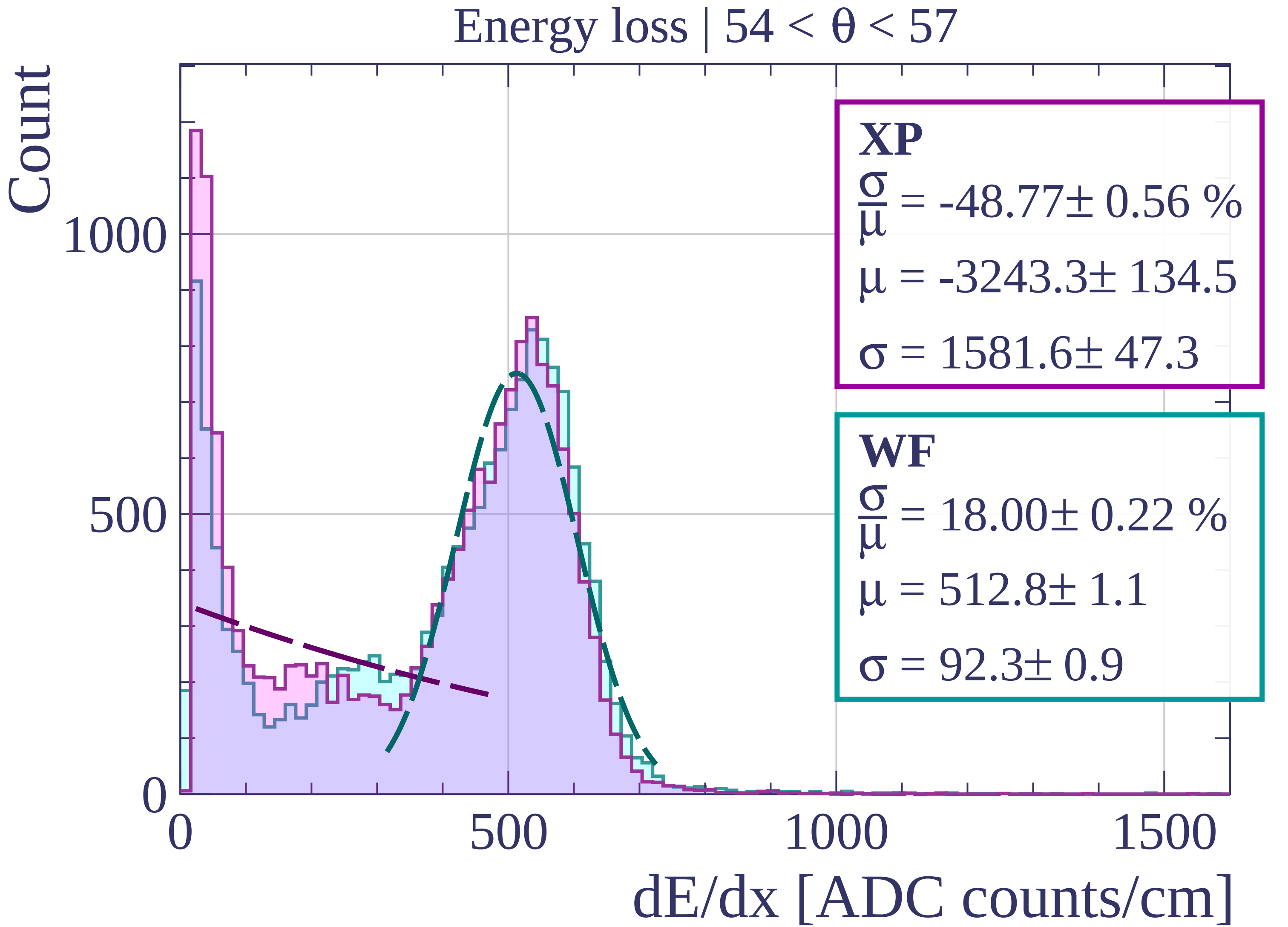


Energy loss | 45 < θ < 48
Count
dE/dx [ADC counts/cm]
XP
σ/μ = 16.74± 0.14 %
μ = 505.9± 0.7
σ = 84.7± 0.6
WF
σ/μ = 17.45± 0.14 %
μ = 513.2± 0.7
σ = 89.6± 0.6
0
500
1000
1500
2000
0
500
1000
1500

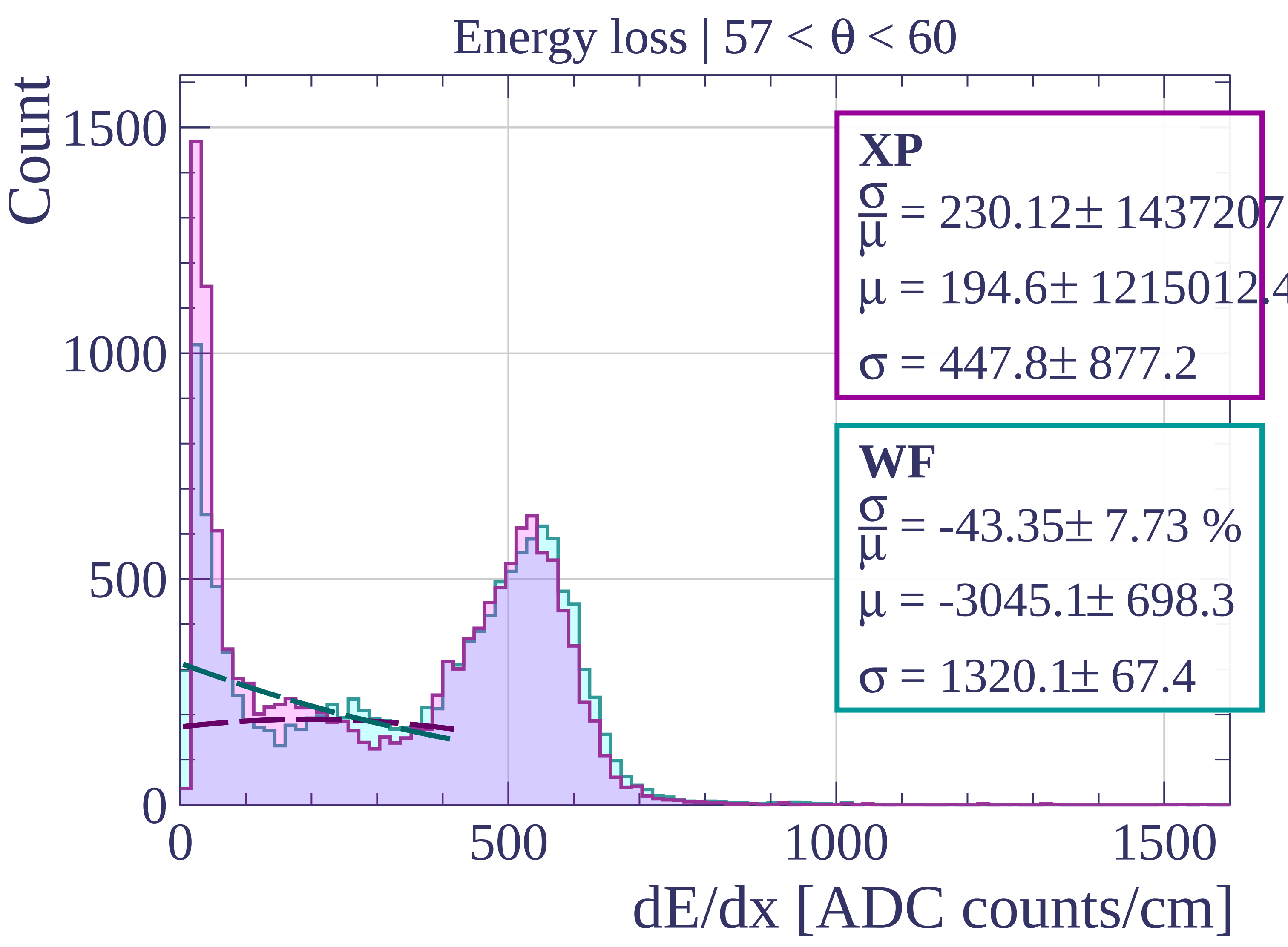

Energy loss | 48 < θ < 51
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 17.17± 0.17 %
μ = 507.3± 0.8
σ = 87.1± 0.7
WF
σ/μ = 17.77± 0.17 %
μ = 514.3± 0.8
σ = 91.4± 0.7



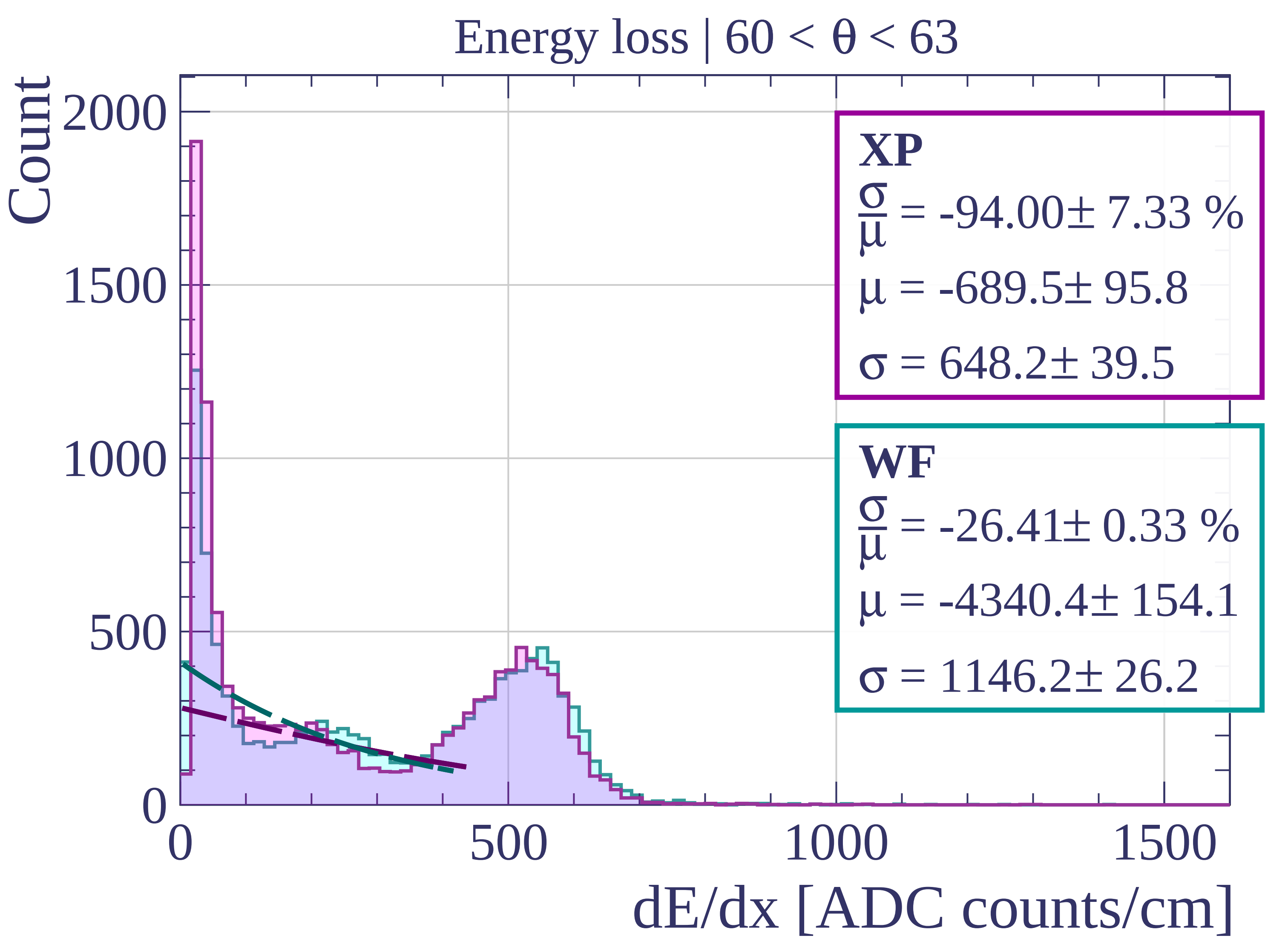
Energy loss | 51 < θ < 54
Count
1000
500
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = -28.42± 0.38 %
μ = -4456.6± 151.3
σ = 1266.4± 26.2
WF
σ/μ = 17.60± 0.20 %
μ = 516.5± 0.9
σ = 90.9± 0.9



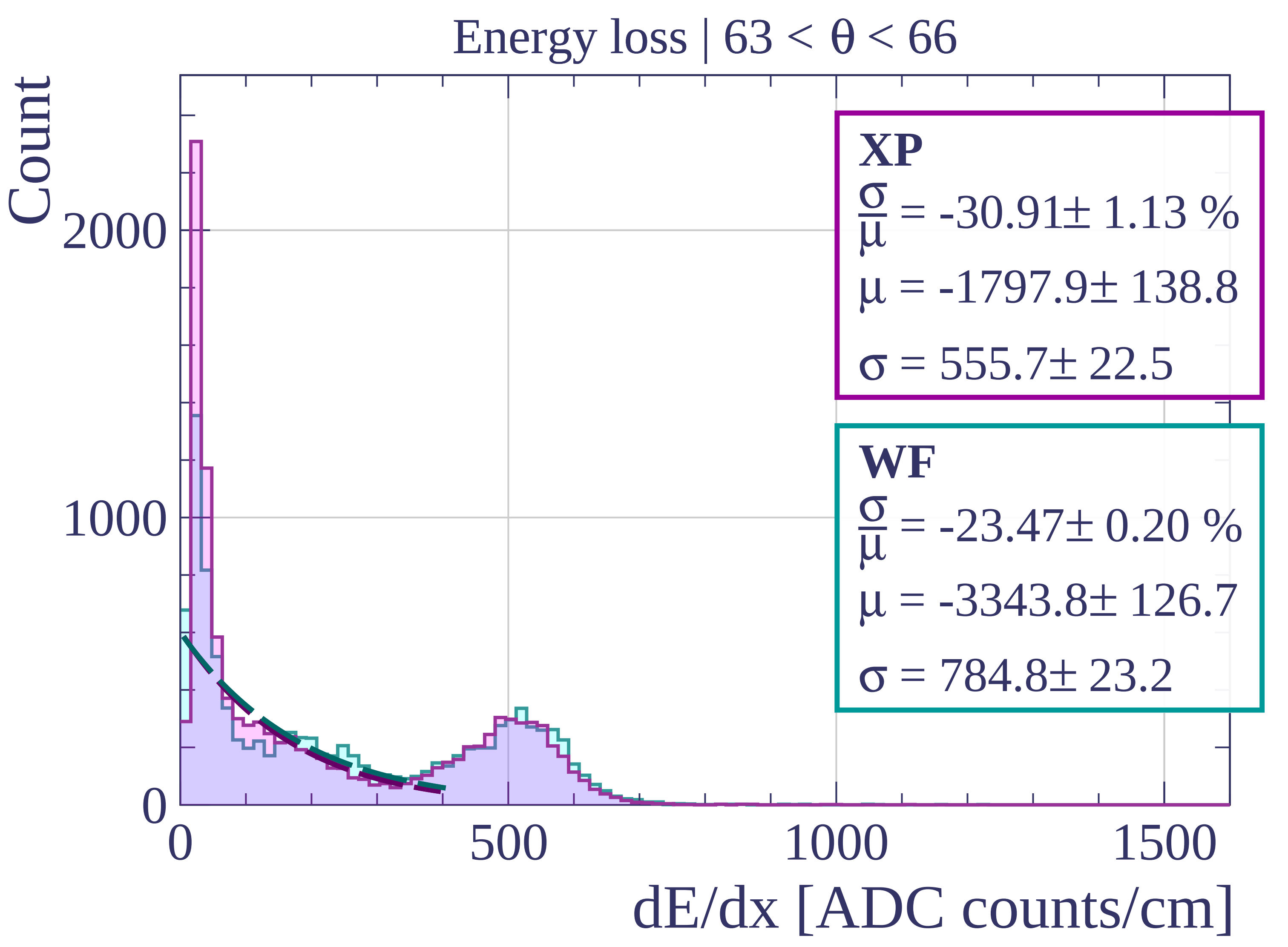
Energy loss | 54 < θ < 57
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = -48.77± 0.56 %
μ = -3243.3± 134.5
σ = 1581.6± 47.3
WF
σ/μ = 18.00± 0.22 %
μ = 512.8± 1.1
σ = 92.3± 0.9



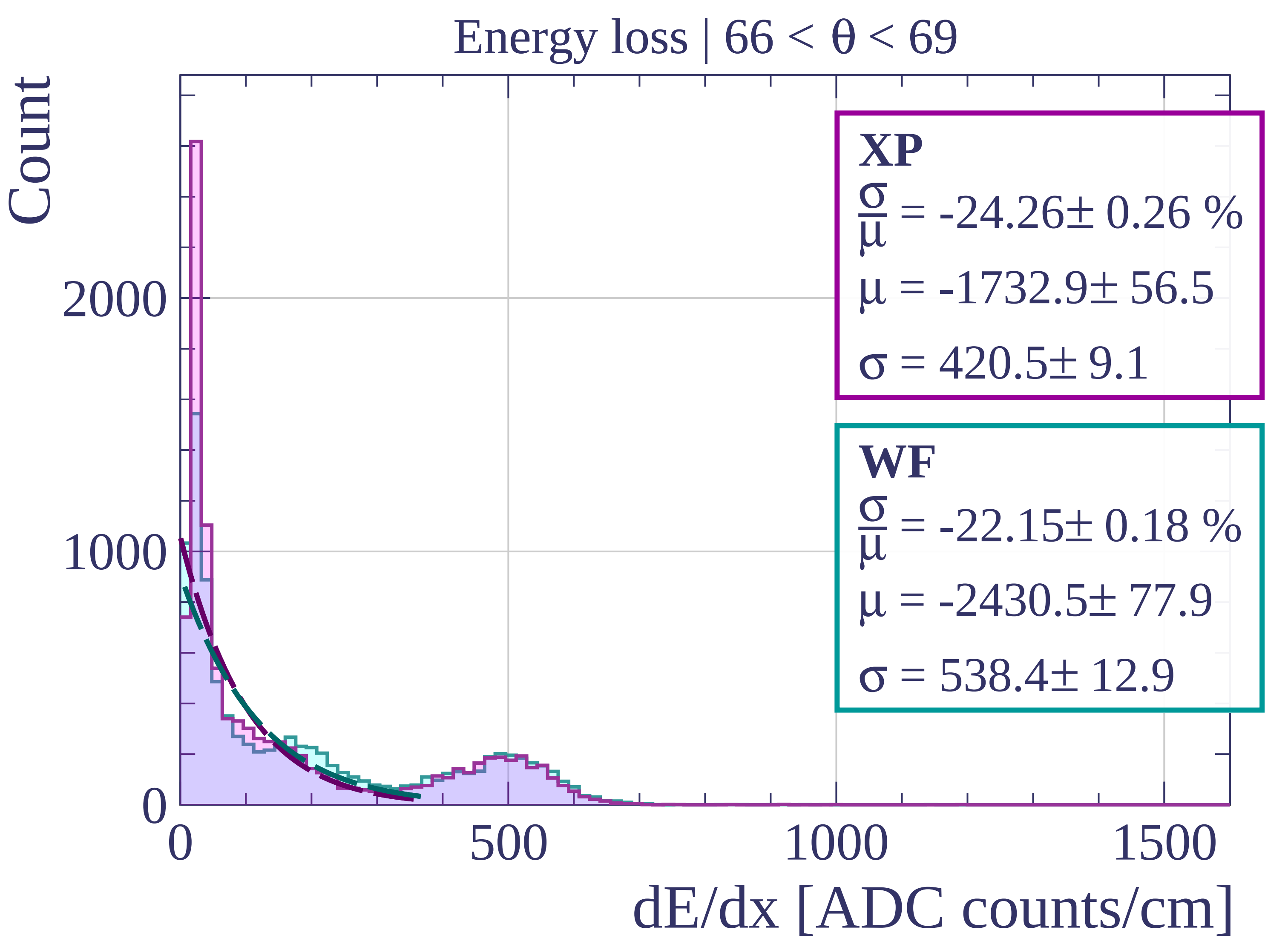
Energy loss | 57 < θ < 60
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
σ/μ = 230.12± 1437207
μ = 194.6± 1215012.4
σ = 447.8± 877.2
WF
σ/μ = -43.35± 7.73 %
μ = -3045.1± 698.3
σ = 1320.1± 67.4



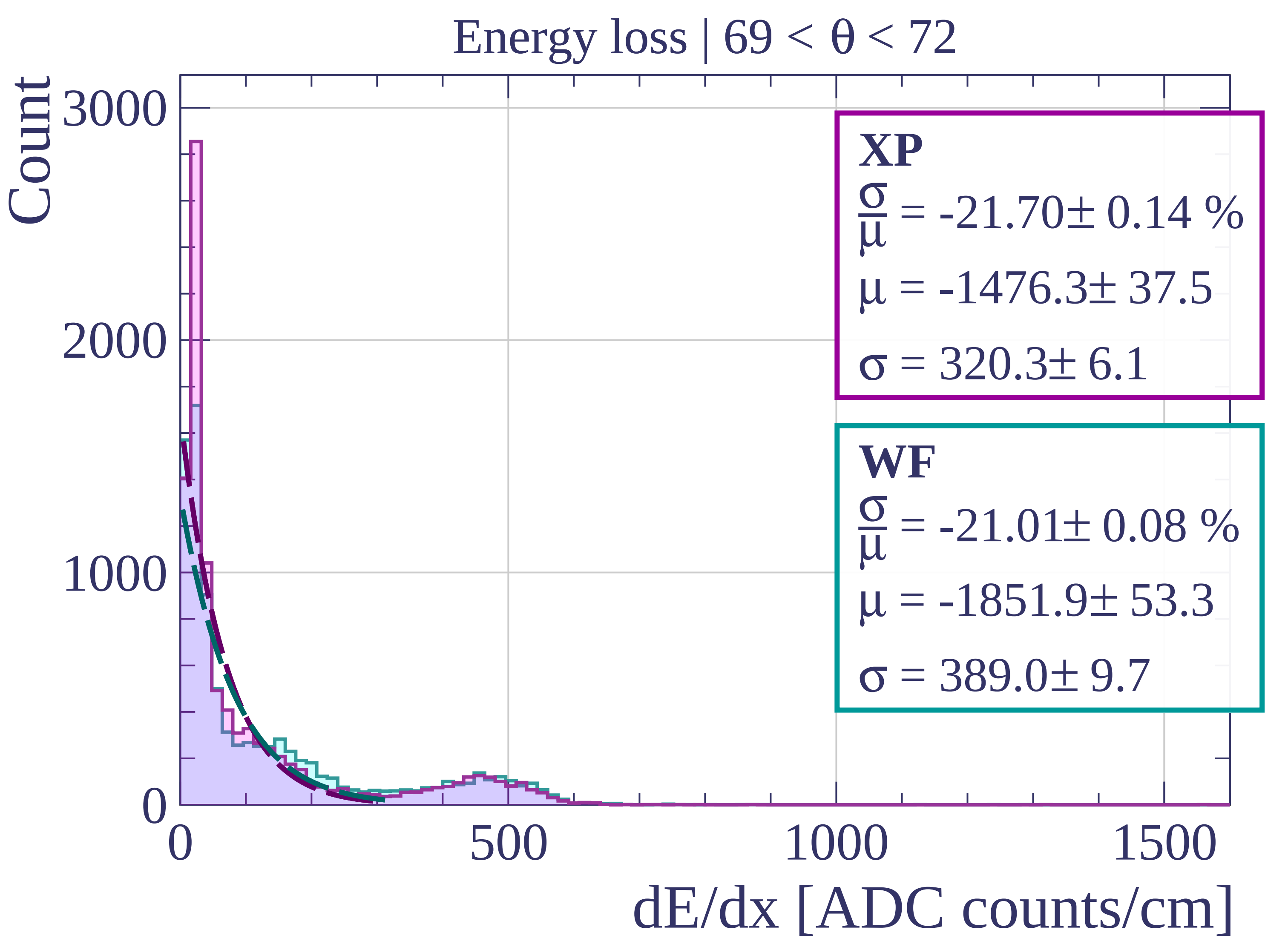
Energy loss | 60 < θ < 63
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = -94.00± 7.33 %
μ = -689.5± 95.8
σ = 648.2± 39.5
WF
σ/μ = -26.41± 0.33 %
μ = -4340.4± 154.1
σ = 1146.2± 26.2



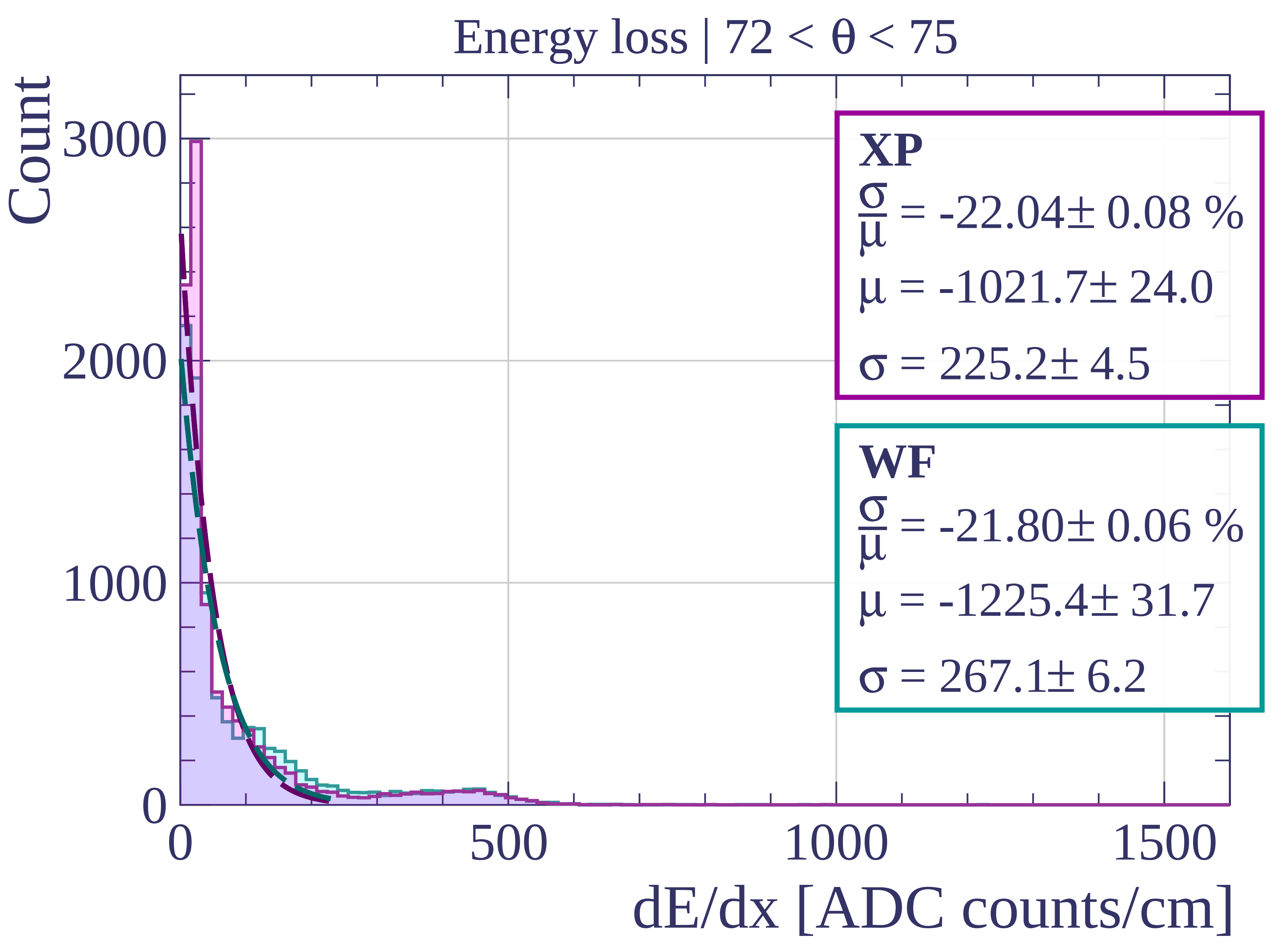
Energy loss | 63 < θ < 66
Count
dE/dx [ADC counts/cm]
0
1000
2000
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = -30.91± 1.13 %
μ = -1797.9± 138.8
σ = 555.7± 22.5
WF
$\frac{\sigma}{\mu}$ = -23.47± 0.20 %
μ = -3343.8± 126.7
σ = 784.8± 23.2

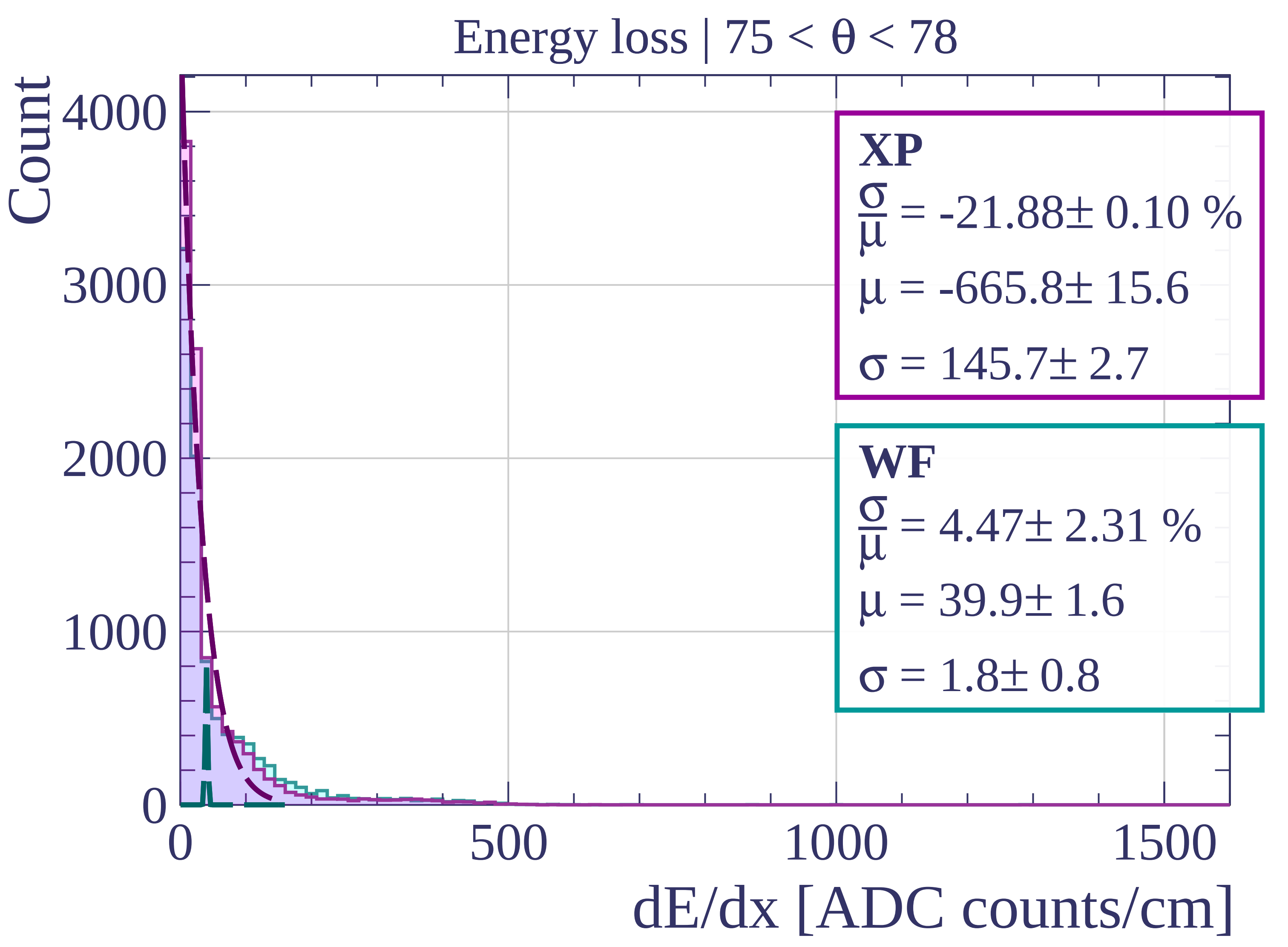


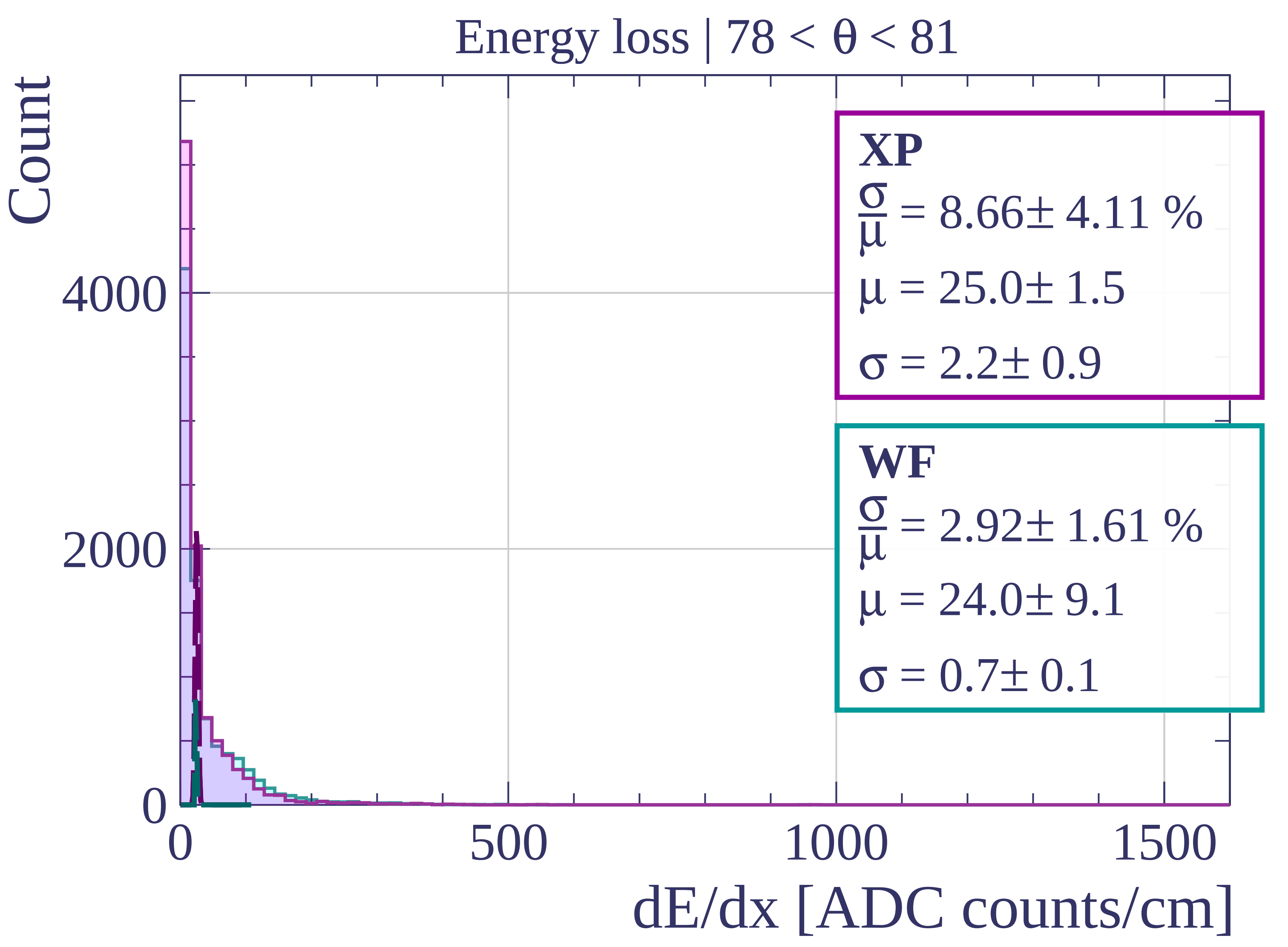
Energy loss | 66 < θ < 69
Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = -24.26± 0.26 %
μ = -1732.9± 56.5
σ = 420.5± 9.1
WF
σ/μ = -22.15± 0.18 %
μ = -2430.5± 77.9
σ = 538.4± 12.9



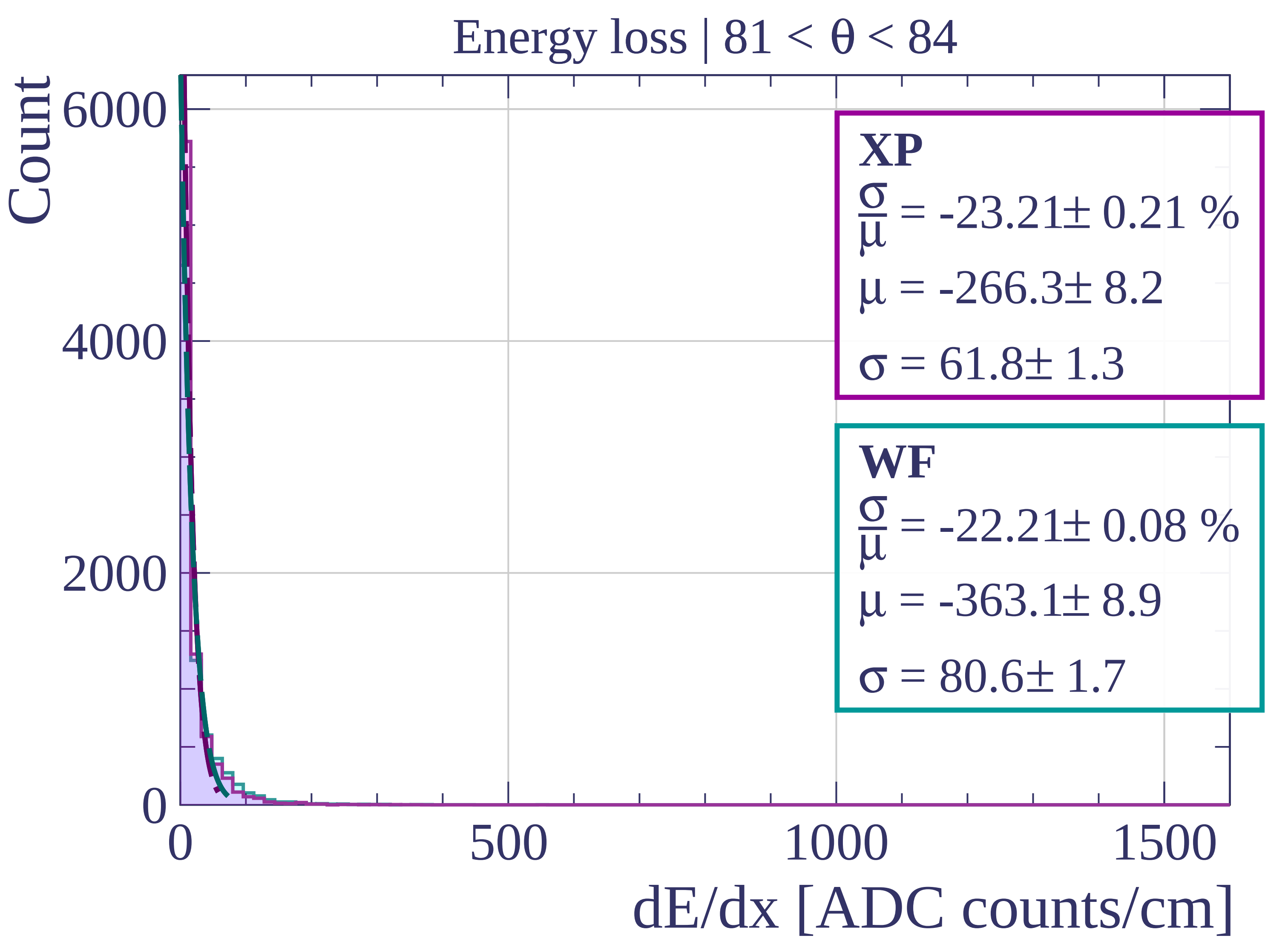
Energy loss | 69 < θ < 72
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = -21.70± 0.14 %
μ = -1476.3± 37.5
σ = 320.3± 6.1
WF
σ/μ = -21.01± 0.08 %
μ = -1851.9± 53.3
σ = 389.0± 9.7


Energy loss | 72 < θ < 75
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = -22.04± 0.08 %
μ = -1021.7± 24.0
σ = 225.2± 4.5
WF
σ/μ = -21.80± 0.06 %
μ = -1225.4± 31.7
σ = 267.1± 6.2

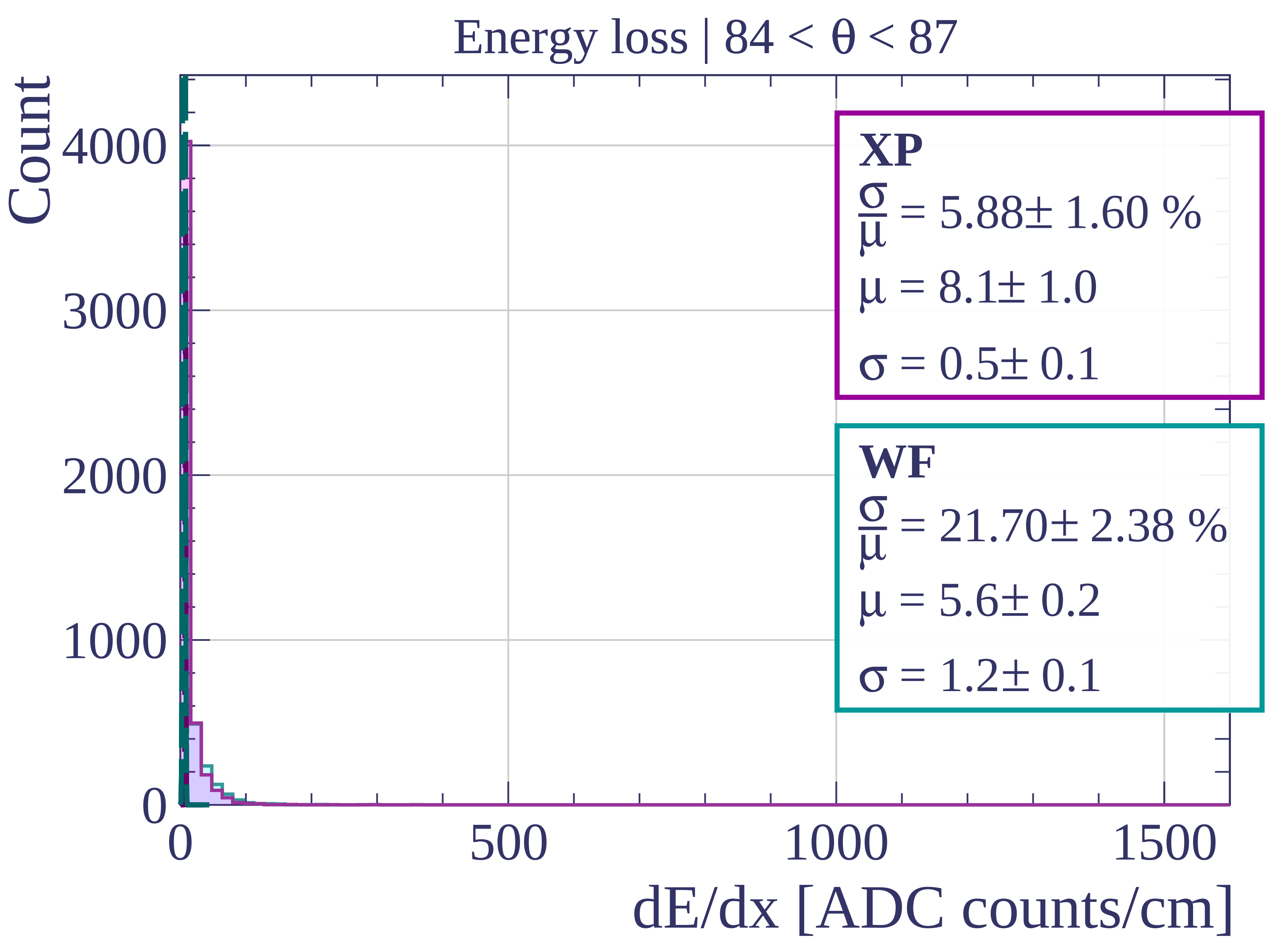

Energy loss | 75 < θ < 78
Count
4000
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = -21.88± 0.10 %
μ = -665.8± 15.6
σ = 145.7± 2.7
WF
σ/μ = 4.47± 2.31 %
μ = 39.9± 1.6
σ = 1.8± 0.8



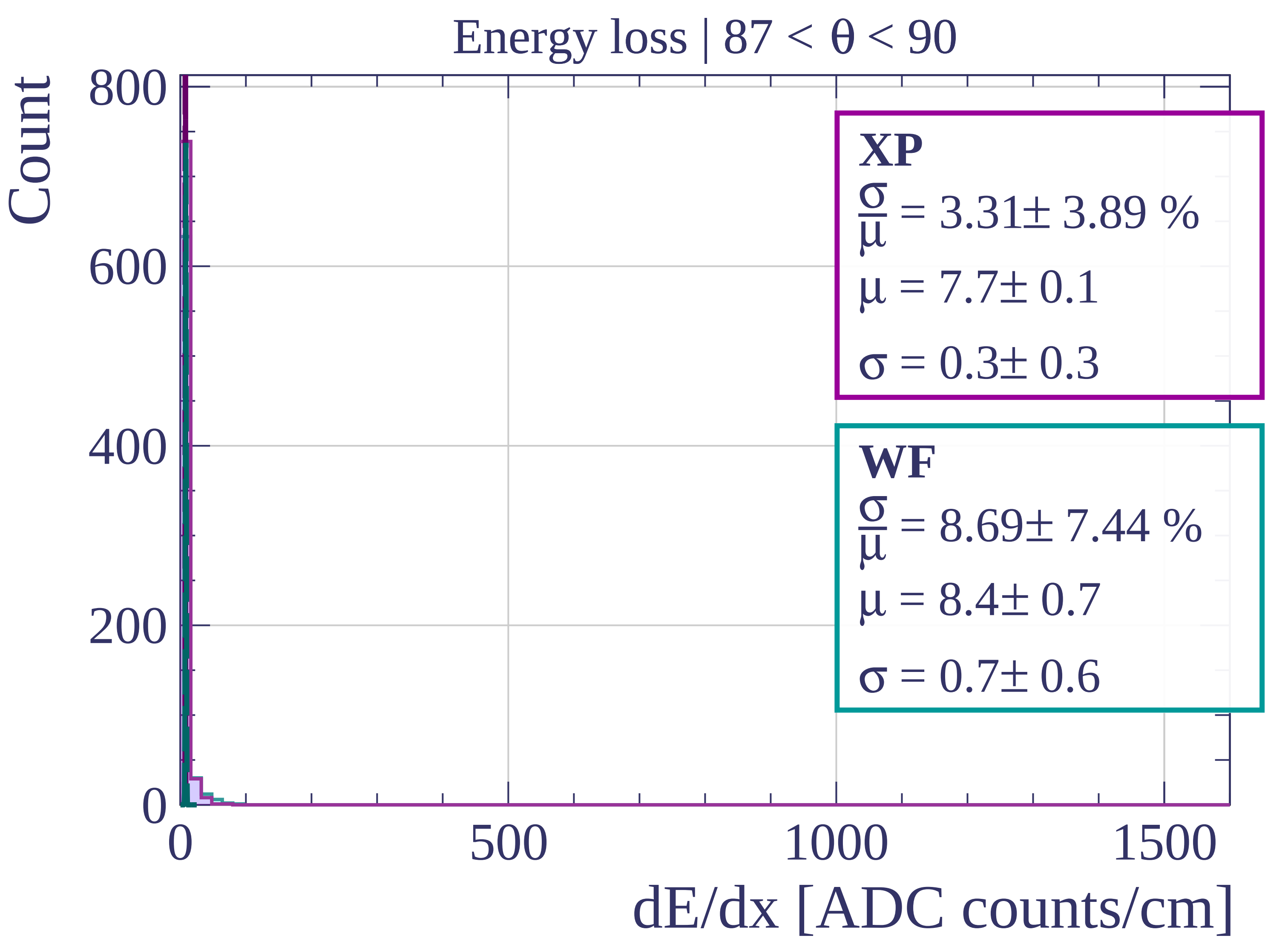
Energy loss | 78 < θ < 81
Count
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 8.66± 4.11 %
μ = 25.0± 1.5
σ = 2.2± 0.9
WF
$\frac{\sigma}{\mu}$ = 2.92± 1.61 %
μ = 24.0± 9.1
σ = 0.7± 0.1



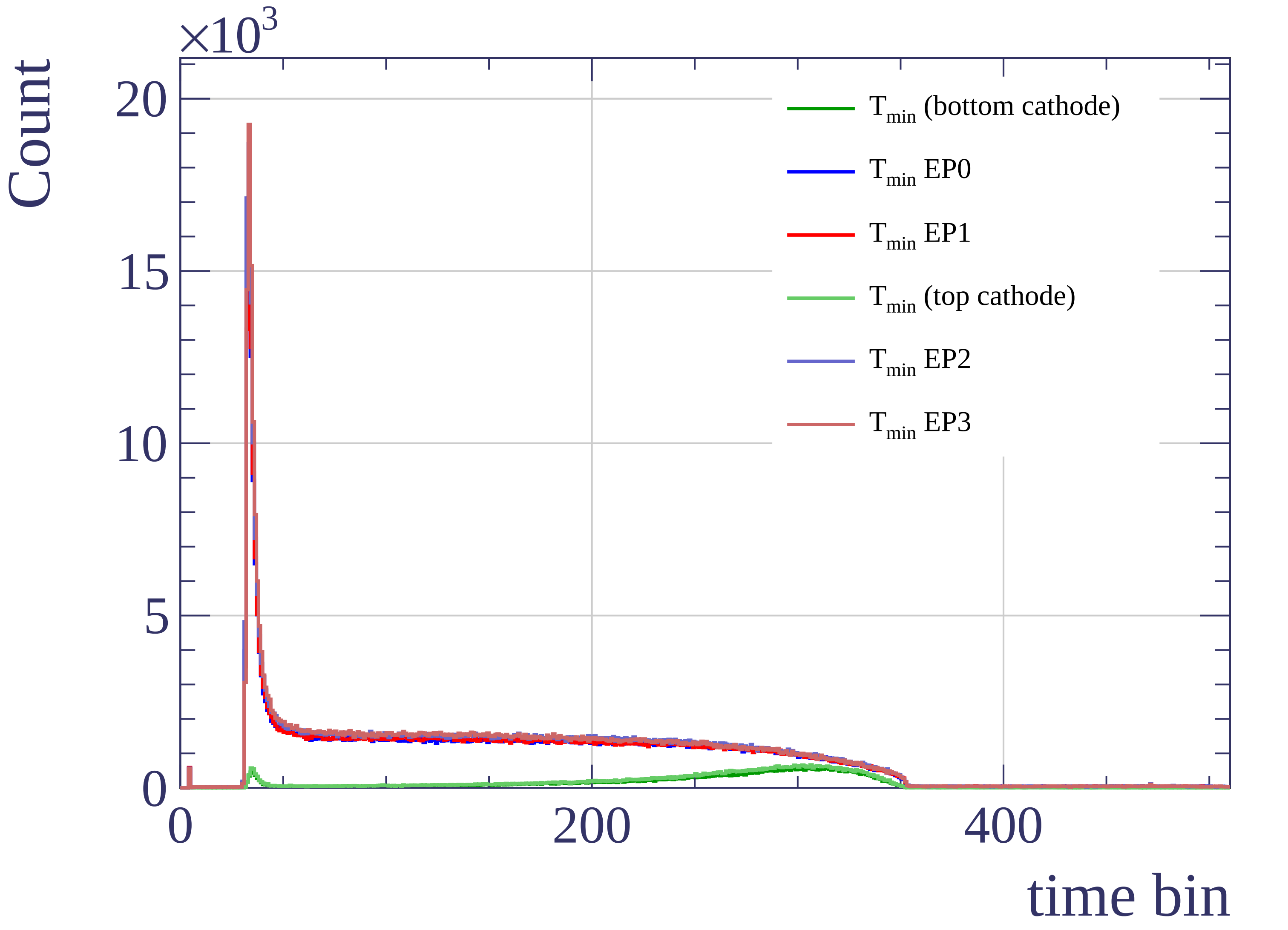
Energy loss | 81 < θ < 84
Count
6000
4000
2000
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = -23.21± 0.21 %
μ = -266.3± 8.2
σ = 61.8± 1.3
WF
σ/μ = -22.21± 0.08 %
μ = -363.1± 8.9
σ = 80.6± 1.7



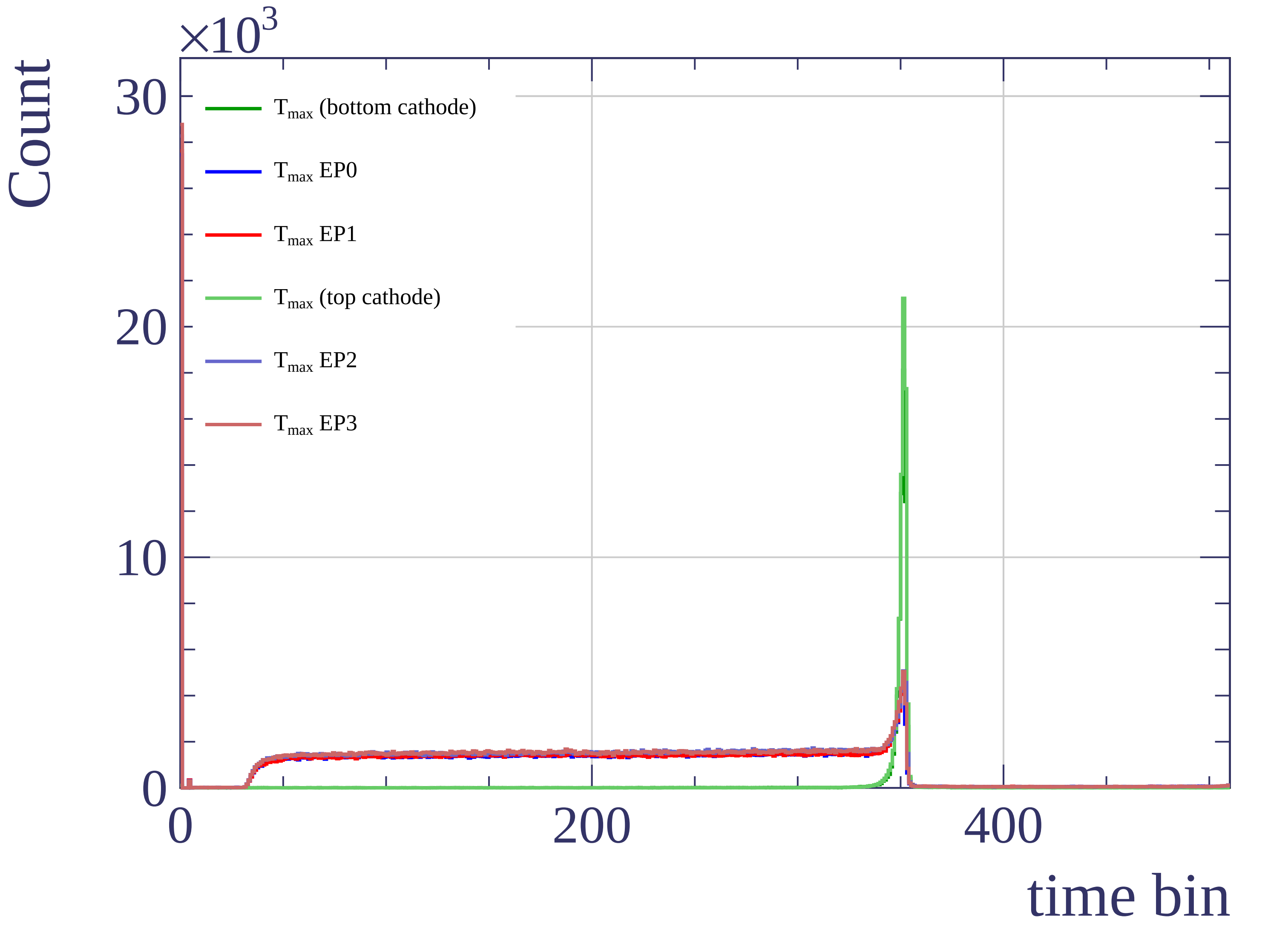
Energy loss | 84 < θ < 87
Count
4000
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 5.88± 1.60 %
μ = 8.1± 1.0
σ = 0.5± 0.1
WF
$\frac{\sigma}{\mu}$ = 21.70± 2.38 %
μ = 5.6± 0.2
σ = 1.2± 0.1



Energy loss | 87 < θ < 90
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 3.31± 3.89 %
μ = 7.7± 0.1
σ = 0.3± 0.3
WF
σ/μ = 8.69± 7.44 %
μ = 8.4± 0.7
σ = 0.7± 0.6



×10^3
Count
20
15
10
5
0
0
200
400
time bin
T_min (bottom cathode)
T_min EP0
T_min EP1
T_min (top cathode)
T_min EP2
T_min EP3



$\times 10^3$
Count
30
20
10
0
0
200
400
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3